MASTER PACKET · GFIBER V8 TRANSFORMED TO AMAZON LEADERSHIP PRINCIPLES · STAR FORMAT · MATH, SOLUTION DESIGN AND PROBE QUESTIONS INTACT

Amazon — Supply Chain Manager III (L6) — the MASTER packet

You. Rajesh Rishiyur Chandrasekaran

What this document is. The GFiber MASTER packet v8 (Sep 13, 2026 — 255 pages) transformed for the Amazon L6 SCM loop: the same eighteen stories in the same uniform shape — **60-second pitch → SITUATION / TASK (the full problem space: what and why, when, scope, scale, and the metrics to improve with their baselines) → ACTION (the solution space: five gated phases, each with entry · delivered · exit bar, plus how it was done and who was influenced, in prose) → RESULTS (the Situation metrics, improved: $ · # · % · time · customer · quality, tied back to the stated baselines) → one PROBING section per story (challenge Q&A, the phase deep-probe bank, the business-impact model with KPI tree and numbers table, the fully worked derivation tables — the problem in numbers, the fix with every value substituted, the metric before → after, and the dollars with every multiplication shown — the solution flow (the system & data map), sensitivity (low/base/high), the spoken total-impact statement and the full "if challenged" answers)** — mapped to Amazon's sixteen Leadership Principles instead of a JD. §3 converts the twelve system-design prompts into design engagements told from the Elementum years, each keeping its stocks-and-flows map, first-principles invariants, solution design, system map, math and deep-dive probes. §4 is the numbers ledger — one canonical value per metric — and the never-say list. **This document is self-contained: nothing requires going back to the source packet.**

Relation to the loop packet. This is the depth book; *amazon-l6-scm-loop-packet.pdf* is the loop-mechanics book (interviewer tracker, AIQB 128-question bank, writing-exercise defense, video execution). The crosswalk — including the places where the two documents currently disagree — is in §1.3. Resolve those conflicts before the loop; never carry both versions of a number into the same building.

Voice, held everywhere. TPM / SAP integration program lead. "I" throughout. SAP teams and configurators built to the signed spec; I owned the data contract, the definitions, the gates, the signatures and the calls. Never claim to configure or deploy SAP. The only platform personally configured: Manhattan DOM. The MIT Supply Chain MicroMasters + Supply Chain Network Design certificate is an acquired toolkit, labeled wherever it appears.

Basis labels, used on every number. *record* — in the program record, say it flat · *user anchor* — a figure a named stakeholder stated, attributed · *industry benchmark* — a published rate, named · *modeled* — computed from stated inputs; say "modeled" in the same breath, never promote it to record.

---

0. How to use this packet in the loop

1. **Reading order = prep order:** §2 is arranged as employer blocks in prep priority — Sephora (current role) first, then Flexport, then Elementum (with the Walmart-client engagements S19–S21 and the twelve design engagements), Manhattan last. Prep front-to-back: the front blocks carry the 3-hour loop day (Thursday), the back blocks round out the 2-hour day (Friday). Story codes are unchanged, so every cross-reference still points where it always did.
2. **Night before:** §1 LP map and roster → §4 never-say list → skim each story's pitch line.
3. **In the room, unprompted:** the 60-SECOND PITCH only. It ends on "Primary metric — X, before → after, $ impact." Stop there. The interviewer's probes are how they fill their STAR worksheet; silence after the pitch works for you.
4. **"Walk me through it":** STAR. The Action is five gated phases — Discovery → Plan & Design → Build & Test → Cutover/Go-Live → Hypercare — each spoken as entry bar · what I delivered · exit bar.

5. **"How did you actually run it":** the PHASE DEEP-PROBE BANK — Q&A in each story's PROBING section. Answer the phase asked, then stop.
6. **"How would you design…":** §3. Open with the stocks-and-flows map, state the invariants, then the design, then the metric. Program methodology lives only in §3's shared section — never inside an employer story.
7. **Numbers:** §4. A record dollar is said flat. Where the record holds no dollar, say the operational metric and then the modeled figure with the word "modeled" in the same breath — each story's *Business impact model* carries the inputs, the math and the low / base / high range. Never promote a modeled number to record.
8. **Amazon loop mechanics:** five 55-minute video interviews, each interviewer assigned 2–3 Leadership Principles, STAR worksheet, notes compared afterward. One question, one story — never the same story twice across the loop. The LP map below is the routing table.

1. The Amazon map — Leadership Principles → stories, the roster, and the crosswalk

1.1 Leadership Principle → story map

Primary = the opener when that LP's generic question lands. Backup/third = same LP, different interviewer. E-numbers are the §3 design engagements — use them when the question is "design…" or "how would you approach…", not for behavioral asks. Four stories open two LPs each (S10, S12, S13, S15): if both of a pair's LPs land in one loop, the second occurrence takes that LP's backup — never the same story twice.

LEADERSHIP PRINCIPLE	PRIMARY	BACKUP	THIRD
Customer Obsession	**S13** Cyber Monday — promise-at-risk allocation	S4 the promise rule fixed for customers	S11 graduation criteria per wave
Ownership	**S16** bus factor 1 → pipeline; owned the security miss	S1 took the definition nobody owned	S14 the seam between OMS and WMS
Invent and Simplify	**S17** lane master beats a solver	S16 four systems → one pipeline	S3 de-aggregation design; any §3 E-design
Are Right, A Lot	**S10** "noise said carriers; the data said measurement"	S8 measured lanes vs brochure	S4 wrong rule, right correction
Learn and Be Curious	**S7** learned CM data by math, not meetings	S16 learned the stack to build the pipeline	MIT toolkit (labeled), E7 forecasting
Hire and Develop the Best	**S15** 20-person team to a 4-week cadence; QA elevated	S11 wave teams grown into owners	—
Insist on the Highest Standards	**S2** robotics held at 94 vs 98 gate	S18 0.5% tolerance published before mock one	S9 ≥95% valid-ASN cohort gate held
Think Big	**S12** flipped the 18-month plan; ~$5M avoided	S9 1,000-lane vision, staged honestly	S1 book-of-record migration
Bias for Action	**S13** 4 AM allocation call	S18 rehearsed fallback fired, date held	S14 two days to the executive review
Frugality	**S12** ~$5M avoided by sequencing	S17 the bid — analyst weeks returned	S16 >$100K/yr priced before build
Earn Trust	**S14** $2.5M discrepancy — transparency over a quiet fix	S1 one signed "available", S3 Finance+Ops on one number	S5 the council; S6 two-signal release
Dive Deep	**S8** 200 lanes ranked; promise at p99	S10 Austria — measured the measurement	S3 aggregates de-aggregated; S7 phantom math
Have Backbone; Disagree & Commit	**S9** 597 vs 1,000, two levels up	S14 refused the quiet fix	S5 council without owning a system; S2 throttle
Deliver Results	**S11** 7 → 68 behind graduation gates	S18 8 months, both hubs, gate cleared	S2 the comingle launch; S6 3 months to live
Strive to be Earth's Best Employer	**S15** QA moved from afterthought to gatekeeper	S2 the floor protected during the ramp	—

LEADERSHIP PRINCIPLE	PRIMARY	BACKUP	THIRD
Success & Scale Bring Broad Responsibility	**S10** one site's fix rolled to 25 sites / ~$8M	S1 oversell is a customer harm, not a ledger error	S9 small carriers kept inside the standard

1.2 Story roster (speak in this order — employer blocks, prep priority; S19–S21 are the Walmart-client Elementum engagements)

#	TITLE	PRIMARY METRIC (BEFORE → AFTER)	$ IMPACT
S1	Sephora S/4HANA ↔ ISM — one signed "available"	Oversell incidents recurring → 0	≈$2.5M/yr: $1M MCS run cost (user anchor) + ≈$1.5M modeled
S2	Sephora UDC — one comingle pool + robotics gate	Processing efficiency → +15%	≈$1.6M/yr modeled (record: none)
S18	Sephora — RP → JDA WMS dual-DC migration	Mocks 7% → 1.8% → 0.3% vs 0.5% tolerance	$20M pipeline shipped on time; ~3% under budget
S3	Sephora SNB de-aggregation — one Financial On-Hand	FOH truths 2 → 1	≈$0.6M/yr modeled (record: none)
S4	Sephora in-transit promise rule (the failure story)	Starved lanes → restored; oversell +0	≈$0.5M/yr modeled (record: none)
S5	Sephora architecture council / signed FSD	Surprise escalations → 0	≈$0.5M one-time modeled (record: none)
S15	Sephora — 20-person team to a 4-week cadence	Critical defects −50%	≈$0.13M/yr modeled
S11	Flexport — Shipment Visibility launch (the wave machine)	CTS $0.66 → $0.03 (−95.5%) across 7 → 68 customers	CTS record; ~$1.5M pre-prod separate
S12	Flexport — FF App export launch (the roadmap flip)	Effort: 18-month plan → 66% less; value month 6	~$5M cost avoidance (record, said once)
S16	Flexport — four systems into one Snowflake pipeline	Accuracy 60 → 90%; parity within 1%	>$100K/yr priced (record of the pricing)
S8	Elementum — p99 lane promise	Lead-time accuracy → 99% on 200 lanes	580K units / ~$1.5M released (record)
S6	Elementum — real-time supply/demand view	Forecast accuracy 70 → 85%	$2M safety stock released (record)
S9	Elementum — 597-lane EDI Pareto	Completeness 60 → 95%	Freight −10% ≈ $2M (record)
S7	Elementum — contract-manufacturer BOM phantoms / revisions	Sites with planning-grade BOM 0 → 6	≈$0.3M/yr modeled; the $1.5M stays with S8
S10	Elementum — Austria routing guide DQ/OTD	Data quality ~55 → ~99%	$500K/yr Austria → ~$8M across 25 sites (record roll-up; ~$320K/site avg)
S17	Elementum — the semi-annual carrier bid	Cycle 6 weeks → ~3 (~50%)	≈$0.19M/yr modeled analyst hours; never award savings

#	TITLE	PRIMARY METRIC (BEFORE → AFTER)	$ IMPACT
S19	Elementum · Walmart Marketplace client — σ-proportional delivery-promise buffers (one governed dial)	Fast-promise share 7.6 → 39.6% (+32 pts) at the held 95% floor (engagement model)	+$255K/wk ≈ $13.3M/yr modeled on the illustrative book; no record dollar — say "the model"
S20	Elementum · Walmart Marketplace client — the pre-purchase WBR and the promise decomposition	Red-week attribution: a blame negotiation → minutes (O1 ≈ I3 × I4 × Φ(b/σ)); detection-to-action 30 days → <24 h (designed Andon)	≈$0.58M per caught episode modeled; Promise Yield 41.5% baseline (doc sample); no record dollar
S21	Elementum · Walmart Marketplace client — SSS adoption rebuilt as a priced insurance contract	SSS share of seller-fulfilled GMV ~38–42% stalled baseline (record-from-docs); contract + wave gates designed, wave-one pilot	≈$3.7M/yr per +10 pts of enrolled GMV, modeled — designed, not banked; no record dollar
S13	Manhattan — Cyber Monday 4 AM allocation	99% fill vs 95% goal	~$6M at risk → ~$8M protected (record)
S14	Manhattan — the $2.5M OMS↔WMS discrepancy	~200 orders reconciled before the review	$2.5M record; $10M initiative won (not claimed as savings)

1.3 Crosswalk to the loop packet — and the conflicts to resolve before the loop

The loop packet (amazon-l6-scm-loop-packet.pdf) tells many of the same programs under its own story codes. Safe mappings first; then the four places the two documents **currently disagree**. Pick one version of each before the loop — interviewers compare notes, and so do documents.

THIS PACKET	LOOP PACKET	STATUS
S1 one "available"	S2	Same program, consistent.
S2 UDC comingle	S11 + S24	Consistent (robots beat = loop S11; log and pacing = loop S24).
S6 supply/demand view	S5	Consistent (70→85%, $2M, two-signal).
S7 CM BOM	S6	Consistent (six sites, 3 waves, 5 months, snapshot-diffing).
S8 p99 lanes	S3 (+S4 math)	Consistent (18d brochure, p99, 580K/$1.5M).
S9 EDI Pareto	S7	Consistent (597/1,000, ≥95% gate, −10% ≈ $2M).
S13 Cyber Monday	S14	Consistent (99% fill, ~$8M).
S17 carrier bid	S8	Consistent (6→~3 wks, <5% gate, no award dollar).
S18 WMS migration	S15	Consistent (0.5%, 7→1.8→0.3, paper-pick fallback).
S3, S4, S5, S10, S14, S15	—	New here; not in the loop packet. Safe to use anywhere.
S11 wave machine	MAIN + S18 + S26	One program: the loop packet splits it into the WBR (MAIN), the pilot launch (its S18) and the lever map (its S26). Consistent apart from the two conflicts below.
S16 Snowflake pipeline	MAIN (BRD/pipeline beat)	Consistent — the security-review miss appears in both. Note: the loop packet's S16 is a *different* story (the Elementum UAT crash).

THIS PACKET	LOOP PACKET	STATUS
S11 automation baseline	MAIN	**CONFLICT.** This packet (GFiber v8 ledger): automation 40% → 90%. Loop packet: 60% → 97%. Same program, two baselines. Decide which is the record and correct the other document.
S12 roadmap flip employer	S12	**CONFLICT.** This packet: Flexport FF App export launch (operator-first vs partner-first). Loop packet: told as a Sephora roadmap. Same numbers (−66% effort, ~$5M, month 6 vs 18) under two employers — the most dangerous inconsistency in the set.
S11 WBR hand-run hours	MAIN	**CONFLICT.** This packet: 1.5–2 → ~4 hrs/wk by hand. Loop packet: WBR prep 30 → 6 h/wk ≈ $103,680/yr. Different claims about the same review's manual effort.
S1 retirement dollars	S2	**DIVERGENT MODEL.** This packet: ≈$2.5M/yr decomposed ($1M MCS run cost, user anchor + ≈$1.5M modeled). Loop packet: ≈$1M/yr support cost. The $1M is the shared core; say the decomposition only from this packet's model.
S18 migration budget	S15	**CONFLICT.** This packet (GFiber v8 ledger): ~$1.8M budget, ~3% under (~$54K — consistent arithmetic). Loop packet (from the SAP playbook): ~$800K budget, ≈$54K under. $54K is 3% of $1.8M, not of $800K — this packet's pair is internally consistent; fix the loop packet.
S18 slip exposure	S15	**CONFLICT.** This packet: $20M pipeline ≈ ~$2.5M/*month* of slip. Loop packet's math box: ~$2.5M/*week* ($20M ÷ 8 weeks). Monthly is the record's framing; fix the loop packet.

> **Rule:** inside this packet, every number follows the §4 ledger (carried from the GFiber v8 record, kept as-is per the transform instruction). Where §1.3 marks a conflict, do not speak either version until you have decided which document is right — then fix the other one. **Code collision warning:** the two documents number stories independently — "S18" here is the WMS migration, "S18" in the loop packet is the pilot launch; always say which packet's code you mean.

Tell me about yourself · Why Amazon — the two scripted openers

Adapted from the loop packet's §2 into this packet's voice and numbers. Where the two documents disagree (§1.3), every figure below is **this packet's §4 value** — the loop packet's copy of these scripts should be corrected to match, never the reverse. The beats map to this packet's codes: Sephora → S1/S18, Flexport → S11, Elementum → S8/S6, Manhattan → S13 — so any follow-up probe has a full story behind it. Say the long TMAY once, early in the loop; interviewers two through five get the 30-second version.

TELL ME ABOUT YOURSELF — NEWEST TO OLDEST, ABOUT 75 SECONDS

"I've spent fifteen years on one problem — making sure inventory is where the demand is and that the promise made on it holds. Four companies, most recent first.

Sephora, today. We moved the inventory book of record onto S/4HANA, and Store, Retail DC and Dotcom each meant something different by 'available' — most oversell bugs are vocabulary bugs. I owned the definitions and got one signed availability number per node per channel, reconciled daily against the book; customer-facing oversell incidents went from recurring to zero through two go-lives. I also ran the dual-DC WMS migration that unblocked a $20M project pipeline — eight months, on time, against a 0.5% reconciliation tolerance published before the first mock.

Flexport, before that. Leadership asked for a weekly review of shipment tracking and nobody owned it. I built it, ran it with the VP every week for a year, then automated the data under it. Cost to serve went from sixty-six cents a container to three — in Finance's loaded rates, not mine — while the base grew from seven customers to sixty-eight behind published graduation gates.

Elementum. Supply-chain software for manufacturers. I replaced carrier-brochure lead times with measured lane distributions on two hundred shipping lanes — planning to the slow day, not the average — which held 99% lead-time accuracy and released about a million and a half dollars of safety stock. I also drove forecast accuracy on the scoped SKUs from 70 to 85 percent — the first forecast the planners trusted enough to act on in their weekly planning meeting.

Manhattan Associates, where I started. Allocation and order promising for a retailer with three mega-DCs and a hundred stores. Through Cyber Monday we held 99% fill against a 95% goal — roughly eight million dollars of orders protected, and that figure is the program record.

I haven't run an Amazon network region, and I won't pretend otherwise. What I have done four times is design the supply chain — the lead times, the buffers, the one definition of inventory — and keep it honest in a weekly review. I want to do that at network scale."

30-second version (interviewers two through five may ask again): Sephora — one signed definition of "available" across every node on S/4HANA; oversells to zero. Flexport — built and ran the weekly review; sixty-six cents to three a container. Elementum — measured lanes, promise at the tail, about $1.5M of safety stock out at 99% lead-time accuracy. Manhattan — 99% fill through peak. Then the last two sentences of the long version, unchanged.

WHY AMAZON — ABOUT 60 SECONDS, USUALLY ONLY THE HIRING MANAGER ASKS

"Three reasons.

First, the forecast. At Elementum, two hundred shipping lanes were planned on the carrier's promised transit time. I replaced that with what actually happened on each lane — the distribution, not the brochure — and promised at the 99th percentile. That released about a million and a half dollars of inventory, and the shortage alert built on that data was the first one planners actually acted on. Amazon plans the whole network off a forecast; I've done the work of making people trust one.

Second, the weekly review. At Flexport, leadership asked for a weekly review of shipment tracking and nobody owned it. I built it and ran it with the VP every week for a year; cost to serve went from sixty-six cents a container to three, in Finance's rates. That input-metric, output-metric weekly review is how Amazon runs. I've already lived that way.

Third, one version of the truth. At Sephora I own how inventory is counted and promised across stores, the warehouses and dotcom. Three node types meant three different things by 'available'; I got it to one signed number per node per channel, checked daily, and customer-facing oversells went to zero. Amazon's promise to the customer rests on that same one number, at a thousand times the scale.

Why now: what I went to Sephora to build is live — the migration, the omni DC, the promising governance. I'm not running from anything; this is the biggest version of the same problem."

PROBING — the short follow-ups these two scripts draw.

What's inside "sixty-six cents to three"?

Cost to serve per tracked container, computed in Finance's loaded rates — their model, not mine. Behind it: tracking automation 40% → 90%, system accuracy 60% → 90%, and the customer base 7 → 68 (9.7×) while the review stayed weekly. The full walk is S11; the ~$1.5M of pre-production measurement savings is a separate number and never merged with it.

How much of that review did you run by hand?

At the start, about 1.5 to 2 hours a week of my own hands across four systems — and heading toward roughly 4 as volume grew, which is exactly why I wrote the BRD and automated it (S16). The value of the automation was scale without headcount, not my hours back.

You said $1.5M at Elementum — your roster also shows a $2M. Same money?

No, and I never add them. The ~$1.5M (580K units) is the safety stock released by the 200-lane measured-lead-time program (S8). The $2M is the separate release from the real-time supply/demand view as forecast accuracy went 70 → 85% (S6). Both are record; they stay apart.

Whose number is the eight million?

The program record from the Cyber Monday event — roughly $6M at risk when backorders piled up, about $8M of orders protected by close at 99% fill against the 95% goal. The decomposition into margin and cancel cost is my model, and I say so when I use it (S13).

Why are you leaving Sephora?

I've delivered what I went there to do — the WMS migration, the omni DC and the promising governance are live. I'm running toward scale, not away from anything: at Sephora a point of promise accuracy is thousands of orders; here it's millions.

Which of the four chapters would you pick for a deep dive?

Sephora, because it's current and I own it end to end — S1 is that walk. If the interviewer owns operating cadence, the Flexport review (S11); if planning or network design, the Elementum lanes (S8).

2. The stories — employer blocks in prep-priority order: pitch, gated STAR, phase probes, and the impact math

Every story now follows the Amazon worksheet shape: **60-second pitch** (say first, unprompted; ends on the primary metric) → **SITUATION / TASK** as the full problem space — what and why, when, the initial scope, the scale, the challenges, the risks if nothing happened, and the metrics to improve with their baseline values → **ACTION** as the solution space — five gated phases (entry · delivered · exit) plus how it was done and who was influenced, in prose → **RESULTS** — the Situation metrics, improved, quantified as $ · # · % · time · customer · quality and tied back to the stated baselines → one **PROBING** section per story: the challenge and results Q&A, the **PHASE DEEP-PROBE BANK** (answer the phase asked, then stop), the **business-impact model** (KPI tree, numbers table), the **fully worked derivation tables** — the problem in numbers, the fix with every value substituted, the metric before → after, the dollars with every multiplication shown — the solution flow, the sensitivity table, the spoken total-impact statement and the full "if challenged" answers. The GFiber fit line is replaced by the Amazon LP mapping and a "where this lands at Amazon" sentence. Dates and counts inside each story are the program record; anything labeled illustrative exists only to make the math complete.

Order. The stories are grouped by employer in prep priority — Sephora (current role) first, then Flexport, then Elementum (closing with the Walmart-client engagements S19–S21 and the twelve §3 design engagements), Manhattan Associates last. Every story code and every word inside each story is unchanged from the previous edition — only the order moved, so the §1.3 crosswalk and every cross-reference (e.g., "the S18 discipline") still point where they always did.

Employer block 1 — Sephora (current role)

S1 S/4HANA ↔ ISM one signed "available" · S2 UDC comingle + robotics gate · S18 RP → JDA WMS migration · S3 SNB de-aggregation · S4 the in-transit rule (failure story) · S5 the architecture council · S15 the 4-week cadence

Prep this block first — it is the current role and the program family interviewers probe hardest for recency. S1, S2 and S18 open the block because they carry the loop's most likely questions (one definition of available, the omni-DC launch, the migration discipline); S18 sits with its Sephora program family even though the roster once read it last.

S1 — Sephora S/4HANA ↔ ISM: one signed meaning of "available" across Store, Retail DC and Dotcom DC — policy before configuration

Amazon LPs: Earn Trust (backup) · Ownership (backup) · Think Big (third — the book-of-record migration) · Broad Responsibility (oversell is a customer harm). **Crosswalk:** loop packet S2 — same program, consistent. **Recency:** current role.

Honesty label. Integration program lead — no system reported to me; every configurator sat in another org. I owned the definitions, the asymmetry, the decrement and reconciliation contract, the signature chain and the council ask; engineers configured and coded to the FSD; planners executed the business acts.

60-SECOND PITCH

"**Situation.** Sephora was moving the inventory book of record from the in-house MCS to SAP S/4HANA, with ISM stood up beside it for the real-time view; Store, Retail DC and Dotcom DC each meant something different by 'available,' and most oversell bugs are vocabulary bugs. **Task.** As the integration program lead for inventory supply — no system reported to me — I had to define every bucket, its owning system, and how available-for-sale is computed and decremented in real time, and get Planning, Finance and operations to sign it before configuration. **Action.** Discovery: a bucket inventory with one proposed owner per bucket; the bar was every bucket owned. Plan & Design: the lower-of-two basis, the AFS equation, the real-time decrement contract with idempotent events, daily reconciliation to S/4 — FSD v1.1 → v1.2 → signed 6/5/25 by three orgs. Build & Test: node × channel × bucket-transition matrix; E2E Inbound 11/22, Dotcom 12/20, Retail 1/30. Cutover: January planning go-live, then May execution; OMS reads published AFS, never computes; MCS retired as book of record. Hypercare: daily ISM-vs-MB52 drift; control owners on Negative AFS. Lesson: policy before configuration. **Result.** Primary metric: customer-facing oversell incidents — recurring across three node types → **zero through two go-lives**; one signed AFS per node per channel. $ impact: ≈$2.5M/yr — $1M measured MCS support cost retired, plus ≈$1.5M modeled (avoided oversell, dual-book reconciliation labor, legacy change premium); the record has no pre-state oversell dollar, so the $1.5M is my model and I say so."

SITUATION / TASK — WHERE IT OCCURRED, WHEN IT HAPPENED, WHY IT IS IMPORTANT

S — Sephora moved the book of record from MCS to S/4HANA, with ISM beside it to see digital demand S/4 cannot see in real time — a web order placed, a BOPIS order packed, a POS sale rung but not yet posted. Three node types, three working definitions of "available." Merchandising wanted merchandise reserve visible, Finance wanted on-hand to tie to the book, dotcom wanted in-transit promisable. Nobody owned the definition; I owned no system.

T — Own the integration design and data contract of inventory supply: which buckets exist, which system owns each, how available-for-sale is computed per node, and how it feeds promising, replenishment and Finance.

Most oversell bugs are vocabulary bugs, and that is why this mattered: the goal, written with the VP Supply Chain in the sponsor memo, was one signed AFS per node per channel, one Financial On-Hand, and zero customer-impacting incidents. Left unfixed, the risks were recurring vocabulary-driven oversells — a customer harm, not a

ledger error — two books reconciled by hand, the MCS run cost continuing, and a legacy change premium on an undocumented stack; vocabulary drift — a bucket without an owner — sat at the top of my risk register.

The metrics on the table, with their baselines: customer-facing oversell incidents — recurring across three node types; target zero. Availability truths — two books reconciled by hand; target one signed AFS per node per channel, with daily ISM-vs-MB52 drift as the standing check. MCS — still running as book of record; target retired.

ACTION — WHAT I PERSONALLY OWNED, HOW I DID IT, WHO ELSE WAS INVOLVED

A — Five gated phases (record dates: kickoff 3/29 → council 5/15 → LOE 7/2 → sign-off 7/31 → FSD 8/14 → peak freeze Oct–Dec; ISM FSD v1.1 3/5/25 → v1.2 5/16/25 → signed 6/5/25; E2E Inbound 11/22 · Dotcom 12/20 · Retail 1/30; issues log 836 rows / 85.4% closed. The dated gate chain, the issues log and the rollback triggers are *program-level artifacts shared with S2/S5 — one umbrella program*; if two of these stories go to one loop, say so):

1. **Discovery** — profiled both books (MCS and S/4) for SKU overlap and order profiles per node; node-by-node sessions with ~15 functions; built the bucket inventory with one proposed owning system per bucket. *Exit bar:* every bucket owned by exactly one system; business sessions closed 4/30; open questions carried with owner and date.
2. **Plan & Design** — wrote the lower-of-two basis, the AFS/FOH equations, the real-time decrement contract (event → bucket delta, idempotency key per event), the daily ISM-vs-MB52 reconciliation into FSD v1.1 → v1.2, the RAID log and the council pack (S5). *Exit bar:* FSD signed 6/5/25 by Supply Chain Strategy, Inventory Management and FSC Inventory Accounting with zero open "who owns this bucket" rows.
3. **Build & Test** — authored the node × channel × bucket-transition matrix with before/after assertions; SIT on the interface contracts before UAT; published the E2E chain dates. *Exit bar:* 0 open contract defects in SIT; E2E green 11/22 · 12/20 · 1/30; Blocker + Critical open = 0 at go/no-go.
4. **Cutover & Go-Live** — wrote the go/no-go recommendation from counted rows, the publish-switch plan (OMS reads ISM AFS, never computes) and the reserve-copy invariants; assigned control-screen roles. *Exit bar:* January go-live (reserve rules live against demand), May go-live (execution); no SKU with two published AFS values; Negative AFS clean at T+1.
5. **Hypercare / Close** — daily drift report (ISM vs MB52), Negative AFS / AFS Correction audit review, hypercare exit memo and FSD as-built note. *Exit bar:* drift = 0 SKUs on consecutive days; blockers/day under threshold; exited into the monthly release train.

Every configurator sat in another org: SAP teams and configurators built to the signed spec, planners confirmed in UAT, and control owners ran the Negative AFS and AFS Correction screens after handoff. Mine: the bucket inventory, the definitions and the asymmetry, the decrement and reconciliation contract, the node × channel × bucket-transition matrix design, the go/no-go recommendation and the signature chain. The unique value was taking the definition nobody owned and getting it signed before configuration — the lower-of-two basis, the AFS equation and the real-time decrement contract written into an FSD three orgs signed on 6/5/25.

The obstacles were three orgs each defending a different truth about “available,” and sixty people to move without authority — overcome by pre-wiring every row of the systems-impacted table with its owner's level-of-effort before the council, running working sessions to one table instead of arbitrating in meetings, and converting objections into FSD rows. Priorities were set by slotting epics by business criticality × effort, with no story entering QA without a matrix cell reference; buy-in came from the pre-wiring — nobody learned their cost in the council room, and the council was asked for three decisions rather than a review. The decision I challenged: Merchandising wanted merch reserve promisable — I priced it (unpickable stock sold) and the no went into the FSD as a row, not a meeting minute; and I held the asymmetry itself — oversell is a customer defect, undersell a planning variance Inventory Control chases through the Negative AFS and AFS Correction screens.

RESULTS — HOW SUCCESS WAS MEASURED: $ · # · % · TIME · CUSTOMER · QUALITY

R — Primary metric: recurring vocabulary-driven oversells → **zero through two go-lives**; one signed AFS per node per channel; MCS retired. $ impact: ≈$2.5M/yr ($1M user anchor + ≈$1.5M modeled — model below).

- **$** — ≈$2.5M/yr — $1M MCS run/support cost retired (user anchor) + ≈$1.5M modeled (avoided oversell, dual-book reconciliation labor, legacy change premium); the record has no pre-state oversell dollar, so the $1.5M is

my model and I say so.

- # — Oversell incidents recurring across three node types → zero through two go-lives (record); dual-book reconciliation ≈4 FTE by hand → ≈1 FTE reading the drift report (modeled).
- **%** — Issues log 85.4% of 836 rows closed (program-level artifact shared with S2/S5); the oversell rate behind the ≈$0.9M row is 0.10% of orders — modeled, benchmark 0.05–0.25%; the record movement is incidents → zero.
- **Time** — Dated gate chain held: FSD v1.1 3/5/25 → v1.2 5/16/25 → signed 6/5/25; E2E Inbound 11/22 · Dotcom 12/20 · Retail 1/30; January planning and May execution go-lives (record dates).
- **Customer** — Customer-facing oversell incidents zero through two go-lives; the asymmetry treats oversell as a customer defect and undersell as a planning variance (record).
- **Quality** — One signed AFS per node per channel; no SKU with two published AFS values; daily ISM-vs-MB52 drift = 0 SKUs on consecutive days at hypercare exit (record).

Against the goal stated above: oversell incidents recurring → zero through two go-lives; two availability truths → one signed AFS per node per channel; MCS retired as book of record. The trade behind it was priced: the asymmetry accepts visible undersell to make oversell zero by construction, and merch reserve got visibility without promisability — unpickable stock sold was never worth the sale.

PROBING — QUESTIONS, PHASE DEEP-DIVES, AND THE MATH

Challenge — Why did you choose this story to illustrate Earn Trust?

Because the deliverable was one signed truth across three orgs that did not report to me — Supply Chain Strategy, Inventory Management and FSC Inventory Accounting each signed the FSD that carried the definitions. One signature would have left two orgs free to disagree later.

Challenge — What other stories demonstrate Earn Trust?

S14 — the $2.5M OMS↔WMS discrepancy, transparency over a quiet fix — is my primary Earn Trust story; S3 (Finance and Ops on one number) and S5 (the council) carry the same LP. This story is the Earn Trust backup in my map.

Challenge — Could you come up with an example that is more recent?

This is from my current role at Sephora — the FSD was signed 6/5/25 and the two go-lives ran January and May — so it is the most recent record I have for this principle.

Probing — Why did you choose to focus on these results? What other results were important?

Oversell was the customer-facing defect the whole program existed to kill, so incidents-to-zero and one signed AFS per node per channel lead. The other results that mattered: MCS retired as book of record, reconciliation moved from argument to a daily drift report with a named owner, and the E2E chain landing on its dates.

Probing — You mentioned dollars — what percentage change is that year over year?

There is no year-over-year revenue record here — the ≈$2.5M/yr is $1M of measured MCS run cost retired (user anchor) plus ≈$1.5M modeled against the pre-state baseline, not a YoY revenue change. The record movement is incidents: recurring → zero through two go-lives.

Probing — I'm concerned about the 0.10% oversell rate — tell me more.

I don't know it to the decimal — the record says incidents were recurring and went to zero; the rate is a benchmark set deliberately low, a twentieth of a typical e-commerce inventory-cancel rate. First thing I'd measure: the “item unavailable after order” cancel-reason code for the twelve months before January.

Challenge — What were the lessons learned? What would you have done differently?

Policy before configuration is the lesson that held. Differently: the in-transit rule (S4) — back-test a ruleset against history before publishing it.

Challenge — How would you implement this at Amazon?

Every fulfillment promise rests on one published availability number per node that downstream systems read and never recompute — at Amazon scale that is the ATP/promise seam between IPC-style planning and the order pipeline: one definition, one owner, policy signed before code.

Challenge — How did these results compare to your actual goals?

The signed criteria were one AFS per node per channel, one Financial On-Hand and zero customer-impacting incidents — all three held through both go-lives, with drift at zero SKUs on consecutive days at hypercare exit and MCS retired as book of record.

PHASE DEEP-PROBE BANK — Q&A (15). Answer the phase asked, then stop.

Discovery — How did you know discovery was done?

When every bucket on the table had exactly one proposed owning system and no open "who owns this" row. Sessions closed 4/30; the exit memo listed remaining questions with an owner and a date, not a hope.

Who wrote the success criteria and who signed?

I wrote them with the VP Supply Chain in the sponsor memo — one signed AFS per node per channel, one Financial On-Hand, zero customer-impacting incidents. Supply Chain Strategy, Inventory Management and FSC Inventory Accounting signed the FSD that carried them.

What did you profile before designing?

Both books — MCS and S/4 — for SKU overlap and order profiles per node type, so the design argument started from their own data, not opinions about it.

Plan & Design — What was in your risk register at design sign-off?

Three at the top: vocabulary drift (a bucket without an owner), feed latency from WMS/OMS/POS, and override abuse on AFS-changing screens. Each had a trigger, a mitigation and an owner — mine was drift.

How did you handle an objection at the design review?

Merchandising wanted merch reserve promisable. I priced it — unpickable stock sold — and offered visibility without promisability. The no went into the FSD as a row, not into a meeting minute.

Why three signatories, not one?

Each owned a different truth: Strategy owned policy, Inventory Management owned execution, Inventory Accounting owned the book. One signature would have left two orgs free to disagree later.

Build & Test — How did you sequence stories in Jira?

Epics per module — reserve module, AFS engine, SNB, control screens — slotted by business criticality × effort; a story could not enter QA without a matrix cell reference.

How were UAT entrance criteria set and who could fail a build?

Entrance: SIT contract defects closed, test data handed off, environments ready. QA owned the gate and could fail a build; planners confirmed in UAT because QA had already demoed each matrix family during build.

Describe your bug-bash equivalent.

The dated E2E days — Inbound 11/22, Dotcom 12/20, Retail 1/30. Business and engineering ran the full flow in one room; every finding got a severity and owner before we broke.

Cutover — What made you willing to say go?

Counted rows: Blocker + Critical at zero, E2E green, the monitor for the interim copy built and tested, rollback triggers armed. Never confidence in the room.

What were your rollback triggers and did any fire?

Sync below 99% for four consecutive hours, or order-flow p95 above twice budget for an hour — consumers revert to the prior publish. Armed for both go-lives; never fired.

Hypercare — How did you decide hypercare was over?

Daily ISM-vs-MB52 drift and blockers/day under threshold on consecutive days, and the control owners — Inventory Control, Internal Audit, Finance — running the Negative AFS report without me. Then the design entered the monthly release train.

What did hypercare hand over, and to whom?

The control-owner map: Negative AFS to Inventory Control, Internal Audit and Finance; AFS Correction to the ISM app team; the daily drift report to Inventory Control. Every AFS-changing act had a screen, a role and an audit row before I stepped back.

Cross-phase — You owned no system; how did you get sixty people to sign?

Pre-wired every row of the systems-impacted table with its owner's level-of-effort before the council, ran working sessions to one table instead of arbitrating in meetings, and asked the council for three decisions rather than a review. A no that lives in a document survives; a no that lives in a meeting gets re-litigated weekly.

Why lower-of-two and not an average? · What would you do differently?

Oversell is a customer defect; undersell is a planning variance Inventory Control chases through the Negative AFS and AFS Correction screens — the asymmetry is the policy. Differently: the in-transit rule (S4) — back-test a ruleset against history before publishing it.

Where this lands at Amazon: every fulfillment promise rests on one published availability number per node that downstream systems read and never recompute — the exact contract I held the OMS to. At Amazon scale the same discipline is the ATP/promise seam between IPC-style planning and the order pipeline: one definition, one owner, policy signed before code.

BUSINESS IMPACT MODEL — KPI TREE, NUMBERS, THE MATH

```
Gross margin + customer trust (primary outcome)
+- Oversell / cancel cost (flow, $/yr) -> 0 incidents
|    oversell = orders x vocab-driven oversell rate x cost/incident
|    (loop) oversell -> NPS down -> repeat rate down -> orders down
+- Dual-book reconciliation labor (stock: FTE) -> 1 FTE + drift report
|    (loop) more drift -> more manual recon -> slower fixes -> more drift
+- Legacy run cost (MCS support, $/yr) -> retired
+- Tech-debt change premium (changes/yr x extra hrs x rate) -> S/4 + ISM
+- Book accuracy -> unexplained shrink / audit-finding risk
+- Time-to-market for a new node or channel (weeks) -> a matrix row
```

DRIVER	BASELINE	AFTER	$/YR	BASIS
MCS run / support cost	≈$1.0M/yr	$0 — retired	$1.0M	user anchor
Oversell / cancel incidents	recurring, ~0.10% of orders	zero through two go-lives	≈$0.9M	modeled (rate benchmark 0.05–0.25%; record: → zero)
Dual-book reconciliation labor	≈4 FTE by hand	≈1 FTE reading the drift report	≈$0.33M	modeled (3 × $110K)
Tech-debt change premium	≈40 changes/yr × +60 h on MCS	platform cost	≈$0.3M	modeled (40 × 60 × $125)
Book accuracy → shrink · time-to-market	—	—	≈$0.3M + $0.2M	modeled upside, not in headline

If challenged. *"How do you know it was a tenth of a percent?"* — "I don't, to the decimal. The record says incidents were recurring and went to zero; the rate is a benchmark set deliberately low — a twentieth of a typical e-commerce inventory-cancel rate. First thing I'd measure: the 'item unavailable after order' cancel-reason code for the twelve months before January." *"Does the $1M already include your change premium?"* — "It might. If enhancement labor sits inside it, I drop the row and the headline is about $2.2M. I'd split the MCS cost center into run vs change tickets for one fiscal year." *"Three FTE sounds like a guess."* — "It's a benchmark count. What's in the record is the mechanism: reconciliation moved from argument to a daily drift report with tolerance zero and a named owner. I'd size it with a two-week time study — how I sized the Austria effort at Elementum."

THE MATH — STEP-BY-STEP DERIVATION

Table 1 — before: the problem in numbers.

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
1	Two books for one SKU	`OOB = S4_Avail − WMS_OnHand = 1,000 − 980 = +20`	S/4 Available 1,000 units; WMS on-hand 980 (illustrative, v105 Ex. 1)	+20 units — the books disagree
2	Units already spoken for	`Σ buckets = 200 + 120 + 60 + 40 = 420`	Picking Reserve 200 · Ecom Pick 120 · Ecom Pack 60 · Shipped-Not-Billed 40 (illustrative)	420 units committed or sold-not-posted
3	Truly promisable	`min(1,000, 980) − 420 = 980 − 420 = 560`	the number the fix will compute — shown here as the yardstick	560 units
4	Definition A — channel reads raw S/4 Available	`promises 1,000; 1,000 − 560 = 440`	the committed buckets are invisible to it (record problem)	440 units over-promised
5	Definition B — channel reads WMS on-hand	`promises 980; 980 − 560 = 420`	same buckets invisible (record problem)	420 units over-promised
6	Oversell outcome	`30,000,000 orders × 0.10% = 30,000`	orders illustrative; rate benchmark 0.05–0.25%; incidents recurring across three node types (record)	≈30,000 incidents/yr
7	Reconciliation burn	`≈4 FTE reconciling two books by hand`	record baseline	≈4 FTE

Table 2 — the fix: step by step, values substituted.

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
1	One supply basis	`Basis = min(S4_Avail, WMS_OnHand) = min(1,000, 980) = 980`	asymmetric lower-of-two (record design)	Basis = 980 units
2	Why the minimum	`OOB = +20 means book above floor; promising the 20 risks oversell`	a positive out-of-balance reduces supply (record design)	the +20 is never promised
3	Available-for-sale	`AFS = 980 − 200 − 120 − 60 − 40 = 560`	AFS = Basis − PickRes − EcomPick − EcomPack − SNB; Merch Reserve and Lost/Found sit outside S4_Avail — no double-subtract path (record design)	AFS = 560 units
4	Floor-above-book case	`Basis = min(1,000, 1,020) = 1,000; AFS = 1,000 − 420 = 580`	WMS 1,020 → OOB = −20 (illustrative)	AFS = 580, not 600 — the 20-unit surplus stays unpromised
5	Real-time decrement	`AFS = 560 − 3 = 557; SNB = 40 + 3 = 43`	store rings 3 units at 10:02 (illustrative); each event maps to exactly one bucket	AFS 557 within the 60 s publish budget
6	Idempotency	`replayed key at 10:04 → no-op; AFS = 557 − 0 = 557`	every event carries an idempotency key; a reversal must reference its original (record design)	AFS stays 557
7	Overnight posting	`SNB 43 → 0; S4_Avail = 1,000 − 43 = 957`	aggregate posting lands in S/4 for 43 units (illustrative)	books move together
8	Daily drift check	`Drift = ISM.S4_Avail − MB52 = 957 − 957 = 0`	tolerance 0 — a non-zero drift is a Sev, not a planning variance (record design)	drift = 0

Table 3 — after: the metric, before → after.

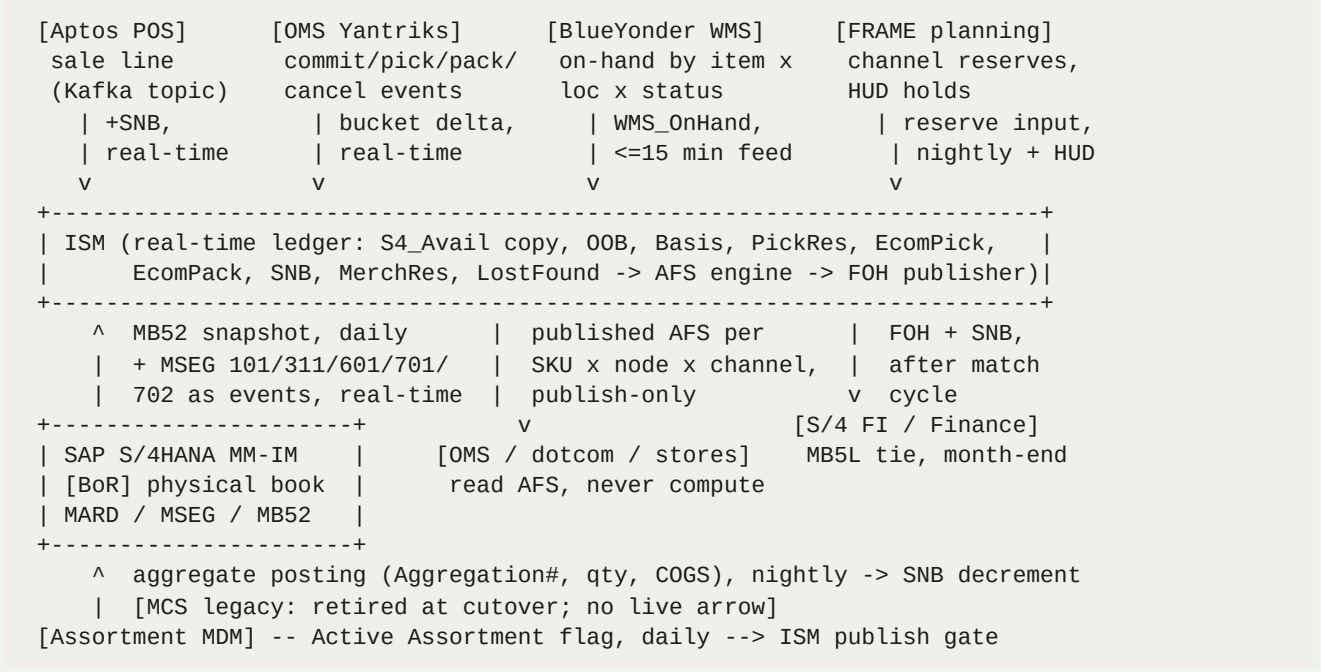
STEP	METRIC	BEFORE	AFTER	BASIS
1	Working definitions of “available”	3	1 signed AFS	record

STEP	METRIC	BEFORE	AFTER	BASIS
2	Oversell incidents	recurring across three node types	zero through two go-lives	record
3	Dual-book reconciliation	≈4 FTE by hand	≈1 FTE reading the drift report	modeled on the record mechanism
4	Drift at hypercare exit	unmeasured	0 SKUs drift on consecutive days	record
5	Legacy MCS run cost	≈$1.0M/yr	retired	user anchor

Table 4 — the dollars, every multiplication shown.

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
1	MCS run / support retired	`≈$1.0M/yr, taken as stated`	user anchor	$1.0M/yr
2	Incidents avoided	`30,000,000 × 0.10% = 30,000/yr`	orders illustrative; rate benchmark 0.05–0.25%; record movement: → zero	30,000 incidents/yr
3	Cost per incident	`$6 + $4 + $19 + $1 = $30`	CS contact $6 · refund $4 · lost margin ~$19 · goodwill ~$1 (benchmark)	$30 each
4	Oversell dollars	`30,000 × $30 = $900,000`	rows 2 × 3 (modeled)	≈$0.9M/yr
5	Reconciliation labor	`4 − 1 = 3 FTE; 3 × $110,000 = $330,000`	$110K loaded (benchmark)	≈$0.33M/yr
6	Tech-debt change premium	`40 × 60 h = 2,400 h; 2,400 × $125 = $300,000`	≈40 changes/yr × +60 h each on MCS × $125/h (modeled)	≈$0.3M/yr
7	Headline	`1.0 + 0.9 + 0.33 + 0.3 = $2.53M`	rows 1, 4, 5, 6	≈$2.5M/yr
8	Not in headline	`shrink ≈$0.3M + time-to-market ≈$0.2M, excluded`	modeled upside	excluded

SOLUTION FLOW — THE SYSTEM & DATA MAP



SENSITIVITY — LOW / BASE / HIGH

INPUT	LOW	BASE	HIGH
Vocabulary-driven oversell rate	0.05%	0.10%	0.25%
Cost per oversell incident	$25	$30	$40
Oversell cost avoided	$0.4M	$0.9M	$3.0M
Reconciliation FTE released × loaded	2 × $95K = $0.19M	3 × $110K = $0.33M	4 × $130K = $0.52M
Change premium (changes × hrs × rate)	24 × 40 × $100 = $0.1M	40 × 60 × $125 = $0.3M	60 × 80 × $150 = $0.72M
MCS run cost (user anchor, fixed)	$1.0M	$1.0M	$1.0M
Total modeled $/yr (headline, excl. upside)	≈ $1.7M	≈ $2.5M	≈ $5.2M

TOTAL IMPACT STATEMENT — SAY IT LIKE THIS

"Conservatively, about two and a half million dollars a year. One million of that is measured: the MCS support cost that went away when S/4 became the book of record. About nine hundred thousand is avoided oversell: on roughly thirty million dotcom orders, a tenth of a percent going wrong at about thirty dollars each, and the record shows those incidents went to zero through both go-lives. About three hundred thousand is reconciliation labor, three analysts who stopped reconciling two books by hand, and about three hundred thousand is the change premium we stopped paying on an undocumented stack. The first number is measured; the other three are my model, and I can show you every input."

S2 — Sephora UDC: two warehouses in one building → one comingle pool, coupled with a BlueYonder upgrade and robotics behind a published gate

Amazon LPs: Insist on the Highest Standards (primary — the 98% gate) · Deliver Results · Have Backbone (the throttle) · Earth's Best Employer (the floor protected). **Crosswalk:** loop packet S11 + S24 — consistent. **Recency:** current role.

Honesty label. I owned discovery through FSD for the location and inventory model, the staging and the gates; BlueYonder, Locus and the platform teams built and configured; DC Ops ran the floor. The +15% is Sephora's signed number; every dollar is my model on an illustrative 100K-units/week DC.

60-SECOND PITCH

"**Situation.** Sephora's Utah DC ran as two systemic warehouses in one building, with a BlueYonder upgrade and Locus robotics on the same path and zero customer-facing incidents allowed. **Task.** I owned discovery through FSD for the location and inventory model, the staging, and the gates — a dozen core systems, 20+ touched, 60+ stakeholders. **Action.** Discovery: both books profiled; systems-impacted table with owners. Plan & Design: one comingle pool, three planning identities, two go-lives so each carried one defect class. Build & Test: node × channel × bucket matrix; robotics gate published at 98% throughput with a throttle built in advance. Cutover: six-step May choreography — clear negatives, transfer AFS to Uncommitted, scan-and-reserve, zero-then-publish so nothing was sellable in two places. Hypercare: week two hit 94; I throttled bots to ~30% of volume, root-caused zone/slotting logic, full volume week three. Lesson: a gate that cannot stop the program is decoration — mine did. **Result.** Primary metric: DC processing efficiency — baseline → **+15%** after one comingle pool replaced two systemic warehouses; customer-facing incidents zero through both go-lives. $ impact: ≈$1.6M/yr modeled (labor at +15%, pooling carry, fewer split shipments, self-transfer handling removed) — the +15% is the signed number; the dollars are my model."

SITUATION / TASK — WHERE IT OCCURRED, WHEN IT HAPPENED, WHY IT IS IMPORTANT

S — The Utah DC ran as two systemic warehouses — Dotcom 801 (selling location 800) and Retail 901 — transferring stock to itself and paying duplicate handling for zero customer value. The ask: one omni/comingle facility, with a BlueYonder WMS 8.2 → 2023.1 upgrade and Locus robotics on the same critical path, a dozen core systems (20+ touched), an Oct–Dec peak freeze, zero customer-facing incidents allowed.

T — Discovery through FSD for the location and inventory model, the staging, and the acceptance gates; 60+ stakeholders on the steering path.

The challenges sat on the critical path itself: the WMS upgrade and the robotics ramp shared it, the Oct–Dec peak freeze bounded it, and the interim would run two live truths. Doing nothing meant continuing to pay duplicate handling on self-transfers; running the program carried the register's top three risks — two live truths during the interim, a robotics ramp with no stop condition, and a peak collision — each given a trigger, a mitigation and an owner before sign-off.

The metrics on the table, with their baselines: processing efficiency — index 100 (units per labor-hour), the pre-UDC baseline DC Ops would measure against post-stabilization; customer-facing incidents — zero allowed, the constraint every other number had to move inside; the robotics ramp — a 98% gate published in advance, set from the pre-ramp baseline and the incident budget before the event; self-transfers and duplicate handling — the standing cost one pool exists to remove.

ACTION — WHAT I PERSONALLY OWNED, HOW I DID IT, WHO ELSE WAS INVOLVED

A — Five gated phases (record: kickoff 3/29 → sessions 4/30 → council 5/15 → LOE 7/2 → sign-off 7/31 → FSD 8/14 → peak freeze → E2E 11/22 · 12/20 · 1/30 → Locus RTL Feb–Mar, slipped 3 weeks, re-slotted openly → January go-live → May go-live; cutover plan 343 tasks / 48,885 task-minutes ≈ 815 hrs; issues log 836 rows / 85.4% closed, Blocker 164 + Critical 277 = 52.8%):

1. **Discovery (3/29 → 4/30)** — profiled 801/901 (SKU overlap, order profiles); sponsor objectives and success criteria (zero customer-facing incidents; one comingle facility); systems-impacted table. *Exit bar:* every impacted

system named to an owner; success criteria signed by steering; open questions dated.

2. **Plan & Design (5/15 → 8/14)** — council pack (three decisions + out-of-scope fence, S5); the 900/901 location/inventory model with three planning identities; staged two-go-live timeline; RAID log; test-matrix skeleton; FSD to 8/14. *Exit bar:* council sign-off 7/31, FSD 8/14, peak-freeze boundary decided, LOE-backed plan across ~30 build tracks.
3. **Build & Test (Aug → Jan/Apr)** — node × channel × bucket-transition matrix as the regression asset; SIT contract-defect list; dated E2E chain; UAT entrance memo; Locus gate definition (98%) with throttle spec; three mock cutovers against the 0.5% tolerance — the S18 discipline, reused across the same Sephora program family. *Exit bar:* E2E green three dates; Blocker + Critical = 0 at each go/no-go; mock 3 inside tolerance; Locus RTL retested after the slip.
4. **Cutover & Go-Live (Jan; May 1 → Jul 1)** — cutover runbook (343 tasks, owner + duration + predecessor per task); six-step May choreography; go/no-go recommendation; rollback plan per interface. *Exit bar:* six invariants held (Σ AFS conserved; no SKU sellable on 800 and 900 at once; negatives cleared first); rollback never fired.
5. **Hypercare / Close (Jun → Jul)** — daily burn-down read (271 → 65 → 21 → 14 → 4), Locus weekly gate read, hypercare exit memo, template handed to the next DC. *Exit bar:* blockers/day < 2 sustained → exit week 3 of 4; Locus at full volume week 3; +15% measured post-stabilization by DC Ops.

BlueYonder, Locus and the platform teams built and configured; DC Ops ran the floor and measured the +15% post-stabilization. Mine specifically: the location/inventory model, the staging, the gates, the council pack, the cutover runbook (343 tasks with owner, duration and predecessor) and the go/no-go recommendation. The contribution that made one pool workable: three planning identities over one physical location — one physical pool serving both channels without blurring anyone's lens — plus the published 98% robotics gate with the throttle dial built in advance, and the staged two-go-live timeline so each go-live carried one defect class.

The obstacles: the Locus RTL slipped three weeks — escalated the week it surfaced, re-slotted openly, and I refused to compress UAT to absorb it; Digital worried a virtual Retail DC would blur their forecasting lens — the three-identity design made the concern the design, not an argument. Priorities ran as ~30 build tracks under one log, stories slotted by business criticality × effort, Blocker + Critical open = 0 as the gate; buy-in came from steering (DC Ops, IT, Supply Chain Strategy) signing the success criteria in the council pack, with DC Ops owning the robotics gate and the dial before the ramp — the rule fired, I didn't improvise. I challenged the single-go-live instinct — two go-lives so we never hunted two defect classes in the same week — and held the comingle-reserve improvements out of the release to protect zero incidents, priced as a later enhancement at steering.

RESULTS — HOW SUCCESS WAS MEASURED: $ · # · % · TIME · CUSTOMER · QUALITY

R — Primary metric: processing efficiency **+15%** (record, Sephora signed); zero customer-facing incidents through both go-lives; robotics gate held at 94 vs 98, full volume week three. $ impact: ≈$1.6M/yr modeled + ≈$2.2M one-time safety-stock release (model below).

- **$** — ≈$1.6M/yr modeled (labor at +15%, pooling carry, fewer split shipments, self-transfer handling removed) + ≈$2.2M one-time safety-stock release modeled via √2 pooling; the +15% is the signed number, the dollars are my model.
- **#** — Cutover runbook 343 tasks / 48,885 task-minutes ≈ 815 hrs; issues log 836 rows / 85.4% closed; post-go-live burn-down 271 → 65 → 21 → 14 → 4 (record).
- **%** — Processing efficiency +15% (record, Sephora signed); robotics gate published at 98%, week two at 94, bots throttled to ~30% of volume, full volume week three (record).
- **Time** — Council 5/15 → sign-off 7/31 → FSD 8/14; E2E 11/22 · 12/20 · 1/30; January and May go-lives; hypercare exited week 3 of 4 (record).
- **Customer** — Zero customer-facing incidents through both go-lives (record) — the constraint every other number was allowed to move inside.
- **Quality** — Six cutover invariants held (Σ AFS conserved; no SKU sellable on 800 and 900 at once; negatives cleared first); rollback armed for both go-lives, never fired (record).

Against the goal stated above: one omni facility with self-transfers eliminated; zero customer-facing incidents through both go-lives; peak untouched; processing efficiency index 100 → 115 (+15%, record, Sephora signed) measured by DC Ops post-stabilization; robotics at full volume by week three. The trades were explicit: the comingle-reserve improvements went out of the release (a later enhancement, priced at steering), the throttle traded a week of robot volume for a root-caused ramp, and for four months every report had to declare which system of record it read — the interim's price.

PROBING — QUESTIONS, PHASE DEEP-DIVES, AND THE MATH

Challenge — Why did you choose this story to illustrate Insist on the Highest Standards?

The 98% robotics gate was set from the pre-ramp baseline and the incident budget before the event — and when week two hit 94, it fired: I throttled bots to ~30% of volume rather than let the ramp erode the floor. A gate that cannot stop the program is decoration; mine did.

Challenge — What other stories demonstrate Insist on the Highest Standards?

S18 — the 0.5% reconciliation tolerance published with Finance before mock one — is this LP's backup, and S9's ≥95% valid-ASN cohort gate, held twice, is the third.

Challenge — Could you come up with an example that is more recent?

This is my current role — the January and May go-lives and the Locus ramp are the most recent record I have for this principle.

Probing — Why did you choose to focus on these results? What other results were important?

Processing efficiency +15% is the signed record and the point of one pool; zero customer-facing incidents was the constraint every other number had to move inside. Also important: the robotics gate holding at 94 vs 98 with full volume by week three, and the burn-down 271 → 65 → 21 → 14 → 4.

Probing — You mentioned dollars — what percentage change is that year over year?

No YoY revenue figure — the +15% is processing efficiency (units per labor-hour) against the pre-UDC baseline, Sephora-signed, and the ≈$1.6M/yr is my model on an illustrative 100K-units/week DC, not a revenue change.

Probing — I'm concerned about the labor number behind the modeled ≈$0.9M — tell me more.

The fifteen percent is measured; the headcount and the $28 loaded rate are my assumptions. Pull the DC's actual direct hours for the two quarters before and after and multiply by payroll — that turns the model into a record in one afternoon.

Challenge — What were the lessons learned? What would you have done differently?

A gate that cannot stop the program is decoration — mine did, at 94. Differently: staff the interim reporting workstream on day one, not month three.

Challenge — How would you implement this at Amazon?

This is an FC launch/retrofit with automation on the critical path — publish the ramp gate and the throttle before the robots arrive, stage go-lives so each carries one defect class, and hold peak sacred. Zero-then-publish is the same discipline as never letting one unit be promisable from two nodes at once.

Challenge — How did these results compare to your actual goals?

The steering charter asked for one omni facility, self-transfers eliminated, zero customer-facing incidents and peak untouched — all four delivered, with +15% efficiency measured by DC Ops post-stabilization and the robotics at full volume week three.

PHASE DEEP-PROBE BANK — Q&A (15). Answer the phase asked, then stop.

Discovery — How did you know discovery was done?

When the systems-impacted table had a named owner and a written LOE for every row — a dozen core systems, 20+ touched — and the success criteria were signed by steering. The LOE landed 7/2.

Who wrote the success criteria and who signed?

I drafted them from the steering charter — one omni facility, self-transfers eliminated, zero customer-facing incidents, peak untouched — and steering (DC Ops, IT, Supply Chain Strategy) signed them in the council pack.

Plan & Design — What was in your risk register at design sign-off?

Two live truths during the interim, a robotics ramp with no stop condition, and a peak collision. Each had a trigger, a mitigation and an owner — the shadow monitor, the 98% gate with a throttle dial, and the freeze in the pack.

How did you handle a business owner who objected at the design review?

Digital worried a virtual Retail DC would blur their forecasting lens. I kept three planning identities over one physical location so each lens survived — the concern became the design, not an argument.

Why two go-lives?

So we never hunted two defect classes in the same week. January proved reserve rules against live demand while the floor was untouched; May left warehouse execution as the only remaining class.

Build & Test — How did you sequence work?

~30 build tracks under one log; stories slotted by business criticality × effort; per-environment workbooks (Dev 310, QA 255, Perf 200, Prod 221 rows) told each track where it could test. Blocker + Critical open = 0 was the gate.

How were UAT entrance criteria set and who could fail a build?

SIT contract defects closed, conversion build done, test data handed off, pre-prod and perf environments ready. QA owned the gate; DC Ops confirmed in UAT because QA had already demoed to them during build.

Describe a bug bash — what came out of it?

The dated E2E days: Inbound 11/22, Dotcom 12/20, Retail 1/30 — every function ran the full flow in one room, every finding logged with severity and owner before we left. The Locus 3-day workshop worked the same way for the robotics contract.

Cutover — What made you say go — or no-go?

The burn-down, not the room: Blocker + Critical at zero, mock 3 inside the 0.5% tolerance (the discipline S18 established, reused here), runbook rehearsed, rollback armed. I would have said no-go on one open Blocker; I never had to.

What were your rollback triggers and did any fire?

Sync < 99% for four consecutive hours, or order-flow p95 > 2× budget for one hour — per-interface rollback behind them. Armed for both go-lives; never fired.

How big was the cutover runbook?

343 tasks, 48,885 task-minutes — about 815 hours — May 1 to Jul 1, each task with an owner, duration and predecessor, and step 4 (zero-then-publish) as the one irreversible instant.

Hypercare — How did you decide hypercare was over?

Blockers per day under two, sustained, and the post-go-live spike decaying 271 → 65 → 21 → 14 → 4. Exited week 3 of 4. A calendar exit declares victory over open defects.

Which wave did you hold and how did the business take it?

The Locus ramp at week 2: 94 against a published 98. I throttled to ~30% for a week. DC Ops took it well because the gate and the dial were theirs before the ramp — the rule fired, I didn't improvise.

Cross-phase — Where did you trade scope for date? What did you escalate?

The comingle-reserve improvements — out of the release to protect zero incidents, priced as a later enhancement at steering. Escalated: the staged timeline and peak-freeze boundary as decisions; the Locus RTL slip the week it surfaced. I refused to compress UAT to absorb it.

Why 98%? · Why not one go-live? · Do differently?

98% was set from the pre-ramp baseline and the incident budget before the event — a gate that cannot stop the program is decoration; mine did, at 94. One go-live would mean hunting two defect classes in one week. Differently: staff the interim reporting workstream on day one, not month three — for four months every report had to declare which system of record it read.

Where this lands at Amazon: this is an FC launch/retrofit with automation on the critical path — publish the ramp gate and the throttle before the robots arrive, stage go-lives so each carries one defect class, and hold peak sacred. The zero-then-publish invariant is the same discipline as never letting one unit be promisable from two nodes at once.

BUSINESS IMPACT MODEL — KPI TREE, NUMBERS, THE MATH

```
DC cost-to-serve per unit (primary outcome)  [record: +15% efficiency]
+- Direct labor $ = units / (units per labor-hour) x loaded rate
|    +- units/labor-hour (+15%, record) <- self-transfers removed · one pick pool
|    +- Locus bot share of volume (+)
|         <- gate T >= 98% -> throttle 30% -> RCA -> ramp again (balancing loop)
+- Inventory carry $ = safety stock x cost of capital
|    +- two pools -> one pool (sqrt-n pooling) (-)
+- Split shipments $ = dotcom orders x split rate x cost per split (-)
+- Customer-facing incidents = 0 (constraint, record) -- the fence:
     every row above was allowed to move only inside this constraint
```

DRIVER	BASELINE	AFTER	$/YR	BASIS
Processing efficiency (units per labor-hour)	index 100	115	—	record (Sephora signed)
Direct labor hours for same volume	240K hrs	209K hrs	≈$0.9M	modeled (120 FTE × 2,000 h × $28 × 13%)
Safety stock, overlapping SKUs (2 pools → 1)	$7.5M	$5.3M	≈$0.2M carry	modeled (√2 pooling × 10% capital)
Split-shipment rate	~4.5M orders/yr	−1 pt	≈$0.45M	modeled (45K × $10)
Self-transfer handling	~260K units/yr	0	≈$0.1M	modeled (260K × $0.80 × 50%)
Total modeled	—	—	**≈$1.6M/yr**	+ $2.2M one-time working capital

If challenged. *"The labor number is the biggest — how firm?"* — "The fifteen percent is measured; the headcount and the $28 loaded rate are my assumptions. Pull the DC's actual direct hours for the two quarters before and after and multiply by payroll — that turns the model into a record in one afternoon." *"Pooling sounds like a textbook claim."* — "It is, and I labeled it. √2 pooling assumes two roughly equal independent pools on the same SKUs. Measure SKU-level on-hand on the last day of 801 vs the first stable month of 901; the release is whatever the difference actually is." *"Split shipments — evidence?"* — "The softest row. Parcels per dotcom order from the ship system before and after May, times the carrier's per-parcel cost. If the point didn't move, I drop the row and the total is about $1.1M."

THE MATH — STEP-BY-STEP DERIVATION

Table 1 — before: the problem in numbers.

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
1	Two warehouses in one building	`planning pools = 2 (801 retail + 800 dotcom)`	two planning identities over one roof (record)	2 pools
2	Baseline efficiency	`index = 100 units per labor-hour`	pre-UDC baseline (record)	100
3	Direct labor to run it	`120 FTE × 2,000 h/yr = 240,000 h/yr`	modeled staffing on an illustrative 100K-units/week DC	240,000 h/yr
4	Self-transfers inside one roof	`≈260,000 units/yr moved on paper between 801 and 800`	modeled	260K units/yr of non-work
5	Duplicated safety stock	`$50M × 25% = $12.5M SS; $12.5M × 60% = $7.5M`	avg inventory (illustrative) × SS share × share held in SKUs both pools carry	$7.5M duplicated poolable SS
6	Split shipments	`4,500,000 orders × 1 pt = 45,000 extra splits/yr`	~4.5M dotcom orders/yr; split rate 1 pt above the comingled state (modeled)	45,000 splits/yr

Table 2 — the fix: step by step, values substituted.

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
1	One pool, three planning identities	`900/901 model → self-transfers = 0 by construction`	three planning identities over one physical location (record design)	the move is a posting, not a truck
2	Robotics gate (published first)	`T = pick-accurate completed / attempted ≥ 98%`	gate published before the ramp (record)	the dial was built in advance
3	Gate evaluated, week 2	`T = 94%; 94 < 98 → gate failed by 4 pts`	record	throttle fires
4	Throttle	`bot share 100% → 30%; manual covers (1 − 0.30) × 100,000 = 70,000 units/wk`	volume illustrative	RCA, then ramp again — full volume week 3 (record)
5	Efficiency after	`index 100 × 1.15 = 115`	+15% (record, Sephora signed, post-stabilization)	115
6	Hours for the same volume	`240,000 / 1.15 = 208,696 ≈ 208,700 h/yr`	from Table 1 row 3 and the +15%	≈208,700 h/yr
7	Hours saved	`240,000 − 208,700 = 31,300; 31,300 / 240,000 = 13.0%`	arithmetic on the two rows above	31,300 h/yr = 13.0% of baseline
8	Pooling (2 equal pools → 1)	`factor = 1 − 1/√2 = 1 − 0.707 = 0.293; $7.5M × 0.293 = $2.2M`	√2 pooling (toolkit label); $7.5M from Table 1 row 5	≈$2.2M released one-time
9	Zero-incident fence	`two staged go-lives — one defect class live at a time`	customer-facing incidents allowed: 0 (record constraint)	0 incidents through both

Table 3 — after: the metric, before → after.

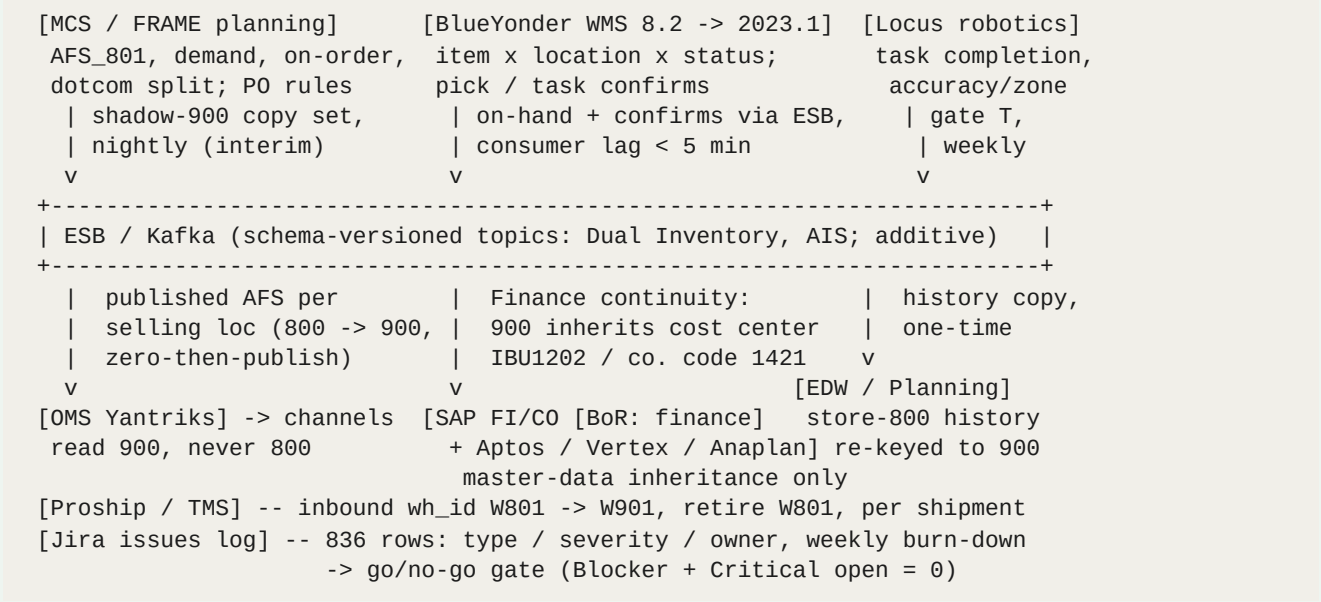
STEP	METRIC	BEFORE	AFTER	BASIS
1	Processing efficiency (units/labor-hour)	index 100	115	record (Sephora signed)
2	Direct labor hours, same volume	240,000 h/yr	≈208,700 h/yr	modeled on the record +15%
3	Self-transfers	≈260K units/yr	0	record design
4	Poolable safety stock	$7.5M	$5.3M ($7.5M − $2.2M)	modeled (√2 pooling)
5	Split-shipment rate	baseline on ~4.5M orders	−1 pt	modeled
6	Customer-facing incidents	0 allowed	0 through both go-lives	record

Table 4 — the dollars, every multiplication shown.

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
1	Labor at +15%	`31,300 h × $28/h = $876,400`	hours from Table 2 row 7; $28 loaded (modeled)	≈$0.88M/yr
2	Pooling release	`$7.5M × 0.293 = $2.2M`	Table 2 row 8 (one-time working capital)	≈$2.2M one-time
3	Carry on the release	`$2.2M × 10% = $220,000`	cost of capital (benchmark 8–12%)	≈$0.22M/yr
4	Split shipments	`45,000 × $10 = $450,000`	splits from Table 1 row 6; $10 per split (modeled)	≈$0.45M/yr

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
5	Self-transfer handling	`260,000 × $0.80 = $208,000; × 50% = $104,000`	50% — the other half already sits in the labor row (modeled)	≈$0.10M/yr
6	Total recurring	`0.88 + 0.22 + 0.45 + 0.10 = $1.65M`	rows 1, 3, 4, 5	≈$1.6M/yr
7	One-time working capital	`= row 2, stated separately`	safety-stock release	≈$2.2M

SOLUTION FLOW — THE SYSTEM & DATA MAP



SENSITIVITY — LOW / BASE / HIGH

INPUT	LOW	BASE	HIGH
Direct DC FTE (illustrative)	100	120	150
Loaded labor rate (benchmark)	$24/h	$28/h	$32/h
Labor $ saved	$0.63M	$0.88M	$1.25M
Cost of capital (benchmark)	8%	10%	12%
Carry $ saved on $2.2M release	$0.18M	$0.22M	$0.26M
Split rate reduction × cost per split	0.5 pt × $8	1 pt × $10	1.5 pt × $12
Split $ saved	$0.18M	$0.45M	$0.81M
Self-transfer handling (fixed share)	$0.1M	$0.1M	$0.1M
Total modeled $/yr	≈$1.1M	≈$1.6M	≈$2.4M

TOTAL IMPACT STATEMENT — SAY IT LIKE THIS

"Conservatively, about one point six million a year: roughly nine hundred thousand of labor, because the same volume takes thirteen percent fewer hours at plus fifteen percent efficiency; about two hundred thousand a year of carry on the two point two million of safety stock one pool releases versus two; about four hundred fifty thousand from fewer split shipments; and about a hundred thousand from the DC no longer shipping to itself. The plus fifteen percent and the zero incidents are measured and signed by Sephora; every dollar figure is my model on an illustrative hundred-thousand-units-a-week DC, and I will tell you which input each one rests on."

S18 — Sephora: RP → JDA WMS dual-DC migration — 0.5% tolerance, three mocks, and the RF-device miss (the last story)

Amazon LPs: Insist on the Highest Standards (backup — the published tolerance) · Deliver Results (backup) · Bias for Action (the rehearsed fallback fired) · Learn and Be Curious (never run a WMS migration before). **Crosswalk:** loop packet S15 — **§1.3 CONFLICT on the budget**: this record says ~$1.8M, ~3% under (~$54K), and ~$2.5M/*month* of slip exposure; the loop packet says ~$800K and ~$2.5M/*week*. This page is this packet's canon. *Never "400 devices failed" — ~100 of ~400.*

Honesty label. Full lifecycle owner, accountable to sponsor and Finance; I had never run a WMS migration and learned the floor in month one. Month markers are illustrative unless stated; the RF miss is mine — I scored the task by effort when impact was what mattered.

60-SECOND PITCH

"**Situation.** Two DCs on a legacy WMS, no documentation, an eight-month deadline, ~$1.8M budget, a $20M ROI pipeline gated behind go-live — about $2.5M a month of slip; I had never run a WMS migration. **Task.** Full lifecycle, accountable to sponsor and Finance. **Action.** Discovery: wrote the success criteria in week one; learned the floor in month one. Plan & Design: interface inventory with an owner and a test cell per contract; the 0.5% reconciliation tolerance published with Finance *before* mock 1. Build & Test: three mocks — 7% (units of measure, location names, order-status vocabularies) → 1.8% (in-transit timing) → 0.3%. Cutover: a 72-hour in-transit freeze; the legacy system read-only for 30 days; ~100 of ~400 RF devices failed the conversion — I invoked the rehearsed paper-pick fallback rather than roll back, and told leadership the training criterion slips 1.5 weeks. Hypercare: exit week 3 at under two blockers a day. Lesson: single-point dependencies get a what-if plan regardless of size. **Result.** Primary metric: inventory reconciliation variance at cutover — **7% → 1.8% → 0.3% against the 0.5% tolerance published with Finance**. $ impact: on time, ~3% under the ~$1.8M budget (~$54K), and the $20M ROI pipeline unblocked the next month — slip avoided, not banked."

SITUATION / TASK — WHERE IT OCCURRED, WHEN IT HAPPENED, WHY IT IS IMPORTANT

S — Two DCs on a legacy WMS; zero inherited documentation; a hard eight-month deadline with a $20M ROI pipeline of four programs gated behind go-live (~$2.5M/month of slip); budget ~$1.8M; and I had never run a WMS migration. Inventory accuracy across two DCs could not wobble at cutover.

T — Own the full lifecycle — discovery charter, design sign-off, the ERP ↔ WMS ↔ 3PL ↔ carrier interface inventory, UAT, cutover, hypercare — accountable to the sponsor and Finance.

The goal, written in the charter in week one: on the date, inside a published reconciliation tolerance, zero accuracy incidents. The risks on the register: undocumented interface surprises, conversion mismatch above tolerance, and 3PL/carrier freeze non-compliance — each with a trigger, a data-first mitigation and an owner; the one I missed: RF conversion sat on the log as a task, not a risk.

The metrics on the table, with their baselines: conversion mismatch — gated at the 0.5% tolerance signed with Finance before mock 1, with mock 1 opening at 7%; accuracy incidents — zero allowed; the date — eight months out, hard; the budget — ~$1.8M.

ACTION — WHAT I PERSONALLY OWNED, HOW I DID IT, WHO ELSE WAS INVOLVED

A — Five gated phases (team 14 core + 6 DC Ops; month markers illustrative unless stated):

1. **Discovery (month 1)** — sponsor charter written in week one (objectives, assumptions, success criteria); phase entrance/exit criteria; interface inventory v0 from message logs; the systems-impacted table with LOE per owner. *Exit bar:* charter signed by sponsor and Finance; every interface has a named owner; I could walk receive → pick → ship unaided.

2. **Plan & Design (months 2–3)** — design sign-off pack; interface inventory v1 with a test cell per contract; the conversion approach (masters → open orders → on-hand → in-transit as its own batch); the mapping-table standard (data, not code); **the 0.5% tolerance published with Finance before mock 1**. *Exit bar:* council sign-off; tolerance signed; stories slotted by business criticality × effort.
3. **Build & Test (months 4–7)** — build tracking; the QA demo schedule to DC Ops; the UAT entrance memo (function testing done, conversion build done, test data handed off, pre-prod + perf ready); three mock cutovers with written root causes; after the RF miss, the training plan and device readiness table. *Exit bar:* mock 3 ≤ 0.5% (0.3%) → written go-live recommendation; training sign-off (slipped 1.5 weeks, date held).
4. **Cutover & Go-Live (month 8)** — the cutover runbook; the 72-hour transfer freeze; conversion batches; post-load reconciliation; the legacy read-only switch; the go/no-go call. When ~100 of ~400 RF devices failed the conversion script, I invoked the rehearsed paper-pick fallback rather than roll back. *Exit bar:* post-load mismatch inside tolerance; ship confirms posting; zero accuracy incidents.
5. **Hypercare / Close (4 weeks planned)** — blockers/day read daily; the 30-day failback watch; budget closed; the exit memo with the single-point-dependency lesson. *Exit bar:* <2 blockers/day sustained → exit week 3; ~3% under budget; pipeline programs started the next month.

I wrote the charter in week one and learned the floor and the vendor docs in month one; the team was 14 core plus 6 DC Ops, with Finance co-owning the tolerance. Mine: the charter, the entrance/exit criteria, the interface inventory, the conversion approach, the mapping-table standard (data, not code), the tolerance, the go/no-go recommendation and the fallback call; the team built, converted, tested and ran the floor. The contribution: writing the success criteria before anyone asked — the date, the tolerance, zero accuracy incidents, the numeric hypercare exit — and the three-mock discipline read by cause class: 7% → 1.8% → 0.3%, each mock removing a named defect category.

The obstacles: no documentation — the interface inventory was built from message logs, every interface with a named owner and a test cell; my own inexperience — the floor walked unaided by the end of month one; and ~100 of ~400 RF devices failing at cutover — answered by the rehearsed paper-pick fallback, not a rollback, with the training criterion slipping 1.5 weeks while the date held. Stories were slotted by business criticality × effort, with UAT entrance criteria written in the charter — function testing done, conversion build done, test data handed off, pre-prod and perf ready — QA could fail a build, and DC Ops confirmed because QA had demoed during build; buy-in: sponsor and Finance signed the charter week one, so nobody negotiated the exit later. The decision challenged was rolling back when the RF devices failed — challenged because inventory was inside tolerance and the failure was device conversion, not data: the rehearsed fallback fired instead, and leadership heard it the same day with the slip priced and my name on the cause. The RF miss is mine: I scored the task by effort when impact was what mattered.

RESULTS — HOW SUCCESS WAS MEASURED: $ · # · % · TIME · CUSTOMER · QUALITY

R — Primary metric: mock variance **7% → 1.8% → 0.3% vs the 0.5% tolerance**; on time; ~3% under ~$1.8M (~$54K); the $20M pipeline unblocked the next month.

- **$** — ~3% under the ~$1.8M budget (~$54K, record); the $20M ROI pipeline unblocked the next month — ~$2.5M/month of slip avoided, not banked; ≈$45K modeled paper-pick fallback cost paid deliberately to hold the date.
- **#** — ~100 of ~400 RF devices failed the conversion script — the rehearsed paper-pick fallback fired; team 14 core + 6 DC Ops across two DCs (record).
- **%** — Mock variance 7% → 1.8% → 0.3% against the 0.5% tolerance published with Finance before mock 1 (record).
- **Time** — Eight-month deadline held; 72-hour in-transit freeze; legacy read-only for 30 days; training slipped 1.5 weeks while the date did not move; hypercare exited week 3 of 4 (record).
- **Customer** — Zero accuracy incidents at cutover; ship confirms posting; post-load mismatch inside tolerance (record exit bars).
- **Quality** — Mocks read by cause class, each removing a named defect category — units of measure, location names, order-status vocabularies, then in-transit timing; failback rehearsed, never used (record).

Against the goal stated above: on the date; 0.3% against the 0.5% tolerance published with Finance before mock 1; zero accuracy incidents; ~3% under the ~$1.8M budget; the $20M pipeline unblocked the next month. The trades: sequence, not scope — training slipped 1.5 weeks so the go-live date held, told to leadership the same day with the slip priced — and ≈$45K of modeled paper-pick productivity paid deliberately to protect the critical path.

PROBING — QUESTIONS, PHASE DEEP-DIVES, AND THE MATH

Challenge — Why did you choose this story to illustrate Insist on the Highest Standards?

Because the standard was published before it could be met: the 0.5% tolerance signed with Finance before mock 1, so the number belonged to Finance and me before the test. Mock 1's 7% was the price; mock 3's 0.3% was worth it — and the tolerance was never renegotiated.

Challenge — What other stories demonstrate Insist on the Highest Standards?

S2 — the robotics gate held at 94 vs 98 — is this LP's primary; S9's ≥95% valid-ASN cohort gate, held twice, is the third.

Challenge — Could you come up with an example that is more recent?

This is a Sephora record, and its discipline is what the current UDC program reuses — S2's three mock cutovers run against the same 0.5% tolerance. S2 is the most recent same-LP record.

Probing — Why did you choose to focus on these results? What other results were important?

The mock trajectory against the published tolerance — 7% → 1.8% → 0.3% vs 0.5% — is the record that made “go” a written recommendation rather than confidence in the room; on time, ~3% under budget and the $20M pipeline unblocked the next month are what the sponsor and Finance measured.

Probing — You mentioned dollars — what percentage change is that year over year?

Not YoY — the ~$54K is ~3% under the ~$1.8M budget, and the ~$2.5M/month is slip exposure avoided, not banked; the pipeline's value belonged to the four programs it gated.

Probing — I'm concerned about the $875K avoided-slip figure — tell me more.

It is record arithmetic on a slip that did not happen, so I never bank it — it is the price that made the fallback the obvious call. What I'd measure is the pipeline's first program start date against plan, which was the next month.

Challenge — What were the lessons learned? What would you have done differently?

Any single-point dependency in front of an exit criterion gets a what-if plan, regardless of estimated effort. Differently: dual-path device conversion on day one, and a device model × OS × script pass/fail table before training week.

Challenge — How would you implement this at Amazon?

Cutover discipline in its most literal form — a tolerance published with Finance before the first rehearsal, mocks read by cause class rather than headline, a freeze, a tested failback, and a numeric hypercare exit nobody can negotiate.

Challenge — How did these results compare to your actual goals?

The charter asked for the date, the tolerance and zero accuracy incidents — delivered on time, 0.3% against 0.5%, zero accuracy incidents, ~3% under budget, and hypercare exited week 3 of 4 under the charter's own blockers-per-day number.

PHASE DEEP-PROBE BANK — Q&A (13). Answer the phase asked, then stop.

Discovery — You had never run a WMS migration; how did you know discovery was done?

When I could walk receive → put-away → pick → ship on the floor unaided and every interface I saw in the message logs had a row, an owner and a test cell. That took month one; the charter had been written week one.

Who wrote the success criteria and who signed?

I wrote them in week one — on the date, inside a published reconciliation tolerance, zero accuracy incidents, hypercare exit under two blockers a day — and the sponsor and Finance signed the charter.

Plan & Design — What was in your risk register at design sign-off?

Undocumented interface surprises, conversion mismatch above tolerance, and 3PL/carrier freeze non-compliance — each with a trigger, a data-first mitigation and an owner. RF conversion was on the log as a task, not a risk; that was my miss.

Why publish the tolerance before you knew you could hit it?

So the number belonged to Finance and me before the test, not after. Mock 1's 7% was the price; mock 3's 0.3% was worth it.

Build & Test — How were UAT entrance criteria set and who could fail a build?

Function testing done, conversion build done, test data handed off, pre-prod and perf ready — written in the charter. QA could fail a build; DC Ops confirmed in UAT because QA had demoed to them during build.

Describe your bug-bash equivalent.

The three mocks. Everyone in one room reading the mismatch by cause class — units of measure, location drift, status vocabularies, then in-transit timing. Nothing left the room without a class, an owner and a data fix.

How did you read a mock?

Never by the headline number alone. Each delta got a root-cause class; if the total shrank without a named cause, the mock was not evidence. 7 → 1.8 → 0.3 each removed a category.

Cutover — What made you say go?

A written recommendation off mock 3 at 0.3% against 0.5%, the freeze agreed in writing, failback tested, training signed off — 1.5 weeks late, inside the window.

What were your rollback triggers and did any fire?

Post-go-live mismatch above 0.5% for two days, or ship-confirm posting failing beyond two hours — rollback per interface, the legacy system read-only for 30 days as the failback. Rehearsed; never used. The paper-pick contingency, also rehearsed, did fire — for the RF miss.

Hypercare — How did you decide hypercare was over?

Blockers per day under two, sustained — exited week 3 of 4. The number was in the charter before go-live, so nobody negotiated the exit.

What did you hand over at hypercare exit?

The interface inventory with owners and test cells, the mapping tables owned by DC Ops and ERP, the failback plan still armed for the rest of the 30 days, and the device readiness table. The RF lesson went into the sponsor read-out with my name on it.

Cross-phase — Where did you trade scope for date? What did you escalate?

Nowhere on scope — I traded sequence: training slipped 1.5 weeks so the go-live date held, and I told leadership the same day with the slip priced. Escalated the RF miss as a priced decision with my name on the cause. The rule I left: any single-point dependency in front of an exit criterion gets a what-if plan, regardless of size.

You'd never run a WMS migration — why you? · Do differently?

I wrote the success criteria before anyone asked and learned the floor and the vendor docs in month one; the mocks were where I proved it. Differently: dual-path device conversion on day one; a device model × OS × script pass/fail table before training week.

Where this lands at Amazon: the cutover discipline in its most literal form — a tolerance published with Finance before the first rehearsal, mocks read by cause class rather than headline, a freeze, a tested failback, and a numeric hypercare exit nobody can negotiate. The RF miss is the honest beat: a rehearsed contingency fired, the date held, and the lesson became a rule.

BUSINESS IMPACT MODEL — KPI TREE, NUMBERS, THE MATH

```
$20M ROI pipeline starts on time (record: ~$2.5M/month of slip at stake)
+- Go-live date held (8 months)
|    +- cutover variance 7% -> 1.8% -> 0.3% vs 0.5% tolerance
|    |    +- mapping tables as data (UoM, location, status) · 72-h freeze ·
|    |       in-transit as its own batch · tolerance signed BEFORE mock 1
|    +- RF miss ~100 of ~400 -> paper-pick fallback, not rollback;
|         training criterion slips 1.5 wk; the date does not move
+- Budget ~$1.8M, ~3% under (~$54K) (record)
+- Inventory $ in dispute at cutover = on-hand $ x variance (modeled)
(loop) mock read by cause -> data fix -> next mock lower -> Finance trust up ->
       tolerance never renegotiated
(loop) slip priced per day (~$83K) -> escalations decided same day ->
       critical path protected -> pipeline on time
```

DRIVER	BASELINE	AFTER	$	BASIS
Cutover variance (three mocks)	7%	1.8% → 0.3% vs 0.5%	—	record
Budget	~$1.8M	~3% under (~$54K)	—	record
$20M pipeline	gated	unblocked next month	~$875K critical-path slip avoided	record math (~$2.5M/mo) — avoided, not banked
Inventory in dispute at cutover	0.5% of $50M	0.3%	≈$0.1M one-time	modeled (illustrative $50M on-hand)
Paper-pick fallback cost	—	paid to hold the date	≈$45K	modeled

If challenged. *"Why not roll back when the RF devices failed?"* — "Because the fallback was rehearsed and the rollback wasn't needed: inventory was inside tolerance; the failure was device conversion, not data. I paid ~$45K of paper-picking to protect ~$2.5M a month of pipeline — and took the miss upstairs the same afternoon with my name on the cause."

THE MATH — STEP-BY-STEP DERIVATION

Table 1 — before: the problem in numbers.

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
1	Mock 1 variance	`M = Σ|Δ| / Σ ERP on-hand = 140,000 / 2,000,000 = 7.0%`	illustrative units on record percentages	7.0% — 14× the tolerance
2	Dominant cause — UoM	`a case booked as 12 eaches creates 12 − 1 = 11 units of error per case; factor 12 on 1% of SKUs`	record mechanism	one mapping class dominates the 7%
3	Second cause — in-transit timing	`transfers in flight at snapshot counted at neither end or both`	record mechanism	the timing class
4	The stakes	`$20M pipeline gated; $2.5M per month of slip`	record	every day priced

Table 2 — the fix: step by step, values substituted.

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
1	Tolerance signed first	`gate: M ≤ 0.5%, signed with Finance BEFORE mock 1`	never renegotiated (record)	the yardstick fixed in advance

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
2	UoM fix as data	`mock 2: M = 36,000 / 2,000,000 = 1.8%`	mapping tables (UoM, location, status) as data, not code (record % on illustrative units)	7.0% → 1.8%
3	Freeze + own batch	`freeze F = 72 h ≥ max lane transit; in-transit converted as its own batch with Σ legacy = Σ JDA; mock 3: M = 6,000 / 2,000,000 = 0.3%`	record % on illustrative units	1.8% → 0.3%
4	Gate evaluated	`0.3% ≤ 0.5% → go`	as a written recommendation (record)	go-live recommended
5	RF-device miss	`~100 / ~400 = 25% failed the conversion script → rehearsed paper-pick fallback, not a rollback`	training +1.5 wk; the date does not move (record)	go-live on the date
6	Hypercare exit	`blockers/day < 2 sustained — the charter's own number`	record	exited week 3 of 4

Table 3 — after: the metric, before → after.

STEP	METRIC	BEFORE	AFTER	BASIS
1	Cutover variance (three mocks)	7.0%	1.8% → 0.3% vs 0.5% tolerance	record
2	Go-live date	at risk from the RF miss	held — 8 months end to end	record
3	Budget	~$1.8M	~3% under	record
4	$20M pipeline	gated	starts on time	record

Table 4 — the dollars, every multiplication shown.

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
1	Budget result	`$1,800,000 × 3% = $54,000`	record	≈$54K under budget
2	Slip price	`$2,500,000 / 30 = $83,333/day`	$20M pipeline gated → ~$2.5M/month of slip (record math)	≈$83K/day
3	Critical-path slip avoided	`1.5 wk × 7 = 10.5 days; 10.5 × $83,333 = $875,000`	the training slip kept OFF the critical path (record math)	≈$875K — avoided, not banked
4	Fallback cost paid	`≈$45K deliberately; $875,000 / $45,000 ≈ 19 ≈ 20× cheaper than the slip it protected`	modeled	≈$45K
5	Dispute delta	`(0.5% − 0.3%) = 0.2%; × $50,000,000 = $100,000`	illustrative on-hand (modeled)	≈$0.1M one-time less in dispute

SENSITIVITY — LOW / BASE / HIGH

INPUT	LOW	BASE	HIGH
Average unit cost (illustrative)	$15	$25	$40
Paper-pick slowdown (benchmark)	15%	25%	40%
DC labor rate (benchmark)	$25/hr	$30/hr	$35/hr
Inventory dispute avoided (0.5% → 0.3%)	~$60K	~$100K	~$160K

INPUT	LOW	BASE	HIGH
Paper-pick fallback cost	~$23K	~$45K	~$84K
Total modeled $/yr	~$40K net	~$55K net	~$75K net

TOTAL IMPACT STATEMENT — SAY IT LIKE THIS

"The record numbers first: on time, about $54K under a $1.8M budget, and the $20M pipeline started the next month, where every month of slip was worth about $2.5M. When a quarter of the RF devices failed, the same slip on the critical path would have cost about $875K, so I paid roughly $45K of paper-pick productivity instead and told leadership the training criterion moved, not the date. The modeled part is smaller and I'll label it: landing at 0.3 percent against a 0.5 tolerance meant about $100K less inventory in dispute at cutover on an illustrative $50M on-hand, and at mock one's 7 percent that would have been $3.5M. The date, the budget, the variance and the pipeline are measured; the on-hand value and the fallback cost are my model."

If challenged — the full answers.

Where does $50M of on-hand come from?

It's illustrative, two million units at $25, chosen so the percentages become dollars; the record gives me 0.3 percent, not the balance sheet. I'd firm it up with the actual ERP on-hand valuation at the freeze snapshot, which Finance already had for the reconciliation.

The $875K avoided slip is a counterfactual.

Yes, it's record arithmetic on a slip that did not happen, so I never bank it; it's the price that made the fallback the obvious call. What I'd measure is the pipeline's first program start date against plan, which was the next month.

Paper-pick cost is a guess.

It's a benchmark, 15 to 40 percent slower on a hundred devices for a week and a half, so somewhere between $23K and $84K. I'd measure it from the WMS: picks per labor hour on the paper lanes versus the RF lanes during those ten days.

S3 — Sephora S/4HANA: Shipped-Not-Billed and de-aggregation — Finance and Operations on one Financial On-Hand number

Amazon LPs: Earn Trust (backup — Finance and Ops on one number) · Dive Deep (the grain problem) · Invent and Simplify. **Crosswalk:** not in the loop packet — safe anywhere. **Recency:** current role.

Honesty label. The flow, controls and ownership map were mine; S/4 and OMS engineers implemented. The record has no canonical dollar for the pre-state drift; the ≈$0.6M/yr is my model and I say so.

60-SECOND PITCH

"**Situation.** S/4 posts sales as aggregates while POS sends lines, so between sale and posting units were gone but the book was not decremented — Finance and operations reconciled by argument. **Task.** Define how the real-time layer tracks that gap so promising, replenishment and Finance agree, under a zero-incident constraint. **Action.** Discovery: profiled the posting-vs-line grain and named the phantom-availability window. Plan & Design: SNB incremented by POS/OMS/WMS events, decremented only on matched S/4 aggregate; de-aggregation pro-rates COGS per line; failure direction under-states, never over-states. Build & Test: Aggregation# join in the matrix; control owner per bucket. Cutover: publish Financial On-Hand and updated SNB back to S/4. Hypercare: unmatched postings age on a report, not as phantom AFS. Lesson: design the failure direction before the happy path. **Result.** Primary metric: Financial On-Hand definitions — **two (Finance book vs operations count) → one shared FOH**, availability decremented at sale/ship rather than at posting; zero incidents. $ impact: ≈$0.6M/yr modeled (phantom-availability cancels avoided + Finance reconciliation labor + month-end close time) — one FOH is the signed result; the record has no pre-state drift dollar and I am not claiming one."

SITUATION / TASK — WHERE IT OCCURRED, WHEN IT HAPPENED, WHY IT IS IMPORTANT

S — S/4 posts sales as aggregates — one posting per Aggregation# with total quantity and COGS; the Aptos POS sends transaction lines with an Order#. Between the sale and the posting, units are gone but the book is not decremented — the phantom-availability window — and Finance and Operations read different on-hand numbers and reconcile by argument.

T — Define how ISM tracks the gap in real time and reconciles it so promising, replenishment and Finance agree, with the program's zero-incident constraint applying to the book as much as to the customer.

The goal: one Financial On-Hand, AFS decremented at sale rather than at posting, with a safe failure direction. The challenges sat in the mechanics: the grain mismatch (one posting per Aggregation# vs transaction lines), a possible double-subtraction path, and unmatched postings accumulating silently. Left unfixed, the phantom-availability window stays open — units sold but still showing available — with two books reconciled by argument, a bridge spreadsheet at every month-end close, and the gap hiding as oversell if an estimate ever decremented the book.

The metrics on the table, with their baselines: Financial On-Hand truths — two; target one. The AFS decrement point — at the S/4 posting; target the sale/ship event. Over-sell from unmatched postings — a live risk in the window; target zero by construction, with the aging report empty or explained at T+1 as the exit bar.

ACTION — WHAT I PERSONALLY OWNED, HOW I DID IT, WHO ELSE WAS INVOLVED

A — Five gated phases (SNB rode the ISM program's dated chain — FSD signed 6/5/25; E2E 11/22 · 12/20 · 1/30; May execution go-live):

1. **Discovery** — problem statement (the phantom-availability window), the event inventory (POS flash sale, omni pending, ship-to-home confirm, S/4 Aggregation# posting), the who-reads-which-number map. *Exit bar:* Finance and Ops agree on the description of the gap and which events move units before posting; Internal Audit named as reviewer.
2. **Plan & Design** — SNB bucket definition (increment on events, decrement only on matched posting); join spec Aggregation# ↔ Aptos header ↔ lines; per-line COGS pro-rata de-aggregation; the failure-direction statement

(unmatched ⇒ SNB stays high); control-owner map into the FSD. *Exit bar:* signed inside the FSD 6/5/25; Finance accepts one FOH; WAC$ confirmed S/4-owned, ISM visibility-only.

3. **Build & Test** — de-aggregation test family (n lines; missing line; duplicate Aptos header); the SNB-never-negative assertion; unmatched-posting aging report spec. *Exit bar:* in SIT every posting matches or ages visibly; Σ per-line COGS = posting COGS to the cent in every cell; UAT = confirmation by Inventory Accounting.
4. **Cutover & Go-Live** — riding the May go-live: SNB opening balance reconciled to unposted sales at the snapshot; SNB publish back to S/4 enabled. *Exit bar:* aging report empty or explained at T+1; no negative SNB; Finance reads FOH from ISM.
5. **Hypercare / Close** — daily aging review with Inventory Accounting; control-owner handoff in the exit memo. *Exit bar:* aging report owned by Finance without me; month-end FOH ties to MB5L (mechanism canonical; days-to-tie not in record).

The flow, controls and ownership map were mine; S/4 and OMS engineers implemented, and Inventory Accounting confirmed in UAT and took over the aging report. Mine specifically: the SNB bucket definition, the join spec, the de-aggregation rule, the failure-direction statement, the control-owner map and the test families. The biggest contribution was designing the failure direction before the happy path — an unmatched posting keeps SNB high, so AFS under-states and Replenishment over-orders visibly instead of overselling invisibly — and making the aging report, with a named Finance owner, the real product.

The obstacles: Finance objected at design review — I showed the same number arriving from the physical side first and gave them the aging report to own, because a parallel spreadsheet dies when its owner owns the real report; the grain problem was solved by the Aggregation# ↔ Aptos header ↔ lines join and exact pro-rata COGS allocation, with the last line absorbing rounding so the sum ties to the cent. Work was sequenced by event source — POS flash sales, then omni pending, then ship-to-home confirmations — each added to SIT only when its increments and matched decrements reconciled; buy-in was FSC Inventory Accounting signing inside the FSD on 6/5/25, with UAT as confirmation because QA had demoed each case during build. Two decisions I refused: real-time per-line posting to S/4 — throughput and posting cadence are why S/4 posts aggregates; ISM is the real-time ledger and S/4 stays settlement authority — and a Finance-derived SNB, because that is two numbers and the whole point was one.

RESULTS — HOW SUCCESS WAS MEASURED: $ · # · % · TIME · CUSTOMER · QUALITY

R — Primary metric: FOH truths **2** → **1**; AFS decremented at sale/ship; over-sell from unmatched postings = 0 by construction; zero incidents. $ impact: ≈$0.6M/yr modeled.

- **$** — ≈$0.6M/yr modeled (phantom-availability cancels avoided + Finance reconciliation labor + month-end close time); the record has no pre-state drift dollar and I am not claiming one.
- **#** — Financial On-Hand truths 2 → 1 (record, FSD signed 6/5/25); over-sell from unmatched postings = 0 by construction.
- **%** — Design KPI: match rate ≥ 99.5% within 24h; the modeled phantom-cancel rate is ~0.10 pt of orders → ~0 (modeled, labeled).
- **Time** — AFS decrement point moved from the S/4 posting to the sale/ship event (record); rode the May execution go-live; aging report empty or explained at T+1 (record exit bar).
- **Customer** — The failure direction protects the customer: an unmatched posting under-states availability — under-sell, never over-sell (record mechanism).
- **Quality** — Σ per-line COGS = posting COGS to the cent in every cell; SNB never negative; month-end FOH ties to MB5L through the ISM number, not a bridge spreadsheet (record mechanism; days-to-tie not in record).

Against the goal stated above: FOH truths two → one; the decrement point moved from posting to sale/ship; over-sell from unmatched postings zero by construction; the aging report empty or explained at T+1 — with zero incidents. The deliberate trade: decrementing only on a matched posting moves cost to Replenishment — an unmatched posting under-states AFS and shows up as an aged row Finance clears the same day — small, visible, and priced that way on purpose.

PROBING — QUESTIONS, PHASE DEEP-DIVES, AND THE MATH

Challenge — Why did you choose this story to illustrate Earn Trust?

Because the result is Finance and Operations trusting one number — Financial On-Hand truths went from two to one, signed inside the FSD on 6/5/25. Finance wanted their own reconciliation; I gave them the real report to own instead — a parallel spreadsheet dies when its owner owns the real report.

Challenge — What other stories demonstrate Earn Trust?

S14 — the $2.5M discrepancy — is my primary Earn Trust story, and S1, one signed “available,” is the other backup from the same program family.

Challenge — Could you come up with an example that is more recent?

Current role — SNB rode the ISM program's dated chain to the May execution go-live, so this is among the most recent records I have for this principle.

Probing — Why did you choose to focus on these results? What other results were important?

FOH truths 2 → 1 is the headline because reconciliation-by-argument was the disease; the decrement point moving to sale/ship and over-sell at zero by construction are what the customer and the book feel. Also important: the aging report owned by Finance without me, and month-end tying to MB5L without a bridge.

Probing — You mentioned dollars — what percentage change is that year over year?

No YoY figure exists — the ≈$0.6M/yr is modeled (cancels avoided, reconciliation labor, close time) against the pre-state, and the record has no pre-state drift dollar; I am not claiming one. The record movement is two on-hand truths becoming one.

Probing — I'm concerned about the cancel rate — it looks borrowed from S1 — tell me more.

Different window: S1's rate is the vocabulary problem across node types; this is the posting-lag window inside one node. If the room wants one number, I keep S1's and drop this row — the mechanism still stands on the record.

Challenge — What were the lessons learned? What would you have done differently?

Design the failure direction before the happy path. Differently: Internal Audit in discovery, not review — the day Inventory Accounting asked me why a posting had aged before I noticed it, the control had an owner.

Challenge — How would you implement this at Amazon?

Every fulfillment event moves units before Finance posts them — the same window at a vastly larger grain. The transferable design: one real-time ledger, the financial system as settlement authority, a safe failure direction chosen before the happy path, and a control report with a named owner instead of a bridge spreadsheet.

Challenge — How did these results compare to your actual goals?

The signed criterion was one Financial On-Hand with AFS decremented at sale, not posting, and a safe failure direction — delivered: FOH 2 → 1, over-sell from unmatched postings zero by construction, aging report empty or explained at T+1, zero incidents.

PHASE DEEP-PROBE BANK — Q&A (14). Answer the phase asked, then stop.

Discovery — How did you know discovery was done?

When Finance and Operations agreed on the description of the gap — units gone, book not decremented, between sale and posting — and on the exact events that move units before posting. Agreement on the description came before any design.

Who wrote the success criteria and who signed?

I wrote "one Financial On-Hand for Finance and Operations; AFS decremented at sale, not posting; safe failure direction." FSC Inventory Accounting signed it inside the FSD on 6/5/25 with Strategy and Inventory Management.

Plan & Design — What was in your risk register at sign-off?

A double-subtraction path, unmatched postings accumulating silently, and late or duplicate POS lines. Mitigations: designed out by construction, an aging report Finance owns, idempotency keys with a late-arrival flag.

How did you handle Finance objecting at the design review?

They wanted their own reconciliation. I showed the same number arriving from the physical side first and gave them the report to own. A parallel spreadsheet dies when its owner owns the real report.

Why decrement only on a matched posting?

An unmatched posting keeps SNB high, so AFS under-states and Replenishment over-orders visibly. Decrementing on an estimate would hide the gap as oversell — the failure direction was designed before the happy path.

Build & Test — How did you sequence the stories?

By event source: POS flash sales, then omni pending, then ship-to-home confirmations — each added to SIT only when its increments and matched decrements reconciled. The de-aggregation family ran on every source.

What was your definition of done?

Matrix cell green, the "SNB never negative" assertion, and Σ per-line COGS equal to the posting COGS to the cent. A story without the failure-direction assertion was not done.

Who could fail a build in UAT?

Inventory Accounting. Their entrance criterion was "every SIT case ties to the cent"; they confirmed rather than discovered because QA had demoed each case during build.

Cutover — What was the cutover risk specific to SNB?

The opening balance — sales made but not yet posted at the snapshot. I reconciled SNB's opening balance to unposted sales before the publish switch; the T+1 aging report was the go/no-go read for the matching cycle.

What were your rollback triggers?

Aging report growing without explanation, or any negative SNB at T+1 — rollback was cheap because S/4 stayed settlement authority; ISM republishes SNB from the event log. Neither fired.

Hypercare — How did you decide hypercare was over?

When Finance ran the aging and Negative AFS reports without me and month-end FOH tied to MB5L through the ISM number, not a bridge spreadsheet. I don't have a canonical days-to-tie figure and won't invent one.

What did the first weeks teach you?

That the aging report is the real product — the day Inventory Accounting asked me why a posting had aged before I noticed it, the control had an owner. Do-differently: Internal Audit in discovery, not review.

Cross-phase — Where did you trade scope for date?

Real-time per-line posting to S/4 — never in scope, and I held that line. The price was a matching cycle to maintain; the benefit was one book.

Why not post every line to S/4? · Why not let Finance derive SNB themselves?

Throughput and posting cadence — S/4 posts aggregates for a reason; ISM is the real-time ledger, S/4 stays the settlement authority. And a Finance-derived SNB is two numbers; the whole point was one.

Where this lands at Amazon: every fulfillment event moves units before Finance posts them — the same window at a vastly larger grain. The transferable design: one real-time ledger, the financial system as settlement authority, a safe failure direction chosen before the happy path, and a control report with a named owner instead of a bridge spreadsheet.

BUSINESS IMPACT MODEL — KPI TREE, NUMBERS, THE MATH

```
One Financial On-Hand for Finance and Operations (primary outcome)
+- Oversell / cancel cost = orders x phantom-cancel rate x cost/cancel
|    +- window W = t_post - t_sale -> SNB increments at sale/ship, AFS decremented now
|    +- failure direction: unmatched => under-sell, never over-sell
|         <- aging report -> Finance fixes match -> SNB falls (loop, -)
+- Finance reconciliation labor $ = analyst hrs/wk x 52 x loaded rate
|    +- bridge spreadsheet retired; de-aggregation ties COGS per line to the cent
+- Month-end close hours = people x days x hrs x 12 -> no bridge at close
+- Replenishment noise (stock: over-orders on under-stated AFS)
     <- match rate >= 99.5% in 24h keeps this stock near zero (loop)
```

DRIVER	BASELINE	AFTER	$/YR	BASIS
Financial On-Hand truths	2 (Finance vs Ops)	1 shared FOH	—	record (FSD 6/5/25)

DRIVER	BASELINE	AFTER	$/YR	BASIS
AFS decrement point	at S/4 posting	at sale / ship	—	record
Phantom-availability cancels	~0.10 pt of orders	~0	≈$0.45M	modeled (30M × 0.10% × $15)
Finance reconciliation hours	32 hrs/wk bridge	~0	≈$0.14M	modeled (32 × 52 × $85)
Month-end close, inventory tie-out	+1 day bridge	ties to MB5L	≈$0.05M	modeled (6 × 8 h × $85 × 12)
Total modeled	—	—	**≈$0.6M/yr**	match rate ≥99.5% in 24h = design KPI

If challenged. *"The cancel rate is borrowed from S1 — double-counted?"* — "Different window: S1's rate is the vocabulary problem across node types; this is the posting-lag window inside one node. If the room wants one number, I keep S1's and drop this row — the mechanism still stands on the record." *"32 hours of Finance time — source?"* — "Benchmark, not record. The record is the mechanism: the bridge spreadsheet died when its owner owned the real report. A two-week time study on Inventory Accounting prices it properly."

THE MATH — STEP-BY-STEP DERIVATION

Table 1 — before: the problem in numbers.

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
1	Financial On-Hand truths	`truths = 2 (Finance's number vs Ops' number)`	reconciled by argument (record)	2
2	Phantom window	`W = t_post − t_sale ≈ 1 day on the overnight batch`	AFS decremented at posting, not at sale (record design problem)	sold units still shown available for up to a day
3	Exposure per SKU	`12 units/day × 1 day = 12 units`	illustrative sales rate (matches the posting example below)	up to 12 phantom-available units per SKU-day
4	Cancels it causes	`30,000,000 × 0.10% = 30,000/yr`	orders illustrative; rate modeled	≈30,000 phantom cancels/yr
5	Bridge labor	`32 h/wk of Finance analyst time`	the bridge spreadsheet between the two books (modeled)	32 h/wk
6	Month-end	`6 people × 8 h × 12 mo = 576 h/yr`	+1 day of close spent bridging (modeled)	576 h/yr

Table 2 — the fix: step by step, values substituted.

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
1	SNB ledger rule	`SNB += each sale/ship event; SNB −= matched posting only`	increments on POS / OMS / WMS events; decrement needs the Aggregation# match (record design)	walked on 12 units below
2	Walk — sales land	`12 units sold through the day → SNB 0 → 12; AFS falls at each sale`	illustrative	AFS honest all day
3	Window closed	`exposure = 12 units/day × 0 days = 0`	decrement moved from t_post to t_sale	phantom window = 0
4	De-aggregation	`COGS_i = $300 × q_i / 12: 3→$75.00 · 2→$50.00 · 4→$100.00 · 1→$25.00 · 2→$50.00`	posting q_P = 12, C_P = $300.00; five lines (illustrative); last line absorbs rounding	Σ = $300.00 to the cent

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
5	Matched posting	`12 posted with Aggregation# match → SNB 12 → 0`	illustrative	books tie
6	Late line	`one line (q = 2) late → SNB = 12 − 10 = 2`	illustrative	aged row visible on the 24h aging report
7	Failure direction	`unmatched ⇒ SNB stays high ⇒ AFS_reported ≤ AFS_true`	AFS = Basis − … − SNB (record design)	under-sell, never over-sell
8	Match-rate KPI	`matched postings / received ≥ 99.5% within 24h`	design target (record)	≥99.5%

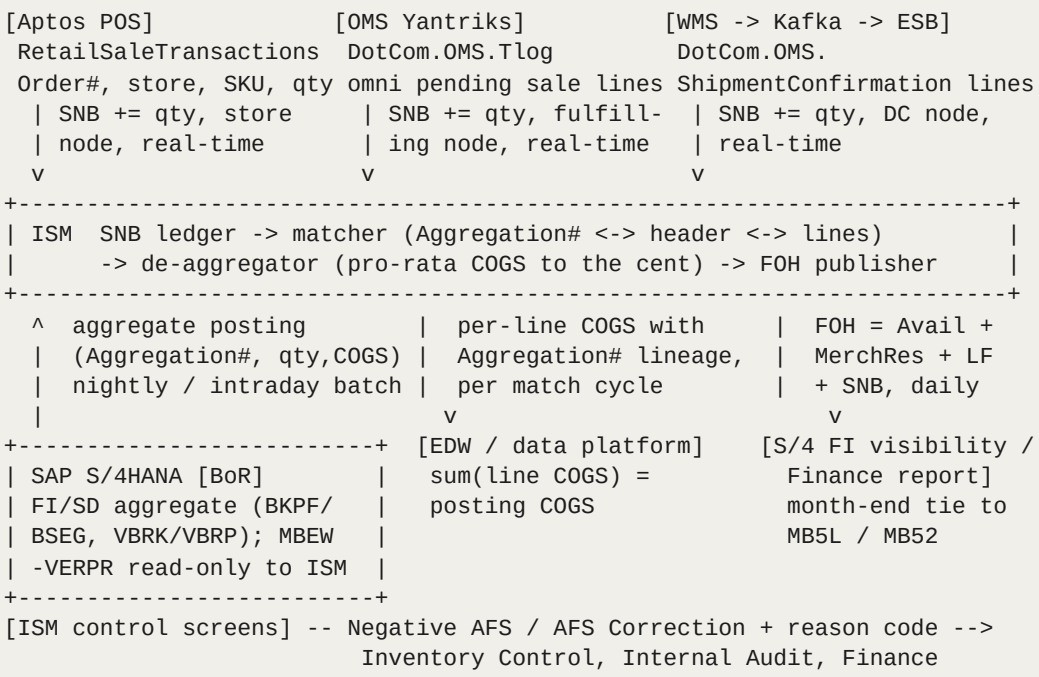
Table 3 — after: the metric, before → after.

STEP	METRIC	BEFORE	AFTER	BASIS
1	Financial On-Hand truths	2 (Finance vs Ops)	1 — signed inside the FSD 6/5/25	record
2	AFS decrement point	at S/4 posting	at sale / ship	record
3	Phantom-availability cancels	~0.10 pt of orders	~0	modeled
4	Finance bridge hours	32 h/wk	~0	modeled
5	Month-end close	+1 day bridge	ties to MB5L	record design

Table 4 — the dollars, every multiplication shown.

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
1	Phantom cancels avoided	`30,000,000 × 0.10% = 30,000; 30,000 × $15 = $450,000`	$15 all-in per cancel, range $10–20 (modeled)	≈$0.45M/yr
2	Reconciliation labor	`32 h/wk × 52 = 1,664 h; 1,664 × $85 = $141,440`	modeled	≈$0.14M/yr
3	Month-end close	`576 h × $85 = $48,960`	hours from Table 1 row 6 (modeled)	≈$0.05M/yr
4	Total	`0.45 + 0.14 + 0.05 = $0.64M`	all modeled; the record has no pre-state drift dollar	≈$0.6M/yr

SOLUTION FLOW — THE SYSTEM & DATA MAP



SENSITIVITY — LOW / BASE / HIGH

INPUT	LOW	BASE	HIGH
Phantom-cancel rate avoided (modeled)	0.05 pt	0.10 pt	0.15 pt
Cost per cancel (benchmark)	$10	$15	$20
Cancel $ avoided	$0.15M	$0.45M	$0.90M
Bridge hours per week (illustrative)	20	32	48
Loaded analyst rate (benchmark)	$75/h	$85/h	$95/h
Reconciliation $	$0.08M	$0.14M	$0.24M
Close days saved per month	0.5	1	2
Close $	$0.02M	$0.05M	$0.11M
Total modeled $/yr	≈$0.25M	≈$0.6M	≈$1.2M

TOTAL IMPACT STATEMENT — SAY IT LIKE THIS

"Conservatively, about six hundred thousand a year: roughly four hundred fifty thousand of cancels that no longer happen because availability drops at the sale instead of at the posting; about a hundred forty thousand of Finance hours that were going into a bridge spreadsheet; and about fifty thousand a year of month-end close time. What is measured is that Finance and Operations went from two on-hand numbers to one, decremented at sale or ship, with zero incidents and the failure direction pointing at under-sell. Every dollar is my model on an illustrative thirty-million-order scale; the record has no pre-state drift dollar and I am not claiming one."

If challenged — the full answers.

The cancel number is three quarters of the total and you say the record has no drift figure.

Right, so it is the hardest number. It rests on a tenth of a point of orders being cancelled for phantom availability before the fix. What I would measure is the cancel-reason code for out-of-stock-after-order, per week, before and after the May go-live, times Digital's own cost per cancel; whatever that shows replaces my assumption.

Finance hours are soft, nobody logs spreadsheet time.

Agreed. Four people at eight hours a week is what Inventory Accounting described, not what a timesheet says. I would firm it up by counting the manual journal entries and bridge lines per month before and after; that is in the posting log and it is not opinion.

Isn't the safe failure direction just moving cost to Replenishment?

Yes, and I priced it that way on purpose: an unmatched posting under-states AFS and shows up as an aged row Finance clears the same day, so the over-order is small and visible. The measure is the aging report's p95 in hours and the match rate inside twenty-four hours; if match rate holds above ninety-nine and a half, the Replenishment side rounds to zero.

S4 — Sephora S/4HANA: the in-transit promise rule I got wrong — fixed through the signed policy, not a side configuration

Amazon LPs: Are Right, A Lot (the failure question — third) · Customer Obsession (backup) · Earn Trust (owning the miss) · Learn and Be Curious. **Crosswalk:** not in the loop packet — safe anywhere. The designated *policy-failure* story. **Recency:** current role.

Honesty label. The v1 rule was mine; the indictment came from my own program's reports. Lane σ ≈ 0.3 day and the ~0.49-day window are the record's promise math; the record carries no lane counts and no dollar for the conversion lost while I was wrong — anything of that kind is labeled illustrative, and I will not invent the loss figure.

60-SECOND PITCH

"**Situation.** My first ISM ruleset barred every channel from promising units in transit; Aging In-Transit showed dotcom starved of fast movers one day from the building. **Task.** Correct my own policy without weakening oversell protection, through the same three signatories. **Action.** Discovery: conversion-loss evidence from the aging and replenishment reports — that was the entry bar to reopen the policy. Plan & Design: promise math — lane σ ≈ 0.3 day, ~0.49-day window ≈ 95% coverage; Promisable = in-transit × 1[ETA ≤ window] × 1[on-time ≥ floor]; Aging In-Transit as kill switch. Build & Test: back-test against receiving history; regression through the node × channel matrix. Cutover: released on the normal monthly cadence as a signed FSD change, not a side config. Hypercare: kill switch armed; never fired. Lesson: both sides of a rule get a number before it ships. **Result.** Primary metric: sellable units on qualifying in-transit lanes — starved → **restored inside a per-lane promise window, oversell increase = zero**, kill switch never fired. $ impact: ≈$0.5M/yr modeled margin recovered — the record has no dollar for the conversion lost while I was wrong, and I will not invent one."

SITUATION / TASK — WHERE IT OCCURRED, WHEN IT HAPPENED, WHY IT IS IMPORTANT

S — In the first ruleset I authored for the ISM availability model, no channel could promise against units in transit. It felt safe. The Aging In-Transit and Replenishment reports showed dotcom starved of fast-moving units reliably one day from the building, with conversion loss on those SKUs.

T — The policy was mine and I had written it conservatively; the task was to correct it without weakening oversell protection, and to take the miss back through the same signatories.

It mattered because both blanket answers were wrong — “always counts” oversells unreliable lanes, “never counts” strands good units. Left standing, the rule kept starving dotcom of fast movers with conversion loss; fixed carelessly, the risks were oversell on an unreliable lane, aging shipments getting promised, and on-hold receipts being counted — each given an automatic mitigation and a named owner.

The metrics on the table, with their baselines: dotcom availability of fast movers in transit — starved, reliably one day from the building; target sellable inside a defensible window. Oversell — zero under v1; the constraint was an increase of zero. The fix itself — to be released as a signed FSD change on the normal monthly cadence, not a hotfix.

ACTION — WHAT I PERSONALLY OWNED, HOW I DID IT, WHO ELSE WAS INVOLVED

A — Five gated phases:

1. **Discovery** — built the evidence pack from the Aging In-Transit and Replenishment reports (fast movers reliably one day out, conversion loss), stated that the v1 rule was mine, reframed the problem as promise math rather than "make in-transit count." *Exit bar:* Dotcom, Inventory Planning and I agree the undersell is real and is the rule's doing.
2. **Plan & Design** — wrote the rule — Promisable = In-transit × 1[ETA ≤ window(lane)] × 1[on-time ≥ floor], window per lane from receiving history, Aging In-Transit as kill switch — rejected both blanket options in writing, carried it as an FSD change. *Exit bar:* signed by the same three signatories as v1.

3. **Build & Test** — owned the back-test of the ruleset against receiving history, four regression cells added to the matrix (qualifying lane / non-qualifying lane / aging breach / on-hold receipt), the release note. *Exit bar:* back-test shows no oversell on qualifying lanes; matrix green; QA gate passed.
4. **Cutover & Go-Live** — released on the normal cadence with the lane qualification list published and monitor thresholds set — no hotfix, no side configuration. *Exit bar:* sellable units restored on qualifying lanes; zero oversell traced to the window.
5. **Hypercare / Close** — kill-switch watch (aging per lane), oversell watch, close-out to the signatories with the miss named as mine plus the mechanism note. *Exit bar:* kill switch never fired; lesson institutionalized — the Locus 98% gate (S2) and the 0.5% tolerance (S18) were both set from data before the event.

The fix was mine end to end — I built the evidence pack, wrote the corrected rule, owned the back-test against receiving history, and carried it as an FSD change through the same three signatories, with the four regression cells mine as well; Dotcom and Inventory Planning agreed the undersell was real and the rule's doing; the signatories re-signed; Inventory Control owns the standing kill-switch review weekly. The contribution that made re-signing possible: naming the miss as mine before proposing a fix, and reframing the problem as promise math rather than "make in-transit count."

The obstacle was "don't touch a signed policy" — answered with the undersell number and the promise math, then changed through the same signatures rather than a side configuration; the design risks — oversell on an unreliable lane, aging shipments, on-hold receipts — got the on-time floor, the kill switch and the reserve logic. The release went lane by lane, not all-or-nothing — a lane whose on-time rate sat below the floor stayed unpromisable, and Dotcom accepted because the alternative was all-or-nothing; buy-in came from the evidence pack and from rejecting both blanket options in writing. The decision challenged was my own signed policy — and I refused to ship the fix as a hotfix or side configuration: the error was in the policy, so the fix had to be too, in the governed artifact with regression.

RESULTS — HOW SUCCESS WAS MEASURED: $ · # · % · TIME · CUSTOMER · QUALITY

R — Primary metric: starved lanes → restored inside the window; **oversell increase = 0**; kill switch never fired. $ impact: ≈$0.5M/yr modeled.

- **$** — ≈$0.5M/yr modeled margin recovered (inputs illustrative: 300 starved units/day × 60% conversion × 40% lost-sale share × $20 margin); the record carries no dollar for the conversion lost while the v1 rule stood, and I will not invent one.
- **#** — Four regression cells added to the matrix — qualifying lane / non-qualifying lane / aging breach / on-hold receipt (record).
- **%** — Lane σ ≈ 0.3 day → a ~0.49-day window ≈ 95% coverage on qualifying lanes (record promise math); oversell increase = 0 (record).
- **Time** — Released on the normal monthly cadence as a signed FSD change — no hotfix, no side configuration (record).
- **Customer** — Starved fast movers one day from the building → sellable inside the window; oversell increase = zero (record).
- **Quality** — Back-test against receiving history before release — the step v1 skipped; kill switch armed, never fired (record).

Against the goal stated above: starved lanes restored to sellable inside a back-tested half-day window; oversell increase zero; the kill switch armed and never fired — re-signed by the same three signatories as v1. The trades were priced: the window accepts a per-lane oversell exposure (~95% coverage, the on-time floor, the kill switch) in exchange for restoring sellable units, lanes below the floor stayed unpromisable, and speed was traded for governance — the normal release train, not a hotfix.

PROBING — QUESTIONS, PHASE DEEP-DIVES, AND THE MATH

> **Challenge** — **Why did you choose this story to illustrate Are Right, A Lot?**

Because the LP is about being right through data, not never being wrong — my own program's reports indicted my rule, and I corrected it with promise math (lane σ ≈ 0.3 day, a ~0.49-day window ≈ 95% coverage) through the same signatures. It is my designated policy-failure story.

Challenge — What other stories demonstrate Are Right, A Lot?

S10 is the primary — “the noise said carriers; the data said measurement” — and S8, measured lanes vs brochure, is the backup.

Challenge — Could you come up with an example that is more recent?

This is current-role Sephora — the fix released on the ISM program's normal monthly cadence — so it is a recent record, and the lesson it produced set the 98% robotics gate (S2) and the 0.5% tolerance (S18).

Probing — Why did you choose to focus on these results? What other results were important?

Starved lanes restored with oversell increase = 0 is the pair that matters — the fix had to recover sales without weakening protection, and the kill switch never firing is the proof. The institutional result matters as much: both sides of every rule now get a number before it ships.

Probing — You mentioned dollars — what percentage change is that year over year?

No YoY — the ≈$0.5M/yr is a modeled recovery run-rate on illustrative volumes, and the record carries no dollar for what my first rule cost while it stood, so I do not claim one. The record movement is starved → restored with oversell at zero.

Probing — I'm concerned about the 300 starved units a day — tell me more.

Correct — it is my hardest input; it assumes the Aging In-Transit report's pattern held across the qualifying lanes every day. To firm it up I would pull the daily count of units within the window on qualifying lanes for the first quarter after the release; that count is in the report, and it replaces my three hundred.

Challenge — What were the lessons learned? What would you have done differently?

Both sides of a rule get a number before it ships — the 98% robotics gate (S2) and the 0.5% tolerance (S18) were both set from data before the event because of this. Differently: back-test against history before publishing v1 — one week of measurement would have skipped the starvation.

Challenge — How would you implement this at Amazon?

Inventory in transit between FCs, or inbound from a vendor, is exactly this promising decision at network scale — and this is the failure answer every Bar Raiser wants: a real policy error, named as mine to the same signatories, fixed with math, institutionalized.

Challenge — How did these results compare to your actual goals?

The task was to correct the policy without weakening oversell protection — delivered: sellable units restored inside a back-tested half-day window, oversell increase zero, kill switch never fired, signed by the same three signatories as v1.

PHASE DEEP-PROBE BANK — Q&A (13). Answer the phase asked, then stop.

Discovery — How did you know this was a policy defect and not a data defect?

The reports showed fast movers reliably one day out with conversion loss — the lanes were on time; the rule forbade promising against them. Data was fine; my rule was the cause.

Who did you tell first?

The three signatories of the v1 rule, with my name on the miss, before proposing a fix. Owning it first is what made re-signing possible without a fight.

Plan & Design — What was in your risk register for the fix?

Oversell on an unreliable lane, aging shipments getting promised, on-hold receipts being counted. Each had an automatic mitigation — floor exclusion, kill switch, reserve logic — and a named owner.

How did you handle "don't touch a signed policy"?

By showing the undersell number and the promise math: σ about 0.3 day, a ~0.49-day window buys ~95% coverage. Then I changed the policy through the same signatures rather than a side configuration.

Why reject both blanket answers?

"Always counts" oversells unreliable lanes; "never counts" strands good units. Both sides of a rule get a number now — that was the lesson.

Build & Test — What was your definition of done?

Back-test against receiving history attached, regression cells green for all four cases, QA's gate passed. No back-test, no QA entry.

How did you test it without a bug bash?

A rule change doesn't need a room; it needs history. I back-tested against receiving history first — the step I had skipped in v1 — then ran the matrix.

Cutover — Why the normal release and not a hotfix?

A policy fix belongs in the governed artifact with regression. A hotfix would have made the fix as untraceable as the error.

What were your rollback triggers?

Any oversell traced to the window, or aging above threshold on a qualified lane — rollback is per lane: window to zero. Neither fired.

Hypercare — How did you decide it was closed?

One release cycle with the kill switch silent and no oversell increase, then a one-page close-out to the signatories. The lesson outlived the fix: the 98% robotics gate and the 0.5% tolerance were both set from data before the event.

What stayed on the RAID log after close?

One standing row: the kill-switch threshold per lane, owned by Inventory Control, reviewed on the normal cadence. A rule that qualifies lanes automatically needs one human who reads the aging report every week.

Cross-phase — Which wave did you hold? What did you refuse?

Lanes, not waves — a lane whose on-time rate sat below the floor stayed unpromisable; Dotcom accepted lane-by-lane because the alternative was all-or-nothing. Refused: shipping it as a side configuration. The error was in the policy; the fix had to be too.

Why didn't you catch it in design? · What changed in how you work?

I optimized for the defect I feared (oversell) and never priced the one I couldn't see (undersell). Now both sides of every rule get a number, and I test a ruleset against history before publishing — one week of measurement would have skipped the starvation.

Where this lands at Amazon: inventory in transit between FCs, or inbound from a vendor, is exactly this promising decision at network scale. And the failure answer every Bar Raiser wants: a real policy error, named as mine to the same signatories, fixed with math, institutionalized as "both sides of a rule get a number."

BUSINESS IMPACT MODEL — KPI TREE, NUMBERS, THE MATH

```
Dotcom margin on fast movers (primary outcome)
+- Conversion recovered $ = starved units x conversion x lost-sale share x unit margin
|    +- Promisable = InTransit x 1[ETA <= w] x 1[OTR >= floor]
|    +- w = z x sigma_lane = 1.645 x 0.3 d ~ 0.49 d  [record]
+- Oversell cost = late units x P(late) x cancel/expedite cost
|    +- OTR floor excludes unreliable lanes · kill switch drops a lane
|         <- aging > k x p90 -> lane out -> planner reviews -> back in (balancing loop)
|    result: oversell increase = 0, kill switch never fired  [record]
+- Governance: FSD change via same 3 signatories, back-test + matrix regression,
     normal release train -> every later rule gets both sides priced (reinforcing loop)
```

DRIVER	BASELINE	AFTER	$/YR	BASIS
Sellable units on qualifying in-transit lanes	starved (excluded)	restored inside window	—	record
Lane σ and promise window	σ ≈ 0.3 d	w ≈ 0.49 d ≈ 95% cover	—	record (promise math)
Oversell increase from the window	—	0	$0	record (kill switch never fired)

DRIVER	BASELINE	AFTER	$/YR	BASIS
Conversion recovered	$0	≈$1,440/day	≈$0.5M	modeled (300 × 0.6 × 0.4 × $20 × 365; inputs illustrative)
Under-sell cost while the v1 rule stood	unknown	—	—	record: no $ in the record; **not claimed**

If challenged. *"What did the error cost?"* — "The record has no dollar for it and I won't invent one. What the record has: fast movers starved one day from the building, for weeks, by my rule. The $0.5M/yr is the recovery run-rate, modeled, labeled." *"Why believe the window is safe?"* — "Because the back-test ran before release — the step v1 skipped — and the kill switch plus on-time floor make the failure direction per-lane and reversible. Zero oversell traced to the window is record."

THE MATH — STEP-BY-STEP DERIVATION

Table 1 — before: the problem in numbers.

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
1	The v1 rule	`Promisable = 0 for ALL in-transit inventory`	equivalent to claiming σ → ∞ or on-time rate → 0 — infinite variability (record)	everything in transit hidden
2	One concrete lane	`Promisable = 120 × 0 = 0`	lane with InTransit = 120 units, ETA 0.4 day away (illustrative)	120 sellable units hidden
3	Across qualifying lanes	`≈300 fast-mover units/day × 0 promisable`	illustrative volume	≈300 units/day sat unsellable one day from the building
4	What starving costs (one SKU)	`100 × 60% × $20 = $1,200/day`	100 units one day out; 60% would convert; $20 margin (illustrative, not program data)	$1,200/day under-sell
5	The ignored other side	`100 × (1 − 0.97) = 3 late units; 3 × $15 = $45/day`	on-time rate 97%; $15 expedite (illustrative)	$45/day over-sell exposure — vs $1,200: the “safe” rule priced one side only

Table 2 — the fix: step by step, values substituted.

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
1	Measure lane variability	`σ = 0.3 day`	trailing receiving history (record)	σ = 0.3 day
2	Choose coverage	`z at 95% = 1.645`	normal approximation on lane transit (toolkit label)	z = 1.645
3	Promise window	`w = z × σ = 1.645 × 0.3 = 0.4935`	rows 1–2	w ≈ 0.49 day
4	Coverage check	`Φ(w / σ) = Φ(0.49 / 0.3) = Φ(1.63) ≈ 0.95`	share of arrivals inside the window	≈95% covered
5	Corrected rule	`Promisable = InTransit × 1[ETA ≤ w] × 1[OTR ≥ floor]`	OTR = on-time rate over trailing receipts; floor = per-lane admission threshold, 95% here (floor illustrative)	lane by lane, not all-or-nothing
6	Lane that qualifies	`ETA 0.4 ≤ 0.49 → 1; OTR 97% ≥ 95% → 1; Promisable = 120 × 1 × 1 = 120`	the Table 1 lane, re-evaluated (ETA and OTR illustrative)	its 120 units become promisable
7	Lane that fails	`ETA 1.2 > 0.49 → 0; Promisable = 80 × 0 = 0`	a slower lane with 80 units in transit (illustrative)	stays at 0 — correctly out

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
8	Sample-size gate	`n = z²p(1−p)/e² = 3.84 × 0.95 × 0.05 / 0.05² = 3.84 × 0.0475 / 0.0025 ≈ 73`	OTR estimated within ±5% at 95% CI (illustrative)	a lane qualifies after ~73 receipts
9	Kill switch, values in	`trigger = 2 × p90 = 2 × 2.0 = 4.0 days; any unit in transit > 4.0 days → lane out`	lane p90 = 2.0 days (illustrative); armed, reviewed weekly by Inventory Control (record)	never fired (record)
10	Back-test	`rule replayed against receiving history; 4 regression cells`	before release (record)	no oversell in the back-test

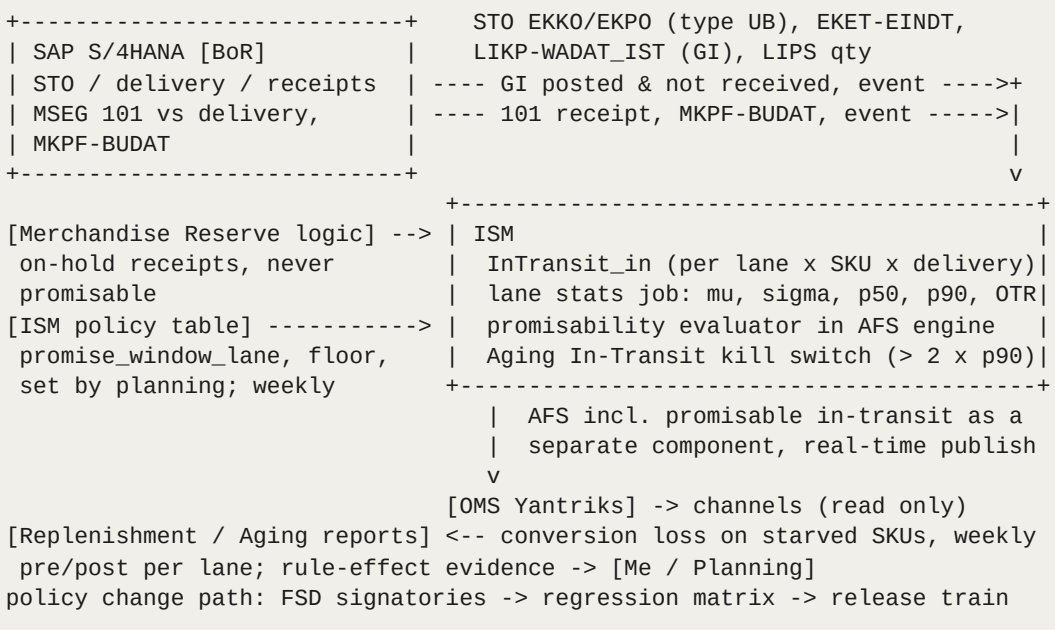
Table 3 — after: the metric, before → after.

STEP	METRIC	BEFORE	AFTER	BASIS
1	Sellable units on qualifying in-transit lanes	0	≈300 units/day restored inside the window	record mechanism; volume illustrative
2	Promise window	none — blanket exclusion	w ≈ 0.49 day at ~95% cover	record (promise math)
3	Oversell increase from the window	—	0	record
4	Kill switch	not armed	armed, never fired	record
5	Rule changes after	ad hoc	same 3 signatories, back-test + matrix regression	record — both sides priced from then on

Table 4 — the dollars, every multiplication shown.

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
1	Would-be sales	`300 × 60% = 180 sales/day`	units restored (illustrative) × conversion (illustrative)	180 sales/day
2	Truly lost without the fix	`180 × 40% = 72 sales/day`	40% lost-sale share — the rest would have waited (illustrative)	72 lost sales/day
3	Margin recovered	`72 × $20 = $1,440/day`	$20 unit margin (illustrative)	≈$1,440/day
4	Annualized	`$1,440 × 365 = $525,600`	modeled; the mechanism is record	≈$0.5M/yr
5	Cost while v1 stood	`no dollar in the record — not claimed`	record	not claimed

SOLUTION FLOW — THE SYSTEM & DATA MAP



SENSITIVITY — LOW / BASE / HIGH

INPUT	LOW	BASE	HIGH
Starved fast-mover units/day (illustrative)	150	300	500
Conversion on a shown unit (illustrative)	60%	60%	60%
Lost-sale share (benchmark)	30%	40%	50%
Unit margin (illustrative)	$15	$20	$25
Oversell increase (record)	$0	$0	$0
Total modeled $/yr	≈$0.15M	≈$0.5M	≈$1.4M

TOTAL IMPACT STATEMENT — SAY IT LIKE THIS

"Conservatively, about half a million a year of margin recovered: roughly three hundred fast-mover units a day that were sitting one day from the building and unsellable, of which about sixty percent would have converted and about forty percent of those were truly lost rather than substituted, at about twenty dollars of margin each. What is measured is that those units became sellable inside a back-tested half-day window and that oversell did not move, the kill switch never fired. The dollars are my model on illustrative volumes, and the record carries no figure for what my first rule cost while it stood, so I do not claim one."

If challenged — the full answers.

Three hundred units a day is your number, not the report's.

Correct, it is the hardest input. It assumes the Aging In-Transit report's pattern held across the qualifying lanes every day. To firm it up I would pull the daily count of units within the window on qualifying lanes for the first quarter after the release; that count is in the report, and it replaces my three hundred.

How do you know those sales were lost and not delayed a day?

I do not, which is why I only take forty percent of them, the middle of the published range for a stockout turning into a lost sale. The measure is conversion on the affected SKUs the week before and the week after the release, from

Digital's own funnel; if it moved less than that, the number shrinks with it.

You net oversell at zero, is that convenient?

It is the record: zero oversell incidents traced to the window and a kill switch that never fired. But I would keep measuring it, because a lane's on-time rate can decay. The standing check is the aging report per lane against two times its ninetieth percentile, owned by Inventory Control weekly.

S5 — Sephora: the architecture council without owning any system — three decisions and a fence

Amazon LPs: Have Backbone; Disagree & Commit (third) · Earn Trust (influence without authority) · Think Big (architecture governance). **Crosswalk:** not in the loop packet — safe anywhere. **Recency:** current role. *Same program family as S1/S2 — the governance story; don't pair with either for one interviewer.*

Honesty label. I owned none of the systems; the design decision was the deliverable. Zero escalations and the dated gate chain are the record; the dollars are my model of what governance avoided.

60-SECOND PITCH

"**Situation.** The UDC location and inventory model had to clear architecture council — a challenge gate — with ~80 named participants across ~15 functions, and I owned none of those systems. **Task.** Take the design from discovery through council to sign-off and FSD without a surprise escalation. **Action.** Discovery: both inventory books profiled before any design argument; business sessions closed 4/30. Plan & Design: every impacted row pre-wired with signed LOE (7/2); council asked for three separable decisions plus an out-of-scope fence. Build & Test: the 'build' was the pack — architecture, staged timeline, peak-freeze boundary. Cutover: 5/15 challenge → 7/31 sign-off → 8/14 FSD. Hypercare: fence held in writing for the whole program; a second live warehouse became a monitored shadow copy, not an argument. Lesson: a no that lives in a document survives. **Result.** Primary metric: surprise escalations on a 60+ stakeholder design — the norm on prior programs → **zero**; the dated gate chain held. $ impact: ≈$0.5M one-time modeled (avoided rework on ~3 escalations that never happened + six weeks of benefit not deferred + a fenced second reserve engine not built)."

SITUATION / TASK — WHERE IT OCCURRED, WHEN IT HAPPENED, WHY IT IS IMPORTANT

S — The UDC location and inventory model had to clear the architecture council — a challenge gate, not a formality — with a dozen core systems (20+ touched) and ~80 named participants across ~15 functions on the impact table, and I owned none of those systems.

T — Take the design from discovery (kickoff 3/29, sessions 4/30) through the council (5/15) to sign-off (7/31) and FSD (8/14) without a surprise escalation.

It mattered because the prior norm was designs this size being re-litigated. The register's top three risks: surprise LOE in the room, a second live warehouse becoming a second truth, and scope re-litigated after sign-off — each with a mitigation: the private pre-wire with the 7/2 LOE deadline, the monitored shadow copy, and the fence.

The metrics on the table, with their baselines: surprise escalations — the recognized norm for designs this size; target zero for the program's life. Pre-wire coverage — none existed at kickoff; target every impacted row with a signed LOE before the council. Scope — two written refusals (reserve math untouched, MCS decommission out of scope) that had to hold without a fence breach.

ACTION — WHAT I PERSONALLY OWNED, HOW I DID IT, WHO ELSE WAS INVOLVED

A — Five gated phases — here the phases are governance milestones; the cutover is the council decision itself:

1. **Discovery (3/29 → 4/30)** — profiled both books (SKU overlap, order profiles); sessions per function with the profile on the table — "design" deliberately not argued yet; systems-impacted table with a provisional owner on every row; stakeholder map (~80 names / ~15 functions). *Exit bar:* every impacted team in a session, every row owned.
2. **Plan & Design → council (5/15)** — council pack as three separable decisions (three planning identities over one location; January planning / May execution timeline; peak-freeze boundary) plus the out-of-scope fence (reserve math untouched; MCS decommission belongs to Global ERP); every owner pre-briefed privately. *Exit bar:* council challenge held 5/15; decisions taken or dated; no surprise objection.

3. **Build & Test → LOE (7/2)** — one LOE template to one deadline across every impacted owner; each open concern converted into design or a dated RAID item (the second-live-warehouse concern → a monitored shadow copy). *Exit bar:* LOE signed 7/2 by every impacted owner; no owner learns their cost after this date.
4. **Cutover → sign-off (7/31), FSD (8/14)** — assembled the sign-off pack (decisions, fence, LOE, risks); published FSD 8/14 with discovery results attached. *Exit bar:* signed 7/31; FSD 8/14; fence lines in writing.
5. **Hypercare / Close** — enforced the fence (scope requests answered from the pack, in writing); applied the peak-freeze boundary Oct–Dec; handed the program into the build (S2). *Exit bar:* zero surprise escalations for the program's life; both scope lines held in writing.

The design decision itself was the deliverable. Mine: the profiling of both books, the systems-impacted table, the council pack, the three-decision structure, the fence, the LOE campaign and the written scope answers afterwards; each system owner produced and signed their own LOE, and the council took the three decisions. The biggest contribution was the pre-wire — one LOE template, one deadline, one private conversation per owner before the council, so nobody learned their cost in the room — against a re-litigation norm everyone in that building recognized.

The obstacle was getting sixty-plus people to sign LOE I didn't control — solved by the template, the deadline and the private conversations, with signatures landed 7/2; the “second truth” objection was converted into the design as a monitored shadow copy, because converting a legitimate concern costs less than winning it. Priorities ran profiling-first — their numbers, not my opinion — with design deliberately not argued in the sessions; buy-in was manufactured before the room: every row owned, every owner pre-briefed privately, and the council asked for three separable decisions it could take one at a time. I challenged the “approve the design” bundle itself — separability was the design — and refused retuning the reserve math and folding in the MCS decommission; both refusals written, signed, held, with later scope requests answered from the pack in writing.

RESULTS — HOW SUCCESS WAS MEASURED: $ · # · % · TIME · CUSTOMER · QUALITY

R — Primary metric: surprise escalations → **0**; gate chain 5/15 → 7/31 → 8/14 held with zero re-litigation. $ impact: ≈$0.5M one-time modeled.

- **$** — ≈$0.5M one-time modeled — avoided rework on ~3 escalations that never happened (benchmark 2–4), six weeks of benefit not deferred, and a fenced second reserve engine not built; the dollars are my model of what did not happen.
- **#** — ~80 named participants across ~15 functions; a dozen core systems (20+ touched); three separable decisions plus the out-of-scope fence (record).
- **%** — Pre-wire coverage 100% — every impacted row with a signed LOE by 7/2, before the council sign-off (record).
- **Time** — Gate chain kickoff 3/29 → sessions 4/30 → council 5/15 → LOE 7/2 → sign-off 7/31 → FSD 8/14 — held to the day, council-to-sign-off with zero re-litigation (record).
- **Quality** — Surprise escalations → 0 for the program's life; fence breaches in the log: 0; both scope lines held in writing (record).

Against the goal stated above: zero surprise escalations for the program's life; 100% pre-wire coverage by 7/2; the chain held to the day with zero re-litigation; fence breaches zero, and the program handed clean into the build (S2). The fence itself was the trade: reserve-math retuning and the MCS decommission were refused to keep the release to one defect class, and the second-live-warehouse concern cost a monitored shadow copy instead of an argument.

PROBING — QUESTIONS, PHASE DEEP-DIVES, AND THE MATH

Challenge — Why did you choose this story to illustrate Have Backbone; Disagree & Commit?

Because the refusals were written and held — reserve math untouched, MCS decommission out of scope — and the council was asked for three separable decisions rather than a blessing. Saying no in a signed document and holding it for the program's life is the backbone; committing to the council's decisions is the commit.

Challenge — What other stories demonstrate Have Backbone; Disagree & Commit?

S9 — 597 vs 1,000, taken two levels up — is the primary; S14, refusing the quiet fix, is the backup.

Challenge — Could you come up with an example that is more recent?

Current role — the 2025 gate chain (council 5/15, sign-off 7/31, FSD 8/14) — so this is among the most recent records I have for this principle.

Probing — Why did you choose to focus on these results? What other results were important?

Zero surprise escalations and the dated chain holding — 5/15 → 7/31 → 8/14 with zero re-litigation — are the record for a governance program whose deliverable was the decision itself. Also important: fence breaches at zero and the program handed clean into the build (S2).

Probing — You mentioned dollars — what percentage change is that year over year?

No YoY — the ≈$0.5M is one-time and entirely modeled: avoided rework on escalations that never happened, six weeks of benefit not deferred, and a fenced engine not built. I hold those loosely and say so.

Probing — I'm concerned that you're pricing escalations that never happened — tell me more.

Correct, and labeled: the ~3 is a benchmark for designs this size and the rework hours are illustrative. The record is zero escalations against a norm everyone in that building recognized, and a gate chain that held to the day — if the room won't price prevention, keep the record and drop the dollars; the mechanism is the point.

Challenge — What were the lessons learned? What would you have done differently?

A no that lives in a document survives; a no that lives in a meeting gets re-litigated weekly. The fence held because it was signed — and answered from, in writing, every time.

Challenge — How would you implement this at Amazon?

Influence without authority is the L6 job description — pre-wire every owner's cost before the review, ask the forum for separable decisions rather than a blessing, fence what you will not do, and answer scope creep from the signed document.

Challenge — How did these results compare to your actual goals?

The task was discovery through council to sign-off and FSD without a surprise escalation — delivered exactly: zero escalations for the program's life, both scope lines held in writing, and the chain held to the day.

PHASE DEEP-PROBE BANK — Q&A (13). Answer the phase asked, then stop.

Discovery — How did you know discovery was done? Why make it a data exercise?

Every impacted team had been in a session on its own profiled data; every row had a provisional owner; sessions closed 4/30. Profiling is disarming — it is their numbers, not my opinion; SKU overlap and order profiles settled arguments before they started.

Plan & Design — What was in your risk register at the council?

Surprise LOE in the room, a second live warehouse becoming a second truth, and scope re-litigated after sign-off. Mitigations: private pre-wire with a 7/2 LOE deadline, the monitored shadow copy, the fence.

How did you handle a partner who objected at the review?

The "second truth" objection became the design. When a concern is legitimate, converting it costs less than winning it.

Why exactly three decisions?

Separability. Architecture, staged timeline, freeze boundary — a no on any one leaves the others standing. A single "approve the design" would have been all-or-nothing.

Build (LOE) — How did you get sixty-plus people to sign LOE you didn't control?

One template, one deadline, one private conversation per owner before the council so nobody learned their cost in the room. Signatures landed 7/2.

What was your definition of done for a row?

Owner named, LOE signed, concern converted to design or dated. A row with an open concern did not go into the sign-off pack.

Cutover — What would have made you delay the 7/31 sign-off?

An open row. There were none — which is what the pre-wire buys.

What was your rollback if the council said no?

Separability. A no on the timeline would have sent us back with the architecture intact; nothing had been configured yet by design — policy before configuration.

Hypercare — How did you keep scope from creeping back?

The fence lived in the signed pack; every later request was answered from it in writing. A no in a document survives; a no in a meeting gets re-litigated weekly.

What did "hypercare" mean for a governance program?

Answering every scope request from the signed pack, in writing, for the program's life — and applying the peak-freeze boundary Oct–Dec as the ruling. Zero surprise escalations was the exit metric, and it held.

Cross-phase — What did you escalate? What did you refuse?

Escalated nothing — the pre-wire is the escalation-prevention mechanism; the freeze boundary went to the council as a decision. Refused: retuning reserve math and folding in the MCS decommission. Both refusals written, signed, held.

What if the council had said no?

The three decisions were separable — a no on the timeline would not have unwound the architecture. That's why I asked for three, not one.

Where this lands at Amazon: influence without authority is the L6 job description. The mechanism transfers whole — pre-wire every owner's cost before the review, ask the forum for separable decisions rather than a blessing, fence what you will not do, and answer scope creep from the signed document.

BUSINESS IMPACT MODEL — KPI TREE, NUMBERS, THE MATH

```
Program delivered on the dated gate chain, zero surprise escalations
  (record: council 5/15 -> sign-off 7/31 -> FSD 8/14)
+- Avoided rework $ = escalations avoided x rework weeks x team $/wk
|    +- pre-wire coverage = rows with signed LOE / rows = 100%
|    +- 3 separable decisions: a no on one cannot unwind all
+- Schedule protection $ = slip weeks avoided x monthly benefit
+- Fenced scope $ = second reserve engine not built
     <- requests answered from the pack in writing -> fewer requests next cycle (loop)
```

DRIVER	BASELINE	AFTER	$	BASIS
Surprise escalations (60+ stakeholder design)	prior norm ~3	0	—	record; ~3 is industry benchmark (2–4)
Pre-wire coverage before council	partial	100% by 7/2	—	record
Avoided rework	—	—	≈$0.3M	modeled (3 × 960 h × $110)
Slip weeks avoided	~6 wks	0	≈$0.1M	modeled (6/52 × S2's ≈$0.9M labor value)
Second reserve engine (fenced, not built)	proposed	not built	≈$0.1M	modeled (2 eng × 12 wks × 40 h × $110)
Total modeled	—	—	**≈$0.5M one-time**	fence breaches in the log: 0 (record)

If challenged. *"You're pricing escalations that never happened."* — "Correct, and labeled: the ~3 is a benchmark for designs this size, the 960 hours per escalation is illustrative. The record is zero escalations against a norm everyone in that building recognized, and a gate chain that held to the day. If the room won't price prevention, keep the record and drop the dollars — the mechanism is the point."

THE MATH — STEP-BY-STEP DERIVATION

Table 1 — before: the problem in numbers.

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
1	Design size	`~80 named participants × ~15 functions; 20+ systems touched`	record	the blast radius
2	Norm for this size	`~3 surprise escalations per design`	industry benchmark 2–4	~3 expected
3	Price of one escalation	`960 rework hours × $110/h = $105,600`	modeled	≈$0.1M each
4	Bundle decision risk	`P(all pass) = 0.9 × 0.9 × 0.9 = 0.73`	one up/down vote on the whole design; p = 0.9 per decision (illustrative)	27% chance one objection unwinds everything

Table 2 — the fix: step by step, values substituted.

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
1	Three separable decisions	`architecture · timeline · peak boundary → 2³ = 8 outcomes`	any partial approval preserves progress (record design)	a no on one cannot unwind all
2	Expected approvals	`E = 0.9 + 0.9 + 0.9 = 2.7 of 3`	vs the bundle's 0.73 chance of everything (illustrative p)	2.7 vs 0.73
3	Pre-wire coverage	`rows with signed LOE / impacted rows = 100% by 7/2`	one template, one deadline, one private conversation per owner (record)	100% before the room
4	Gate chain held	`3/29 → 4/30 → 5/15 → 7/2 → 7/31 → 8/14 = 77 days council→sign-off`	held to the day (record)	0 re-litigation

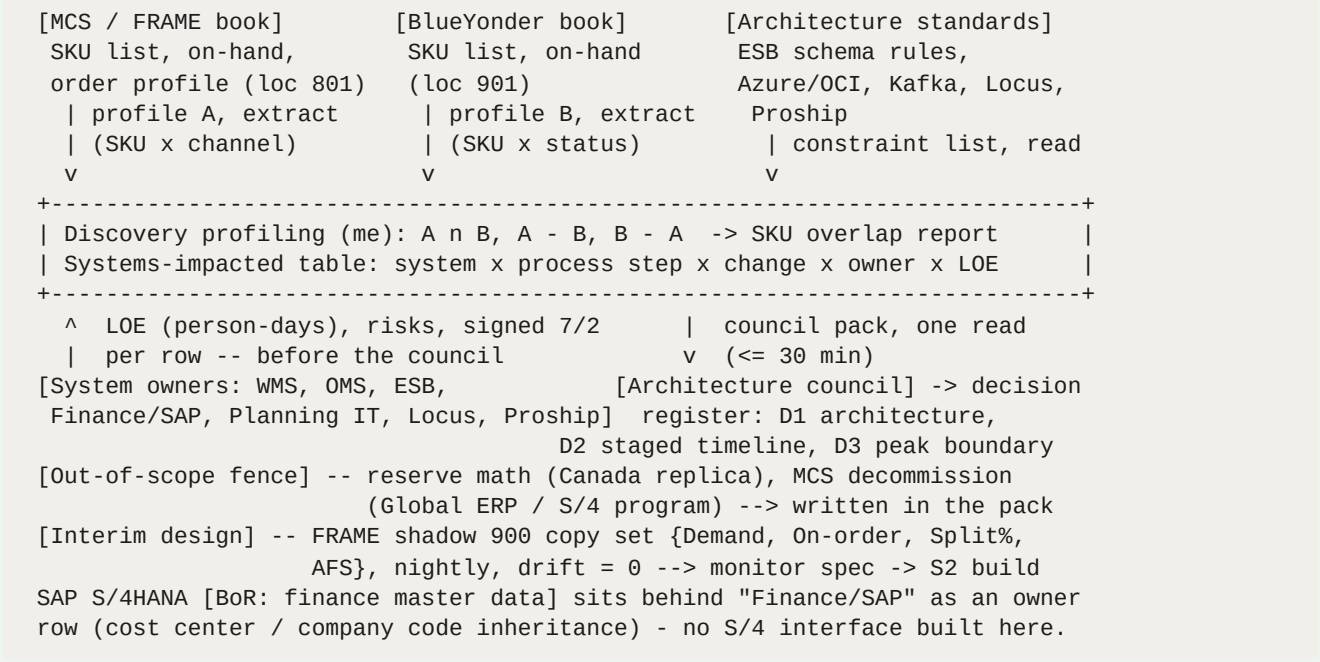
Table 3 — after: the metric, before → after.

STEP	METRIC	BEFORE	AFTER	BASIS
1	Surprise escalations	~3 (the norm)	0 for the program's life	record; norm is benchmark
2	Pre-wire coverage before council	partial	100% by 7/2	record
3	Fence breaches in the log	—	0	record
4	Re-litigation after sign-off	—	0 across 77 days	record

Table 4 — the dollars, every multiplication shown.

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
1	Avoided rework	`3 × 960 h = 2,880 h; 2,880 × $110 = $316,800`	~3 escalations (benchmark) × rework hours × rate (modeled)	≈$0.3M
2	Slip weeks avoided	`6 / 52 = 0.115 yr; 0.115 × $900,000 = $103,800`	six weeks of S2's ≈$0.9M/yr labor value not deferred (modeled)	≈$0.1M
3	Fenced engine not built	`2 eng × 12 wk × 40 h = 960 h; 960 × $110 = $105,600`	second reserve engine (modeled)	≈$0.1M
4	Total one-time	`0.3 + 0.1 + 0.1 = $0.5M`	all modeled — the dollars are what did not happen	≈$0.5M

SOLUTION FLOW — THE SYSTEM & DATA MAP



SENSITIVITY — LOW / BASE / HIGH

INPUT	LOW	BASE	HIGH
Escalations avoided (benchmark)	2	3	4
Rework weeks per escalation (illustrative)	1	2	3
Loaded blended rate (benchmark)	$90/h	$110/h	$130/h
Avoided rework $	$0.09M	$0.32M	$0.75M
Slip weeks avoided	4	6	8
Schedule protection $	$0.07M	$0.10M	$0.14M
Fenced engine build $	$0.09M	$0.11M	$0.12M
Total modeled $/yr	≈$0.25M one-time	≈$0.5M one-time	≈$1.0M one-time

TOTAL IMPACT STATEMENT — SAY IT LIKE THIS

"Conservatively, about half a million, one time, of cost this governance avoided: roughly three hundred thousand of rework on the two or three escalations that never happened because every owner had signed their effort before the council; about a hundred thousand of the comingle benefit arriving six weeks sooner than it would have after those re-litigations; and about a hundred thousand for the second reserve engine the fence kept out of the release. What is measured is zero surprise escalations and the dated chain holding, council to sign-off to spec. The dollars are my model of what did not happen, so I hold them loosely and I say so."

If challenged — the full answers.

You are pricing things that did not happen.

Yes, and that is the hardest part of any governance number. It rests on a norm of two to four re-litigations for a design with sixty-plus stakeholders and two weeks of core-team rework each. To firm it up I would take the prior two programs

of this size at Sephora and count the scope re-openings and the weeks each cost from their own RAID logs; that replaces my norm with theirs.

A hundred thousand of schedule protection borrows S2's number.

It does, deliberately, and it is labelled that way. The assumption is six weeks of slip avoided against a benefit worth about seventeen thousand a week. The measure is simple: the FSD landed on the fourteenth of August against the plan, so slip avoided is the plan minus the actual, and the weekly value comes from the DC's measured efficiency once it is priced.

The fence saving is just work you chose not to do.

Right, so I count it once and small. The assumption is two engineers for twelve weeks. What would firm it up is the actual estimate the engineering team gave when they proposed the US-tuned engine; if their number was different, mine changes, but the point stands that a second engine was a second defect class in the same release.

S15 — Sephora: converting a 20-person Product/Eng/QA team to a 4-week cadence — and the QA lead who now owns the gate

Amazon LPs: Hire and Develop the Best (primary) · Earth's Best Employer (primary — QA elevated, not blamed) · Insist on the Highest Standards (the gate). **Crosswalk:** not in the loop packet — safe anywhere. *Never invent a direct-report headcount.*

Honesty label. VP air cover, no line authority over anyone in the room; the QA lead authored the definition-of-done — I coached the frame, she owned the gate. The −50% and the six weeks are record; the dollars are my model on illustrative defect counts.

60-SECOND PITCH

"**Situation.** The VP asked me to move a 20-person Product/Eng/QA team from a six-week hybrid cadence to four-week agile; QA feared compressed test windows — and they were right. **Task.** Convert in weeks without line authority. **Action.** Discovery: one-on-ones; influencers per function. Plan & Design: the QA lead authored the definition-of-done — I coached the frame, she owned the gate. Build & Test: Jira benefit-ranking and review-status fields; a QA drop date with reason codes. Cutover: 4-week cadence live in 6 weeks. Hypercare: QA on product-business calls with pre-approved scenarios; she defended the gate without me. Lesson: development that depends on me staying is not development. **Result.** Primary metric: critical defects per release — baseline → **−50%** after the 20-person team moved to a four-week cadence in six weeks. $ impact: ≈$130K/yr modeled — the −50% and the six weeks are the record; the dollars are my model; no headcount invented."

SITUATION / TASK — WHERE IT OCCURRED, WHEN IT HAPPENED, WHY IT IS IMPORTANT

S — Newly joined, the VP of Supply Chain asked me to move a 20-person Product/Engineering/QA team from a six-week hybrid cadence to four-week agile. QA (a contracting firm) and engineering (in-house) were heading for a standoff: QA feared compressed test windows — and they were right.

T — Convert the cadence in weeks, not quarters, without the standoff becoming the story. VP air cover, no line authority over anyone in the room.

The success criteria: the four-week cadence with no rise in escaped criticals, and the QA window protected by rule. The risks if the conversion went wrong: the standoff becoming the story, dev consuming the shrunken test window, Sev-1s mis-attributed to QA — and a faster cadence shipping more escapes, the exact failure the VP could not afford.

The metrics on the table, with their baselines: release cadence — six-week hybrid; escaped critical defects per release — the baseline the −50% would be measured against, with the constraint of no rise; the QA test window — being compressed by arithmetic, target protected by rule.

ACTION — WHAT I PERSONALLY OWNED, HOW I DID IT, WHO ELSE WAS INVOLVED

A — Five gated phases (week markers illustrative):

1. **Discovery (weeks 1–2)** — one-on-ones with every person (objections, motivations); an influencer named per function; the problem statement — QA's fear of compressed test windows is correct — and the success criteria (four-week cadence, no rise in escaped criticals). *Exit bar:* every function's real objection heard privately; the QA lead agreed to author the gate.
2. **Plan & Design (weeks 2–3)** — the ceremony calendar for a four-week cycle; the definition-of-done *frame* (what a gate needs to survive pressure — the QA lead wrote the content); the Jira field spec (benefit ranking, review status); the user-guide outline. *Exit bar:* DoD with entry criteria written by QA and reviewed with engineering; fields live in Jira.
3. **Build & Test (weeks 3–5)** — the user guide (grooming → completion); the live dashboard; the QA drop date with reason codes; QA seated on the product-business calls with pre-approved scenarios. *Exit bar:* cycle 1

completed on the new cadence; drop-date rule in force; Sev-1 cause classification running.

4. **Cutover (week 6)** — facilitated the retro once; gave the VP a data read-out; handed the gate to the QA lead to defend herself. *Exit bar:* converted in 6 weeks; critical defects −50% under the drop-date rule.
5. **Hypercare / Close** — stepped back from the review line; neighbouring teams copied the ceremony from the user guide. *Exit bar:* the gate survived my leaving; Sev-1 trend from story-accuracy defects reversed.

The QA lead authored the definition-of-done — I coached the frame, she owned the gate; the −50% and the six weeks are record, the dollars are my model. Mine: the one-on-ones, the ceremony calendar, the DoD frame, the Jira field spec (benefit ranking, review status), the drop-date rule and the retro — facilitated once; hers: the DoD content and the gate, defended by her; engineering shaped the entry criteria, and neighbouring teams copied the ceremony from the user guide. The contribution: handing the load-bearing artifact to the skeptic — a gate the skeptic owns survives pressure; one I write dies when I leave — and the drop-date rule that converted silent late lands into visible, reason-coded re-scopes.

The obstacle was the QA–engineering standoff — defused by one-on-ones with every person first and by making QA the author rather than the blamed party; engineering's objection to the DoD was converted to co-ownership through review, not veto, with me covering the low-impact stories they had to drop. Stories entered QA only on met entry criteria, tracked by the review-status field, and late heavy stories got a drop date with a reason code instead of a silent slide into the last two days; buy-in came from every function's real objection being heard privately before anything was scheduled, and from the team signing by writing the DoD themselves. The assumption challenged was that QA's fear was resistance — it was correct arithmetic, and saying so is what made the QA lead willing to author the gate; I also named the rollback (reverting to six weeks) as a two-way door at kickoff and never used it.

RESULTS — HOW SUCCESS WAS MEASURED: $ · # · % · TIME · CUSTOMER · QUALITY

R — Primary metric: critical defects per release **−50%**; 20-person team on a four-week cadence in **six weeks**; the gate defended by its owner, not by me.

- **$** — ≈$130K/yr modeled (escaped-critical cost avoided + late-land rework removed; range $80–270K); none of it is a record figure, and no headcount was invented to get there.
- **#** — 20-person Product/Eng/QA team; releases per year 8.7 → 13 (modeled from the cadence: 52 ÷ cycle weeks).
- **%** — Critical defects per release −50% (record) — while shipping 50% more releases.
- **Time** — Six-week hybrid cadence → four-week agile, converted in six weeks (record); the QA lead defending the gate by cycle 2.
- **Quality** — The QA-authored definition-of-done with entry criteria that can say no; the drop date with reason codes; Sev-1 cause classification — and the gate survived my leaving (record).

Against the goal stated above: the cadence converted six-week → four-week in six weeks; critical defects per release −50% while shipping 50% more releases; the QA window protected by a gate its owner defended by cycle 2 — and the gate survived my leaving. The visible price: low-impact stories dropped under the drop-date rule — dev's date protected, QA's window protected, some drops covered by me — a visible re-scope instead of a silent slide into the last two days.

PROBING — QUESTIONS, PHASE DEEP-DIVES, AND THE MATH

Challenge — Why did you choose this story to illustrate Hire and Develop the Best?

Because the development mechanism is the story: the most skeptical person — the QA lead — authored the definition-of-done; I coached the frame, she owned the gate, and she was defending it to engineering without me. Development that depends on me staying is not development.

Challenge — What other stories demonstrate Hire and Develop the Best?

S11 — the wave teams grown into owners — is the backup for this LP.

Challenge — Could you come up with an example that is more recent?

This is my current role at Sephora — a recent record, and the most recent one I have for this principle.

Probing — Why did you choose to focus on these results? What other results were important?

Critical defects per release −50% while shipping 50% more releases is the result that answers the obvious objection — faster cadence usually means more escapes. The conversion in six weeks and the gate surviving my leaving are the development results.

Probing — You mentioned dollars — what percentage change is that year over year?

No YoY — the ≈$130K/yr is entirely modeled (escaped-critical cost on an illustrative base count, late-land rework at benchmark rates); the record is the −50%, the cadence and the six weeks, and I did not invent a headcount to get the dollars.

Probing — I'm concerned about the four-criticals-per-release baseline — tell me more.

It is illustrative, not record — the record gives me the fifty percent drop, not the base count. I'd firm it up by pulling the Sev-1 log for the last four six-week releases and using the actual mean.

Challenge — What were the lessons learned? What would you have done differently?

Development that depends on me staying is not development. Differently: go to QA first, and put her name on the review line from week one.

Challenge — How would you implement this at Amazon?

The Hire & Develop mechanism — develop people by giving them the load-bearing artifact and the frame, not by writing it for them, and measure development by whether the gate survives your leaving. The Earth's-Best angle: QA moved from blamed afterthought to owner of the standard, with the data protecting them.

Challenge — How did these results compare to your actual goals?

The asks were a four-week cadence in weeks, not quarters, with no rise in escaped criticals — delivered in six weeks with criticals per release down 50%, the drop-date rule in force, and the gate defended by its owner, not by me.

PHASE DEEP-PROBE BANK — Q&A (14). Answer the phase asked, then stop.

Discovery — How did you know discovery was done?

When I had heard every function's real objection privately and the most skeptical person — the QA lead — had agreed to author the gate. Nothing was scheduled before it.

Who wrote the success criteria and who signed?

I wrote them with the VP: four-week cadence, no rise in escaped criticals, QA window protected by rule. The VP sponsored them; the team signed by writing the DoD themselves.

Plan & Design — What was in your risk register?

A QA–engineering standoff becoming the story, dev consuming the test window, and Sev-1s mis-attributed to QA. Mitigations: the skeptic owns the gate, a drop date with reason codes, cause classification with QA on product calls.

Why hand the load-bearing artifact to the skeptic?

A gate the skeptic owns survives pressure; one I write dies when I leave. I coached the frame — what a gate needs to say no — and she wrote the content.

How did you handle engineering objecting to the DoD?

Review, not veto — they shaped the entry criteria and I covered the low-impact stories they had to drop. Objection became co-ownership.

Build & Test — How did you sequence stories in Jira?

By the DoD: a story entered QA only when its entry criteria were met, tracked by the review-status field. Late heavy stories got a drop date with a reason code instead of a silent slide into the last two days.

What was your definition of done?

Hers, not mine — entry criteria per story type, evaluated at the Ready-for-QA transition. My frame: criteria that can say no and survive a release-day argument.

What replaced a bug bash?

The drop-date rule and demo-during-build. When dev landed heavy stories on the last two days, the reason codes made the pattern visible and criticals fell 50%.

Cutover — What made you say the conversion was done?

Two cycles on four weeks, the drop-date rule in force, criticals down 50%, and the QA lead defending the gate to engineering herself. Six weeks.

What was the rollback?

Reverting to six weeks — a two-way door I named at kickoff and never used.

Hypercare — How did you decide you could step back?

When the gate was defended without me and neighbouring teams copied the ceremony. Development that depends on me staying is not development.

What did you watch after the conversion to know it held?

Stories entering QA meeting their entry criteria, drop-date reason codes trending toward genuine re-scopes rather than late lands, and Sev-1 causes classified. Two cycles of that and I stopped attending the review line.

Cross-phase — Where did you trade scope for date? What did you escalate?

Low-impact stories dropped under the drop-date rule — dev's date protected, QA's window protected, and I covered some of the drops myself. Escalated nothing up; hard feedback went sideways through the issues-log rows — some of it turned out to be genuine overload, and the evidence got those people help.

Have you managed direct reports? · Do differently?

Yes — onshore/offshore delivery teams, SOWs and estimates earlier in my career. I have not run a large HR org with comp cycles and won't claim that muscle. Differently: go to QA first, and put her name on the review line from week one.

Where this lands at Amazon: the Hire & Develop answer with a mechanism: develop people by giving them the load-bearing artifact and the frame, not by writing it for them — and measure development by whether the gate survives your leaving. The Earth's-Best angle: QA moved from blamed afterthought to the owner of the standard, with the data protecting them.

BUSINESS IMPACT MODEL — KPI TREE, NUMBERS, THE MATH

```
Release quality at a faster cadence (escaped criticals/yr, $ avoided)
+- Releases/yr (flow): 52 / cycle weeks: 6w -> 4w = 8.7 -> 13 (+50%)
|    +- test window per cycle (stock of QA days) shrinks ->
|         DoD (QA-authored) -> stories arrive ready · drop date -> no last-2-day lands
+- Critical defects per release (record -50%)
|    +- story-accuracy Sev-1s -> QA on product calls with scenarios
+- Release-end rework (late-land re-test + hotfix) -> ~0 under the drop rule
(loop broken) Sev-1 blamed on QA -> standoff -> gate weakens -> more Sev-1s
   broken by: the skeptic owns the gate; cause classification, not blame
(loop built)  gate says no -> dev drops low-impact story -> window held ->
   fewer criticals -> gate trusted -> gate survives my leaving
```

DRIVER	BASELINE	AFTER	$/YR	BASIS
Cycle length / releases per year	6 wks / 8.7	4 wks / 13	—	record / modeled (52 ÷ weeks)
Critical defects per release	4 (illustrative)	2	—	record (−50%); count illustrative
Escaped-critical cost avoided	~$350K	~$260K	≈$90K	modeled (criticals/yr × $10K; benchmark $8–15K)
Late-land rework per release	~$4.8K	~$0	≈$40K	modeled (6 person-days × $800 × 8.7)
Total modeled	—	—	**≈$130K**	range $80–270K; no headcount

If challenged. *"Faster cadence usually means more escapes — why did yours fall?"* — "Because the gate got stronger as the window got shorter: entry criteria authored by the person with the most to lose, and a drop date that converted silent late lands into visible re-scopes. The −50% is the record; the defect counts behind the dollars are illustrative and labeled."

THE MATH — STEP-BY-STEP DERIVATION

Table 1 — before: the problem in numbers.

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
1	Cadence	`releases/yr = 52 / 6 = 8.67 ≈ 8.7`	six-week hybrid cycle (record)	8.7 releases/yr
2	Criticals	`4 per release × 8.7 = 34.8 ≈ 35/yr`	count illustrative; the record is the −50%	≈35/yr
3	The real constraint	`15 eng × 18 d = 270 eng-days → 270/3 = 90 stories; 4 QA × 18 d = 72 QA-days → 72/1 = 72 stories`	story ≈ 3 eng-days + 1 QA-day (illustrative; theory-of-constraints, toolkit label)	QA caps throughput at 72 — QA is the constraint
4	Late-land rework	`6 person-days × $800 = $4,800 per release`	re-test + hotfix after last-2-day lands (modeled)	≈$4.8K per release

Table 2 — the fix: step by step, values substituted.

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
1	QA-authored DoD	`entry criteria at Ready-for-QA — stories arrive ready`	the skeptic owns the gate (record)	the gate has an owner
2	Drop date	`not QA-ready by the drop date → dropped with a reason code`	silent late lands become visible re-scopes (record)	no last-2-day lands
3	Sev-1 attribution	`share caused by wrong stories vs wrong tests, classified by cause not blame`	QA put on product calls with pre-approved scenarios (record)	the trend reverser
4	Cadence converted	`releases/yr = 52 / 4 = 13; 13 / 8.7 = +50%`	converted in six weeks (record)	13 releases/yr
5	Criticals per release	`4 → 2 = −50%`	record	2 per release

Table 3 — after: the metric, before → after.

STEP	METRIC	BEFORE	AFTER	BASIS
1	Cycle / releases per year	6 wks / 8.7	4 wks / 13 (+50%)	record
2	Critical defects	≈35/yr (4 × 8.7)	2 × 13 = 26/yr — while shipping 50% more	record −50%; counts illustrative
3	Late-land rework	≈$4.8K per release	~0 under the drop rule	modeled
4	The gate itself	mine to defend	QA lead defending it by cycle 2	record — survives my leaving

Table 4 — the dollars, every multiplication shown.

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
1	Escaped criticals avoided	`35 − 26 = 9/yr; 9 × $10,000 = $90,000`	per-critical cost benchmark $8–15K (modeled)	≈$90K/yr
2	Late-land rework removed	`$4,800 × 8.7 releases = $41,760`	→ ~0 under the drop rule (modeled)	≈$40K/yr
3	Total	`90 + 40 = $130K`	range $80–270K; no headcount invented	≈$130K/yr

SENSITIVITY — LOW / BASE / HIGH

INPUT	LOW	BASE	HIGH
Criticals per release, baseline (illustrative)	3	4	6
Cost per escaped critical (benchmark)	$8K	$10K	$15K
Late-land rework per release (benchmark rates)	$3K	$4.8K	$8K
Escaped-critical cost avoided	~$55K	~$90K	~$200K
Late-land rework avoided	~$25K	~$40K	~$70K
Total modeled $/yr	~$80K	~$130K	~$270K

TOTAL IMPACT STATEMENT — SAY IT LIKE THIS

"Conservatively, about $130K a year, and I want to be clear that none of it is a record figure: roughly $90K is escaped-critical cost avoided, because criticals per release halved even as we shipped fifty percent more releases, and about $40K is release-end rework that the drop-date rule simply removed. What is measured is the cadence, six weeks to four, the conversion in six weeks, and the fifty percent drop in criticals per release. The dollars are my model on an illustrative four criticals per release and a benchmark ten thousand dollars per escaped critical, and I did not invent a headcount to get there."

If challenged — the full answers.

Where does four criticals per release come from?

It's illustrative, not from the record; the record gives me the fifty percent drop, not the base count. I'd firm it up by pulling the Sev-1 log for the last four six-week releases and using the actual mean.

Ten thousand dollars per critical is a soft number.

Agreed, it's a benchmark for hotfix plus regression plus incident time plus ops disruption; the range is eight to fifteen. I'd measure it by costing three real Sev-1s end to end with the team's loaded rates.

Fifty percent more releases could mean more total defects.

That's exactly why I count per year, not per release; halving per release at thirteen releases still nets nine fewer a year. I'd watch escaped criticals per quarter, which is the number the VP actually feels.

Employer block 2 — Flexport

S11 Shipment Visibility launch (the wave machine — super important) · S12 FF App export launch (the roadmap flip — super important) · S16 four systems into one Snowflake pipeline

S11 and S12 are flagged super-important for this loop: the WBR-and-waves machine and the roadmap flip are the two Flexport records most likely to carry Deliver Results, Think Big and Frugality. S16 is the Ownership/failure-adjacent companion.

S11 — Flexport: Shipment Visibility launch — 7 → 68 behind graduation gates, the WBR, and pre-prod measurement

Amazon LPs: Deliver Results (primary) · Ownership · Customer Obsession (graduation criteria per wave) · Earn Trust (the discarded 99.4% week) · Hire and Develop (backup — wave teams grown into owners). **Crosswalk:** loop packet MAIN + S18 + S26 — **carries the §1.3 conflicts** (automation 40→90 here vs 60→97 there; WBR hours 1.5–2 → ~4 here vs 30→6 there). Resolve before the loop; inside this packet, this page is canon. *Never "10×" — say 7 → 68 (9.7×). Never "$1.00 → $0.03" or "70% baseline."*

Honesty label. Program owner across nine teams / ~100 people; the WBR format was copied from the LSA review, not invented; week counts for the build are illustrative; the ~$1.5M pre-prod saving is a separate canonical figure, never merged with CTS.

60-SECOND PITCH

"**Situation.** Flexport's visibility product had public targets — 68 monthly active customers, 170 MAU, 32K shipments — off seven pilots at $0.66 a container; Sales wanted the whole pool boarded at once, and leadership said the program was not showing value. **Task.** As program owner across nine teams / ~100 people, I owned the boarding path, the measurement system, and the pre-prod proof. **Action.** Discovery: ~50-container calibration entry bar per client; WBR input/output OS copied from an existing review, seven metrics frozen. Plan & Design: six graduation criteria published to Sales before wave 1; launch gate 100 real containers × 7 checks; sample size 300 containers, ±1.6% at 95% confidence. Build & Test: weekly ops audit; WBR by hand 1.5–2 then ~4 hrs/wk; pre-prod ingest, per-hop monitors, kill switches. Cutover: wave 1 (2 clients) → wave 2 (4 of 7) → wave 3 (20+) held in public on an ETA-accuracy miss from one bad source → full rollout. Hypercare: discarded a flattering 99.4% three-touch week; the VP unlocked the 68-client rollout two months early off the WBR. Lesson: board no faster than automation matures; a gate that only holds in private is a suggestion. **Result.** Primary metric: cost-to-serve per container — **$0.66 → $0.03 (−95.5%)** across customers 7 → 68 (9.7×) with zero rollbacks and no added ops headcount; automation 40% → 90%; the ~$1.5M pre-prod saving is a separate record figure."

SITUATION / TASK — WHERE IT OCCURRED, WHEN IT HAPPENED, WHY IT IS IMPORTANT

S — Senior leadership told me the visibility program wasn't demonstrating value at the anticipated rate — it would drop from #3 to #14 of 16 initiatives.

T — I had been reporting activity, not evidence. Funding and the 68-customer rollout hung on whether value was legible.

The public targets: 68 monthly active customers, 170 MAU, 32K shipments, off seven pilots at $0.66 a container — and I was program owner across nine teams / ~100 people, owning the boarding path, the measurement system and the pre-prod proof, with Sales wanting the whole pool boarded at once. The risks: boarding faster than automation matured — a bad first impression at scale, with rollbacks — and the program losing its funding slot; the critique was about evidence, not effort, and I accepted that first.

The metrics on the table, with their baselines: customers — 7 pilots; target 68. Cost-to-serve — $0.66 a container. Automation — 40%; ETA accuracy — 60%. All of it to move with zero rollbacks and no added ops headcount.

ACTION — WHAT I PERSONALLY OWNED, HOW I DID IT, WHO ELSE WAS INVOLVED

A — Five gated phases (canonical: WBR copied from the LSA review — Mon–Sun week, five-week window, WoW = (B−A)/A, ±5% green/red, every input tagged to an output; seven frozen metrics; run by hand 1.5–2 hrs/wk then ~4 for over a year; wave roster W1 Q4'21 (2 clients) → W2 Q1'22 (4 of a 7-pool) → W3 Q2'22 (20+, held on ETA accuracy) → full H2'22):

1. **Discovery** — wrote the honest diagnosis (reporting activity, not evidence); inventoried what leadership actually decides on (rollout pace, funding); chose the LSA WBR as the pattern to copy rather than invent. *Exit bar:*

leadership agrees on the questions the review must answer weekly.

2. **Plan & Design** — froze seven metric definitions (coverage auto, coverage w/ human, tickets, tickets per active container, 4PL SLA %, 3PL SLA %, delta); input→output tagging; the ±5% WoW rule; the 15-day inactive rule as the honest denominator; the four-system pull procedure; six graduation criteria published to Sales before wave 1. *Exit bar:* definitions signed by the owners of each input; first Monday deck dry-run.
3. **Build & Test** — built the deck manually every week (1.5–2 hrs → ~4); pre-decomposed every red with an owner and a dated fix; enforced the data-honesty rule (a truncated week is a data miss, not a result); pre-prod: 300-container measurement, per-hop monitors, kill switches. *Exit bar:* two consecutive Mondays where every red arrives with an owner and the number verifies the following week.
4. **Cutover & Go-Live** — the Monday WBR became the operating forum, with the rollout decision taken in the review; waves boarded only through the six graduation criteria; wave 3 held in public on ETA accuracy traced to one bad source. *Exit bar:* VP unlocks the 68-client rollout two months early.
5. **Hypercare / Close** — handed the manual pull to the Snowflake pipeline (S16) with the definitions in code; left the critic steering by the numbers. *Exit bar:* automation 40% → 90%; CTS $0.66 → $0.03; the same critic uses the review to steer.

The WBR format was copied from the LSA review, not invented — copying a trusted format was the point — and I ran it by hand every Monday for over a year, 1.5–2 hours and then about 4. Mine: the honest diagnosis, the frozen definitions, the WBR built by hand every week, the graduation criteria, the hold decisions and the pre-prod measurement design; nine teams built and operated, each input owner signed the definition they would be red against, and the VP took the rollout decisions in the review. The contribution was making value legible: seven frozen metric definitions signed by their owners, input→output tagging, and the data-honesty rule — including discarding a flattering truncated 99.4% week and printing a data miss instead.

The obstacles: Sales wanting the whole pool boarded at once — answered with six graduation criteria published before wave 1, and wave 3 held in public on an ETA-accuracy miss traced to one bad source — the rule fired, and the held customers became the cleanest wave; bus factor one on the pull — retired by handing it to the Snowflake pipeline (S16). The priority rule was board no faster than automation matures — waves only through the six criteria, every red pre-decomposed with an owner and a dated fix; buy-in came from leadership agreeing the questions the review must answer weekly, and the Monday WBR becoming the operating forum where the rollout decision was taken. The decisions challenged: the “board everything now” push — met with published gates — and the temptation to print the 99.4% week: three touches is not a week, and a number that cannot survive the next Monday teaches the VP the wrong thing.

RESULTS — HOW SUCCESS WAS MEASURED: $ · # · % · TIME · CUSTOMER · QUALITY

R — Primary metric: CTS **$0.66 → $0.03 (−95.5%)**; customers 7 → 68 (9.7×), zero rollbacks; accuracy 60% → 90%; both public targets two months early.

- **$** — CTS $0.66 → $0.03 per container (record); the ~$1.5M pre-prod measurement saving is a separate record figure, never merged with CTS; direct CTS dollars ≈$50K/yr modeled on an illustrative container count.
- **#** — Customers 7 → 68 (9.7×) with zero rollbacks and no added ops headcount; targets 68 customers / 170 MAU / 32K shipments (record).
- **%** — CTS −95.5%; automation 40% → 90%; accuracy 60% → 90% (record).
- **Time** — Both public targets hit two months early; the rollout unlocked two months early off the WBR (record).
- **Customer** — Six graduation criteria published to Sales before wave 1 — including CSAT ≥ consignee average; wave 3 held in public on an ETA-accuracy miss, and the held customers became the cleanest wave (record).
- **Quality** — A flattering truncated 99.4% week discarded and printed as a data miss; seven metric definitions frozen and later moved into code (record).

Against the goal stated above: customers 7 → 68 (9.7×) with zero rollbacks and no added ops headcount; CTS $0.66 → $0.03 (−95.5%); automation 40 → 90%; accuracy 60 → 90%; both public targets hit two months early — and the VP unlocked the 68-client rollout off the review's own numbers. The trades: new metrics were parked to

keep the seven stable, and boarding speed was traded for a rollout that never rolled back — wave 3 held publicly rather than take a bad first impression at scale.

PROBING — QUESTIONS, PHASE DEEP-DIVES, AND THE MATH

Challenge — Why did you choose this story to illustrate Deliver Results?

Because both public targets landed two months early with zero rollbacks: CTS $0.66 → $0.03 across customers 7 → 68 on the same ops team. Results delivered behind graduation gates, not by boarding faster than the automation could carry.

Challenge — What other stories demonstrate Deliver Results?

S18 — 8 months, both hubs, gate cleared — is the backup; S2's comingle launch and S6's three-months-to-live are the thirds.

Challenge — Could you come up with an example that is more recent?

Flexport — recent, though not current role. The current-role Deliver Results record is S2: the comingle launch, zero incidents, +15% signed.

Probing — Why did you choose to focus on these results? What other results were important?

CTS −95.5% across 7 → 68 with zero rollbacks is the unit-economics proof that scale came without headcount; automation 40 → 90% is the mechanism; two months early is the time result. The ~$1.5M pre-prod saving matters too — kept separate because it is a separate record.

Probing — You mentioned dollars — what percentage change is that year over year?

Not YoY — the $0.66 → $0.03 is cost-to-serve per container, before vs after, and the direct dollars are honestly small: ≈$50K/yr modeled on an illustrative container count. The value is scale without headcount, and the ~$1.5M pre-prod figure is a separate record I never merge with CTS.

Probing — I'm concerned about your containers-per-shipment assumption — tell me more.

It is a guess and I said so — the record gives me thirty-two thousand shipments, not containers; at one and a half to four containers per shipment the saving runs thirty to eighty thousand. I would firm it up by pulling the active-container count straight from the 15-day active rule in the WBR.

Challenge — What were the lessons learned? What would you have done differently?

Board no faster than automation matures; a gate that only holds in private is a suggestion. Differently: start the WBR at kickoff, not after the critique.

Challenge — How would you implement this at Amazon?

A WBR culture story told to the company that invented the WBR — copy the trusted format, freeze definitions, tag inputs to outputs, and let the review make the rollout decision; the graduation criteria are Amazon-shaped launch gates, published before the wave and held in public.

Challenge — How did these results compare to your actual goals?

The public targets were 68 customers, 170 MAU and 32K shipments — both targets hit two months early, 7 → 68 with zero rollbacks, automation 40 → 90%, and the same critic who said value wasn't legible steering by the review.

PHASE DEEP-PROBE BANK — Q&A (13). Answer the phase asked, then stop.

Discovery — How did you know discovery was done?

When leadership had told me what they actually decide on — rollout pace and funding — and agreed the questions the review must answer weekly. Their critique was about evidence, not effort, and I accepted that first.

Why copy a format instead of designing one?

The LSA review already had leadership's trust — five-week window, WoW at ±5%, inputs tagged to outputs. Inventing a format would have spent credibility I didn't have.

Plan & Design — Who wrote the metric definitions and who signed?

I wrote the seven; the owner of each input — Data Services, Support, GTM — signed the definition they would be red against. Frozen after that; new metrics out of scope.

What was in your risk register?

Measurement-caused reds, a flattering wrong number, and bus factor one. Mitigations: decompose before the meeting, print data misses honestly, automate the pull — which became S16.

How did you handle the objection to freezing the list?

Stability over coverage — a number that changes definition can't steer a rollout. I offered a parking lot for new metrics and kept the seven.

Build & Test — What was your definition of done for a red week?

Decomposed before the meeting, an owner named, a dated fix, and the same number verified the following Monday. A red without an owner was my miss, not theirs.

Describe a hard call in the build.

Discarding a truncated 99.4% week. It flattered the program and would have taught the VP a number that couldn't survive the next Monday. I printed a data miss.

Cutover — When did the review become "live"?

The Monday the VP took the rollout decision in the room — unlocking the 68-client rollout two months early off the numbers, not a narrative.

Which wave did the review hold and how did Sales take it?

Wave 3, on ETA accuracy traced to one bad source. Sales had the six graduation criteria before wave 1, so the hold was the rule firing — and the customers we held became the cleanest wave.

Hypercare — How did you know the review would outlive your hands?

When the pipeline (S16) produced the same seven numbers within 1% for consecutive weeks and the critic who started this steered by them.

What did you keep doing after the review was credible?

Running it by hand every Monday for over a year — 1.5–2 hours, then about 4 — until the pipeline matched within 1%. A review that skips a week teaches leadership it can be skipped.

Cross-phase — Where did you trade scope for date? What did you escalate?

New metrics for stability, and boarding speed for a rollout that never rolled back — held wave 3 publicly rather than take a bad first impression at scale. Escalated nothing outside the review — the review was the escalation forum, with the VP deciding in it every Monday.

Why discard 99.4%? · Do differently?

Three touches is not a week; printing it would have taught the VP a number that couldn't survive the next Monday. Differently: start the WBR at kickoff, not after the critique.

Where this lands at Amazon: this is a WBR culture story told to the company that invented the WBR — the respect move is that I copied a trusted format rather than inventing one, froze definitions, tagged inputs to outputs, and let the review make the rollout decision. The graduation criteria are Amazon-shaped launch gates: published before the wave, held in public when they fired.

BUSINESS IMPACT MODEL — KPI TREE, THE WAVE-MACHINE MATH

```
Program value legible to leadership (funding + 68-client rollout)
+- Cost-to-serve per container $0.66 -> $0.03 (record)
|    +- Human interactions/container 3.7 -> 0.17 (record cross-walk: $0.18/interaction)
|    +- Automation coverage 40% -> 90% (record)
|         (loop) fewer touches -> more calibration data -> better ETA/DQ ->
|                fewer exceptions -> fewer touches
+- Customers boarded 7 -> 68 (9.7x), zero rollbacks (record)
|    +- Six graduation criteria published before W1 (gate)
|    +- W3 held publicly on ETA accuracy -> cleanest wave
+- Time-to-value: rollout unlocked 2 months early off the WBR (record)
+- Pre-prod measurement path ~$1.5M saved (record, SEPARATE canonical)
```

If challenged. *"The CTS saving is only $50K?"* — "Directly, yes — and I say that out loud. The point of −95.5% was never the direct dollars; it's that 7 customers became 68 with zero added ops headcount. The modeled figure is honest

because the volume is; the scaling is the record." *"Why ±1.6%?"* — "n = 300 at an expected 98% accuracy gives a 95% confidence interval of about ±1.6 points — big enough to certify the gate, small enough to afford. That measurement path cut rollout from 6 months to 1.5 and carries the separate ~$1.5M record."

THE MATH — STEP-BY-STEP DERIVATION

Table 1 — before: the problem in numbers.

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
1	Cost-to-serve	`$0.66 per container`	off seven pilots (record)	$0.66
2	Residual human interactions	`$0.66 / $0.18 = 3.67 ≈ 3.7 per container`	$0.18 per eliminated interaction (record cross-walk)	≈3.7 touches per container
3	Automation / accuracy	`coverage 40%; accuracy 60%`	record	most containers still touched by hand
4	The legibility problem	`value invisible to leadership — funding at risk`	the problem the WBR solves	no instrument

Table 2 — the fix: step by step, values substituted.

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
1	Six graduation criteria, all must pass	`DQ ≥ 90% · ETA ≥ 80% · SLA 90% reviewed <24h · ≥100 containers + 5 features · stability · CSAT ≥ consignee average`	published to Sales before wave 1; entry bar ~50 containers calibration (record)	wave 3 held in public on ETA accuracy — became the cleanest wave
2	Launch gate	`100 real containers × 7 checks; load proof 500 containers / 20 min / 99.6%`	before the first paying customer (record)	passed
3	Honest denominator	`active = touched within 15 days; a 99.4% week on 3 touches discarded — Wilson CI ≈ [0.44, 1.0]`	a 95% CI on N = 3 is wider than the metric (record decision; illustrative math)	printed as a data miss, not a win
4	WoW instrument	`WoW = (B − A)/A = (21.9 − 40.0)/40.0 = −45% → red`	A = 40.0%, B = 21.9% (illustrative on the real decision); ±5% red/green	root cause: a carrier format change — measurement, not operators
5	Every input tagged	`each metric tagged to the output line it moves ("RollUp")`	record design	cause and effect on one page

Table 3 — after: the metric, before → after.

STEP	METRIC	BEFORE	AFTER	BASIS
1	Cost-to-serve per container	$0.66	$0.03 = ($0.03 − $0.66)/$0.66 = −95.5%	record
2	Residual interactions per container	≈3.7	$0.03 / $0.18 = 0.17	record cross-walk
3	Automation / accuracy	40% / 60%	90% / 90%	record
4	Customers	7	68 = 9.7×; zero rollbacks; no added ops headcount; both public targets 2 months early	record

Table 4 — the dollars, every multiplication shown.

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
1	Containers at the 32K-shipment target	`32,000 × 1.5 = 48,000; 32,000 × 4 = 128,000`	1.5–4 containers per shipment (illustrative)	48–128K containers/yr
2	Unit saving	`$0.66 − $0.03 = $0.63 per container`	record endpoints	$0.63
3	Direct CTS dollars	`48,000 × $0.63 = $30,240; 128,000 × $0.63 = $80,640`	modeled	$30–80K/yr → stated ≈$50K
4	Pre-production measurement path	`n = 300 containers → ±1.6% at 95% CI; rollout 1.5 vs 6 months`	separate record figure — never merged with CTS	~$1.5M saved
5	The value	`scale 7 → 68 on the same ops team`	record	scale without headcount — not the direct dollars

SENSITIVITY — LOW / BASE / HIGH

INPUT	LOW	BASE	HIGH
Containers per shipment (benchmark)	1.5	2.5	4.0
Containers/yr at 32K shipments	48K	80K	128K
Direct CTS saving ($0.63 × containers)	$30K	$50K	$81K
ARR per customer (benchmark, illustrative)	$10K	$25K	$50K
Revenue pull-forward (one-time)	$102K	$254K	$508K
WBR labor released ($/hr 80–120)	$17K	$21K	$25K
Revenue pull-forward is one-time; $1.5M record excluded	—	—	—
Total modeled $/yr	≈$47K	≈$71K	≈$106K

TOTAL IMPACT STATEMENT — SAY IT LIKE THIS

"Honestly, the cost-to-serve number is a unit-economics result, not a big dollar number: ninety-five percent off per container is measured, and at the thirty-two-thousand-shipment target that is only about fifty thousand dollars a year of direct cost, which is my model on an illustrative container count. What it bought is scale: sixty-eight customers boarded on the same ops team with zero rollbacks, about two and a half people we never had to hire, and a rollout unlocked two months early, which I model at roughly a quarter million of one-time revenue pull-forward. The million and a half on the pre-prod measurement path is a separate record figure and I keep it separate. So: the ninety-five percent and the seven-to-sixty-eight are record; every dollar figure except the one-point-five is my model."

If challenged — the full answers.

Your containers-per-shipment is a guess.

It is, and I said so. The record gives me thirty-two thousand shipments, not containers; I used two and a half FCL containers per shipment, and at one and a half to four the saving runs thirty to eighty thousand. I would firm it up by pulling the active-container count straight from the fifteen-day active rule in the WBR, which is the same denominator the review already used.

Fifty thousand a year is small — why lead with it?

Because it is honest, and because the point was never the dollars, it was boarding nine point seven times the customers without nine point seven times the ops headcount. The hardest part of that claim is the one-minute-per-interaction assumption behind the two-and-a-half FTE; I would measure it from the ticket-grain automation data, five hundred thirty-eight auto versus four hundred fifty-six human.

> **Where does the million and a half come from, then?**
> That is the pre-prod measurement path, a separate canonical: three hundred containers at plus or minus one point six percent let us prove the product in six weeks instead of six months. It is a record figure, it is not cost-to-serve, and I never add it to the fifty thousand. To firm it up I would want the finance sign-off memo that priced the four and a half months saved.

S12 — Flexport: FF App export launch — operator-first vs partner-first, CTS −25% vs −50%, the 8-week rebuild/reuse rule

Amazon LPs: Think Big (primary — the flip) · Frugality (primary — ~$5M avoided) · Have Backbone (changing the manager's plan) · Earn Trust (privately first). **Crosswalk:** loop packet S12 — **§1.3 CONFLICT: the loop packet tells this as a Sephora roadmap; this packet's record is Flexport FF App.** Resolve before the loop. *Say the ~$5M once — never "avoidance" and "annual reduction" as two results.*

Honesty label. The roadmap was my manager's; I owned the sequencing recommendation. Engineering produced the re-estimates; the ~$5M is the record's cost-avoidance figure, said once; the run-rate decomposition is my model.

60-SECOND PITCH

"**Situation.** The Freight Forwarding App export roadmap was operator-first, 18 months, already socialized, and my manager owned it; baselines 339.5 min/shipment and 2,200 operator interactions per departed shipment. **Task.** With no authority over the roadmap, re-sequence it on cost-to-serve evidence. **Action.** Discovery: time study × loaded rates. Plan & Design: operator-first −25% CTS in 9 months vs partner-first −50% in 6; the 8-week rebuild/reuse rule; the reversal trigger written. Build & Test: engineering re-estimated with confidence buffers; I showed my manager privately first. Cutover: the Bellevue onsite ratified the flip; burn-up 55.8% → 83.2%. Hypercare: operators lived in old screens two more quarters — priced and published. Lesson: bring the model, name the owner, write the reversal trigger. **Result.** Primary metric: engineering effort to first value — the 18-month operator-first plan → **66% less effort on the partner-first path, value in 6 months instead of 18**. $ impact: ~$5M cost avoidance (record, said once); CTS −50% vs −25% at the work-item grain; 48 of 53 partners migrated within a month."

SITUATION / TASK — WHERE IT OCCURRED, WHEN IT HAPPENED, WHY IT IS IMPORTANT

S — The inherited launch plan for the freight-forwarding app was operator-persona first, 18 months, already socialized, and my manager owned it. Baselines: 339.5 min/shipment time-to-serve, 2,200 operator interactions per departed shipment.

T — I owned the sequencing recommendation — and changing a plan with my manager's name on it.

The real question was time-to-value and unit cost, not persona preference. The challenges: re-opening a committed, socialized plan; getting honest re-estimates with confidence levels; and not abandoning operators while partners went first. Left as inherited, the plan meant eighteen months to first value on the operator path, with operators still spending 20% of their time in the old app at the end of it — half the unit saving (−25% vs −50% at the work-item grain), reached a year later.

The metrics on the table, with their baselines: time-to-serve — 339.5 minutes per shipment; operator interactions — 2,200 per departed shipment; time to first value — month 18 on the inherited path; cost-to-serve at the work-item grain — the −25% vs −50% comparison the recommendation would turn on.

ACTION — WHAT I PERSONALLY OWNED, HOW I DID IT, WHO ELSE WAS INVOLVED

A — Five gated phases:

1. **Discovery** — baseline memo (339.5 min/shipment; 2,200 interactions); bare-minimum scope per persona cut with the business; the question restated as time-to-value and unit cost, not persona preference. *Exit bar:* business agrees the bare minimums; engineering agrees to re-estimate with confidence levels.
2. **Plan & Design** — re-estimates (9 months operator vs 6 months partner) with confidence buffers (×1.0/×1.4 + 15%); the 20%-still-in-old-app finding; work-item-level cost-to-serve translation (−25% vs −50%); the rebuild-vs-reuse rule per module (>8 dev-weeks → reuse); read privately to my manager first; engineering got two days to break it. *Exit bar:* my manager puts his name on the flip at the Bellevue onsite; sequencing signed.
3. **Build & Test** — 15 domains × 6 modules (Booking Validated → Customs Declaration → …); Jira burn-up with the scope tripwire ("new work estimated and swapped, never absorbed"); operator-interactions-eliminated as the

success metric; the tenet "no go-live without every work item assigned to one operator." *Exit bar:* burn-up 55.8% → 83.2% across Q3–Q4; every work item owned.

4. **Cutover & Go-Live (waves)** — MVP with 11 partners in 10 countries; partner migration plan published; operators' two further quarters on old screens priced and published. *Exit bar:* 48 of 53 import partners migrated within a month; MVP live.
5. **Hypercare / Close** — tracked interactions eliminated; sequenced FCL 2022 → Air/LCL 2023; close-out with ~$5M cost avoidance and 66% less effort. *Exit bar:* value in 6 months instead of 18; operators moved onto the shared chassis on the published schedule.

Engineering produced the re-estimates; the cost-avoidance figure is the record's, said once; the run-rate decomposition is my model. Mine specifically: the baseline memo, the comparison model, the rebuild/reuse rule and the scope tripwire; the business signed the bare minimums per persona, and my manager signed the sequencing and owned the flip. The contribution: restating the question as time-to-value and unit cost, building the work-item-grain comparison, and reading it privately to my manager first so the flip could carry his name at the Bellevue onsite — engineering got two days to break it.

The obstacle was changing a plan with my manager's name on it — handled privately first, with the numbers: 9 versus 6 months, 20% still in the old app, −25% versus −50%; he argued two points, I adjusted one, and the onsite ratified it. Optimistic estimates got confidence buffers, operators got a priced, published two-quarter schedule, and silent scope absorption got the burn-up tripwire. Priorities were ranked by benefit × impact divided by dev-weeks per module, with new work estimated and swapped, never absorbed; buy-in ran private-first, two days for engineering to break the model, then the room ratifies — I have never won a disagreement by seniority. The decision challenged was my manager's own operator-first sequencing — on cost-to-serve evidence at the work-item grain, not persona preference — and the influence mechanism was making the better decision his decision, with his name on it at the onsite.

RESULTS — HOW SUCCESS WAS MEASURED: $ · # · % · TIME · CUSTOMER · QUALITY

R — Primary metric: effort to first value **−66%; value month 6 instead of 18**. ~$5M cost avoidance (record, said once); CTS −50% vs −25%; 48/53 partners in a month.

- **$** — ~$5M cost avoidance (record, said once — never “avoidance” and “annual reduction” as two results); the ≈$1.5M/yr run-rate underneath is my model at −50% CTS on an illustrative 20K export shipments.
- **#** — 48 of 53 import partners migrated within a month; MVP with 11 partners in 10 countries; baselines 339.5 min/shipment and 2,200 operator interactions per departed shipment (record).
- **%** — Engineering effort to first value −66%; CTS −50% partner-first vs −25% operator-first at the work-item grain; burn-up 55.8% → 83.2% across two quarters (record).
- **Time** — Value in month 6 instead of month 18 (record); FCL 2022 → Air/LCL 2023 sequenced off the interactions read.
- **Customer** — Partners completed end-to-end in the new app; operators' two further quarters on old screens were priced and published, not hidden (record).
- **Quality** — New scope estimated and swapped, never absorbed; the tenet — no go-live without every work item assigned to one operator (record).

Against the goal stated above: first value in month 6 instead of month 18, at 66% less engineering effort; −50% versus −25% cost-to-serve at the work-item grain against the stated baselines; 48 of 53 partners migrated within a month. The trades were priced and published: operators stayed on old screens two more quarters so partners finished end-to-end in six months, and estimates carried confidence buffers — a slower, honest plan over a fast optimistic one.

PROBING — QUESTIONS, PHASE DEEP-DIVES, AND THE MATH

> **Challenge — Why did you choose this story to illustrate Think Big?**
>
> Because the flip re-opened a committed 18-month plan on evidence and changed the destination: value in month 6 instead of 18 at 66% less engineering effort — the goal mattered more than the plan. The reversal trigger and the scope

tripwire are what made the big swing safe.

Challenge — What other stories demonstrate Think Big?

S9 — the 1,000-lane vision, staged honestly — is the backup; S1's book-of-record migration is the third.

Challenge — Could you come up with an example that is more recent?

Flexport — recent. For a current-role Think Big record, S1 — the book-of-record migration — is the one I would pair with it.

Probing — Why did you choose to focus on these results? What other results were important?

Effort −66% and value in month 6 instead of 18 are the record pair that justified the flip; the cost avoidance is the program's signed figure, said once. 48 of 53 partners migrating within a month proved the partner path was real.

Probing — You mentioned dollars — what percentage change is that year over year?

Not YoY — the avoidance is one-time (engineering never spent, plus a year of unit savings pulled forward), and I never present it as an annual reduction; the recurring run-rate underneath is a separate, smaller modeled number on an illustrative shipment volume.

Probing — I'm concerned about the engineer count and rate behind your decomposition — tell me more.

Twelve engineers at $20K per month are my numbers — the record does not itemize the avoidance. At ten to fifteen engineers and fifteen to twenty-five thousand the effort piece runs $1.8M to $4.5M, and the record's figure sits inside that band; I would firm it up with the two scoping envelopes — 66.2 and 99.8 engineer-months — and Finance's actual loaded rate.

Challenge — What were the lessons learned? What would you have done differently?

Bring the model, name the owner, write the reversal trigger. The scope tripwire — new work estimated and swapped, never absorbed — is what kept the burn-up honest, 55.8% → 83.2%.

Challenge — How would you implement this at Amazon?

Think Big with Frugality attached — re-open a committed plan when the goal matters more than the plan, price it in unit economics, and have the owner whose name was on the original ratify the change; the reversal trigger and the scope tripwire make the swing safe.

Challenge — How did these results compare to your actual goals?

The inherited plan promised operator value in 18 months; the flip delivered first value in month 6 at 66% less effort, −50% versus −25% cost-to-serve at the work-item grain, and 48 of 53 partners migrated within a month.

PHASE DEEP-PROBE BANK — Q&A (13). Answer the phase asked, then stop.

Discovery — How did you know discovery was done?

When each persona had a bare-minimum scope the business had signed and engineering had agreed to re-estimate with confidence levels. Baselines — 339.5 minutes and 2,200 interactions per shipment — were on paper first.

Who wrote the success criteria?

I did: time-to-value and cost-to-serve at work-item level, with monthly operator interactions eliminated as the metric. The business signed the minimums; my manager signed the sequencing.

Plan & Design — How did you handle your manager's plan being wrong?

Privately first, with the numbers — 9 versus 6 months, 20% still in the old app, −25% versus −50%. He argued two points, I adjusted one, and the onsite ratified it with his name on it.

What was in your risk register at ratification?

Optimistic estimates, operators feeling abandoned, and scope absorbed silently. Mitigations: confidence buffers, a priced two-quarter schedule, and the burn-up tripwire.

What was the rebuild-vs-reuse rule?

Over eight dev-weeks to rebuild — reuse; under — rebuild; rank by benefit × impact divided by dev-weeks. Agreed with product, UX and engineering per module.

Build & Test — How did you sequence in Jira?

Fifteen domains by six modules, tracked as a burn-up against planned engineering days — 55.8% to 83.2% across two quarters. New work was estimated and swapped, never absorbed.

What was your definition of done for a module?

Every work item assigned to one operator and its interactions-eliminated count recorded. The tenet: no go-live without every work item owned.

Cutover — What did each wave deliver?

MVP: 11 partners in 10 countries end-to-end on the new backend. Migration: 48 of 53 import partners within a month. Then operators onto the same chassis, then Air/LCL in 2023.

What was your rollback?

Partners could keep working in the old flow during migration, and operators were still on old screens by design. Reversibility was built into the sequencing.

Hypercare — How did you know the flip was right?

Value in six months instead of eighteen, 66% less engineering effort, about $5M cost avoidance — said once — and operators moved onto the chassis on the published schedule.

What did you track after the partners were live?

Monthly operator interactions eliminated against the 2,200-per-shipment baseline, by work item, and the burn-up on the remaining modules. Air/LCL 2023 was scheduled off that read, not off a date on a slide.

Cross-phase — Where did you trade scope for date? What did you escalate?

Operators stayed on old screens two more quarters so partners finished end-to-end in six months — priced, published, scored. Escalated nothing past my manager. I have never won a disagreement by seniority — I rank, privately first, and let the room ratify.

How did your manager react?

He asked for the estimation details, argued two points, and put his name on the flip at the onsite. That's the point of private-first.

Where this lands at Amazon: Think Big with Frugality attached — re-opening a committed plan because the goal mattered more than the plan, priced in unit economics, ratified by the owner whose name was on the original. The reversal trigger and the scope tripwire are the mechanisms that made a big swing safe.

BUSINESS IMPACT MODEL — KPI TREE, NUMBERS, THE MATH

```
~$5M cost avoidance (record, said once) = effort avoided + value pulled in
+- Engineering effort to first value -66% (record)
|    +- rebuild-vs-reuse: >8 dev-weeks -> reuse · confidence x1.0/x1.4 + 15% buffer
|    +- burn-up 55.8% -> 83.2%; scope swapped, never absorbed
+- CTS per shipment: 339.5 min x ~$0.45/min ~= $153 derived
|    +- partner-first -50% vs operator-first -25% (record, work-item grain)
|    +- shared chassis loop: partner backend -> less rework -> faster operator move
+- Time-to-value 18 -> 6 months (record); 12 months of unit saving pulled forward
+- Cost of the trade: operators on old screens 2 more quarters (priced, published)
```

DRIVER	BASELINE	AFTER	$	BASIS
Time-to-value	18 months	6 months	—	record
Engineering effort to first value	100%	34%	≈$2.9M one-time	modeled (12 eng × 18 mo × 66% × $20K/eng-mo; benchmark $15–25K)
CTS per shipment	≈$153	≈$76 (−50%)	≈$1.5M/yr	record inputs, derived $ × 20K shipments (benchmark volume)
Value pulled forward 12 months	0	12 mo × $1.5M/yr	≈$1.5M one-time	modeled
Partners migrated in a month	0	48 of 53 (90.5%)	—	record

If challenged. *"Is the $5M savings or avoidance?"* — "Avoidance — engineering never spent and value pulled forward — and I say it once. The recurring run-rate is a separate, smaller modeled number; I never present the avoidance as an annual reduction." *"Your manager's plan — why not escalate?"* — "Because the fastest path to a better decision was making it his decision. Private first, two days for engineering to break it, then the room ratifies. I've never won a disagreement by seniority."

THE MATH — STEP-BY-STEP DERIVATION

Table 1 — before: the problem in numbers.

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
1	Inherited plan	`operator-first: 18 months to value, socialized`	record	month 18
2	Unit baselines	`339.5 min/shipment; 2,200 operator interactions per departed shipment`	record	the two numbers everything prices from
3	CTS per shipment	`339.5 min × $0.45/min = $152.78 ≈ $153`	blended loaded rate from FCL $0.56 · Network $0.61 · Data $0.24 (record inputs; derived, labeled)	≈$153/shipment
4	Operator-first path, work-item grain	`9 months → −25%; operators still 20% of time in the old app`	engineering re-estimates (record)	−25%, capped by the legacy residue

Table 2 — the fix: step by step, values substituted.

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
1	Partner-first path	`6 months → −50%; the partner backend is the chassis operators later ride`	engineering re-estimates with confidence levels (record)	−50%, 12 months earlier
2	Unit saving per path	`$153 × 50% = $76.50 ≈ $76 vs $153 × 25% = $38.25 ≈ $38`	from Table 1 row 3	double the unit saving
3	Effort to first value	`−66% ⇒ 100 − 66 = 34% of the inherited path`	record	34%
4	Confidence rule	`(10 × 1.0 + 10 × 1.4) × 1.15 = (10 + 14) × 1.15 = 27.6 wks planned vs 20 naive`	est × {1.0, 1.4} by confidence + 15% testing buffer; two 10-wk modules (illustrative)	buffers only inflate low-confidence items
5	Rebuild-vs-reuse rule	`dev_weeks > 8 → reuse; rank by benefit × impact / dev-weeks`	the 8-week rule was the engineering median (record rule)	module-by-module, not ideology
6	Scope tripwire	`burn-up 55.8 → 63.3 → 66.0 → 69.3 → 76.0 → 83.2% across two quarters`	new work estimated and swapped, never absorbed (record)	scope held in daylight

Table 3 — after: the metric, before → after.

STEP	METRIC	BEFORE	AFTER	BASIS
1	Time-to-value	month 18	month 6	record
2	Engineering effort to first value	100%	34%	record
3	CTS per shipment	≈$153	≈$76 (−50%)	record inputs, derived
4	Partners migrated	0	48 of 53 = 90.5% within a month	record
5	Burn-up (completed/pointed)	55.8%	83.2%	record

Table 4 — the dollars, every multiplication shown.

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
1	Unit saving	`$153 × 50% = $76/shipment`	from Table 2 row 2	≈$76
2	Run-rate	`$76 × 20,000 export shipments = $1,520,000`	volume illustrative (my model)	≈$1.5M/yr
3	Engineering avoided	`12 eng × 18 mo = 216 eng-mo; × 66% = 142.6 eng-mo; × $20,000 = $2,851,200`	counts and rate mine (benchmark $15–25K/eng-mo)	≈$2.9M one-time
4	Value pulled forward	`12 mo × $1.5M/yr = $1.5M`	from row 2 (modeled)	≈$1.5M one-time
5	Decomposition check	`2.9 + 1.5 = $4.4M`	within ~12% of the record's ~$5M avoidance (said once); the record number is the headline, the decomposition is mine	$4.4M

SENSITIVITY — LOW / BASE / HIGH

INPUT	LOW	BASE	HIGH
Engineers on the plan (illustrative)	10	12	15
Loaded $/eng-month (benchmark)	$15K	$20K	$25K
Eng $ avoided (one-time)	$1.8M	$2.9M	$4.5M
Export shipments/yr (benchmark)	15K	20K	30K
Unit saving at −50% ($76 × vol)	$1.1M	$1.5M	$2.3M
Value pulled forward 12 mo (one-time)	$1.1M	$1.5M	$2.3M
Decomposition total (one-time, vs record ~$5M)	$2.9M	$4.4M	$6.8M
Total modeled $/yr	$1.1M	$1.5M	$2.3M

TOTAL IMPACT STATEMENT — SAY IT LIKE THIS

"About five million of cost avoidance, and I say that once. Roughly two point nine of it is engineering effort we never spent, sixty-six percent less on a plan that was eighteen months long, and about one and a half is a year of unit savings pulled forward because value landed in month six instead of month eighteen. Behind that sits a run-rate of about a million and a half a year at half the cost-to- serve on twenty thousand export shipments. The sixty-six percent, the six versus eighteen months, and the minus fifty versus minus twenty-five are the record; the five million is the program's signed figure; the split between engineering and pulled-forward value, and the shipment volume, are my model."

If challenged — the full answers.

Twelve engineers and twenty thousand dollars a month are your numbers, not the program's.

Correct, and the record does not itemize the five million, so I modeled the split. Ten to fifteen engineers at fifteen to twenty-five thousand puts the effort piece anywhere from one point eight to four and a half million; the record's five million sits inside that band. I would firm it up with the two scoping envelopes, sixty-six point two and ninety-nine point eight engineer-months, and finance's actual loaded rate.

Where do twenty thousand shipments come from?

It is an illustrative FCL export volume, and it is the softest input: at fifteen to thirty thousand the run-rate moves from one point one to two point three million a year. The three-thirty-nine-and-a-half minutes and the per-minute rates are the time study; I would replace the volume with the departed-shipment count from the same system that produced the two thousand two hundred interactions baseline.

> **Are you double counting the pulled-forward value and the five million?**
>
> No. The five million is the total; the pulled- forward year is one slice of how it decomposes, and the million and a half a year is the run-rate underneath, not a second result. To make that airtight I would ask for the finance close-out that priced the avoidance and show the two slices summing to it.

S16 — Flexport: four systems into one Snowflake pipeline — and the security-review miss I owned upward the day it surfaced

Amazon LPs: Ownership (primary — bus factor 1, and owning the miss) · Invent and Simplify (backup) · Frugality (the >$100K priced baseline) · Learn and Be Curious. **Crosswalk:** loop packet MAIN's BRD/pipeline beat + the security-review miss — consistent. *Same program family as S11 — never both to one interviewer.*

Honesty label. I wrote the BRD and owned the cutover; the infra team owned the build. The >$100K/yr is a *priced baseline* in the infra team's own loaded rates — priced, not claimed as savings. The two-week security-review miss is mine, owned upward the same day.

60-SECOND PITCH

"**Situation.** The WBR ran on four systems stitched in Sheets by me — bus factor 1, accuracy 60%, tickets with no container ID. **Task.** I wrote the BRD and owned the cutover; infra owned the build. **Action.** Discovery: froze seven metric definitions; priced the status quo at >$100K/yr in infra's own rates. Plan & Design: ETL → S3 → SQS → Snowflake → Looker; parallel-run gate within 1%. Build & Test: a cross-reference table first, then a real join key pushed upstream. Cutover: consecutive weeks inside 1%. Hypercare: my miss — I did not plan the data-access security review; it cost two weeks; I owned it upward the same day; zero review deliveries skipped. Lesson: every kickoff asks whose approval the data needs and its lead time. **Result.** Primary metric: WBR data accuracy — **60% → 90%** once four systems became one Snowflake pipeline. $ impact: the manual status quo was priced at >$100K/yr (a priced baseline, not a claimed saving); first automated catch was a 4% ticket-sync drop."

SITUATION / TASK — WHERE IT OCCURRED, WHEN IT HAPPENED, WHY IT IS IMPORTANT

S — The Monday review ran on four systems that didn't talk, stitched in Google Sheets by one person — me: bus factor 1, accuracy 60%, 1.5–2 hours per snapshot, tickets carried no container ID. A VP was steering on a number one person could break.

T — Nobody had written requirements for a program's operating metrics. I wrote the BRD and owned the cutover; the infra team owned the build.

The goal: one pipeline producing the same seven numbers, accurate and owned by more than one human. The challenges: the ticket join gap, earning parity before anyone would switch, and — as it surfaced — a data-access security review I had not planned. The risk of leaving it: a wrong number steering the review, and silent data loss nobody would catch — the 4% ticket-sync drop was invisible in the manual era.

The metrics on the table, with their baselines: WBR accuracy — 60%; bus factor — 1; snapshot effort — 1.5–2 hours per Monday; ticket-to-container joinability — none.

ACTION — WHAT I PERSONALLY OWNED, HOW I DID IT, WHO ELSE WAS INVOLVED

A — Five gated phases (week counts illustrative except the 2-week miss):

1. **Discovery** — documented the as-is (manual pulls + extracts in Sheets, 1.5–2 hrs per snapshot); named the join gap (tickets without container ID); priced the status quo at >$100K/yr in the infra team's own rates including VP decision risk. *Exit bar:* infra prioritizes it themselves off the price; scope = the seven metrics, nothing new.
2. **Plan & Design** — the BRD: seven frozen definitions, daily/hourly feed, failure alerts, a cross-reference table for the ticket join then a real key upstream; acceptance = parallel run within 1% for consecutive weeks. The RAID log — **missing the data-access security review**. *Exit bar:* BRD accepted by infra; acceptance gate published.
3. **Build & Test** — extract/load specs per system, parity checks, the parallel-run procedure; when the security review surfaced mid-build I re-sequenced and owned the miss upward the same day. *Exit bar:* review closed (2 weeks); parallel run within 1% for consecutive weeks; mismatches explained (mostly late-arriving tickets on the manual side).

4. **Cutover & Go-Live** — the cutover memo; the Monday deck moved onto the pipeline; the manual pull kept as rollback. *Exit bar:* first Monday off the pipeline, same seven numbers; zero WBR deliveries skipped through the build.
5. **Hypercare / Close** — watched the failure alerts; triaged the first automated catch (a 4% ticket-sync drop invisible in the manual era); definitions into code; institutionalized the kickoff question. *Exit bar:* accuracy 60% → 90%; the pipeline steers the review without me.

The build was the infra team's — the ETL → S3 → SQS → Snowflake → Looker pipeline and the failure alerts; mine were the as-is documentation, the >$100K pricing, the BRD's seven frozen definitions, the acceptance gate, the parallel-run procedure and the cutover memo. The contribution: pricing the status quo in the owning team's own loaded rates — including VP decision risk, not just hours — so infra prioritized the build themselves (the >$100K/yr is a priced baseline, not claimed savings), and the acceptance gate: parallel run within 1% per frozen metric for consecutive weeks, every mismatch explained by cause.

The obstacles: the join gap — shipped with a documented cross-reference table while the real key was pushed upstream to Support — and the unplanned security review: two weeks lost, re-sequenced, owned upward the same day, with the WBR kept alive on the manual pull so zero Mondays were skipped. Scope held to the seven metrics, nothing new — new metrics traded for stability; buy-in came from the price in their own rates, and the parity gate made the cutover unarguable — the VP saw the same seven numbers on the first Monday off the pipeline. What I challenged was my own category error — treating the security review as an engineering concern rather than a program dependency with a lead time; the fix is institutional: every kickoff now asks whose approval the data needs and what its lead time is.

RESULTS — HOW SUCCESS WAS MEASURED: $ · # · % · TIME · CUSTOMER · QUALITY

R — Primary metric: WBR accuracy **60%** → **90%**; bus factor 1 → ≥2; first automated catch: a 4% ticket-sync drop. >$100K/yr priced baseline (record of the pricing).

- **$** — >$100K/yr priced baseline in the infra team's own loaded rates (record of the pricing) — priced, not claimed as savings; my model splits it ≈$30K hours + ≈$70K decision risk, both modeled.
- **#** — Four systems → one Snowflake pipeline; bus factor 1 → ≥2 (record); zero WBR deliveries skipped through the build (record).
- **%** — WBR accuracy 60% → 90%; parallel-run parity within 1% for consecutive weeks; first automated catch a 4% ticket-sync drop invisible in the manual era (record).
- **Time** — The security-review miss cost two weeks — owned upward the same day, build re-sequenced, review closed (record).
- **Customer** — The VP steered on the same seven numbers — the first Monday off the pipeline showed the same seven, and the manual pull stayed as rollback until parity had been earned and held a week beyond (record).
- **Quality** — Seven frozen definitions moved into code (no drift); every parallel-run mismatch explained by cause — mostly late-arriving tickets on the manual side (record).

Against the goal stated above: WBR accuracy 60 → 90%; bus factor 1 → ≥2; the seven frozen definitions moved into code; and the first automated catch — a 4% ticket-sync drop invisible in the manual era — with the pipeline now steering the review without me. The trades: two weeks of date for a security review I should have planned, and new metrics for stability — the BRD froze the seven; the manual pull stayed as rollback until parity had been earned and held a week beyond.

PROBING — QUESTIONS, PHASE DEEP-DIVES, AND THE MATH

Challenge — Why did you choose this story to illustrate Ownership?

Because both halves are ownership: taking the bus-factor-1 system I had built and writing it out of my own hands, and owning the security-review miss upward the same day it surfaced — with zero review deliveries skipped while I paid for it.

Challenge — What other stories demonstrate Ownership?

S1 — taking the definition nobody owned — is the backup; S14, the seam between OMS and WMS, is the third.

Challenge — Could you come up with an example that is more recent?

Flexport — recent. The current-role Ownership record is S1: taking the “available” definition nobody owned through three signatures.

Probing — Why did you choose to focus on these results? What other results were important?

Accuracy 60 → 90% and bus factor 1 → ≥2 are the record results the build existed for; the first automated catch — a 4% ticket-sync drop invisible in the manual era — is the proof the pipeline sees what the Sheets could not.

Probing — You mentioned dollars — what percentage change is that year over year?

Not YoY — the >$100K/yr is a priced baseline (the cost of the manual status quo in infra's own rates, including VP decision risk), not a claimed saving; my $30K/$70K split of it is modeled, and I will not call any of it realized savings.

Probing — I'm concerned about the decision-risk dollar — tell me more.

Yes — it is the soft part: one wrong number in ten becoming a wrong steer worth a squad-week, range $12K to $117K. I'd firm it up by tagging every WBR action item for a quarter and counting how many were reversed when the number was corrected.

Challenge — What were the lessons learned? What would you have done differently?

Security, privacy and data-access reviews are program dependencies with lead times — I paid for that lesson once. The standing question at every kickoff: whose approval does this data need, and what is its lead time?

Challenge — How would you implement this at Amazon?

The Ownership pattern — price the status quo in the owning team's own rates so they prioritize it themselves, keep the old path alive until the new one has earned parity, and treat security and data-access reviews as dependencies with lead times.

Challenge — How did these results compare to your actual goals?

The BRD's acceptance gate was parity within 1% for consecutive weeks — met, with most mismatches traced to the manual side; accuracy went 60 → 90%, the bus factor above one, and the pipeline steers the review without me.

PHASE DEEP-PROBE BANK — Q&A (12). Answer the phase asked, then stop.

Discovery — How did you know discovery was done?

When the as-is was documented — four systems, Sheets, 1.5–2 hours per snapshot, tickets without a container ID — and the status quo was priced in the infra team's own rates. They prioritized it off that number.

Who wrote the requirements?

I did — nobody had written requirements for a program's operating metrics. The BRD carried the seven frozen definitions and the acceptance gate.

Plan & Design — What was in your risk register at BRD acceptance — and what wasn't?

On it: the join gap, parity failure, bus factor one. Not on it: the data-access security review. That omission cost two weeks and is the item I now add at every kickoff.

How did you handle the join problem in design?

A documented cross-reference table so the pipeline could ship, then a real key pushed upstream to Support. Ship with the bridge, fix the source.

Build & Test — What was your acceptance test?

Manual versus pipeline within 1% for consecutive weeks, every mismatch explained by cause. Most mismatches turned out to be the manual side — late-arriving tickets.

How did you handle the security review mid-build?

Owned it upward the same day as a scoping miss, re-sequenced the build, and kept the WBR alive on the manual pull. Zero deliveries skipped; two weeks lost.

Cutover — What made you say go?

Parity within 1% for consecutive weeks and the review closed. The deck source switched; the VP saw the same seven numbers.

What was your rollback?

The manual pull — kept running until the pipeline had earned consecutive weeks of parity, and a week beyond.

Hypercare — How did you know it was working?

The first automated catch — a 4% ticket-sync drop invisible in the manual era — and accuracy moving from 60% to 90%. The alerts came to me, not silence.

What was your definition of "stable" after cutover?

Failure alerts arriving to me, not silence; the seven numbers matching the retired manual pull for a week beyond the parity gate; and the first real catch triaged to a source cause. Then the manual pull was retired.

Cross-phase — Where did you trade scope for date? What did you escalate?

New metrics for stability, and two weeks of date for a security review I should have planned. I would trade the second again; I would plan it now. Escalated the two-week miss, to my leadership, the day it surfaced.

Why didn't you know about the review?

I treated it as an engineering concern rather than a program dependency with a lead time. That's the category error I fixed — the standing question now: whose approval does this data need, and what is its lead time?

Where this lands at Amazon: security, privacy and data-access reviews are program dependencies with lead times — I paid for that lesson once and now ask the question at every kickoff. The rest is the Ownership pattern: price the status quo in the owning team's own rates so they prioritize it themselves, and keep the old path alive until the new one has earned parity.

BUSINESS IMPACT MODEL — KPI TREE, NUMBERS, THE MATH

```
VP steers the program on a number that is right and survives me
+- Status quo priced >$100K/yr (record; priced, not claimed)
|    +- manual hours (stock of my time) ~$30K + decision-risk term ~$70K
|         decision risk = P(wrong number) x cost of a wrong steer
+- WBR accuracy 60% -> 90% (record)
|    +- seven definitions frozen in code (no drift) · join rate -> ~100%
|    +- parity gate <1% consecutive weeks before cutover
+- Bus factor 1 -> >=2 (risk stock retired)
+- Silent data loss -> alerts; first catch: 4% ticket-sync drop
(loop broken) sheet built by hand -> only I can fix it -> more asks -> more errors
(loop held)   security review unplanned -> 2-wk slip -> WBR kept alive by hand ->
              zero skipped Mondays -> trust held -> cutover accepted
```

DRIVER	BASELINE	AFTER	$/YR	BASIS
WBR accuracy	60%	90%	—	record
Status quo priced	>$100K/yr	retired	—	record of the pricing (infra's own rates)
— hours inside that price	~4 hrs/wk	~0	≈$30K	modeled
— decision risk inside that price	P(wrong) × cost	≈$50K retired by accuracy gain	≈$70K	modeled
Bus factor	1	≥2 + alerts	≈$15K	modeled
Security-review slip	—	2 weeks	—	record (the named miss)

If challenged. *"You're claiming $100K of savings?"* — "No — pricing, not claiming. The number existed to make the infra team prioritize the build off their own rates, and it worked. The claimed record is the accuracy move and the first automated catch; the decomposition of the price is my model."

THE MATH — STEP-BY-STEP DERIVATION

Table 1 — before: the problem in numbers.

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
1	WBR accuracy / fragility	`accuracy 60%; bus factor 1; 1.5–2 h per snapshot rising to ~4 h/wk`	record	one person, wrong 2 weeks in 5
2	Status quo priced	`hours term + decision-risk term > $100K/yr`	priced in the infra team's own rates (record of the pricing)	>$100K/yr — priced, not claimed
3	Join problem	`tickets carried no container ID → low join rate`	record	ticket metrics unlinkable
4	Silent data loss	`a sync drop had no alarm in the manual era`	record problem	silence read as health

Table 2 — the fix: step by step, values substituted.

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
1	Parity gate	`|pipeline − manual| / manual ≤ 1% per frozen metric, consecutive weeks`	the BRD's acceptance gate (record)	cutover unarguable
2	Parity evaluated — pass	`|70.9 − 71.2| / 71.2 = 0.42% ≤ 1% → pass`	coverage: manual 71.2% vs pipeline 70.9% (illustrative)	0.42% gap — inside the gate
3	Parity evaluated — investigate	`|461 − 456| / 456 = 1.1% > 1% → investigate`	tickets: manual 456 vs pipeline 461 (illustrative)	5 late tickets — the manual side wrong, pipeline right
4	Join fix	`documented cross-reference now; real key pushed upstream`	record	join rate → ~100%
5	Silence alert	`alert if 1 − sync_rate > 1%; first catch: 1 − 0.96 = 4% > 1% → alert`	threshold illustrative; a 4% ticket-sync drop (record catch)	silence never mistaken for health
6	Definitions into code	`seven frozen metrics moved into code with owners`	record	no drift; bus factor ≥ 2

Table 3 — after: the metric, before → after.

STEP	METRIC	BEFORE	AFTER	BASIS
1	WBR accuracy	60%	90%	record
2	Bus factor	1	≥2	record
3	Manual hours	~4 h/wk	~0	modeled
4	Silent data loss	invisible	alerted; first catch a 4% drop	record
5	Security-review miss	unplanned	2-wk slip re-sequenced; zero skipped Mondays	record

Table 4 — the dollars, every multiplication shown.

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
1	Hours term	`4 h/wk × 52 = 208 h; 208 × ≈$145/h = $30,160`	rate illustrative, consistent with the infra team's priced ≈$30K	≈$30K/yr
2	Decision-risk term	`52 wk × 10% = 5.2 wrong-number weeks; 5.2 × $13,500 = $70,200`	P(wrong number) ≈ 1 in 10 weeks; cost of a wrong steer ≈ a squad-week ≈ $13.5K (both illustrative; range $12–117K)	≈$70K/yr
3	Priced baseline	`30 + 70 = >$100K/yr`	priced — NOT claimed as savings (record of the pricing)	>$100K/yr

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
4	Bus factor / alerts	`≈$15K/yr, modeled`	modeled	≈$15K/yr

SENSITIVITY — LOW / BASE / HIGH

INPUT	LOW	BASE	HIGH
Loaded rate, hours (benchmark)	$120/hr	$150/hr	$200/hr
Cost of a wrong steer (benchmark squad-week)	$15K	$30K	$50K
P(wrong number → wrong steer) (illustrative)	5%	10%	15%
Key-person attrition (benchmark)	15%	20%	25%
Hours value	~$22K	~$27K	~$36K
Decision-error exposure retired	~$12K	~$47K	~$117K
Bus-factor exposure	~$8K	~$17K	~$31K
Total modeled $/yr	~$40K	~$90K	~$185K

TOTAL IMPACT STATEMENT — SAY IT LIKE THIS

"The record number is a price, not a saving: the manual status quo was priced at more than $100K a year in the infra team's own loaded rates, and that number is what got it prioritized. My model shows what sits inside it: about $30K is my hours, roughly four a week, and about $70K is decision risk, a VP steering on a number that was wrong forty percent of the time. Taking accuracy from sixty to ninety retires about $50K of that risk, and getting the bus factor above one is worth maybe $15K a year in expected loss. So the >$100K is record and priced; the $30K, $50K and $15K splits are modeled, and I will not call any of it realized savings."

If challenged — the full answers.

The decision-risk number is the soft part.

Yes, it rests on one wrong number in ten becoming a wrong steer worth a squad- week; the range runs from $12K to $117K. I'd firm it up by tagging every WBR action item for a quarter and counting how many were reversed when the number was corrected.

You never claimed the $100K, so why is it here?

Because it's the number the infra team used to prioritize the build; it's a priced baseline in their rates. What I'd measure now is the hours actually retired and whether any Monday number was restated after cutover, which is the honest realized version.

Bus factor is not a dollar.

Agreed, so I priced it as expected loss: twenty percent attrition times the rebuild plus blind weeks. The real test is whether the seven numbers publish when I'm on vacation, and after cutover they did.

Employer block 3 — Elementum

S8 measured lanes at p99 · S6 supply/demand view · S9 597-lane EDI Pareto · S7 CM BOM · S10 Austria routing guide · S17 carrier bid · S19–S21 the Walmart Marketplace client engagements · then the twelve design engagements (§3)

S8, S6 and S9 lead the block (the lane math, the shortage alert and the EDI Pareto are the Elementum records this loop will most likely probe). S19–S21 are the Walmart Marketplace client engagements — delivery-promise, WBR and adoption work in the pre-purchase delivery-experience domain; every number in them carries a record-from-docs / illustrative / modeled / benchmark label and their canon lives in §4.4. The block closes with the twelve §3 design engagements, told from the same client years.

S8 — Elementum: replacing brochure lead times with measured lane distributions — promise at p99, safety stock released by real variability

Amazon LPs: Dive Deep (primary) · Are Right, A Lot (backup — measured vs brochure) · Frugality (the release).
Crosswalk: loop packet S3 (+S4's math) — consistent. *A different program from S6's $2M — never add them.*

Honesty label. I owned the requirement (measured lanes, not brochures), the p99 promise policy and the release rule; the data team built the distributions; planners sized the buffers; Finance signed the release. The safety-stock formula is how I explain it now with the MIT inventory-theory toolkit — an illustration, not a program artifact.

60-SECOND PITCH

"**Situation.** Planning ran on carrier-brochure lead times — buffers sized on a variability nobody had measured. **Task.** I owned the requirement, the promise policy, and the release rule; the data team built the distributions; planners sized the buffers. **Action.** Discovery: the ask was a 'variability report'; I reframed it as a promise-and-buffer policy. Plan & Design: promise at p99 per lane — the smallest d with $P(L \le d) \ge 0.99$ — not average-plus-padding; release rule as a signed planning act with Finance. Build & Test: actual transit and production cycle measured on 200 lanes; drift flag and refresh cadence. Cutover: I would not endorse the cut until lead-time accuracy hit 99% on that set. Hypercare: post-release stockout-and-expedite watch; fat-tail lanes keep buffer until the source process moves. Lesson: fat tails keep their buffer; I owned the rule, not the model as a day job. **Result.** Primary metric: lead-time accuracy on 200 lanes — brochure lead times → **99% at the p99 promise**. $ impact: 580K units / ~$1.5M of safety stock released without a stockout (record, one-time working capital), with ≈$0.6M/yr modeled around it — a different program from S6's $2M; never add them."

SITUATION / TASK — WHERE IT OCCURRED, WHEN IT HAPPENED, WHY IT IS IMPORTANT

S — The six-site manufacturer's logistics manager asked for a "lead-time variability report." The real problem: inventory planning ran on carrier-brochure lead times — every buffer sized on a variability nobody had measured; too much buffer on reliable lanes, too little on fat-tailed ones.

T — Nobody owned the seam between transportation reality and planning parameters; I took it.

The goal: measured lanes replace catalog values, a promise at a defensible tail, and buffer released only on evidence — scoped at 200 lanes by site, lane and mode. The risk cut both ways: keep planning on brochures and you hold too much on reliable lanes and stock out on fat-tailed ones; cut buffer without evidence and the stockout is yours — and an unmaintained promise decays into a new brochure, which is why every parameter got a drift flag.

The metrics on the table, with their baselines: lead-time parameters — carrier-brochure values, variability unmeasured; target measured distributions with lead-time accuracy gated at 99% on the 200-lane set. Buffer capital — sized on catalog values; target released only under a signed evidence rule, with service held.

ACTION — WHAT I PERSONALLY OWNED, HOW I DID IT, WHO ELSE WAS INVOLVED

A — Five gated phases (dependency: the six-site canonical model (S7) — six definitions cannot produce one σ; lane-cohort counts and week markers are not in the record):

1. **Discovery** — reframed the report request as a parameter problem; scoped 200 lanes by site/lane/mode with success criteria (measured lanes replace catalog values; buffer released only on evidence); showed one right-skewed lane where the average hid the tail. *Exit bar:* planning and logistics agree the problem is the parameter, not the report.
2. **Plan & Design** — wrote the requirement to the data team (mean, spread, tail per lane; production cycle + inbound transit vs actual), the p99 promise policy, the release rule as a signed planning act (fat-tail lanes keep buffer until the source process moves — refused as a blanket release, agreed as a rule), the stockout-and-expedite watch, the drift flag and refresh cadence. On p95 vs p99 I showed the asymmetry per lane and left the

percentile choice to planning — their choice, my frame. *Exit bar:* policy signed by planning; Finance agrees the release rule in the room; data team accepts the spec.

3. **Build & Test** — data team built the lane distributions and tracked lead-time accuracy per lane; I showed the asymmetry per lane (line-down cost vs carrying cost). *Exit bar:* lead-time accuracy 99% on the 200-lane set — the bar I would not endorse a cut below.
4. **Cutover & Go-Live** — planners executed the buffer release under the signed rule, fat-tail lanes excluded until their source process moved, the watch armed first. *Exit bar:* 580K units / ~$1.5M released; no stockout in the watch window.
5. **Hypercare / Close** — read the stockout-and-expedite watch, set drift flags per lane, handed the refresh cadence to planning. *Exit bar:* watch clean; the promise stays measured rather than decaying into a new brochure.

I owned the requirement (measured lanes, not brochures), the p99 promise policy and the release rule; the data team built the distributions and tracked accuracy; planners sized the buffers and executed the release; Finance signed each lane. The contribution was reframing a report request into a promise-and-buffer policy, and writing the release rule as a signed planning act with the exception inside it — fat-tail lanes keep buffer until the source process moves — so the rule did the refusing when Finance pushed for a broader release.

The obstacle was that push — answered with the rule Finance had signed and the lanes' asymmetry; the percentile fight never happened, because I showed each lane's line-down cost against carrying cost and left the choice to planning, and they chose p99. Scope was 200 lanes by site, lane and mode with success criteria written at discovery; the release went lane by lane under the signed rule with the stockout-and-expedite watch armed first, and planning signed the policy with Finance agreeing the release rule in the same room, before any cut. I challenged the brochure itself — showed one right-skewed lane where the average hid the tail — and refused both averaging away a tail and endorsing a cut below 99% accuracy; both refusals were written into the act before anyone asked.

RESULTS — HOW SUCCESS WAS MEASURED: $ · # · % · TIME · CUSTOMER · QUALITY

R — Primary metric: lead-time accuracy → **99% on 200 lanes** at the p99 promise. $ impact: **580K units / ~$1.5M released without a stockout** (record, one-time); ≈$0.6M/yr modeled recurring around it.

- **$** — 580K units / ~$1.5M of safety stock released without a stockout (record, one-time working capital); ≈$0.6M/yr modeled recurring around it (carry at a benchmark 20% all-in rate + expedites avoided) — a different program from S6's $2M; never add them.
- **#** — 200 lanes measured by site, lane and mode; 580K units released (record).
- **%** — Lead-time accuracy 99% on the 200-lane set at the p99 promise — the bar I would not endorse a cut below (record).
- **Customer** — Service held: zero stockouts on released lanes through the watch window (record).
- **Quality** — Release only under the signed rule — fat-tail lanes kept buffer until the source process moved; drift flags and a refresh cadence keep the promise measured, never a new brochure (record).

Against the goal stated above: measured lanes replaced catalog values at 99% lead-time accuracy on the 200-lane set; 580K units / ~$1.5M of safety stock released without a stockout; fat tails kept their buffer — all three discovery criteria held. The trade: capital stayed tied up where the tail said it must, and release speed was traded for a release with zero stockouts in the watch window.

PROBING — QUESTIONS, PHASE DEEP-DIVES, AND THE MATH

Challenge — Why did you choose this story to illustrate Dive Deep?

Because the whole story is refusing the surface request — a report — and measuring the underlying distributions: mean, spread and tail per lane, production cycle plus inbound transit vs actual, with p99 defined cumulatively because a right-skewed lane's normal shortcut understates the tail.

Challenge — What other stories demonstrate Dive Deep?

S10 — Austria, where I measured the measurement — is the backup; S3 (the grain problem) and S7 (the phantom math) are thirds.

Challenge — Could you come up with an example that is more recent?

Elementum years — older. The current-role story with the same LP is S3 — aggregates de-aggregated to the cent — and S10 is the same era's backup.

Probing — Why did you choose to focus on these results? What other results were important?

The release — 580K units / ~$1.5M without a stockout — is the record that proves the promise, and 99% lead-time accuracy on 200 lanes is the bar that made it safe. Service held is the result planners feel; a release that causes a stockout is worse than no release.

Probing — You mentioned dollars — what percentage change is that year over year?

No YoY — the ~$1.5M is a one-time working-capital release, booked once as a signed planning act, not an annual saving; what recurs is my model (≈$0.3M carry at a benchmark rate plus avoided expedites), and I never add this program to S6's $2M.

Probing — I'm concerned about the expedite number — tell me more.

The softest — it assumes ten thousand shipments a year on the two hundred lanes and one late arrival in ten forcing an expedite before p99. To firm it I would pull the expedite ledger by lane for the year before and after the release and compare directly.

Challenge — What were the lessons learned? What would you have done differently?

Fat tails keep their buffer, and I owned the rule, not the model as a day job. And give every parameter a drift flag so the promise never decays into a new brochure.

Challenge — How would you implement this at Amazon?

Inbound lead-time variability is half of every buffer and every promise — measure the lane's own distribution, promise at the tail you can defend, release capital only on evidence with the exception written into the rule, and give every parameter a drift flag.

Challenge — How did these results compare to your actual goals?

The success criteria from discovery were: measured lanes replace catalog values, buffer released only on evidence, fat tails keep buffer. All three held — 99% accuracy at the p99 promise, 580K units / ~$1.5M out, no stockout in the watch window.

PHASE DEEP-PROBE BANK — Q&A (13). Answer the phase asked, then stop.

Discovery — How did you know discovery was done?

When planning and logistics agreed the problem was the parameter, not the report — brochure lead times sizing every buffer — and the 200-lane scope by site, lane and mode was written. The canonical model from the six-site program made one σ possible.

Who wrote the success criteria and who signed?

I wrote them — measured lanes replace catalog values; buffer released only on evidence; fat tails keep buffer. Planning signed the policy; Finance agreed the release rule in the same room.

Plan & Design — What was in your risk register at design sign-off?

The average hiding the tail, a buffer cut before evidence, and the promise decaying into a new brochure. Mitigations: cumulative p99 per lane, a 99% accuracy bar before endorsement, drift flags with a refresh cadence.

How did you handle Finance pushing for a broader release?

With the rule they had signed — fat-tail lanes keep buffer until the source process moves. I showed the lanes and the asymmetry; the rule did the refusing.

Why p99 and not p95?

Per lane, the cost of a line-down on a fat-tail lane dwarfed the carrying cost of the extra day. I showed the asymmetry and let planning choose; they chose p99.

Build & Test — What was your definition of done for a lane?

Distribution built from actuals — mean, spread, tail — accuracy tracked, p99 published, drift flag set. Below 99% accuracy on the set, I did not endorse a release.

Who built the distributions?

The data team, to my requirement. I owned the requirement, the promise policy and the release rule; planners sized the buffers; Finance signed. The formula is how I explain it now with the MIT toolkit — not a program artifact.

Cutover — What made you willing to endorse the release?

Lead-time accuracy at 99% on the 200-lane set and the watch armed. Then planners released under the rule, lane by lane, with fat tails excluded.

What was your rollback trigger?

Any stockout or expedite on a released lane during the watch — buffer restored. It never triggered.

Hypercare — How did you know it was closed?

The watch stayed clean, 580K units and about $1.5M were out without a stockout, and planning owned the refresh cadence and drift flags. The promise stayed measured.

What did you hand to planning at close?

The drift flag per lane with its refresh cadence, the p99 promise per lane, and the release rule as a signed act they could re-apply. The stockout-and-expedite watch became a standing report.

Cross-phase — Which lanes did you hold and how did the business take it?

The fat-tail lanes — buffer kept until the source process moved. Finance accepted because the exception was in the rule they signed, not a judgment I made afterwards.

What did you refuse?

Averaging away a tail, and endorsing a cut below 99% accuracy. Both were written into the act before anyone asked.

Where this lands at Amazon: inbound lead-time variability is half of every buffer and every promise. The transferable discipline: measure the lane's own distribution, promise at the tail you can defend, release capital only on evidence with the exception written into the rule, and give every parameter a drift flag so the promise never decays into a new brochure.

BUSINESS IMPACT MODEL — KPI TREE, NUMBERS, THE MATH

```
Service held at lower inventory: 580K units / ~$1.5M released (record)
+- Safety stock (stock) = z * d * sigma_L per lane
|    +- sigma_L: brochure folklore -> measured per lane (200 lanes)
|    +- release rule: 99% LT-accuracy bar + fat-tail lanes keep buffer; Finance
|         signed each lane (S6's two-signal rule is that program's own control)
|    +- fat-tail lanes: sigma_L worse than brochure -> buffer KEPT
+- Promise integrity (flow): p99 per lane, LT accuracy 99% (record)
|    +- cost asymmetry: line-down >> one day of holding -> p99 not p95
+- Drift loop: refresh cadence -> p99 recomputed -> drift flag ->
     buffer decision re-opened -> promise stays measured, never a new brochure
```

DRIVER	BASELINE	AFTER	$/YR	BASIS
Lead-time accuracy, 200 lanes	brochure	99% at p99	—	record
Safety stock released (one-time)	brochure-sized	measured-sized	—	record: 580K units / ~$1.5M
Stockouts on released lanes	—	0 in watch	—	record
Carry on released $1.5M	held	free	$0.3M	modeled ($1.5M × 20% all-in carry; benchmark 15–25%)
Expedites on late arrivals	~1,000 late × 10%	~100 × 10%	$0.3M	modeled (90 × $3K)

DRIVER	BASELINE	AFTER	$/YR	BASIS
Total modeled recurring	—	—	**≈$0.6M**	around the ~$1.5M record; never added to S6's $2M

If challenged. *"Is the $1.5M measured?"* — "The release is record: 580K units out, no stockout in the watch window, Finance signed each lane. The ~$2.59/unit is the derived average on low-value components. What's modeled is the recurring value around it — carry and expedites — and each input is labeled." *"Why did you own the percentile choice?"* — "I didn't — I owned the frame. I showed each lane's asymmetry, line-down cost against carrying cost, and planning chose p99. Their choice surviving an audit is worth more than my choice winning a meeting."

THE MATH — STEP-BY-STEP DERIVATION

Table 1 — before: the problem in numbers.

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
1	Lead-time parameters	`carrier-brochure values; variability unmeasured on 200 lanes`	record	folklore, not data
2	One lane sized on folklore	`SS = z × d × σ_L = 2.33 × 500 × 5 = 5,825 units`	d = 500 units/day; brochure σ = 5 days; z = 2.33 at 99% (illustrative, toolkit form)	5,825 units of buffer
3	Tail blindness	`normal shortcut μ + 2.33σ understates a right-skewed lane's true p99 by 3.4 days`	illustrative from the record	the promise misses the tail
4	Late arrivals	`200 lanes × ~50 shipments = 10,000/yr; × 10% = 1,000 late/yr`	volumes illustrative	~1,000 late/yr

Table 2 — the fix: step by step, values substituted.

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
1	Measure per lane	`μ, σ, p50, p90, p99 from shipment history, 200 lanes`	production cycle + inbound transit vs actual (record)	measured distributions
2	p99 defined cumulatively	`p99 = min{d : P(L ≤ d) ≥ 0.99}`	not μ + 2.33σ unless the lane is normal (record definition)	the tail promise
3	Accuracy gate	`LTacc = 1 − |actual − promised| / promised; bar = 99% on the 200-lane set`	before endorsing any cut (record)	99% held
4	Reliable lane re-sized	`SS = 2.33 × 500 × 1.2 = 1,398; release = 5,825 − 1,398 = 4,427 units`	measured σ = 1.2 days vs brochure 5 (illustrative)	4,427 units off this lane
5	Fat-tail lane	`SS = 2.33 × 500 × 6.5 = 7,573 units — buffer KEPT`	measured σ = 6.5 > brochure 5 (illustrative)	kept until the source process moves
6	Drift flag	`p99 recomputed on refresh; buffer decision re-opened if accuracy falls`	record	the promise stays measured, never a new brochure

Table 3 — after: the metric, before → after.

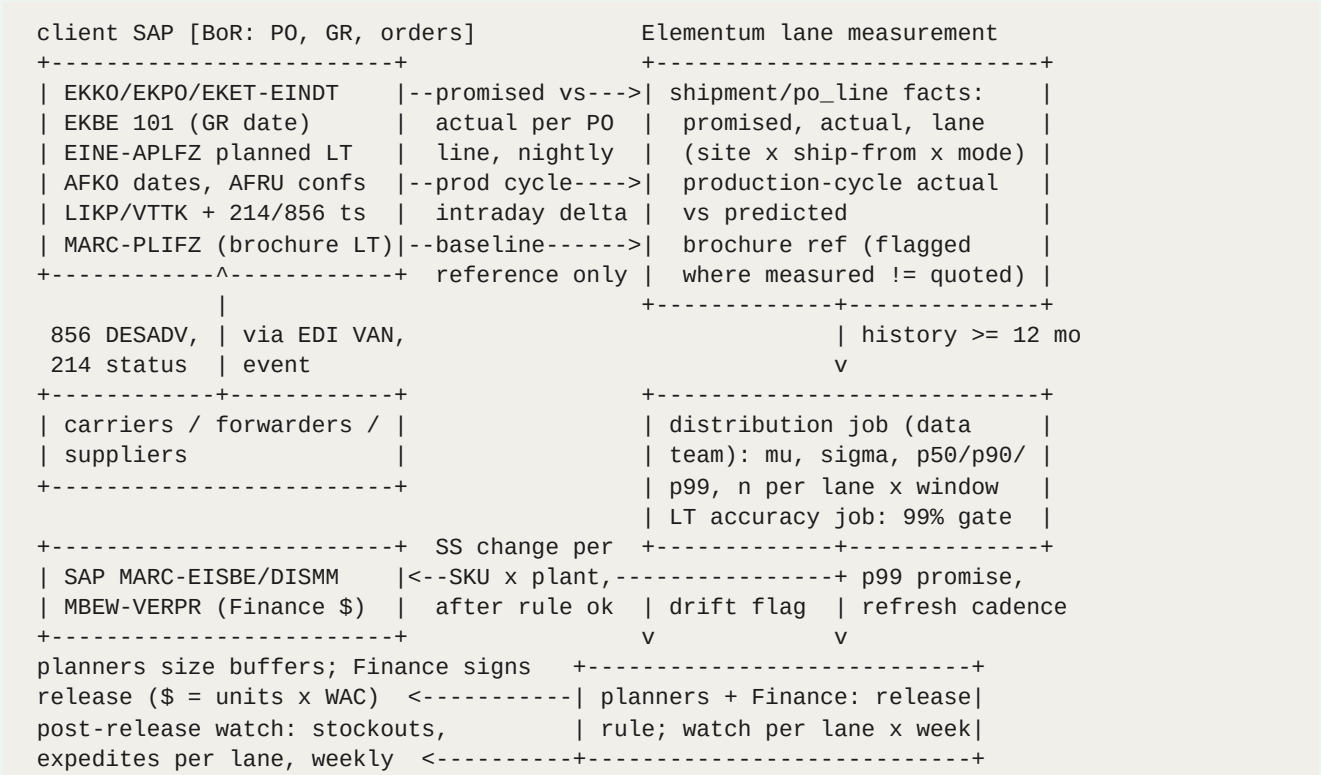
STEP	METRIC	BEFORE	AFTER	BASIS
1	Lead-time parameters	brochure folklore	measured distributions, 200 lanes	record
2	Lead-time accuracy	unmeasured	99% at p99	record
3	Inventory	brochure-sized	580K units / ~$1.5M released one-time	record pair

STEP	METRIC	BEFORE	AFTER	BASIS
4	Stockouts on released lanes	—	0 in the watch window	record
5	Late arrivals	~10%	~1% at the 99% bar	record; volumes illustrative

Table 4 — the dollars, every multiplication shown.

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
1	Unit value	`$1,500,000 / 580,000 = $2.59/unit`	derived from the record pair; low-value components	≈$2.59/unit
2	Carry on the release	`$1,500,000 × 20% = $300,000`	all-in carry (benchmark 15–25%)	≈$0.3M/yr
3	Late reduced	`10,000 × 10% = 1,000 before; 10,000 × 1% = 100 after; 1,000 − 100 = 900 fewer late`	volumes illustrative; the 99% is record	900 fewer late/yr
4	Expedites avoided	`900 × 10% = 90 expedited; 90 × $3,000 = $270,000`	expedited share and premium modeled	≈$0.3M/yr
5	Total recurring	`0.3 + 0.3 = $0.6M`	modeled; never added to S6's $2M	≈$0.6M/yr

SOLUTION FLOW — THE SYSTEM & DATA MAP



SENSITIVITY — LOW / BASE / HIGH

INPUT	LOW	BASE	HIGH
Carrying rate (benchmark)	15% → $0.23M	20% → $0.30M	25% → $0.38M
Late share before p99 (illustrative)	5% → 400 fewer	10% → 900 fewer	15% → 1,400 fewer
Share of late arrivals expedited (illustrative)	5%	10%	15%

INPUT	LOW	BASE	HIGH
Expedite premium (benchmark)	$2K → $0.04M	$3K → $0.27M	$5K → $1.05M
Total modeled $/yr	≈$0.3M	≈$0.6M	≈$1.4M

TOTAL IMPACT STATEMENT — SAY IT LIKE THIS

"The measured result is five hundred eighty thousand units, about a million and a half dollars of safety stock, released at ninety-nine percent lead-time accuracy with zero stockouts on the watch; that is record and Finance signed it. What recurs around it I model at roughly six hundred thousand a year: about three hundred thousand in carrying cost on capital we no longer hold, and about three hundred thousand in expedites we no longer pay because the promise sits at the ninety-ninth percentile instead of a brochure number. The release and the accuracy are measured; the two recurring pieces are my model on benchmark carrying and expedite rates, and I never add this to the two million from the forecast program."

If challenged — the full answers.

The expedite number is the softest

It assumes ten thousand shipments a year on the two hundred lanes and that one late arrival in ten forced an expedite before p99; to firm it I would pull the expedite ledger by lane for the year before and after the release and compare directly.

Twenty percent carrying cost is high

Fifteen to twenty-five is the textbook range and I use the middle; capital alone is eight to twelve, so if Finance wants capital only the carry is about a hundred and fifty thousand and I would say that instead.

Why is the million and a half not an annual saving

Because it is a one-time working-capital release, booked once as a signed planning act; what recurs is the carry and the avoided expedites, and I would rather be right at six hundred thousand a year than wrong at a million and a half.

S6 — Elementum: eight SAP data domains into one real-time supply/demand view — a shortage visible a lead-time early

Amazon LPs: Deliver Results (third) · Earn Trust (the two-signal release rule) · Dive Deep · Think Big (one view over eight domains). **Crosswalk:** loop packet S5 — consistent (70→85%, $2M, both-errors rule). *Different program from S7 (the six-site ingest) and S8 (the $1.5M lanes) — never merge the dollars or the timelines.*

Honesty label. I owned the definitions, the data-integrity program and the delivery; the client's SAP teams and our integration engineers built the feeds; planners owned the forecast — I owned its measurement and the release rule. I have not owned a demand forecast end-to-end and won't claim it.

60-SECOND PITCH

"**Situation.** A six-site manufacturer learned about component shortages when the line stopped; eight SAP domains never met, so 'which orders are at risk' had no data path. **Task.** I took the definitions, the data-integrity program, and the delivery. **Action.** Discovery: profiled all eight domains; signed gap findings. Plan & Design: supply = on-hand + production plan + released planned orders only; grain SKU × location × date; shortage alert = days-of-cover vs lane lead-time p90. Build & Test: inventory and production first — numbers planners already believed; daily reconciliation to MD04/COOIS/VA05/MB52. Cutover: shortage-alert product live in 3 months behind data-quality gates; nobody mandated off a spreadsheet. Hypercare: safety-stock release only where forecast error AND lead-time error both improved, signed with planning and Finance. Lesson: define supply honestly before any screen. **Result.** Primary metric: forecast accuracy on scoped class-A SKUs — **70% → 85%**. $ impact: $2M of safety stock released under the both-errors-improved rule (record, one-time working capital), with ≈$0.8M/yr modeled recurring value around it — separate from the $1.5M lane program; never add them."

SITUATION / TASK — WHERE IT OCCURRED, WHEN IT HAPPENED, WHY IT IS IMPORTANT

S — A strategic manufacturing customer with six sites (mix of make-to-order and make-to-stock): planners learned about a component shortage when the line stopped; S&OP ran off private spreadsheets; shipment data was 60% complete; supplier, material, BOM, inventory, production, transportation, sales-order and planning data all lived in SAP and site systems but never met. Leadership's question — "which customer orders are at risk this quarter?" — had no data path.

T — Nobody owned the seam; there was no spec, no agreed grain, no definition of "supply," no data-quality bar. I chose all four, as owner of the definitions, the data-integrity program and the delivery.

The goal: a real-time supply/demand view with honest definitions and a shortage visible a lead-time early. Beyond the data itself, the challenge was planner trust — and left alone, the line-down surprises continue, with expedites instead of re-plans and buffers sized on noise.

The metrics on the table, with their baselines: forecast accuracy — 70% on scoped class-A SKUs; shortage detection — at line-down, target a lead-time early; shipment-data completeness — 60% at the start; planner effort on reconciliation and firefighting — the baseline the program had to cut.

ACTION — WHAT I PERSONALLY OWNED, HOW I DID IT, WHO ELSE WAS INVOLVED

A — Five gated phases (canonical plan: Discovery 4–6 wks → master-data foundation 4–6 wks parallel → domain waves 8–12 wks → views live 4 wks → scale & hypercare 4 wks; the 3-month figure is the shortage-alert product; the six-site ingest is S7's separate 5-month program):

1. **Discovery (4–6 wks)** — profiled every SAP domain (completeness, latency, joinability) to my own gap templates; wrote the definitions memo (grain SKU × location × date; supply = on-hand + production plan + *released* planned orders only) and the metrics tree. When Planning asked to count unfirmed MRP orders as supply because it looked better, I refused: unfirmed orders breathe every run. *Exit bar:* signed gap findings per domain; scope frozen in writing; planners agree the supply definition before any screen.

2. **Plan & Design (parallel)** — master-data foundation plan; MPN mapping worklist with its coverage gate; BOM extract and phantom clean-up plan; domain-wave sequence by trust; DQ scorecard spec; the 10-step onboarding playbook. *Exit bar:* master-data gates green; unmapped-MPN list under threshold with a named owner; wave order signed by domain owners.
3. **Build & Test (8–12 wks)** — domain waves behind published DQ gates; shortage-alert rule specified (days-of-cover vs lane LT p90); daily reconciliation to MD04/COOIS/VA05/MB52; weekly root-cause loop with false alarms worked as defects. *Exit bar:* per-domain completeness ≥ gate; reconciliation passing daily; false alarms trending down; pilot planners using the alert.
4. **Cutover & Go-Live (4 wks)** — views live for pilot planners with the at-risk list replacing the Monday spreadsheet; go/no-go read from the scorecards. *Exit bar:* planners answer "where is my product" from the platform in the live S&OP meeting — not from a spreadsheet.
5. **Hypercare / Close (4 wks)** — exception workbench staffed; weekly root-cause reviews; the safety-stock release rule signed with Planning and Finance. *Exit bar:* blockers/day under threshold two consecutive weeks; release applied only where forecast AND lead-time error both improved.

The client's SAP teams and our integration engineers built the feeds; planners owned the forecast — I owned its measurement and the release rule, and I have not owned a demand forecast end-to-end and won't claim it. Mine specifically: the definitions memo, the metrics tree, the DQ scorecard spec, the wave sequence, the release rule and the 10-step onboarding playbook; domain owners held sign-off authority over their own gates. The contributions that mattered most: honest definitions — refusing to count unfirmed MRP orders as supply because it looked better — and the two-signal release rule, buffer out only where forecast error and lead-time error both improved, which is why the release survived audit.

The obstacles were planner trust and the MPN mapping worklist, which gated longer than planned. Trust was built by sequencing waves by trust — inventory and production first, numbers planners already believed — and by working false alarms as defects; the MPN slip went to the sponsor as a re-sequenced date with the coverage number attached. Priorities: waves by trust, class-A SKUs before the tail, every feed behind a published DQ gate — below the gate, the feed got fixed, not automated. Buy-in: domain owners could fail their own wave — the business could say no to its own data — and nobody was mandated off the spreadsheet. The decisions challenged: Planning asked to count unfirmed MRP orders as supply because it looked better — refused, since unfirmed orders change every run, amplify noise and inflate supply; and the safety-stock release was refused as a blanket act and agreed as a rule.

RESULTS — HOW SUCCESS WAS MEASURED: $ · # · % · TIME · CUSTOMER · QUALITY

R — Primary metric: forecast accuracy **70%** → **85%** on scoped class-A SKUs (record). $2M safety stock released under the both-errors rule (record, one-time); planner effort −50%; shortage alert live in 3 months; adoption 8×.

- **$** — $2M safety stock released (record, one-time working capital, under the both-errors-improved rule signed with Planning and Finance) + ≈$0.8M/yr modeled recurring (carry, expedites pre-empted, planner capacity) — never added to S8's $1.5M.
- **#** — Eight SAP data domains into one view across six sites; shortage-alert product live in 3 months; adoption 8× (record).
- **%** — Forecast accuracy 70% → 85% on scoped class-A SKUs (record); planner effort −50% (record); shipment data 60% complete at the start.
- **Time** — Shortage alert fires at days-of-cover ≤ lane lead-time p90 — a lead-time early instead of at line-down (record mechanism); views live in 3 months.
- **Customer** — “Which customer orders are at risk this quarter?” answered from the platform in the live S&OP meeting, not a spreadsheet (record exit bar).
- **Quality** — Daily reconciliation to MD04/COOIS/VA05/MB52; false alarms classified weekly to root cause and burned down as defects (record).

Against the goal stated above: forecast accuracy 70 → 85% on scoped class-A SKUs; the shortage alert live in three months, firing a lead-time early instead of at line-down; planner effort −50%; the $2M safety-stock release

under the both-errors-improved rule Planning and Finance signed. The trades: the lane tail and the MPN tail were deferred behind the same gates — priced in coverage, not count — rollback stayed free because the spreadsheet was never banned, and the release rule traded speed of release for a release that survived audit.

PROBING — QUESTIONS, PHASE DEEP-DIVES, AND THE MATH

Challenge — Why did you choose this story to illustrate Deliver Results?

Because the results landed behind gates and signatures: forecast accuracy 70 → 85% on scoped class-A SKUs, the shortage alert live in three months, and $2M of safety stock out under the both-errors-improved rule Planning and Finance signed. Delivering the number honestly — released planned orders only — is the point.

Challenge — What other stories demonstrate Deliver Results?

S11 — 7 → 68 behind graduation gates — is the primary Deliver Results story; S18 (8 months, both hubs, gate cleared) is the backup, and S2's comingle launch the third.

Challenge — Could you come up with an example that is more recent?

This is from the Elementum years — older. The most recent same-LP record is S2's comingle launch in my current role (zero incidents, +15% signed); S11 at Flexport sits between.

Probing — Why did you choose to focus on these results? What other results were important?

Forecast accuracy 70 → 85% is the record headline and the $2M release is the signed, audit-surviving consequence; planner effort −50% and the alert live in three months are what the planners felt. Adoption 8× mattered because trust was the real product.

Probing — You mentioned dollars — what percentage change is that year over year?

No YoY revenue figure — the 70 → 85% is forecast accuracy on scoped class-A SKUs against the pre-state, and the $2M is a one-time working-capital release, not an annual saving; the recurring value I model is ≈$0.8M/yr around it, labeled.

Probing — I'm concerned about the expedites-avoided figure — tell me more.

The softest row — about sixty expedites a year pre-empted at a five-thousand-dollar premium, both illustrative. To firm it I would pull the expedited-freight ledger for the six sites for the twelve months before and after the alert went live and count the premium directly.

Challenge — What were the lessons learned? What would you have done differently?

Define supply honestly before any screen. Differently: start the MPN-mapping worklist on day one — it is never “done” and it gated more than I planned.

Challenge — How would you implement this at Amazon?

Supply/demand visibility built honestly — define supply before the screen, measure MAPE and bias at the decision grain, alert a lead-time early, and release buffer only when both error signals improve. The two-signal release rule is the part most rooms have never seen.

Challenge — How did these results compare to your actual goals?

Leadership's question was “which customer orders are at risk this quarter?” — answered a lead-time early from the platform in the live S&OP meeting, with accuracy 70 → 85%, the alert live in three months, and the $2M release under the signed rule.

PHASE DEEP-PROBE BANK — Q&A (14). Answer the phase asked, then stop.

Discovery — How did you know discovery was done?

Every domain had a signed gap finding on its own profiled data and the metrics tree was agreed. Four to six weeks; scope frozen in writing at the exit.

Who wrote the success criteria and who signed?

I wrote the metrics tree — OTIF, inventory dollars, MAPE/bias, lane lead-time accuracy, completeness, latency, adoption — and the domain owners with sign-off authority signed their gates.

What did you define before any screen?

Grain — SKU × location × date — and supply: on-hand plus production plan plus released planned orders only. Planners accepted the definition first; the view was built to it.

Plan & Design — What was in your risk register at sign-off?

MPN mapping never done, carrier EDI capability spread, planner trust, and SAP extract load — carried openly with a mitigation and owner each. MPN became a standing metric rather than a task.

How did you sequence the waves?

By trust: inventory and production first because planners already believed those numbers, then purchasing/ASN, then transportation with the best lanes first. Class-A SKUs before the tail.

Build & Test — What was your definition of done for a feed?

Its DQ scorecard — completeness, latency, accuracy vs source, joinability — above the published gate, and daily reconciliation passing. Below the gate, the feed got fixed, not automated.

How did you treat false alarms?

As defects: classified weekly to a root cause — mapping gap, master-data defect, carrier capability, pipeline bug — owned and burned down. Adoption followed planners watching their reported defects die.

Who could fail a wave?

The domain owner — they held sign-off authority over their gate. I designed it so the business could say no to its own data.

Cutover — What was the go-live gate for the views?

Planners answering "where is my product" from the platform in the live S&OP meeting rather than from a spreadsheet. The shortage alert was live in three months.

What was your rollback?

The spreadsheet — still there, never banned. Reversibility was free because I never mandated anyone off it.

Hypercare — How did you decide hypercare was over? What did it hand over?

Blockers/day under threshold two consecutive weeks, adoption targets met, the exception workbench staffed. Handover: the weekly root-cause review to domain owners, the onboarding playbook (profile → map → test file → parallel run → scorecard → live), and the MPN exception queue as a standing metric.

Cross-phase — Where did you trade scope for date? What did you escalate?

The lane tail and the MPN tail — deferred behind the same gates, priced in coverage, not count. Escalated the MPN worklist gating longer than planned — to the sponsor as a re-sequenced date with the coverage number attached.

Who owned the forecast?

The planners. I owned the accuracy measurement (MAPE/bias per SKU family), input integrity and the release rule. I have not owned a demand forecast end-to-end and won't claim it.

Why released planned orders only? · Do differently?

Unfirmed MRP orders change every run; planning against them amplifies noise and inflates supply. Differently: start the MPN-mapping worklist on day one — it is never "done" and gated more than I planned.

Where this lands at Amazon: this is supply/demand visibility built honestly — define supply before the screen, measure MAPE and bias at the decision grain, alert a lead-time early, and release buffer only when both error signals improve. The two-signal release rule is the part most rooms have never seen.

BUSINESS IMPACT MODEL — KPI TREE, NUMBERS, THE MATH

```
Customer orders at risk this quarter -> answered a lead-time early
+- Inventory $ (stock): $2M safety stock released (record)
|    +- forecast error 30% -> 15% (accuracy 70 -> 85%, record)
|    +- lead-time error (lane p90 measured, not brochure)
|    +- release rule: BOTH errors improved -> SS = z*sigma*sqrt(L) falls
+- Shortage response (flow): alert when DoC <= lane LT p90
|    +- fix window = LT p90 - DoC: days to re-plan instead of fly
+- Planner effort -50% (record) -> hours to root cause, not reconciliation
+- Data-integrity loop: daily recon -> defects burned down -> planner trust (+)
     -> adoption 8x -> more domains feed the peg -> errors fall further (+)
```

DRIVER	BASELINE	AFTER	$/YR	BASIS
Forecast accuracy, class-A SKUs	70%	85%	—	record
Safety stock released (one-time)	—	$2M out	—	record
Carry on released $2M	tied up	free	$0.2M	modeled ($2M × 10% WACC; benchmark 8–12%)
Expedites avoided via the alert	~150 events/yr	~60 pre-empted	$0.3M	modeled (60 × $5K; premium benchmark $3–8K)
Planner effort on reconciliation	12 × 40%	12 × 20%	$0.3M	modeled (2.4 FTE × $120K; −50% is record)
Total modeled recurring	—	—	**≈$0.8M**	around the $2M one-time record; never added to S8's $1.5M

If challenged. *"The $2M — measured how?"* — "It's the record: the release Finance signed, applied only on SKU families where forecast error and lead-time error had both demonstrably improved. That rule is why the release survived audit — capital came out on evidence, not on a better-looking average." *"150 shortage events?"* — "Illustrative volume — six sites at roughly 25 escalations a year. The record is the mechanism: the alert fires at days-of-cover ≤ lane p90, which is the last moment a re-plan is cheaper than a flight."

THE MATH — STEP-BY-STEP DERIVATION

Table 1 — before: the problem in numbers.

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
1	Forecast accuracy	`accuracy = 1 − MAPE = 1 − 0.30 = 70%`	scoped class-A SKUs (record)	70%
2	What the error math looks like	`MAPE = (30/100 + 10/120 + 20/80 + 10/100)/4 = (0.30 + 0.083 + 0.25 + 0.10)/4 = 18.3%`	4 illustrative weeks: A = 100, 120, 80, 100; F = 130, 110, 100, 90; |e| = 30, 10, 20, 10	18.3% on this sample; bias = (430 − 400)/400 = +7.5%
3	The supply lie	`supply counted unfirmed MRP planned orders — they change every run`	record problem	plans built on sand
4	Shortage discovery	`at line-down: ~150 expedite events/yr`	benchmark-modeled count	~150 fire-drills/yr
5	Planner burn	`12 planners × 40% = 4.8 FTE-equivalent on reconciliation`	record share	4.8 FTE reconciling, not root-causing

Table 2 — the fix: step by step, values substituted.

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
1	Honest supply	`supply(t) = on-hand + in-transit ≤ t + production ≤ t + RELEASED planned orders only`	unfirmed MRP planned orders excluded (record rule)	walked below on ROUTER-100
2	FG netting walk — setup	`FG SKU ROUTER-100 at DC-East: on-hand 400; in-transit 600 lands day 5; production 800 day 9; demand 500 d3 / 700 d7 / 400 d10`	illustrative (v105 §5.12)	projected balance below
3	Day 3	`PAB(3) = 400 − 500 = −100`	only on-hand has arrived by day 3	dated shortage of 100 on day 3
4	Day 7	`PAB(7) = (400 + 600) − (500 + 700) = 1,000 − 1,200 = −200`	day-9 production is too late for day-7 demand	shortage of 200; fix window = 9 − 7 = 2 days
5	Day 10	`PAB(10) = (400 + 600 + 800) − (500 + 700 + 400) = 1,800 − 1,600 = +200`	all supply lands by day 10	+200 — the shortage is timing, not volume
6	Shortage alert rule	`DoC = 1,200 / 100 = 12 days; 12 ≤ p90 14 → alert now`	FG SKU: on-hand 1,200 at 100/day; lane p90 = 14 days (illustrative)	alert a lead-time early, not at line-down
7	Counter-case	`p90 = 9 days: 12 > 9 → no alert`	same SKU, faster lane (illustrative)	3-day slack — no false alarm
8	Two-signal release rule	`SS = z × d × σ_L = 2.33 × 100 × 4 = 932; with σ_L 4 → 2: 2.33 × 100 × 2 = 466; release 932 − 466 = 466 units`	release only if forecast error AND lead-time accuracy both improved; d = 100/day, z = 2.33 at 99% (illustrative, toolkit form)	466 units off this SKU
9	Accuracy after	`error 30% → 15% (halved); accuracy = 1 − 0.15 = 85%`	gated waves; false alarms worked as defects (record)	85%

Table 3 — after: the metric, before → after.

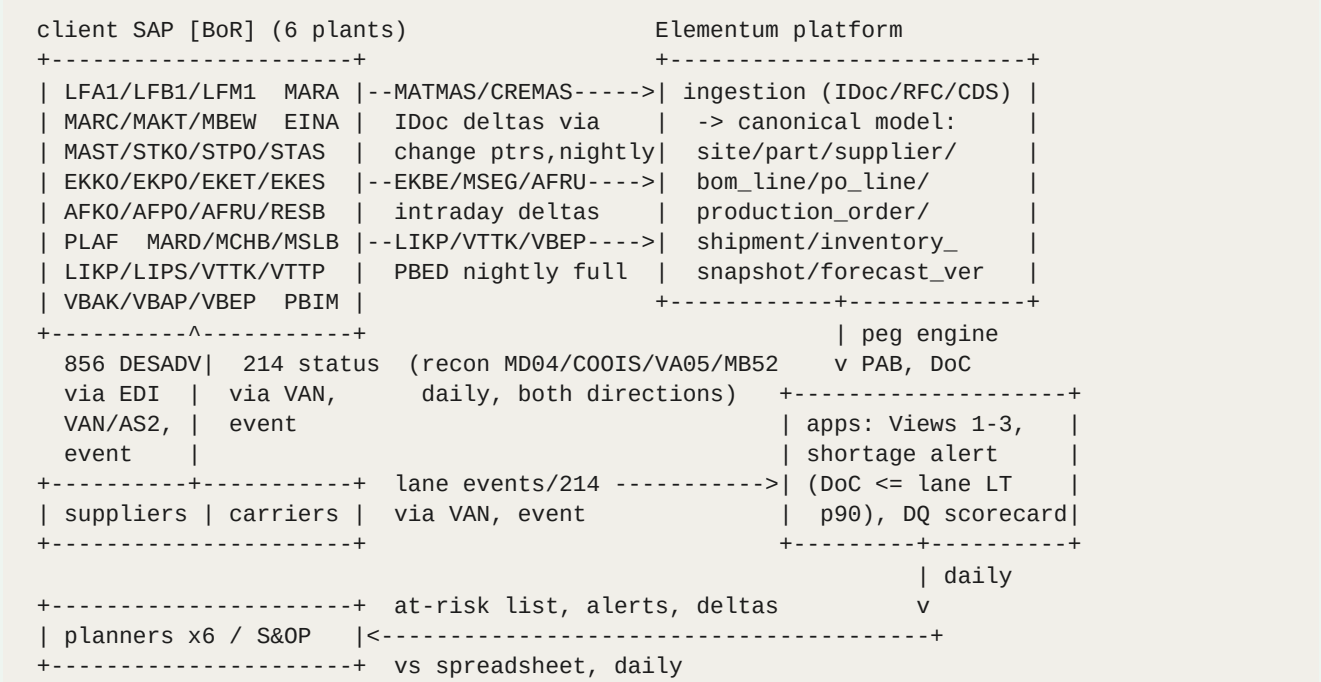
STEP	METRIC	BEFORE	AFTER	BASIS
1	Forecast accuracy, class-A SKUs	70%	85%	record
2	Shortage discovery	at line-down	a lead-time early (DoC ≤ p90 alert)	record design
3	Planner reconciliation share	12 × 40% of time	12 × 20% — halved	record (−50%)
4	Safety stock	sized on bad signals	$2M released one-time under the signed two-signal rule	record

Table 4 — the dollars, every multiplication shown.

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
1	Carry on the release	`$2,000,000 × 10% = $200,000`	$2M release (record, one-time) × WACC (benchmark 8–12%)	≈$0.2M/yr
2	Expedites pre-empted	`150 × 40% = 60 caught early; 60 × $5,000 = $300,000`	events/yr modeled; premium benchmark $3–8K	≈$0.3M/yr
3	Planner capacity	`12 × 40% = 4.8 FTE; halved → 2.4 FTE; 2.4 × $120,000 = $288,000`	−50% is record; capacity redeployed, not headcount removed	≈$0.3M/yr

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
4	Total recurring	`0.2 + 0.3 + 0.3 = $0.8M`	modeled; never added to S8's $1.5M	≈$0.8M/yr
5	One-time	`$2M safety stock released, taken as stated`	record, under the signed two-signal rule	$2M

SOLUTION FLOW — THE SYSTEM & DATA MAP



SENSITIVITY — LOW / BASE / HIGH

INPUT	LOW	BASE	HIGH
Cost of capital (benchmark)	8% → $0.16M	10% → $0.20M	12% → $0.24M
Expedites avoided (illustrative)	45	60	75
Expedite premium (benchmark)	$3K → $0.14M	$5K → $0.30M	$8K → $0.60M
Loaded planner cost (benchmark)	$100K → $0.24M	$120K → $0.29M	$140K → $0.34M
Total modeled $/yr	≈$0.5M	≈$0.8M	≈$1.2M

TOTAL IMPACT STATEMENT — SAY IT LIKE THIS

"The measured number is two million dollars of safety stock released, and that is record, signed with planning and Finance under the both-errors-improved rule. Around it I model roughly eight hundred thousand a year of recurring value: about two hundred thousand in carry cost on the capital that came out, about three hundred thousand in expedite premium the shortage alert pre-empts because it fires a lead-time early instead of at line-down, and about three hundred thousand of planner capacity from the fifty percent effort cut. The two million and the fifty percent are measured; the three recurring pieces are my model on benchmark rates, and I keep them separate from the one-and-a-half million in the lane program."

If challenged — the full answers.

The expedite figure is the softest

It rests on about sixty expedites a year pre-empted at a five thousand dollar premium, both illustrative; to firm it I would pull the expedited-freight ledger for the six sites for the twelve months before and after the alert went live and count the premium directly.

Planner capacity is not cash

Agreed, it is two point four FTE of time redeployed, not headcount removed; the fifty percent is measured, the forty percent share of the week is my assumption, and a two-week time study on the S&OP prep cycle would replace it.

Why not claim the whole two million every year

Because a release is a one-time working-capital event; the recurring value is the carry at eight to twelve percent, and I would rather say two hundred thousand a year honestly than two million a year wrongly.

S9 — Elementum: 1,000 supplier/carrier routes asked, 597 built first — X12 EDI onboarding behind a ≥95% ASN cohort gate

Amazon LPs: Have Backbone; Disagree & Commit (primary — 597 vs 1,000, two levels up) · Deliver Results · Insist on the Highest Standards (the gate held twice) · Think Big (the staged 1,000-lane vision) · Broad Responsibility (the portal floor kept small carriers inside the standard). **Crosswalk:** loop packet S7 — consistent. *Cohort sizes and the number of cohorts are not in the record — do not quote them.*

Honesty label. I owned the scope analysis and the onboarding standard, influencing an executive two levels above my sponsor; engineers built the maps and harness to the standard; Transportation owned the carrier relationships.

60-SECOND PITCH

"**Situation.** The Global Head of Supply Chain wanted inbound visibility on 1,000 lanes in two months; shipment-data completeness was 60% and high-value lanes were blind. **Task.** I owned the scope analysis and the onboarding standard, influencing an executive two levels above my sponsor. **Action.** Discovery: spend Pareto — 597 lanes = 80% of freight spend — pre-validated with Transportation and Procurement. Plan & Design: X12 850/855/856/940/945/214/997 with an XML/portal floor; cohort gate ≥95% valid ASN. Build & Test: mapping-and-test harness; scorecards per lane. Cutover: 597 live in month one; one cohort failed twice on a timestamp format — I held it. Hypercare: completeness scorecards to 95%; ~200 next wave behind the same gate. Lesson: automating dirty data manufactures confident wrong answers — the gate is the product. **Result.** Primary metric: inbound shipment-data completeness — **60% → 95%**. $ impact: freight spend −10% ≈ $2M in six months (record); my model splits it into routing-guide compliance, mode discipline and expedites avoided, and adds ≈$0.4M/yr of modeled document-handling labor on top; 597 lanes = 80% of spend live in month one; adoption 8×."

SITUATION / TASK — WHERE IT OCCURRED, WHEN IT HAPPENED, WHY IT IS IMPORTANT

S — The customer's Global Head of Supply Chain wanted end-to-end inbound visibility on 1,000 global lanes in two months — scope, schedule and quality all fixed. Lanes differed by orders of magnitude in spend, volume and criticality; carriers differed in EDI capability; completeness was 60% and high-value lanes were effectively blind.

T — Own the scope analysis and the onboarding standard, influencing an executive two levels above my sponsor.

The goal: visibility that was actually trustworthy, at the spend that mattered first. The risk of taking the ask literally: automating dirty data manufactures confident wrong answers — 1,000 lanes at 60% completeness in two months would have shipped exactly that — and without a floor for carriers without EDI, the long tail gets priced out of the standard.

The metrics on the table, with their baselines: shipment-data completeness — 60%; high-value lane visibility — effectively blind, target 100% on the lanes carrying the spend; the activation standard — none existed, target a ≥95% valid-ASN cohort gate a lane had to clear before going live.

ACTION — WHAT I PERSONALLY OWNED, HOW I DID IT, WHO ELSE WAS INVOLVED

A — Five gated phases:

1. **Discovery** — built the spend Pareto and lane scoring model (spend × criticality × mode × volume) and the carrier EDI-capability survey; pre-validated the 597 / ~200 / remainder scope with Transportation and Procurement before taking it up. *Exit bar:* lane owners cannot contradict the cut; the executive accepts the phased plan in spend share, not lane count.
2. **Plan & Design** — wrote the onboarding standard (document set + XML/portal floor — "the standard was the data, not the transport"), the ≥95% valid-ASN cohort gate, the per-lane scorecard spec (events received / expected; 997 rate; tracking-number joinability), the harness design. *Exit bar:* standard signed by Transportation; gate published to carriers before cohort 1; harness templates exist.

3. **Build & Test** — held engineers to the harness (per-partner mapping templates, test scripts, DQ checklist), test files per carrier, a parallel run per lane, scorecards live. *Exit bar:* each cohort's lanes clear ≥95% valid ASN in test before activation.
4. **Cutover & Go-Live (per cohort)** — activated cohorts only off the published gate; published the scorecards; recorded the hold decision when a cohort failed. *Exit bar:* cohort active only after the gate; a failing cohort held — twice, on a timestamp format.
5. **Hypercare / Close** — weekly completeness scorecards; harness reused for later waves; method written up as the sourcing SOP (spend file in → phased plan out). *Exit bar:* completeness 60% → 95%; 100% visibility on high-value lanes; remaining waves behind the same gate.

The scope analysis and the standard were mine end to end — the spend Pareto and lane scoring model, the carrier EDI-capability survey, the onboarding standard, the gate, the scorecard spec and the hold decisions; engineers built the mapping templates, test scripts and harness, and Transportation held the carrier relationships and accepted the holds. The contribution: pre-validating the 597 cut with Transportation and Procurement so no lane owner could contradict it in the leadership meeting, and publishing the ≥95% gate to carriers before cohort one — a gate that only holds in private is a suggestion.

The obstacles: carriers unable to produce valid ASNs — answered with the XML/portal floor ("the standard was the data, not the transport") and a hold rule; a cohort that failed the gate twice on a timestamp format — held both times, fixed, then activated; bespoke maps per partner — avoided by the parameterized harness. Priorities ran on spend share, not lane count — 597 lanes carrying 80% of spend first, ~200 next, the remainder after, each behind the same gate; buy-in came from the pre-validation, then the executive accepted the phased plan off the curve. The decision challenged was the 1,000-in-two-months ask itself — taken two levels up with the spend curve, deferring ~400 lanes carrying ~20% of spend, priced — and I refused activating any cohort that failed the gate.

RESULTS — HOW SUCCESS WAS MEASURED: $ · # · % · TIME · CUSTOMER · QUALITY

R — Primary metric: completeness **60%** → **95%**; 597 lanes = 80% of spend live in month one; freight −10% ≈ $2M in six months (record); adoption 8×.

- **$** — Freight spend −10% ≈ $2M in six months on the onboarded lanes (record); my three-way split (routing-guide compliance, mode discipline, expedites and detention) decomposes the record, never adds to it; ≈$0.4M/yr modeled document-handling labor on top.
- **#** — 597 of 1,000 lanes = 80% of freight spend live in month one; ~200 the next wave; the remainder behind the same gate (record).
- **%** — Completeness 60% → 95%; the ≥95% valid-ASN cohort gate — held twice on a cohort that failed on a timestamp format (record).
- **Time** — 597 lanes live in month one; the record $2M landed within six months (record).
- **Customer** — 100% visibility on high-value lanes; small carriers kept inside the standard by the XML/portal floor instead of being priced out (record).
- **Quality** — The gate is the product — automating dirty data manufactures confident wrong answers; adoption 8×; the method written up as the sourcing SOP (record).

Against the goal stated above: completeness 60 → 95%; 100% visibility on high-value lanes; 597 lanes = 80% of spend live in month one at ≥95% valid ASN, the remaining waves behind the same gate — with freight spend −10% ≈ $2M in six months funding the next wave. The trade was explicit and priced: ~400 lanes carrying ~20% of spend deferred so the 597 could go live at the highest data quality — and quality was never traded: a lane that slipped below its scorecard went back to "fix," not "live."

PROBING — QUESTIONS, PHASE DEEP-DIVES, AND THE MATH

> **Challenge — Why did you choose this story to illustrate Have Backbone; Disagree & Commit?**
>
> Because I took the re-scope — 597 lanes first, not 1,000 — two levels up to the Global Head of Supply Chain with the spend curve and the lane owners' validation attached, then committed to delivering the remainder behind the same gate. And the gate held in public: one cohort failed twice on a timestamp format and I held it both times.

Challenge — What other stories demonstrate Have Backbone; Disagree & Commit?

S14 — refusing the quiet fix — is the backup; S5 (the council without owning a system) and S2 (the throttle) are the thirds.

Challenge — Could you come up with an example that is more recent?

Elementum years — older. For current-role records of the same LP, S5 and S2's throttle are the ones I would pair with this.

Probing — Why did you choose to focus on these results? What other results were important?

Completeness 60 → 95% is the record the whole program was for, and freight −10% ≈ $2M in six months is what the executive funded the next wave on. The 597 = 80% of spend live in month one is the proof the re-scope was right.

Probing — You mentioned dollars — what percentage change is that year over year?

Not YoY — the $2M is the record as measured: minus ten percent on the onboarded lanes' spend in six months, kept as stated rather than doubled into a run-rate. My three-way split of it decomposes the record, never adds to it.

Probing — I'm concerned about your split of the $2M — tell me more.

The split is a judgment on how the ten points divide; to firm it I would tag every invoice line on the onboarded lanes by cause — off-guide carrier, mode change, expedite — for the six months and let the ledger set the shares.

Challenge — What were the lessons learned? What would you have done differently?

Automating dirty data manufactures confident wrong answers — the gate is the product. Differently: the harness from cohort one, not cohort two.

Challenge — How would you implement this at Amazon?

Vendor and carrier onboarding at scale is a Pareto plus a gate — spend share first, a published data-quality bar a partner must clear before activation, and a floor (portal/templates) so the long tail is kept inside the standard rather than priced out of it.

Challenge — How did these results compare to your actual goals?

The ask was 1,000 lanes in two months; the delivery was 597 lanes = 80% of spend live in month one at ≥95% valid ASN, completeness 60 → 95%, the remaining waves behind the same gate — and the executive accepted the phased plan in spend share, not lane count.

PHASE DEEP-PROBE BANK — Q&A (13). Answer the phase asked, then stop.

Discovery — How did you know discovery was done?

When the spend Pareto was pre-validated by Transportation and Procurement and the scope proposal — 597 first, ~200 next, remainder after — could not be contradicted by any lane owner in the leadership meeting.

Who signed the scope?

The Global Head of Supply Chain accepted it after seeing the curve; the lane owners had already validated it. I priced the deferral in spend share, not lane count.

Plan & Design — What was in your risk register at design sign-off?

Carriers unable to produce valid ASNs, the scope being contradicted in the room, and bespoke maps per partner. Mitigations: XML/portal floor plus a hold rule, pre-validation, and a parameterized harness.

What was the entry bar for a cohort?

Standard and gate published to its carriers, mapping templates instantiated, test files in hand, a scorecard per lane. Then ≥95% valid ASN in test before activation.

How did you set the standard?

850/855/856/940/945/214/997 on X12, with XML or portal templates as the floor for carriers without EDI. The standard was the data, not the transport.

Build & Test — What was your definition of done for a lane?

Valid-ASN rate ≥95%, 997 acknowledgments flowing, tracking numbers joinable, completeness on its scorecard. Below that, the lane was fixed, not automated.

Describe your bug-bash equivalent.

The cohort test day: every carrier's file through the harness in one sitting; defects classified as format, timestamp, missing segment or join key, each with an owner. That is where the timestamp defect surfaced.

Cutover — Which cohort did you hold and how did the business take it?

One cohort failed twice on a carrier timestamp format; I held it both times. Transportation accepted because the 95% gate was published before cohort one — a gate that only holds in private is a suggestion.

What did each wave deliver?

Wave 1: 597 lanes, 80% of spend, live in month one. Wave 2: ~200 lanes on a harness that made the work mostly configuration. Wave 3: the tail, including non-EDI carriers on templates.

Hypercare — How did you know you were done? What did you watch?

Completeness 60% → 95%, 100% visibility on high-value lanes, the harness reused without me, the method adopted as the sourcing SOP. Watched weekly: per-lane completeness, the 997 rate, tracking-number joinability. A lane that slipped below its scorecard went back to "fix," not "live."

Cross-phase — Where did you trade scope for date? What did you escalate?

Deferred ~400 lanes carrying ~20% of spend so the 597 could go live at the highest data quality in month one — priced, and delivered later behind the same gate. Escalated the re-scope to the Global Head of Supply Chain — two levels up — with the spend curve and the lane owners' validation attached. Refused: activating a cohort that failed the gate.

Did you deliver the remaining 400? · Do differently?

Yes — later waves, same gate; the first 597 carried the spend and the decision. Differently: the harness from cohort one, not cohort two.

Where this lands at Amazon: vendor and carrier onboarding at scale is a Pareto plus a gate: spend share first, a published data-quality bar a partner must clear before activation, and a floor (portal/templates) so the long tail is kept inside the standard rather than priced out of it. The sentence that travels: automating dirty data manufactures confident wrong answers.

BUSINESS IMPACT MODEL — KPI TREE, NUMBERS, THE MATH

```
Inbound freight spend -10% ~= $2M in 6 months (record)
+- Visibility (stock): completeness 60% -> 95%, high-value lanes 100%
|    +- routing-guide compliance: off-guide carrier spend caught
|    +- mode discipline: air -> ocean/ground when ETA is trusted
|    +- expedites + detention: chase calls replaced by 214 events
+- Document handling (flow): 850/855/856/940/945/214 by X12, not by hand
+- DQ gate loop: >=95% valid ASN -> cohort activates -> scorecard per lane ->
     defects fixed as data -> harness parameterized -> cohort 3 mostly config -> adoption 8x
+- Scope loop: 597 = 80% spend first -> $ lands month one -> executive funds the next wave
```

DRIVER	BASELINE	AFTER	$	BASIS
Shipment-data completeness	60%	95%	—	record
Lanes live month one	0	597 = 80% of spend	—	record
Freight spend on onboarded lanes	~$20M/yr (packet-derived)	−10%	≈$2M / 6 mo	record ($2M)
— routing-guide compliance	off-guide unseen	caught by 856/214	~$0.8M	modeled split of the record $2M
— mode discipline	premium modes by habit	ETA trusted	~$0.6M	modeled split
— expedite + detention avoided	blind lanes chased	events replace calls	~$0.6M	modeled split

DRIVER	BASELINE	AFTER	$	BASIS
Manual document handling	~90K docs/yr keyed	X12/XML validated	$0.4M/yr	modeled (90K × 6 min × $45/hr)

If challenged. *"Is the $2M annualized?"* — "No — it's the record as measured: minus ten percent on the onboarded lanes' spend in six months. I keep it as stated rather than doubling it into a run-rate." *"How do you know the split?"* — "I don't, to the dollar — the split is my model of where the ten percent came from, and the three slices sum to the record, never on top of it."

THE MATH — STEP-BY-STEP DERIVATION

Table 1 — before: the problem in numbers.

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
1	Completeness	`480 / 800 = 60%`	expected = 200 shipments × 4 milestones = 800; received 480 (illustrative volumes; 60% record)	60% — 2 of 5 events missing
2	What blindness costs	`off-guide spend unseen · premium modes by habit · chase calls per late shipment`	record problem	no instrument, no lever
3	Manual documents	`597 lanes × ~50 shipments × 3 docs ≈ 90,000 docs/yr keyed by hand`	illustrative volumes	~90K docs/yr

Table 2 — the fix: step by step, values substituted.

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
1	Spend Pareto	`S(k) = cumulative spend share; S(597) = 80%`	lanes sorted by annual freight spend (record)	597 of 1,000 lanes carry 80% of spend — do them first
2	Cohort gate	`validASN rate ≥ 95% in test before activation`	published to carriers before cohort one (record)	one cohort failed twice on a timestamp format — held both times
3	Long-tail floor	`XML / portal when EDI is absent`	“the standard was the data, not the transport” (record)	small carriers kept inside the standard
4	Completeness after	`760 / 800 = 95%`	illustrative volumes; 95% record	95%; high-value lanes 100%

Table 3 — after: the metric, before → after.

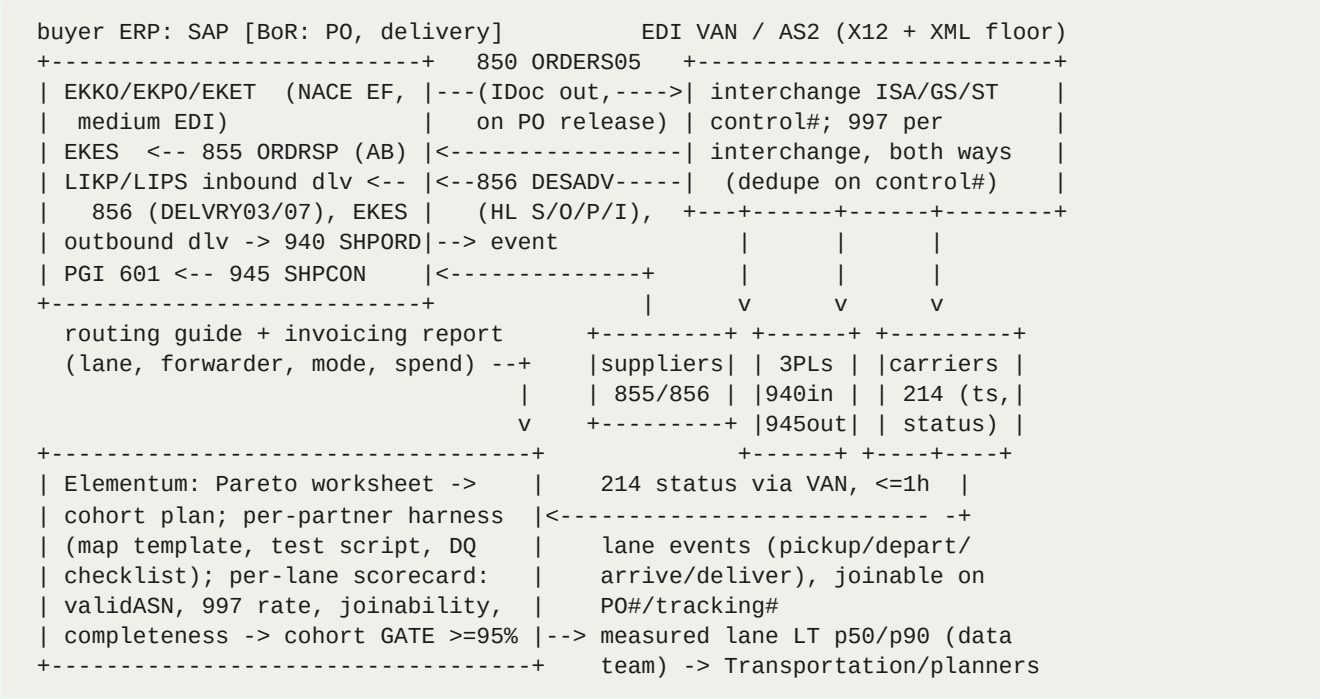
STEP	METRIC	BEFORE	AFTER	BASIS
1	Shipment-data completeness	60%	95%; high-value lanes 100%	record
2	Lanes live month one	0	597 = 80% of spend	record
3	Freight spend on onboarded lanes	base	−10% ≈ $2M in 6 months	record
4	Document handling	~90K docs/yr keyed	X12/XML validated	modeled

Table 4 — the dollars, every multiplication shown.

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
1	Implied addressable base	`$2,000,000 / 10% = $20,000,000/yr`	derived from the record pair, stated as such	≈$20M/yr
2	The record result	`$20M × 10% = $2M in 6 months`	record	≈$2M

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
3	Slice — routing-guide compliance	`~$0.8M of the $2M`	off-guide spend caught by 856/214 (modeled split)	~$0.8M
4	Slice — mode discipline	`~$0.6M of the $2M`	premium modes → ocean/ground when ETA is trusted (modeled split)	~$0.6M
5	Slice — expedites + detention	`~$0.6M of the $2M`	events replace chase calls (modeled split)	~$0.6M
6	Split check	`0.8 + 0.6 + 0.6 = $2.0M`	the slices decompose the record — they never add to it	= the record $2M
7	Document labor	`90,000 × 6 min = 540,000 min; / 60 = 9,000 h; × $45 = $405,000`	illustrative volumes (modeled add-on)	≈$0.4M/yr

SOLUTION FLOW — THE SYSTEM & DATA MAP



SENSITIVITY — LOW / BASE / HIGH

INPUT	LOW	BASE	HIGH
Shipments/yr on 597 lanes (illustrative)	20,000	30,000	40,000
Minutes per manual document (benchmark)	4	6	8
Loaded coordinator rate (benchmark)	$40	$45	$55
Total modeled $/yr	≈$0.2M	≈$0.4M	≈$0.9M

TOTAL IMPACT STATEMENT — SAY IT LIKE THIS

"The measured number is freight spend down ten percent, about two million dollars in six months on the five hundred ninety-seven lanes that carried eighty percent of spend, with completeness sixty to ninety-five percent and adoption eight x. Inside that two million I model roughly eight hundred thousand from catching off-guide carrier spend, six hundred thousand from mode discipline once the ETA was trusted, and six hundred thousand from expedites and detention that chase calls used to cause; those three are my split, the two million is the record. On top of it I model about four hundred thousand a year of document-handling labor that X12 replaced, and that piece is entirely mine."

If challenged — the full answers.

The split of the two million is the softest

It is a judgment on how the ten points divide; to firm it I would tag every invoice line on the onboarded lanes by cause, off-guide carrier, mode change, or expedite, for the six months and let the ledger set the shares.

Ninety thousand documents is a guess

Fifty shipments a lane and three touched documents each is illustrative; the ISA and GS control-number counts on the VAN for the first full quarter would give the exact document volume in an afternoon.

Labor saved is not cash

Correct, nine thousand hours is coordinator capacity moved from keying to exceptions; I would measure it as headcount held flat while lanes went from six hundred to a thousand, which is the number Finance can see.

S7 — Elementum: making a contract manufacturer's BOM safe to plan on — phantoms, revision overlaps, one supply equation

Amazon LPs: Learn and Be Curious (primary — learned CM data by math) · Dive Deep · Earn Trust (external partner).
Crosswalk: loop packet S6 — consistent. *Guardrails: "six sites," never "6 NA sites"; no "500+ product families"; 60→95% belongs to S9; no tolerance value — the record doesn't carry one.*

Honesty label. I owned the ingest requirements, the reconciliation gate, the wave plan and the weekly Flextronics call; engineers built the normalization layer and quarantine to the spec; the commercial relationship belonged to the customer's procurement — my leverage was evidence. The $1.5M/580K belongs to the p99 program this backbone enabled (S8).

60-SECOND PITCH

"**Situation.** Flextronics supplied the BOM our component-demand visibility depended on; phantoms double-time demand if ingested as stocked, and overlapping revisions rewrite history if you auto-load one. **Task.** I owned ingest requirements, the reconciliation gate, the wave plan, and the weekly Flextronics call. **Action.** Discovery: sampled finished goods, exploded BOMs, compared component counts to Flextronics MRP — phantoms found by math. Plan & Design: phantom flatten with an explicit marker; effective-date quarantine, human disposition, never auto-load; supply equation signed. Build & Test: mapping tables owned by named site super-users; per-site reconciliation is the go-live gate. Cutover: three waves / six sites / 5 months; wave 3's hostile ERP used snapshot-diffing — 6 weeks vs a 3-week average, priced up front. Hypercare: the weekly call became standing, framed as their protection. Lesson: reconcile to the source's math, not its documentation. **Result.** Primary metric: contract-manufacturer sites with planning-grade BOM/supply visibility — **0 → 6 in 3 waves / 5 months, zero auto-loaded revision overlaps**. $ impact: ≈$0.3M/yr modeled — the six sites and zero overlaps are measured; the dollars are my model; the $1.5M belongs to S8."

SITUATION / TASK — WHERE IT OCCURRED, WHEN IT HAPPENED, WHY IT IS IMPORTANT

S — Same manufacturer; Flextronics supplied the BOM and planner data that component-demand visibility depended on. Two traps documentation never confesses: phantom assemblies (logical groupings that pass demand through without stocking — ingest them as stocked and component need-dates get double-timed) and revision overlaps (two BOM revisions partially true for weeks — auto-load one and you rewrite demand history). Six site systems carried six definitions of yield, lead time and order status; MTO and MTS had different ATP semantics.

T — Own the ingest requirements, the reconciliation gate, the wave plan — and the chair of the weekly Flextronics call.

The goal: contract-manufacturer sites with planning-grade BOM and supply visibility, with zero auto-loaded overlaps. The challenges beyond the traps themselves: one hostile legacy ERP with flat files and no change-data-capture, and no commercial leverage — evidence was my only lever. The consequence of ingesting naively was operational: false shortages and expedites acted on — both traps named, with evidence, in the discovery memo.

The metrics on the table, with their baselines: contract-manufacturer sites with planning-grade BOM — zero; target six. Auto-loaded revision overlaps — the standing trap; target zero, ever. Manual BOM reconciliation and expedites fired at phantom shortages — the baseline the ingest layer existed to remove.

ACTION — WHAT I PERSONALLY OWNED, HOW I DID IT, WHO ELSE WAS INVOLVED

A — Five gated phases (canonical: wave 1 = two cleanest high-volume sites; wave 2 = three sites on parameterized maps; wave 3 = the hostile legacy ERP, flat files, no change-data-capture):

1. **Discovery** — sampled finished goods and exploded them against Flextronics MRP (the phantom signature found by math); catalogued the six-site definition inventory (yield, lead time, order-status semantics); wrote the MTO/MTS tagging rule and a discovery memo naming the two traps. *Exit bar:* Flextronics engineering confirms the phantom signature; six definitions catalogued; wave order proposed.

2. **Plan & Design** — supply equation (on-hand + production plan + released planned orders); ingest requirements (phantom flatten with marker; effective-date quarantine queue with human disposition — never auto-load); the reconciliation report as the per-site gate; the 2/3/1 wave plan; mapping-table ownership model; weekly call charter. *Exit bar:* requirements signed by customer planning; Flextronics staffs the weekly call; wave economics accepted.
3. **Build & Test** — held engineers to the spec for normalization and quarantine; reconciliation report per site; parameterized mapping templates for wave 2; snapshot-diff design for wave 3 (consecutive full extracts, checksums, a completeness flag on every extract). *Exit bar:* wave-1 reconciliation within tolerance; zero auto-loaded overlaps in test; completeness flag on every extract.
4. **Cutover & Go-Live (per site)** — each site's go-live ran off its matched reconciliation report, mapping tables handed to a named super-user, disposition queue live. The wave-3 duration went to customer leadership up front as a priced wait. *Exit bar:* the reconciliation report was the gate, not a date on a slide — six sites live over three waves; hardest site 6 weeks vs the 3-week average, absorbed in plan.
5. **Hypercare / Close** — weekly Flextronics DQ/disposition call left standing (it had become theirs — the forum that protected their feed's reputation); the wave-2 yield gap fixed in the mapping row, model untouched; completeness flag written as a rule in the ingest layer. *Exit bar:* zero auto-loaded revision overlaps; phantom double-timing eliminated at source; backbone handed to the 200-lane program (S8).

Engineers built the normalization layer and quarantine to the spec; named site super-users owned the mapping tables; the commercial relationship belonged to the customer's procurement — my leverage was evidence. Mine specifically: the sampling-and-explosion analysis, the ingest requirements, the supply equation, the 2/3/1 wave plan, the gate and the call charter. The contribution: finding the phantoms by math rather than meetings, and designing the two mechanisms — the explicit phantom marker and the effective-date quarantine — that made the traps impossible rather than policed; the reconciliation report was the per-site gate, not a date on a slide.

The obstacles: an external supplier with no obligation to care — solved by making the weekly call theirs, since it protected Flextronics from blame for misses their feed didn't cause — and the hostile wave-3 ERP, handled by snapshot-diffing at six weeks against a three-week average, priced up front so leadership took it as the plan, not a slip. Priorities ran waves by cleanliness — the two cleanest high-volume sites first, so a failure in wave 1 cost two sites of credibility, not six; buy-in came from requirements signed by customer planning, Flextronics staffing the call because it was their protection, and the wave economics accepted up front. I refused auto-loading revisions, forcing one lead-time semantics across sites, and going live on a date instead of a matched report — ATP and lead-time meaning genuinely differ between MTO and MTS; one forced semantics lies about both.

RESULTS — HOW SUCCESS WAS MEASURED: $ · # · % · TIME · CUSTOMER · QUALITY

R — Primary metric: CM sites with planning-grade BOM **0** → **6** in 3 waves / 5 months; zero auto-loaded overlaps; hardest site in plan. $ impact: ≈$0.3M/yr modeled.

- **$** — ≈$0.3M/yr modeled (manual BOM reconciliation labor, expedites fired at phantom shortages, overlap rework avoided); the record has no dollar for this story — the $1.5M belongs to S8, the program this backbone enabled.
- **#** — Contract-manufacturer sites with planning-grade BOM 0 → 6, in 3 waves (record); zero auto-loaded revision overlaps — ever (record).
- **Time** — 5 months for six sites; hardest site 6 weeks vs the 3-week average — priced up front and absorbed in plan (record).
- **Customer** — Phantom double-timing eliminated at source, so planners stopped chasing false shortages that were never real; the weekly Flextronics call left standing as their forum (record).
- **Quality** — The reconciliation report was the per-site go-live gate, not a date on a slide; mapping tables handed to named site super-users; completeness flag written as a rule in the ingest layer (record).

Against the goal stated above: planning-grade sites 0 → 6 in three waves and five months; zero auto-loaded overlaps, ever; phantom double-timing eliminated at source — and the backbone handed to the 200-lane program (S8). The trade was speed, never quality: three waves over five months instead of a big bang, so one surprise could not spend six sites — and the hostile site ran six weeks against a three-week average, priced up front.

PROBING — QUESTIONS, PHASE DEEP-DIVES, AND THE MATH

Challenge — Why did you choose this story to illustrate Learn and Be Curious?

Because I learned the CM's data by math, not meetings — sampled finished goods, exploded their BOMs in our model, compared component counts to Flextronics' own MRP, and their engineers confirmed the phantom signature the mismatches revealed. Reconcile to the source's math, not its documentation.

Challenge — What other stories demonstrate Learn and Be Curious?

S16 — learning the stack to build the Snowflake pipeline — is the backup; the MIT toolkit (labeled) and E7 forecasting are the standing third.

Challenge — Could you come up with an example that is more recent?

Elementum years — older. For a current-role record of the same curiosity, S4 is the one: my own program's reports indicted my rule and I learned the promise math to fix it.

Probing — Why did you choose to focus on these results? What other results were important?

Sites 0 → 6 with zero auto-loaded overlaps is the record that matters — planning-grade data where there was none, in five months. The downstream dollar deliberately lives elsewhere: this backbone enabled the p99 lane program, and the $1.5M belongs to S8.

Probing — You mentioned dollars — what percentage change is that year over year?

No YoY — the ≈$0.3M/yr is modeled (reconciliation labor, false-shortage expedites, overlap rework), deliberately small because the big dollar belongs to the program this backbone enabled, and I won't double-count it. The record movement is sites 0 → 6 and overlaps at zero.

Probing — I'm concerned about the 360 false shortages — tell me more.

Illustrative volume at the “phantom shows a shortage of 100” pattern, scaled. To firm it I would join the expedite ledger to the shortage alerts for the year before ingest and count how many alerts had no real component gap behind them.

Challenge — What were the lessons learned? What would you have done differently?

Reconcile to the source's math, not its documentation. Differently: put the completeness flag on the extract from wave 1 — the unflagged partial extract is the failure that looks like thousands of phantom deletions.

Challenge — How would you implement this at Amazon?

Vendor and manufacturer master data at scale carries the same two traps — a phantom level in a kit BOM and an effectivity overlap on an engineering change both manufacture phantom demand. Detect by the source's own math, quarantine with a human disposition, and let the reconciliation report — not a date — open each wave.

Challenge — How did these results compare to your actual goals?

The ask was planning-grade BOM and supply visibility at the CM sites — delivered 0 → 6 in three waves and five months, zero auto-loaded overlaps, the hardest site absorbed in plan, and the backbone handed to the 200-lane program (S8).

PHASE DEEP-PROBE BANK — Q&A (13). Answer the phase asked, then stop.

Discovery — How did you find the phantoms?

By math, not documentation: sampled finished goods, exploded their BOMs in our model, compared component counts to Flextronics' own MRP. The mismatches were the phantom signature; their engineers confirmed.

How did you know discovery was done?

The two traps — phantoms and revision overlaps — named with evidence, the six sites' definitions catalogued, and Flextronics confirming the signature.

Plan & Design — What was in your risk register at sign-off?

Phantoms ingested as stocked, revision overlaps auto-loaded, and partial extracts consumed. The first two had mechanisms from day one — marker and quarantine; the third got its completeness flag properly at wave 3, my do-differently.

How did you get an external supplier to staff a standing call?

By making the call theirs: it protected Flextronics from blame for misses their feed didn't cause. Evidence was my only leverage — I owned no commercial relationship.

What was the entry bar for a site?

Extracts flowing, a named super-user for every mapping table, and the reconciliation report built for that site. No super-user, no wave slot.

Build & Test — What was your definition of done for a site?

Reconciliation report matched within the published tolerance — the report was the gate, not a date on a slide. I won't quote the tolerance value; the record doesn't carry it.

Describe your bug-bash equivalent.

The wave-2 yield gap: several points between their report and ours. Planners, Flextronics and engineers in one session traced it to engineering-sample pulls netted on their side versus gross started-units on ours — fixed in one mapping row.

Cutover — Which wave did you hold and how did the business take it?

Wave 3 ran six weeks against a three-week average because the no-CDC ERP needed snapshot-diffing. I had priced it up front, so leadership took it as the plan, not a slip.

What did each wave deliver?

W1: the normalization layer, quarantine, and the first two sites. W2: parameterized mapping tables and three more sites. W3: snapshot-diffing and the hardest site. Each wave's mapping tables stayed with a named super-user.

Hypercare — How did you decide it was closed? What did you leave behind?

Six sites live behind matched reports, zero auto-loaded overlaps, the weekly call running without me. Left behind: a named super-user per mapping table, the quarantine queue with human disposition, the completeness-flag contract, the call as their forum. The model stayed standard; the mappings stayed theirs.

Cross-phase — Where did you trade scope for date? What did you refuse?

Nowhere on quality; I traded speed — three waves over five months instead of a big bang, so one surprise could not spend six sites. Refused: auto-loading revisions, forcing one lead-time semantics, going live on a date instead of a matched report.

Why not force one lead-time semantics across sites?

ATP and lead-time meaning genuinely differ between MTO and MTS; one forced semantics lies about both.

Do differently?

Put the completeness flag on the extract from wave 1 — the unflagged partial extract is the failure that looks like thousands of phantom deletions.

Where this lands at Amazon: vendor and manufacturer master data at scale carries the same two traps — a phantom level in a kit BOM and an effectivity overlap on an engineering change both manufacture phantom demand. The transferable rules: detect by the source's own math, quarantine with a human disposition, and let the reconciliation report — not a date — open each wave.

BUSINESS IMPACT MODEL — KPI TREE, NUMBERS, THE MATH

```
Planning-grade component demand at six CM sites (0 -> 6, record)
+- Demand-timing integrity (stock of trust)
|    +- phantom double-timing: components pulled early AND twice
|    |     -> false shortages -> expedites acted on (modeled $)
|    +- revision overlaps auto-loaded: 0 (record) -> no rewritten demand history
+- Reconciliation labor (flow): manual BOM-vs-MRP per site -> disposition queue only
+- Backbone enabled: one sigma per lane -> S8 p99 program ($1.5M is S8's)
+- Gate loop: recon in tolerance -> site live -> super-user owns mapping row ->
     defects fixed as data -> next wave mostly configuration -> 5 months for 6 sites
```

DRIVER	BASELINE	AFTER	$/YR	BASIS
CM sites with planning-grade BOM	0	6 (+6 in 5 months)	—	record
Auto-loaded revision overlaps	uncontrolled	0	—	record

DRIVER	BASELINE	AFTER	$/YR	BASIS
Manual BOM reconciliation labor	6 hrs/wk × 6 sites	1.5 hrs/wk × 6	$0.08M	modeled (1,300 hrs × $60)
Expedites acted on false shortages	~72/yr	~0	$0.2M	modeled (72 × $3K premium)
Overlap rework avoided	~24 overlaps/yr	dispositioned	$0.02M	modeled (24 × 16 h × $60)
Total modeled	—	—	**≈$0.3M**	record has no $ for S7; $1.5M is S8's

If challenged. *"360 false shortages — source?"* — "Illustrative volume at the 'phantom shows a shortage of 100' pattern, scaled. The record is the mechanism and the zero: phantom double-timing eliminated at source, zero auto-loaded overlaps ever. The dollars are deliberately small because the big dollar — the $1.5M release — belongs to the program this backbone enabled, and I won't double-count it."

THE MATH — STEP-BY-STEP DERIVATION

Table 1 — before: the problem in numbers.

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
1	Planning-grade CM sites	`0 of 6; manual BOM-vs-MRP reconciliation at every site`	record	0 of 6
2	Manual recon burn	`6 h/wk × 6 sites = 36 h/wk`	modeled	36 h/wk of reconciliation
3	What MRP computes (finished-goods grain)	`NetReq(FG, t) = GrossReq − OnHand − SchedReceipts`	the netting identity; walk below at FG grain only (illustrative)	substituted in rows 4–5
4	Clean row	`NetReq = 500 − 150 − 200 = 150`	Gross 500; on-hand 150; receipts 200 (illustrative)	NetReq = 150; need-date = due − LT_FG
5	Distorted row — phantom FG record	`NetReq = 600 − 150 − 200 = 250; false shortage = 250 − 150 = 100`	a phantom/stale finished-goods record inflates gross requirements by 100 (illustrative)	an expedite fires on 100 units never needed
6	How often	`~5/site/mo × 6 sites × 12 mo = 360 false shortages/yr`	modeled count	360/yr
7	Timing corruption	`the same stale record pulls the need-date a lead-time early`	record mechanism	planners chase demand that was never real
8	Revision overlaps	`two revisions both partially true in [d0, d1]; ~24/yr`	auto-loading would rewrite demand for orders released under the prior revision (record rule; count modeled)	~24 history-rewrite risks/yr

Table 2 — the fix: step by step, values substituted.

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
1	Explicit marker	`distorted row reconciled: NetReq = 500 − 150 − 200 = 150`	the phantom record carries an explicit marker; it no longer inflates gross (record mechanism)	NetReq = 150 — matches the site's own MRP
2	Per-site gate	`count_model vs count_MRP on sampled finished goods must match`	the reconciliation report is the go-live gate, not a date on a slide (record)	a site opens on a matched report

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
3	Quarantine	`overlaps held for human disposition per order/effectivity`	record rule	auto-loaded overlaps = 0 — ever
4	Hostile flat-file site	`completeness = rows received / expected; a 40%-rows file would read as 60% deletions → blocked`	snapshot-diffing with a completeness flag (record mechanism)	partial files never consumed
5	Wave plan	`W1: 2 × 3 = 6 wk; W2: 3 × 3 = 9 wk; W3: 6 wk; 6 + 9 + 6 = 21 weeks`	hostile site at 6 wks vs the 3-wk average, priced up front (record)	≈5 months for six sites
6	Recon labor after	`6 → 1.5 h/wk per site — disposition queue only`	modeled	−4.5 h/wk per site

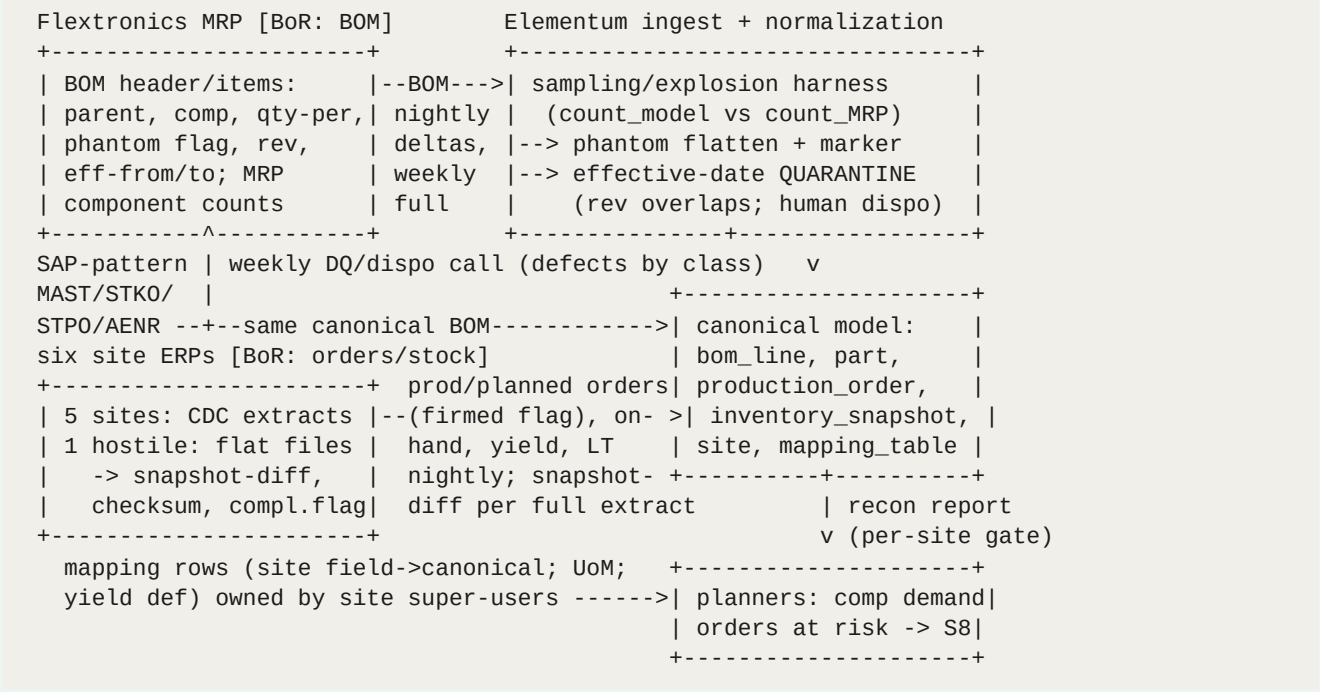
Table 3 — after: the metric, before → after.

STEP	METRIC	BEFORE	AFTER	BASIS
1	Planning-grade CM sites	0 of 6	6 of 6, in 3 waves ≈ 5 months	record
2	Manual reconciliation	6 h/wk × 6 sites	1.5 h/wk × 6	modeled
3	Expedites on false shortages	~72/yr acted on	~0 — the marker removes the distortion	modeled
4	Auto-loaded revision overlaps	uncontrolled	0 — ever	record

Table 4 — the dollars, every multiplication shown.

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
1	Reconciliation labor	`(6 − 1.5) = 4.5 h/wk; × 6 sites = 27 h/wk; × 48 wk = 1,296 ≈ 1,300 h; × $60 = $77,760`	modeled	≈$0.08M/yr
2	False-shortage expedites	`360 × 20% = 72 acted on; 72 × $3,000 = $216,000`	count from Table 1 row 6; premium benchmark $2–5K (modeled)	≈$0.2M/yr
3	Overlap rework avoided	`24 × 16 h = 384 h; 384 × $60 = $23,040`	the quarantine makes the error 0 by construction (modeled)	≈$0.02M/yr
4	Total	`0.08 + 0.2 + 0.02 = $0.3M`	the record has no dollar for S7; the $1.5M belongs to S8	≈$0.3M/yr

SOLUTION FLOW — THE SYSTEM & DATA MAP



SENSITIVITY — LOW / BASE / HIGH

INPUT	LOW	BASE	HIGH
Loaded analyst rate (benchmark)	$55 → $0.07M	$60 → $0.08M	$70 → $0.09M
Share of false shortages expedited (illustrative)	10% → 36	20% → 72	30% → 108
Expedite premium (benchmark)	$2K → $0.07M	$3K → $0.22M	$5K → $0.54M
Overlap rework (illustrative)	$0.01M	$0.02M	$0.03M
Total modeled $/yr	≈$0.2M	≈$0.3M	≈$0.7M

TOTAL IMPACT STATEMENT — SAY IT LIKE THIS

"I do not claim a dollar figure from the record on this one; the million and a half of safety stock belongs to the lane program this backbone made possible. What I can model is about three hundred thousand a year: roughly eighty thousand in manual BOM reconciliation the quarantine queue replaced across six sites, about two hundred thousand in expedites that used to be triggered by phantom shortages that were never real, and a small piece, twenty thousand or so, in plans that no longer get reworked because a revision overlap was auto-loaded. The six sites, the five months and the zero auto-loaded overlaps are measured; every dollar here is my model on benchmark rates."

If challenged — the full answers.

The expedite line is the hardest

It assumes about seventy expedites a year were being fired at phantom shortages, one in five of the false signals; to firm it I would join the expedite ledger to the shortage alerts for the year before ingest and count how many alerts had no real component gap behind them.

Where is the record dollar

There is none for this story, on purpose; the inventory release was measured downstream in the p99 program and I will not book it twice, so what I offer here is the labor and expedite model, labelled as such.

> **Six hours a week per site is a guess**
>
> It is; the honest replacement is the disposition-queue log we left behind, which timestamps every human decision, so a month of that log against the pre-ingest planner timesheets would give you the real before and after.

S10 — Elementum: Austria — "the noise said carriers; the data said measurement" (routing-guide rules engine, 25 sites, ~$8M)

Amazon LPs: Are Right, A Lot (primary) · Dive Deep (backup) · Broad Responsibility (primary — one site's fix rolled to 25 sites) · Earn Trust (the Finance-signed method). **Crosswalk:** not in the loop packet — safe anywhere. *Guardrails: say $8M, never "$1M"; "~99% in 3 months," never "two months"; 1–2 hrs/wk is the after, never the before.*

Honesty label. I owned the analysis and the rules design and pulled the data myself; the data platform team built the engine; Finance signed the savings method before any dollar was claimed. Site-wave counts and dates beyond "Austria first, then 25 sites" are not in the record.

60-SECOND PITCH

"**Situation.** Austria transportation could not measure on-time delivery; reports said data quality was 40%; the team spent 8–10 hours a week reconciling by hand; everyone believed the carriers were late. **Task.** I owned the analysis and the rules design. **Action.** Discovery: three years of shipment history in SQL — true DQ ~55%, not 40; defect classes named. Plan & Design: promised-delivery-date as a routing-guide rules engine with Incoterm in the key (~8% of lanes); Finance-signed savings method before any dollar — contracted guide rate minus paid rate, executed only. Build & Test: alerts by reason code; automated OTD vs a manual sample per cohort before any scorecard. Cutover: Austria scorecard released; manual 8–10 → 1–2 hrs/wk. Hypercare: super-users per region, a one-page runbook; 25 EU sites. Lesson: measure before you blame; sign the method before the number. **Result.** Primary metric: routing-guide data quality — **~55% true (40% reported) → ~99% in 3 months**. $ impact: $500K/yr at Austria (10% of a ~$5M base, Finance-signed executed-only method), then ~$8M across 25 EU sites / 400+ lanes — both record; plus ≈$0.5M/yr of modeled analyst capacity."

SITUATION / TASK — WHERE IT OCCURRED, WHEN IT HAPPENED, WHY IT IS IMPORTANT

S — Transportation at the customer's Austria site could not measure on-time delivery. Reports said data quality was 40%; the team spent 8–10 hours a week reconciling promised vs actual dates by hand; the freight base was ≈$5M/yr with no enforceable OTD. Everyone — the site, the carriers' account teams, my own sponsor — believed the carriers were late.

T — I owned the analysis and the rules design.

The goal: an OTD the carriers could not dispute, and savings claimed only on a method Finance signed first. The real challenge: the data had to overturn the room's explanation. Left alone: blaming carriers off bad measurement — a scorecard on 55% data the carriers would dispute and win — the manual reconciliation continuing at every site, and any savings claim disputed after the fact.

The metrics on the table, with their baselines: routing-guide data quality — 40% reported, ~55% true once measured; manual reconciliation — 8–10 hours a week per site; enforceable OTD — none, on the ≈$5M/yr freight base.

ACTION — WHAT I PERSONALLY OWNED, HOW I DID IT, WHO ELSE WAS INVOLVED

A — Five gated phases:

1. **Discovery** — pulled three years of shipment history in SQL myself; ran the defect census (missing rates, wrong-grain lanes, five spellings of one carrier name, brochure transit times); established true DQ ~55% vs 40% reported; reframed the problem as measurement, not carriers. *Exit bar:* sponsor accepts the reframe on the evidence; defect classes named and counted.
2. **Plan & Design** — rules-engine spec (lookup by ship-from/to, mode, volume, Incoterm, SLA; a no-hit pages a human, never a default carrier); reason-code alert design; **the savings method signed with Finance before any dollar was claimed** (contracted guide rate − paid rate, per shipment, executed only); the scorecard DQ

gate (automated OTD = manual sample per lane cohort). *Exit bar:* spec signed by transportation; Finance method signed; scorecard gate published.

3. **Build & Test** — held engineering to the spec; weekly defect burn-down by reason code; manual-sample comparison per lane cohort. *Exit bar:* DQ trending to ~99%; automated OTD matches the manual sample per cohort.
4. **Cutover & Go-Live (Austria)** — released the carrier scorecard only after the gate passed; retired the manual reconciliation (8–10 → 1–2 hrs/wk); stood up the savings ledger per shipment. *Exit bar:* a scorecard the carriers could not dispute; $500K/yr on the Finance-signed method.
5. **Hypercare / Close → scale** — super-user model per region; one-page runbook (add a lane · read a no-hit · when to escalate); two shadowed rule reviews per site; rolled out site by site. *Exit bar:* 25 EU sites / 400+ lanes ≈ $8M; ~99% DQ; adoption cost per site = a rules file + one DQ cycle, in days.

I pulled the three-year shipment history in SQL myself — the room already had an explanation, and SQL was faster than a request queue and could not be argued with; the data platform team built the engine; regional super-users ran the rules files after two shadowed reviews each. Mine beyond the analysis and rules: the SQL pull, the defect census, the rules-engine spec, the reason-code alert design, the Finance-signed savings method and the scorecard gate — the method (contracted guide rate minus paid rate, per shipment, executed only) signed with Finance before the number existed.

The obstacle was a sponsor unhappy with the delay for weeks — answered with the defect classes trending down weekly and the manual-sample gate the scorecard had to pass; he was right to want speed, and the carriers never disputed the result. Incoterm semantics were put into the lane key rather than papered over. Priorities ran evidence before enforcement — the manual reconciliation stayed alive until automated OTD matched the manual sample per lane cohort — and buy-in came from transportation signing the spec, Finance signing the method, and scale going site by site at a rules file plus one DQ cycle each. The decisions challenged: the room's explanation that the carriers were late — overturned with three years of their own data — and I refused a scorecard on 55% data, a silent default carrier on a no-hit, and any savings number without a Finance-signed method.

RESULTS — HOW SUCCESS WAS MEASURED: $ · # · % · TIME · CUSTOMER · QUALITY

R — Primary metric: DQ **~55% → ~99% in 3 months**; $500K/yr Austria → ~$8M across 25 sites (record); manual reconciliation 8–10 → 1–2 hrs/wk.

- **$** — $500K/yr at Austria — 10% of a ~$5M freight base on the Finance-signed, executed-only method — then ~$8M across 25 EU sites / 400+ lanes (record roll-up; ~$320K/site average, derived); plus ≈$0.5M/yr modeled analyst capacity.
- **#** — 25 EU sites / 400+ lanes on the same rules file and one DQ cycle each; three years of shipment history pulled in SQL for the baseline (record).
- **%** — Routing-guide data quality ~55% true (40% reported) → ~99% in 3 months (record).
- **Time** — ~99% DQ in 3 months; adoption cost per site = a rules file + one DQ cycle, in days (record).
- **Customer** — A carrier scorecard the carriers could not dispute — automated OTD matched the manual sample per lane cohort before release (record).
- **Quality** — Manual reconciliation 8–10 → 1–2 hrs/wk per site; the savings method signed with Finance before any dollar was claimed; negotiated-but-not-executed never counted (record).

Against the goal stated above: DQ ~55 → ~99% in three months; a scorecard the carriers could not dispute; manual reconciliation down from 8–10 to 1–2 hours a week; $500K/yr at Austria on the Finance-signed method and ~$8M across 25 EU sites. The trade was date for evidence: I delayed the scorecard weeks rather than ship it on bad data; scope was never cut.

PROBING — QUESTIONS, PHASE DEEP-DIVES, AND THE MATH

Challenge — Why did you choose this story to illustrate Are Right, A Lot?

“The noise said carriers; the data said measurement” — I pulled three years of shipment history in SQL myself, established true DQ was ~55%, not 40%, and the fix was a rules engine, not a carrier beating. Being right here meant

the data overturning the explanation everyone held.

Challenge — What other stories demonstrate Are Right, A Lot?

S8 — measured lanes vs brochure — is the backup; S4, the wrong rule corrected right, is the third.

Challenge — Could you come up with an example that is more recent?

Elementum years — older. For the same LP in my current role, S4 is the record: my own rule, indicted by my own program's reports, fixed with promise math.

Probing — Why did you choose to focus on these results? What other results were important?

DQ ~55 → ~99% in three months is the record that made everything else possible; $500K/yr at Austria and ~$8M across 25 sites are the Finance-signed consequences. The manual hours falling from 8–10 to 1–2 a week is what the team felt.

Probing — You mentioned dollars — what percentage change is that year over year?

Not YoY — the $500K is 10% of a ~$5M freight base on the executed-only method (contracted guide rate minus paid rate, per shipment), and the ~$8M is the same ledger across 25 sites. Negotiated-but-not-executed never counted.

Probing — I'm concerned about the ~$8M roll-up — tell me more.

It is the Finance-signed, executed-only method applied at each site, not a projection; the assumption underneath is that the ten percent Austria rate held elsewhere, and what firms it is the per-shipment ledger at each site — which is what Finance read.

Challenge — What were the lessons learned? What would you have done differently?

Measure before you blame; sign the method before the number. Differently: start with the industry benchmark for lane transit times instead of proving the brochure wrong lane by lane — same answer, faster.

Challenge — How would you implement this at Amazon?

This is the model for how a savings claim should be built — method signed with Finance before the number, executed-only, then scaled site by site with the adoption cost measured in days. The Broad Responsibility angle: one site's fix became 25 sites' instrument.

Challenge — How did these results compare to your actual goals?

The site needed enforceable OTD; it got a scorecard the carriers could not dispute, DQ at ~99% in three months, $500K/yr on the signed method and ~$8M at network scale — with manual reconciliation down to 1–2 hours a week.

PHASE DEEP-PROBE BANK — Q&A (13). Answer the phase asked, then stop.

Discovery — How did you know discovery was done?

When the three-year pull had every blank classified — missing rate, wrong grain, name variant, brochure time — and the sponsor accepted that true quality was ~55%, not 40%, and that the problem was measurement, not carriers.

Why pull the data yourself?

Because the room already had an explanation. Three years of history in SQL was faster than a request queue and could not be argued with.

Plan & Design — What was in your risk register at design sign-off?

Incoterm changing who owns the clock, a silent default carrier on a no-hit, and savings disputed after the fact. Mitigations: Incoterm in the key, page a human, Finance-signed method before any claim.

How did you handle the sponsor objecting to the delay?

I showed the defect classes trending down weekly and the manual-sample gate the scorecard had to pass. He was unhappy for weeks and right to want speed; the carriers never disputed the result.

Who signed the savings method and when?

Finance, before the first dollar was claimed: contracted rate minus paid rate, per shipment, per lane, executed only. Negotiated-but-not-executed never counted.

Build & Test — What was your definition of done for a lane rule?

It resolves without no-hits across the sample window and its automated OTD matches the manual sample for its cohort. Until then the manual reconciliation kept running.

How did you treat data defects?

As classes, not one-offs: alerts by reason code, trended weekly, burned down — root cause at scale. That is how 55% became ~99% in three months.

Cutover — What made you say go on the scorecard?

Automated OTD matching the manual sample per lane cohort. Anything less would have shipped a scorecard the carriers could dispute — and they would have won.

What was your rollback?

The manual reconciliation stayed alive until the match held. Reversibility was free because I didn't retire the old process early.

Hypercare — How did you decide Austria was closed and scale could start?

DQ ~99%, manual hours at 1–2 a week, the scorecard undisputed, and a regional super-user running the rules file after two shadowed reviews. Then 25 sites at a rules file plus one DQ cycle each.

What did you hand to the regional super-users?

Their rules file, the one-page runbook — add a lane, read a no-hit, when to escalate — and two shadowed rule reviews each. After that, adding a site cost days.

Cross-phase — Where did you trade scope for date? What did you refuse?

I traded date for evidence — delayed the scorecard weeks rather than ship it on bad data; scope was never cut. Refused: a scorecard on 55% data, a silent default carrier, and any savings number without a Finance-signed method.

How do you know the $8M is real? · Do differently?

Finance-signed, executed-only method applied across 25 sites; the $500K pilot is 10% of a ~$5M base; anything merely negotiated is excluded. Differently: start with the industry benchmark for lane transit times instead of proving the brochure wrong lane by lane — same answer, faster.

Where this lands at Amazon: the cleanest "used data to overturn the room's explanation" story — and the model for how a savings claim should be built: method signed with Finance before the number, executed-only, then scaled site by site with the adoption cost measured in days. Broad Responsibility angle: one site's fix became 25 sites' instrument.

BUSINESS IMPACT MODEL — KPI TREE, NUMBERS, THE MATH

```
Enforceable OTD + executed freight savings: $500K Austria -> ~$8M (record)
+- Measurement integrity (stock): true DQ ~55% (40% reported) -> ~99%
|    +- promised date = ship + transit(lane_key), Incoterm in key (~8% of lanes)
|    +- no-hit pages a human, never a default carrier
+- Savings ledger (flow): (guide rate - paid rate) x executed shipments
|    +- Austria: ~$5M base x 10% = $500K/yr (record)
|    +- 25 EU sites / 400+ lanes: rules file + one DQ cycle each -> ~$8M
+- Manual reconciliation: 8-10 hrs/wk -> 1-2 hrs/wk per site (modeled $)
+- Defect loop: alerts by reason code -> weekly burn-down -> DQ rises ->
     fewer no-hits -> scorecard trusted -> carriers fix at source (+)
```

DRIVER	BASELINE	AFTER	$/YR	BASIS
Routing-guide data quality	~55% true (40% reported)	~99% in 3 months	—	record
Austria savings (guide − paid, executed)	$0	10% of ~$5M base	$500K	record
Network roll-up, 25 EU sites / 400+ lanes	$0	~$8M	~$8M	record
— implied avg per site · implied base	—	~$320K/site · ~$80M at 10%	—	modeled (derived; Austria above average — a larger site, consistent)
Manual reconciliation, 25 sites	8–10 hrs/wk each	1–2 hrs/wk	≈$0.5M	modeled (~9,400 hrs × $55; per-site $20K)

If challenged. *"How is the $8M real?"* — "Because the method was signed before the number existed: contracted guide rate minus paid rate, per shipment, executed only. Negotiated-but-not-executed never counted. The $500K Austria pilot is the method's proof; the $8M is the same ledger across 25 sites."

THE MATH — STEP-BY-STEP DERIVATION

Table 1 — before: the problem in numbers.

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
1	Reported data quality	`40% — a field-populated proxy`	record	the number leadership saw
2	True data quality	`valid promised ∧ valid actual ∧ resolvable lane / all shipments ≈ 55%`	three years of shipment history pulled in SQL (record)	~55% — both broader and stricter than the proxy
3	Defect classes	`missing rate · wrong-grain lane · carrier name in five spellings · brochure transit posing as a promise`	record	4 named classes to burn down
4	Manual reconciliation	`8–10 h/wk per site`	record	~9 h/wk × 25 sites
5	OTD unenforceable	`measuring OTD on brochure promises penalizes carriers for our error`	record problem	no basis for a carrier conversation

Table 2 — the fix: step by step, values substituted.

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
1	Promised-date rule	`promised = ship_date + transit(lane_key)`	lane_key = (from, to, mode, volume band, Incoterm, SLA); Incoterm needed on ~8% of lanes; a no-hit pages a human, never a default carrier (record)	a promise per shipment, from the guide
2	OTD definition	`OTD = deliveries with actual ≤ promised / valid promises only`	brochure promises excluded (record)	validated at ~97% agreement (next row)
3	Validation	`automated vs manual sample per lane cohort ≈ 97% agreement`	record	gate per cohort before release
4	Burn-down	`defect classes worked weekly → ~99% in 3 months`	record	~99% DQ
5	Savings method	`saving = (contracted guide rate − paid rate) per executed shipment only`	Finance-signed; never “negotiated” (record)	$500K/yr Austria result follows in Table 4

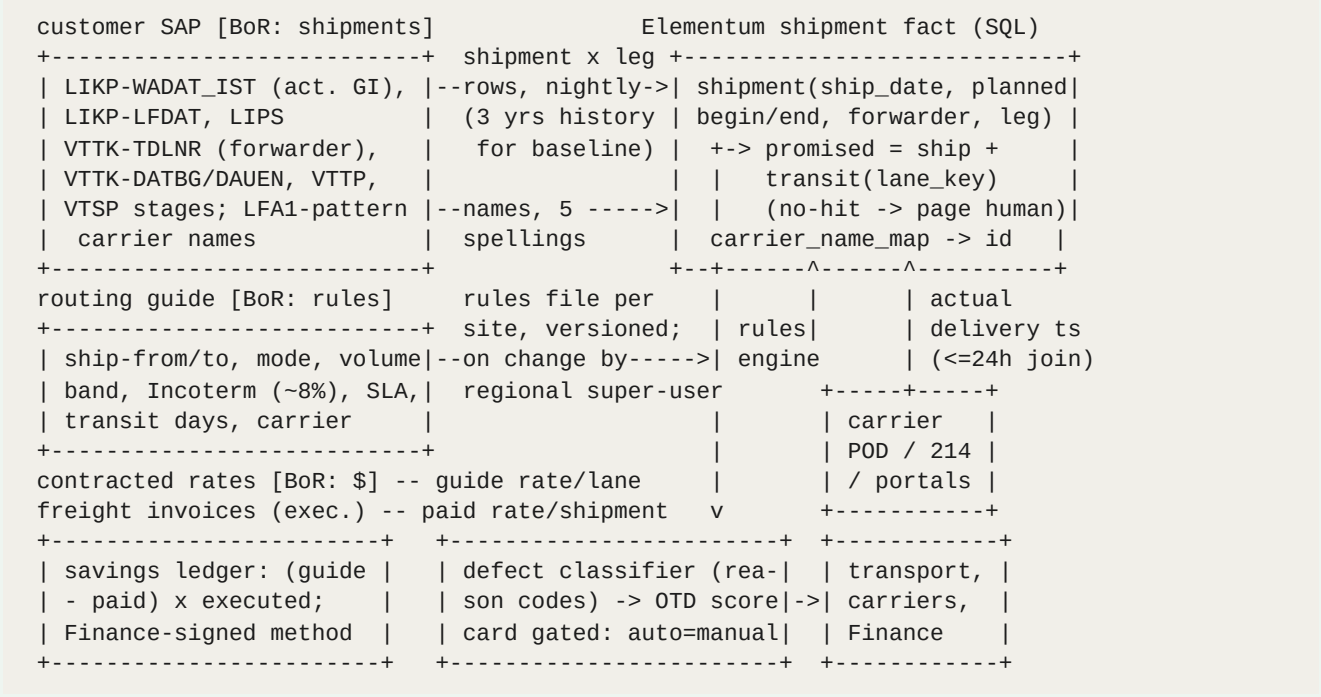
Table 3 — after: the metric, before → after.

STEP	METRIC	BEFORE	AFTER	BASIS
1	Routing-guide data quality	~55% true (40% reported)	~99% in 3 months	record
2	OTD	unenforceable	enforceable; ~97% agreement vs manual sample	record
3	Manual reconciliation per site	8–10 h/wk	1–2 h/wk (−80%)	record
4	Austria savings	$0	$500K/yr, executed-only	record
5	Network roll-up	$0	~$8M across 25 EU sites / 400+ lanes	record

Table 4 — the dollars, every multiplication shown.

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
1	Austria	`$5,000,000 × 10% = $500,000/yr`	~$5M freight base × 10% (record; Finance-signed, executed-only)	$500K/yr
2	Network roll-up	`~$8M (record); / 25 sites = $320K avg; / 10% = ~$80M implied base`	averages derived, stated as such; Austria above average — a larger site, consistent	~$8M
3	Recon capacity returned	`9 − 1.5 = 7.5 h/wk saved (range midpoints); × 25 sites = 187.5 h/wk; × 50 wk = 9,375 h; × $55 = $515,625`	modeled add-on	≈$0.5M/yr

SOLUTION FLOW — THE SYSTEM & DATA MAP



SENSITIVITY — LOW / BASE / HIGH

INPUT	LOW	BASE	HIGH
Hours saved per site per wk (record range)	6 (8→2)	7.5 (9→1.5)	8.5 (10→1.5)
Loaded analyst rate (benchmark)	$50	$55	$65
Sites with the same manual burden (illustrative)	15	25	25
Total modeled $/yr	≈$0.2M	≈$0.5M	≈$0.7M

TOTAL IMPACT STATEMENT — SAY IT LIKE THIS

"The measured numbers are five hundred thousand a year at Austria, ten percent of a five million freight base on a method Finance signed before I claimed a dollar, contracted rate minus paid rate on executed shipments only, and about eight million when the same rules file and one data-quality cycle went to twenty-five sites and four hundred plus lanes. The roll-up is simple: eight million over twenty-five sites is about three hundred twenty thousand a site, which implies roughly eighty million of freight at ten percent, and Austria at five hundred thousand sits above that average because it was a bigger site. On top I model about half a million a year of analyst capacity from cutting manual reconciliation from eight to ten hours a week to one or two at each site; the eight million is record, the half million is my model."

If challenged — the full answers.

Is the eight million really measured

It is the Finance-signed executed-only method applied at each site, not a projection; the assumption underneath is that the ten percent Austria rate held elsewhere, and the thing that firms it is the per-shipment ledger at each site, which is what Finance read.

The half million of labor is the softest

It assumes all twenty-five sites carried Austria's eight to ten hours a week, which I cannot prove; a one-week time log at three sites before their rules file went live would replace that assumption with data.

Why not count negotiated rate cuts too

Because negotiated but not executed is a promise, not a saving; the method counts only shipments that moved at the guide rate, and that discipline is why the carriers never disputed the scorecard or the dollars.

S17 — Elementum: the semi-annual carrier bid — constraint rules and a lane master that cut the cycle from six weeks to about three (no $ claimed)

Amazon LPs: Invent and Simplify (primary) · Frugality (backup) · Dive Deep (the time audit). **Crosswalk:** loop packet S8 — consistent. *Never a dollar on the award — Transportation and Procurement ran it. Never "$500K/yr" or "−25% planning effort."*

Honesty label. I owned the requirements, the constraint rules and the parallel run — not the award; transportation, procurement and engineering did not report to me. Procurement joining late is my named miss.

60-SECOND PITCH

"**Situation.** Semi-annual carrier bid: hundreds of lanes, dozens of rate files, six weeks, mostly reconciliation because every carrier named lanes differently. **Task.** I owned requirements, constraint rules, and the parallel run — not the award. **Action.** Discovery: sat a full cycle logging hours — the problem class was reconciliation, not optimization. Plan & Design: a lane master; constraints as explicit rules; <5% mismatch vs the last two awards before the optimizer could run; scenario comparison, not a black box. Build & Test: parallel-ran one region's real bid. Cutover: weekly rate-file refresh; a stale banner past 10 days. Hypercare: other regions adopted without a mandate; my miss was procurement joining late. Lesson: diagnose the problem class from a time audit before building anything. **Result.** Primary metric: carrier-bid cycle time — **6 weeks → ~3 (~50%)**; pilot region 3 weeks early. $ impact: ≈$190K/yr modeled analyst time returned — the record has no dollar; never savings on the award."

SITUATION / TASK — WHERE IT OCCURRED, WHEN IT HAPPENED, WHY IT IS IMPORTANT

S — The customer's transportation team ran a carrier bid twice a year: hundreds of lanes, dozens of carrier rate files, constraints like incumbency floors and volume commitments — six weeks per cycle, mostly reconciliation, because origin/destination naming and mode codes differed per carrier. Everyone assumed it was an optimization problem.

T — Own the requirements, the constraint rules and the parallel run; transportation, procurement and engineering did not report to me.

The goal: a defensible award in half the cycle. The challenge: the diagnosis had to contradict the room — and the proposed cure, a black-box solver, would have produced awards nobody could defend to an incumbent carrier. Left alone: six weeks of analyst time, twice a year, spent reconciling lane names and mode codes instead of analyzing.

The metrics on the table, with their baselines: bid-cycle time — six weeks, twice a year; award defensibility — the bar any tool had to clear, with the optimizer gated behind a <5% back-test against the last two awards; analyst time — the reconciliation hours the lane master existed to return.

ACTION — WHAT I PERSONALLY OWNED, HOW I DID IT, WHO ELSE WAS INVOLVED

A — Five gated phases:

1. **Discovery (one full bid cycle)** — sat through a whole cycle and built the time audit by activity: the six weeks went to reconciling lane names and mode codes, not analysis; wrote the problem-class statement. *Exit bar:* Transportation agrees the cost is reconciliation. (Procurement not yet in the room — the miss.)
2. **Plan & Design** — the lane-master spec (origin/destination/mode normalization); constraints written one per rule (volume commitments, incumbency floors, mode rules); the validation gate (<5% mismatch vs the last two awards); scenario comparison required as first-class output; the black-box solver rejected in writing. *Exit bar:* rules signed by Transportation; gate published before any optimizer run.
3. **Build & Test** — held engineering to the spec; back-test against the two historical awards; the 10-day stale-data banner required. *Exit bar:* mismatch <5% against both awards; scenarios explainable to an incumbent.
4. **Cutover & Go-Live (pilot region)** — the parallel run — tool vs manual on one region's real bid; acceptance = the tool matches their defensible choices and beats cycle time. *Exit bar:* pilot region finished 3 weeks early;

award defensible in carrier meetings.

5. **Hypercare / Close → adoption** — weekly rate-file refresh cadence; Procurement brought into the constraint definitions (late — my miss); the tool handed over as a standing global tool. *Exit bar:* other regions adopted without a mandate; cycle ~3 weeks; procurement in discovery for the next cycle.

Engineering built the model to the rules; transportation and procurement ran the award, and the regional team's judgment was the acceptance test — the award itself was never mine. Mine specifically: the time audit, the problem-class statement, the lane-master spec, the constraint rules, the validation gate and the parallel-run design. The contribution: sitting through a full bid cycle logging hours — the time audit that contradicted the room — and the lane master plus constraints written one per rule, turning six weeks of reconciliation into configuration; explainability was the spec.

The obstacles: engineering wanted a cheapest-lane solver — answered by making scenario comparison a first-class output and gating the optimizer behind the <5% back-test; constraint semantics differing across functions — "incumbency floor" meant different things to transportation and the category manager — fired because procurement wasn't in the room: my named miss. The back-test read-out classified every mismatch against two historical awards — naming, mode code, constraint semantics — in one session, with owners; buy-in came from rules signed by Transportation, the gate published before any optimizer run, and adoption by envy — other regions came on without a mandate after the pilot finished three weeks early. The decisions challenged: the room's framing that "it's an optimization problem" — overturned by the hour log; the black-box solver, rejected in writing; and the dollar figure on the award, refused — transportation and procurement ran it.

RESULTS — HOW SUCCESS WAS MEASURED: $ · # · % · TIME · CUSTOMER · QUALITY

R — Primary metric: cycle **6 weeks → ~3 (~50%)**; pilot 3 weeks early; adopted without a mandate. ≈$190K/yr modeled analyst hours returned; **no dollar on the award**.

- **$** — ≈$190K/yr modeled analyst time returned (≈2,200 hrs/yr on illustrative team and region counts; range $35–420K); no dollar on the award — Transportation and Procurement ran it, and never savings on the award.
- **#** — Hundreds of lanes, dozens of carrier rate files, twice a year; constraints written one per rule (volume commitments, incumbency floors, mode rules) (record).
- **%** — Back-test mismatch <5% vs the last two awards before the optimizer could run — the published gate (record); cycle time down ~50%.
- **Time** — Carrier-bid cycle 6 weeks → ~3; pilot region finished 3 weeks early; weekly rate-file refresh with a stale banner past 10 days (record).
- **Customer** — Scenario comparison as a first-class output — the award defensible in carrier meetings; other regions adopted without a mandate (record).
- **Quality** — The lane master normalized origin/destination/mode naming — the time audit showed reconciliation, not optimization, ate the six weeks; constraint definitions now written with procurement in the room (record).

Against the goal stated above: the carrier-bid cycle six weeks → ~3, twice a year; the pilot region three weeks early; the mismatch gate below 5% against both historical awards; the award defensible in carrier meetings. The trade was mine: a slower first cycle to write and validate constraints, for a tool that halved every cycle after; nothing was traded on the bid itself.

PROBING — QUESTIONS, PHASE DEEP-DIVES, AND THE MATH

Challenge — Why did you choose this story to illustrate Invent and Simplify?

Because the simplification beat the fancier tool: a lane master plus constraints written one per rule, with a <5% back-test gate against the last two awards before any optimizer ran. The invention was the diagnosis — a time audit that showed the problem class was reconciliation, not optimization.

Challenge — What other stories demonstrate Invent and Simplify?

S16 — four systems into one pipeline — is the backup; S3's de-aggregation design is the third, and the §3 E-designs carry the same muscle for design prompts.

Challenge — Could you come up with an example that is more recent?

Elementum years — older. The more recent same-LP records are S16 at Flexport and S3's de-aggregation design in my current role.

Probing — Why did you choose to focus on these results? What other results were important?

Cycle 6 weeks → ~3 is the record, and the pilot finishing three weeks early is what drove adoption without a mandate. The <5% back-test gate is the quality result: an award that survives the incumbent meeting.

Probing — You mentioned dollars — what percentage change is that year over year?

There are no dollars on the award at all — deliberately: transportation and procurement ran it, and the record has no figure. The only dollars I model are ≈$190K/yr of analyst time returned, on illustrative team and region counts, range $35–420K.

Probing — I'm concerned about the three-analysts, three-regions assumption — tell me more.

They are guesses, which is why the range runs from about $35K to about $420K. I'd replace them with the actual bid roster per region and the hour log I kept in discovery — the same audit that found the problem class.

Challenge — What were the lessons learned? What would you have done differently?

Diagnose the problem class from a time audit before building anything. Differently: procurement in discovery on day one — the constraint-semantics risk fired because they weren't.

Challenge — How would you implement this at Amazon?

Invent and Simplify with the diagnosis first — a time audit that contradicted the room, then a simplification (one lane master plus explicit rules) that beat the fancier tool; refusing the dollar I can't defend is what makes the ones I do defend credible.

Challenge — How did these results compare to your actual goals?

The cycle was six weeks; it runs about three, twice a year, with the pilot region three weeks early, the award defensible in carrier meetings, and the mismatch gate below 5% against both historical awards.

PHASE DEEP-PROBE BANK — Q&A (12). Answer the phase asked, then stop.

Discovery — How did you know discovery was done?

When the time audit from a full cycle showed the six weeks were reconciliation — lane names and mode codes differing per carrier — not analysis, and transportation agreed. It was not fully done: procurement was missing, which I learned later.

Why sit through a whole cycle instead of interviewing?

Interviews tell you what people think they do. Logging hours told me where the time actually went, and the answer contradicted the room.

Plan & Design — What was in your risk register at design sign-off?

Constraint semantics differing across functions, a black-box award, and stale rates. The first one fired — "incumbency floor" meant different things to transportation and the category manager — because I hadn't put procurement in the room.

How did you handle engineering objecting to the design?

They wanted a cheapest-lane solver. I made scenario comparison a first-class output and required <5% mismatch against past awards before the optimizer could run. Explainability was the spec.

Build & Test — What was your definition of done for the model?

<5% mismatch against the last two award cycles, every scenario explainable in one sentence, and a stale-data banner past 10 days. Below that bar the optimizer did not run.

Describe your bug-bash equivalent.

The back-test read-out: every mismatch against two historical awards classified — naming, mode code, constraint semantics — in one session, with owners. That is where most of the reconciliation logic got fixed.

Cutover — What was your go criterion for the pilot?

The tool matched the regional team's defensible choices on their real bid and beat the cycle time. Their judgment was the acceptance test, not mine.

What was the rollback?

The manual process ran in parallel through the pilot. If the tool had disagreed indefensibly, they would have awarded by hand — never needed.

Hypercare — How did other regions come on?

Without a mandate — the pilot finished three weeks early and the tool carried each region's own constraints as rules. Adoption by envy; the refresh cadence kept it honest.

What kept the tool honest after the pilot?

The weekly rate-file refresh with the stale banner past 10 days, and constraint definitions now written with procurement in the room. A model with stale rates or unowned rules decays into a spreadsheet with a logo.

Cross-phase — Where did you trade scope for date? What did you refuse?

Nowhere on the bid itself. The trade was mine: a slower first cycle to write and validate constraints, for a tool that halved every cycle after. Refused: a black-box solver — and a dollar figure; transportation and procurement ran the award.

What did it save? · Do differently?

I don't have a dollar figure and won't invent one; the cycle halved and the pilot finished three weeks early — the savings claim belonged to the negotiation. Differently: procurement in discovery on day one.

Where this lands at Amazon: Invent and Simplify with the diagnosis first — a time audit that contradicted the room ("it's an optimization problem") and a simplification (one lane master + explicit rules) that beat the fancier tool. The refusal to claim the award dollar is the credibility move interviewers remember.

BUSINESS IMPACT MODEL — KPI TREE, NUMBERS, THE MATH

```
Defensible carrier award, delivered in half the cycle (no $ on the award)
+- Cycle time 6 wks -> ~3 (record); pilot region 3 weeks early (record)
+- Reconciliation hours (the time audit's finding) -> lane master + rules
|    +- back-test gate <5% vs last two awards before the optimizer runs
|    +- scenario comparison first-class -> award survives the incumbent meeting
+- Freshness loop: weekly rate-file refresh; stale banner past 10 days
+- Adoption loop: pilot 3 wks early -> regions adopt without mandate ->
     each region's constraints as rules -> tool stays theirs
```

DRIVER	BASELINE	AFTER	$/YR	BASIS
Carrier-bid cycle time	6 weeks	~3 weeks	—	record
Back-test mismatch	—	<5% vs two awards	—	record (the published gate)
Analyst hours returned	—	≈2,200 hrs/yr	≈$190K	modeled (illustrative team and region counts; range $35–420K)
Award savings	—	—	**none claimed**	the award belonged to Transportation and Procurement

If challenged. *"Surely the better tool saved freight dollars?"* — "Possibly — and that claim belongs to the people who ran the award. What I can defend is the cycle halving, the <5% back-test, and the analyst hours returned, modeled and labeled. Refusing the dollar I can't defend is what makes the ones I do defend credible."

THE MATH — STEP-BY-STEP DERIVATION

Table 1 — before: the problem in numbers.

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
1	Bid cycle	`6 weeks, twice a year`	mostly reconciliation of lane names and mode codes — the hour log's finding (record)	6 wks, reconciliation-dominated

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
2	Worked lane set	`3 lanes, volumes 100 / 200 / 300 loads; rates (A, B): L1 (900, 950) · L2 (1,100, 1,000) · L3 (800, 850)`	illustrative, not program data	the example the fix re-evaluates
3	All-cheapest award	`L1→A: 100 × $900 = $90,000; L2→B: 200 × $1,000 = $200,000; L3→A: 300 × $800 = $240,000; Σ = $530,000`	ignores B's volume commitment	$530,000 — and it cannot survive the incumbent meeting

Table 2 — the fix: step by step, values substituted.

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
1	Lane master + explicit rules	`coverage · volume commitments · incumbency floors · mode eligibility — one constraint per rule`	reconciliation turned into configuration (record)	rules as data
2	Back-test gate	`mismatched lanes / lanes evaluated < 5% vs the last two awards`	published before any optimizer run; validates rules, not prices (record)	gate held
3	Constraint evaluated	`B needs ≥ 250 loads; all-cheapest gives B only 200 → short 50`	the illustrative commitment	50 loads must move to B
4	Cheapest repair	`50 loads of L3 to B: +(850 − 800) = +$50 each; 50 × $50 = $2,500`	L3 is the cheapest place to honor the commitment	+$2,500
5	Constrained award	`$530,000 + $2,500 = $532,500; delta = 2,500 / 530,000 = 0.47%`	the price of the relationship constraint, shown explicitly	$532,500 — survives the room
6	Freshness	`banner if rate-file age > 10 days; weekly refresh`	record	stale data never hides

Table 3 — after: the metric, before → after.

STEP	METRIC	BEFORE	AFTER	BASIS
1	Carrier-bid cycle	6 weeks	~3 weeks (−50%)	record
2	Pilot region	—	3 weeks early; adopted without mandate	record
3	Back-test mismatch	—	<5% vs two awards — gate held	record
4	Award dollars	—	none claimed — Transportation and Procurement ran the award	record

Table 4 — the dollars, every multiplication shown.

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
1	Hours returned	`2 cycles × 3 wks × 40 h = 240 h per analyst-region; × 3 analysts × 3 regions = 2,160 ≈ 2,200 h/yr`	team and region counts illustrative	≈2,200 h/yr
2	Analyst value	`2,200 × $85 = $187,000`	loaded analyst rate (benchmark); range $35–420K	≈$190K/yr
3	Award dollars	`none claimed, by rule`	record	—

SENSITIVITY — LOW / BASE / HIGH

INPUT	LOW	BASE	HIGH
Analysts per regional bid (illustrative)	2	3	4

INPUT	LOW	BASE	HIGH
Regions on the tool (illustrative)	1	3	4
Loaded analyst rate (benchmark)	$70/hr	$90/hr	$110/hr
Hours returned per year	~480	~2,160	~3,840
Total modeled $/yr	~$35K	~$190K	~$420K

TOTAL IMPACT STATEMENT — SAY IT LIKE THIS

"I don't have a dollar figure for the award and I won't invent one; transportation and procurement ran it. What I can model is the time: the cycle went from six weeks to about three, twice a year, and on an illustrative three analysts and three regions that is roughly 2,200 analyst hours a year, about $190K of capacity at benchmark rates. The cycle time, the pilot finishing three weeks early and the five percent gate are the record; the analyst count, the region count and the dollars are my model, and none of it is savings on the award."

If challenged — the full answers.

Three analysts and three regions are guesses.

They are, which is why the range runs from about $35K to about $420K. I'd replace them with the actual bid roster per region and the hour log I kept in discovery, which is the same audit that found the problem class.

Returned hours are not cash.

Correct, nobody was let go; the hours became analysis and onboarding the next region. I'd measure it as analyst-weeks per bid, before and after, which is the number transportation itself would recognize.

Surely the faster bid saved freight money?

Maybe, but I did not run the award and the record has no figure, so I don't claim one. The scenario delta, constrained cost minus cheapest, is shown in every award pack; the negotiation owns whatever it saves.

S19 — Elementum (client: Walmart Marketplace): the delivery-promise buffer rebuilt as priced risk — one governed dial, +32 points of fast-promise share at the same 95% floor (all model, no record dollar)

Amazon LPs: Dive Deep (alternate when S8/S10 are spent) · Invent and Simplify (alternate when S17/S16 are spent) · Frugality (speed bought with zero new spend) — §1.1 unchanged; S19 is extra bench for design-flavored probes. **Crosswalk:** not in the loop packet — safe anywhere. **Recency:** Elementum client years. *Every quantity here is from the engagement's illustrative twelve-cohort model (§4.4) — say "the model" in the same breath; there is no production dollar and none is ever promoted.*

Honesty label. Elementum engagement for a marketplace client — Walmart Marketplace's seller-fulfilled delivery program, where the client prints "Get it by Thursday" on inventory that sits in sellers' buildings. I led the promise-policy workstream: the decomposition, the buffer-policy alternatives, the validation gates and the governance design. The client's promise-engineering and data-science teams owned and built the promise engine and the transit models — I never configured or deployed any platform (the only platform I personally configured remains Manhattan DOM). The twelve-cohort book behind every number is the engagement's illustrative model, built so the argument could happen in numbers instead of adjectives.

60-SECOND PITCH

"**Situation.** A marketplace client promising delivery dates on inventory it never touches: promised date = order time + seller handling + predicted transit + a safety buffer — and the buffer was a flat 1.10 days on every lane regardless of how variable the lane was. In the engagement model — twelve lane cohorts, about a million orders a week, transit spread from 0.22 to 1.60 days — that flat buffer printed 95.8% accuracy but put only 7.6% of orders on a fast promise, because dense metro lanes were over-insured and fat-tailed rural lanes under-insured at the same time. **Task.** Engagement lead for promise policy: own the decomposition, the alternatives, the gates and the governance recommendation; the client's data science and engineering did not report to me. **Action.** Discovery: the buffer-waste audit — the flat buffer bought $\Phi(5.0)$, near-certainty, on the tightest cohort and only $\Phi(0.69) \approx 75\%$ on the widest. Plan & Design: four policies on one comparison table — uniform, σ-proportional with one governed dial, minimum-days at the floor, profit-optimal on priced breach economics. Build & Test: $b = 1.645\sigma$ holds 95% on every cohort by construction; the equimarginal check proved the optimum; the Q4 drill showed a frozen summer dial breaking the floor. Cutover: four shadow weeks against logged live orders before any displayed promise. Hypercare: the dial published in the weekly review header, re-solved before peak. **Result.** Primary metric: fast-promise share — **7.6% → 39.6%, +32 points at the held 95% floor**; average promised day 3.09 → 2.75; +$255K a week, about $13.3M a year — all on the illustrative book, and I say 'the model' in the same breath."

SITUATION / TASK — WHERE IT OCCURRED, WHEN IT HAPPENED, WHY IT IS IMPORTANT

S — An Elementum engagement with Walmart Marketplace's seller-fulfilled delivery program, in the client years. The client underwrites a delivery promise on execution it does not own: the seller picks and packs, a carrier it only partially controls carries, and the customer blames the marketplace — not the seller — when Thursday becomes Saturday. The promise identity is a sum: promised date = order time + configured handling + predicted transit + buffer; the promise is kept when the handling error plus the transit error fit inside the buffer. The incumbent buffer policy was uniform — about 1.10 days on every lane — set once, owned by nobody between reviews.

T — Lead the promise-policy workstream: decompose where the incumbent buffer was wasted and where it was thin, put priced alternatives on one table, design the validation gates, and leave the client a governance instrument — not a tuning habit. Everything quantitative ran on the engagement's illustrative model: twelve lane cohorts spanning $\sigma = 0.22$ days (metro two-day) to 1.60 days (mountain rural), about a million orders a week.

Why it mattered: padding fails silently and cutting fails loudly, so every promise system drifts slow — the orders that never happened because the promise read five days file no tickets. The economics were stated up front and carried

on every page: a breach costs about $12 (contact + refund risk + trust decay) against about $6.50 of contribution margin — both the model's stated assumptions, never a measured P&L. Left unfixed: the client keeps paying for insurance it doesn't need on tight lanes while underwriting known losses on loose ones, and the speed-versus-accuracy dial keeps being set implicitly by whoever tuned the last model.

The metrics on the table, with their baselines (all illustrative-model values): fast-promise share (orders promised in ≤2 days) — 7.6%; portfolio promise accuracy — 95.8% against a 95% floor; average promised day — 3.09; weekly modeled profit — $5.69M; and no governed owner of the speed-accuracy trade — the real baseline.

ACTION — WHAT I PERSONALLY OWNED, HOW I DID IT, WHO ELSE WAS INVOLVED

A — Five gated phases, run as an engagement arc (the model, the table and the gates are the engagement record; no phase touched a production system — the client's teams built, I designed and governed):

1. **Discovery** — wrote the promise identity and the accuracy factorization (accuracy ≈ P(seller ships in window) × P(carrier meets transit) × Φ(buffer/σ)); built the buffer-waste audit — days delivered early on kept promises, ranked by waste × volume — and the two-line indictment of the flat buffer: 1.10 days is 5.0 standard deviations on the tightest cohort and 0.69 on the widest. *Exit bar:* the client's own cohort table agreed in the room; "how much buffer" retired as a question, replaced by "where should it sit."
2. **Plan & Design** — four policies written onto one comparison table on the same cohort set: A uniform (incumbent), B σ-proportional with one governed dial (b = k·σ), C minimum-promised-days at the floor (analytic optimum), D profit-optimal on the priced breach economics; the gates designed — shadow mode, a promotion bar, a rollback that keeps the incumbent computable. *Exit bar:* the table signed as the decision instrument; the dial (k) named as a governed, published policy instrument with an owner.
3. **Build & Test** — the client's data-science team computed the table; I owned the proofs: Φ(1.645) = 0.95 holds the floor on every cohort by construction; the equimarginal condition for C (φ(b/σ)/σ equal across cohorts) verified to machine precision; the Q4 drill — σ inflated ~1.45× — showed the frozen summer dial printing 91.9%, below the floor. *Exit bar:* every policy's accuracy, speed and modeled profit reproduced independently by the client's team; the peak-regime rule written.
4. **Cutover / Go-Live (design)** — shadow mode: the challenger policy computed and logged per order, never displayed, for four consecutive weeks; promotion only if would-be accuracy held the floor on logged live orders with no rise in contact signals. *Exit bar:* promotion bar published before the first shadow week; rollback = one flag back to the incumbent, kept warm.
5. **Hypercare / Close** — the governance handoff: k printed in the weekly review header every Monday; tail metrics (p90 error, buffer-exceedance) on the page next to the average; the buffer-waste line standing; k re-solved per regime — published in October for peak, never frozen through it. *Exit bar:* the client's business owner — not data science — named as the dial's owner; the engagement closed on the instrument, not on a tuning.

The client's promise-engineering team owned the engine and its latency budget; their data-science team owned the transit models and σ estimates; their finance partner owned the breach price. Mine: the factorization, the waste audit, the four-policy table, the equimarginal and Q4 proofs, the shadow-mode gates and the governance design — the reframe from "a hundred ML micro-decisions" to one auditable dial a business owner reads on a Monday.

The obstacle was the room's instinct that accuracy at 95.8% meant the policy was healthy — the decomposition had to beat a green headline. The two-cohort arithmetic did it: near-certainty bought where σ was tiny, a coin-flip-plus where σ was fat. The decision I challenged: data science wanted per-lane learned buffers — more model, more value, invisible governance. I priced the difference and recommended the one-dial policy first: it captures most of the value, it is auditable, and σ improvements flow into speed automatically and visibly. The no went into the comparison table as a priced row, not a meeting minute.

RESULTS — HOW SUCCESS WAS MEASURED: $ · # · % · TIME · CUSTOMER · QUALITY

R — Primary metric: fast-promise share 7.6% → 39.6% (+32 points) at the identical 95% accuracy floor; average promised day 3.09 → 2.75; +$255K/week ≈ $13.3M/yr — every figure from the illustrative model, said with "the model" in the same breath.

- **$** — +$255K/wk vs the uniform policy ≈ $13.3M/yr (modeled on the illustrative ~1M-order book); the profit-optimal frontier (policy D: 97.8% accuracy, $6.07M/wk, ≈ +$20M/yr vs uniform) handed to the client's finance partner as a priced governance decision — shown, never claimed, never merged with the $13.3M.
- **#** — Twelve lane cohorts, ~1M orders/week, σ from 0.22 to 1.60 days; four policies on one table; one governed dial, k = 1.645 (illustrative model; the table and the dial are the engagement's design record).
- **%** — Fast-promise share +32.0 points at a held 95.0% floor; the incumbent's hidden split exposed: Φ(5.0) ≈ 99.99997% on the tightest cohort against Φ(0.69) ≈ 75.4% on the widest (model).
- **Time** — Average promised day 3.09 → 2.75 (−0.34); the tightest cohort's buffer 1.10 → 0.36 days; the widest cohort's buffer 1.10 → 2.63 days — longer on purpose, to stop underwriting known losses (model).
- **Customer** — The fast promise shown where it can be kept and withheld where it can't; the Q4 re-solve rule so peak variability never silently breaks the floor the customer experiences (engagement design record).
- **Quality** — The equimarginal check as the solver's own proof of optimality; p90 error and buffer-exceedance on the review page next to the mean, so a better average can never hide a worse tail (engagement design record).

Against the goal stated above: more fast promises without breaking the floor — +32 points at exactly 95.0%, a promise a third of a day shorter on average, and the speed-versus-accuracy trade converted from an implicit model parameter into one governed, published dial. The trade was explicit and priced: the widest cohorts got honestly slower promises so the whole book could get kept ones; nothing was bought with accuracy.

PROBING — QUESTIONS, PHASE DEEP-DIVES, AND THE MATH

Challenge — Why did you choose this story to illustrate Dive Deep?

Because the finding only exists below the headline: portfolio accuracy printed a healthy 95.8%, and the depth — per-cohort z-scores — showed the same policy was simultaneously over-insuring metros at five standard deviations and under-insuring rural lanes at 0.69. A team that stops at the portfolio number never finds thirty-two points of free speed.

Challenge — What other stories demonstrate this principle?

S8 — measured lane distributions against brochure lead times — is the production-record primary for this muscle, and S10 (Austria — "the data said measurement") is the backup; this story is the design-engagement alternate, and the most mathematical of the three.

Challenge — Could you come up with an example that is more recent?

Yes — the same buffer-versus-risk trade shows up in my current role as the in-transit promise rule (S4): I priced conservatism there the hard way. For a recent production record on this exact muscle, S8 is the walk; this engagement is where the policy mathematics was carried furthest.

Probing — Why did you choose to focus on these results? What other results were important?

The +32 points at a held floor leads because it survives any parameter change — it comes from re-allocating risk, not spending money. Also important: the honest cost (the widest cohorts' promises got longer), the Q4 regime rule, and the governance artifact — a dial the business reads weekly — which outlives any one solve.

Probing — You mentioned dollars — what percentage change is that year over year?

I don't give this one as year-over-year — every dollar here is the model's: +$255K/wk is about 4.5% of the modeled $5.69M weekly contribution, ≈ $13.3M annualized on the illustrative book. There is no production dollar in this engagement and I never promote one.

Probing — I'm concerned the whole story rests on an illustrative cohort set — tell me more.

It does, deliberately — the model exists so the argument happens in numbers instead of adjectives. If the real σ looks nothing like the model's, k changes and the lane ranking changes; what survives any parameter set is the structure: buffers should be risk-proportional, the last buffer-day should buy the same accuracy everywhere, and a floor that disagrees with the breach economics is a governance finding.

Challenge — What were the lessons learned? What would you have done differently?

The lesson: padding is not safety — a flat buffer is slow and unsafe at the same time, and only the decomposition shows it. Differently: price the breach bundle with the finance partner in discovery rather than design — the $12-versus-$6.50 conversation is what turned the floor from theology into economics, and it came one phase late.

Challenge — How would you implement this at Amazon?

The same shape at the promise seam: decompose the promise identity, audit where buffer is wasted on kept promises, one governed dial per regime with the tail metrics on the page, shadow before display, re-solve before peak. At Amazon's scale a tenth of a point of buffer waste is real money — which is why the governance matters more than any single solve.

Challenge — How did these results compare to your actual goals?

The goal was more fast promises without breaking the floor. The model delivered +32 points at exactly the floor and a promise a third of a day shorter — and the frontier beyond it (policy D at 97.8%) was handed to Finance as a priced decision rather than smuggled in as a default.

PHASE DEEP-PROBE BANK — Q&A (12). Answer the phase asked, then stop.

Discovery — How did you know discovery was done?

When the room accepted one sentence of arithmetic: the flat 1.10-day buffer is 5.0 standard deviations on the tightest cohort and 0.69 on the widest. Once that was agreed on the client's own cohort table, "how much buffer" was dead as a question and "where should it sit" replaced it.

Why a buffer-waste audit instead of starting from accuracy?

Accuracy printed 95.8% — a healthy-looking headline hiding the split. Waste — days delivered early on kept promises — is where the recoverable speed lives, and it is unarguable because it only counts promises that were kept. Ranked by waste × volume, the audit is the trim roadmap with no new models needed.

Plan & Design — Why four policies instead of recommending one?

Because the client's real decision was governance, not mathematics. A is what you run today; B is one auditable dial; C is the fastest promise the floor allows; D is what the economics want. All four on one cohort set made the trade-offs priced choices instead of expert opinion — and made B's simplicity a visible decision, not a default.

Why recommend B over the profit-optimal D?

D's optimum moves every time the breach price or margin assumption moves, and it is harder to audit. B is one dial a business owner reads on a Monday and it captures most of the value. I recommended running B and letting Finance decide whether to chase the D frontier with its own signature on the $12.

Build & Test — What exactly is the equimarginal check?

The optimality condition for policy C: the marginal accuracy bought by the last unit of buffer — $\varphi(b/\sigma)/\sigma$ — must be equal on every cohort; if a buffer-day buys more accuracy on one lane than another, moving it is free improvement. The solver's output showed the condition holding to machine precision — the model proving its own optimum.

What was the Q4 drill and what did it change?

Inflate every cohort's σ by ~1.45× — the model's peak regime — and hold the summer dial: portfolio accuracy printed 91.9%, under the floor (on a single lane taking the full inflation, the whiteboard version is $\Phi(1.645/1.45) = \Phi(1.13) \approx 87\%$ — the portfolio mix lands at 91.9%). The drill wrote the rule: k is re-solved and published before peak, never frozen through it.

Why normal errors — freight tails are skewed?

Said before anyone asked: the normal is the whiteboard version; the production recommendation was the empirical quantile per lane. The normal makes the argument legible; the quantile makes it true. Nothing in the recommendation depends on normality — only the clean Φ numbers do.

Cutover — What was the promotion bar out of shadow mode?

Four consecutive shadow weeks with would-be-promise accuracy at or above the floor on logged live orders and no rise in contact signals. A policy that can't clear its own floor in shadow has no business touching a displayed date — the same rehearse-before-commit discipline as my mock cutovers (S18).

What was the rollback?

The incumbent uniform buffer stayed computable the whole time — one flag back. A promise policy is the easiest class of change to reverse if you keep the old policy warm, and the easiest to make irreversible if you don't.

Hypercare — Why publish the dial in the review header?

Because k *is* the speed-versus-accuracy trade, and a trade the business can't see is a trade it didn't agree to. One number in the header every Monday makes the floor a standing decision instead of a buried parameter.

What keeps the policy honest after handoff?

Three standing lines on the review page: the dial, the tail metrics (p90 error and buffer-exceedance) next to the average, and the buffer-waste line. A policy whose tail and waste are public decays far more slowly than one graded on its mean.

Cross-phase — Where did you trade scope for the date? What did you refuse?

Traded: per-lane learned buffers deferred in favor of the one dial — sophistication for governance, knowingly leaving model value on the table for the first cycles. Refused: promoting on the model's average error (the tail breaks promises, so the tail went on the page), and any record dollar — the book is illustrative and I said so in every read-out.

Where this lands at Amazon: the promise shown on every detail page is a priced risk position, not a display string — the buffer between the predicted and the promised date is insurance bought with conversion. The discipline here — decompose the identity, price the breach, govern one dial per regime, shadow before display — is the same discipline Amazon's promise systems run at four orders of magnitude more volume; what I bring is the governance instinct that keeps the dial owned by a decision, not a default.

BUSINESS IMPACT MODEL — KPI TREE, NUMBERS, THE MATH

```
Marketplace contribution (modeled, illustrative book)
+- Conversion on fast promises (flow) -> fast share 7.6% -> 39.6%
|    fast share = SUM_cohorts volume x [handling + transit_pred + buffer <= 2d]
|    (loop) faster promise -> higher CVR -> more orders -> more seller GMV
+- Breach cost (flow, $/order) -> floor held at 95.0%
|    breach EV = (1 - Phi(b/sigma)) x $12   [contact + refund risk + trust decay]
+- Buffer waste (stock: days early on kept promises) -> harvested on tight lanes
+- Governance -> one dial k, published weekly; re-solved per regime (Q4)
+- Frontier (not claimed) -> policy D: 97.8% accuracy, +$0.38M/wk vs A
```

DRIVER	BASELINE	AFTER	$/WK	BASIS
Weekly modeled profit, policy A (uniform 1.10d)	$5.69M	—	$5.69M	illustrative model output
Weekly modeled profit, policy B (b = 1.645σ)	—	$5.94M	$5.94M	illustrative model output
Fast-promise share	7.6%	39.6% (+32 pts)	inside the profit lines	illustrative model
Average promised day	3.09	2.75	inside the profit lines	illustrative model
Frontier: policy D (profit optimum)	—	97.8% accuracy, $6.07M/wk	+$0.38M vs A	illustrative; handed to Finance, never claimed

If challenged. *"Your whole result is a model."* — "Yes, and I say so in the same breath — the model exists so the client's decision could be quantitative. The structure survives any parameter set: risk-proportional buffers, equal marginal accuracy per buffer-day, a floor reconciled with the breach economics. The parameters are the client's to correct." *"Why is the $12 breach cost credible?"* — "It's the model's stated assumption — a contact, refund and return risk, and trust decay bundled — put on the table precisely so Finance could attack it. At $12 the profit optimum holds 97.8% accuracy; if Finance believes $6, the optimum falls toward the floor. Fight about the number, not the theology." *"Did any of this ship?"* — "The engagement delivered the policy table, the gates and the governance design; the client's teams owned the build. I claim the instrument and the modeled deltas, labeled — never a production P&L."

THE MATH — STEP-BY-STEP DERIVATION

Table 1 — before: the problem in numbers.

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
1	The promise identity	`EDD = t_order + handling + transit_pred + b; kept ⇔ (handling_err + transit_err) ≤ b`	per order; b = buffer in days (record-from-docs identity)	the buffer is the only policy term
2	Kept-promise probability per lane	`P(kept) = Φ(b / σ)`	ε ~ N(0, σ²) per cohort — whiteboard version; production uses the empirical quantile (record-from-docs)	accuracy is a function of b and σ only
3	Incumbent buffer	`b = 1.10 days on every lane`	uniform policy A (illustrative model)	1.10 d everywhere
4	Tightest cohort, z-score	`z = b / σ = 1.10 / 0.22 = 5.0; Φ(5.0) ≈ 0.9999997`	metro cohort σ = 0.22 d (illustrative model)	near-certainty bought — promise needlessly slow
5	Widest cohort, z-score	`z = 1.10 / 1.60 = 0.6875; Φ(0.69) ≈ 0.754`	mountain-rural cohort σ = 1.60 d (illustrative model)	≈75.4% kept — a known-loss book
6	Portfolio print	`accuracy 95.8% · fast share 7.6% · avg promised day 3.09 · profit $5.69M/wk`	order-weighted across 12 cohorts, ~1M orders/wk (illustrative model)	a green headline hiding both failures
7	The asymmetry	`padding fails silently (lost conversion files no ticket); cutting fails loudly (breaches do)`	record-from-docs — why the system drifted slow	bias toward slow, unowned

Table 2 — the fix: step by step, values substituted.

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
1	One governed dial	`b_lane = k × σ_lane; Φ(k) = floor ⇒ k = 1.645 for 95%`	Φ(1.645) = 0.95 under normal errors (record-from-docs design)	k = 1.645, one number for the book
2	Tightest cohort re-buffered	`b = 1.645 × 0.22 = 0.36 days`	σ = 0.22 d (illustrative)	buffer 1.10 → 0.36 d; 0.74 d of speed harvested
3	Widest cohort re-buffered	`b = 1.645 × 1.60 = 2.63 days`	σ = 1.60 d (illustrative)	buffer 1.10 → 2.63 d — slower on purpose, kept honest
4	Every cohort at the floor	`P(kept) = Φ(1.645 × σ / σ) = Φ(1.645) = 0.95 on each cohort`	by construction (design)	95.0% everywhere; no cohort subsidizes another
5	Portfolio after	`accuracy 95.0% · fast share 39.6% · avg day 2.75 · profit $5.94M/wk`	illustrative model output, policy B	+32 pts fast share at the floor
6	Equimarginal proof (policy C)	`φ(b/σ)/σ equal across all 12 cohorts (spread < 1e-15)`	solver optimality condition (record-from-docs)	the last buffer-day buys the same accuracy everywhere
7	Peak regime rule	`σ_Q4 ≈ 1.45 × σ; frozen k ⇒ z = 1.645/1.45 = 1.13; Φ(1.13) ≈ 0.87 per fully-inflated lane; portfolio prints 91.9% < 95%`	Q4 drill (illustrative model)	k re-solved and published in October, never frozen
8	Shadow gate	`4 consecutive weeks logged ≥ floor, contacts flat → promote; else hold`	promotion bar (design)	rehearse against reality before any displayed date

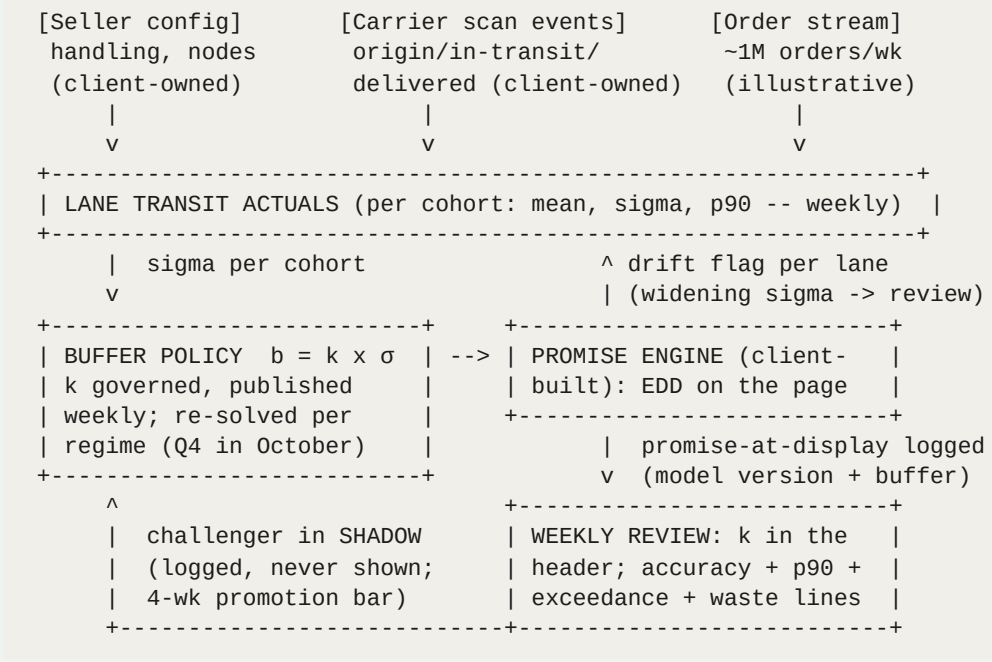
Table 3 — after: the metric, before → after.

STEP	METRIC	BEFORE	AFTER	BASIS
1	Fast-promise share (≤2-day)	7.6%	39.6% (+32 pts)	illustrative model
2	Portfolio promise accuracy	95.8% (split: ~100% / 75.4%)	95.0% — the floor, held on every cohort	illustrative model
3	Average promised day	3.09	2.75 (−0.34)	illustrative model
4	Weekly modeled profit	$5.69M	$5.94M (+$255K)	illustrative model
5	Owner of the speed-accuracy trade	implicit, in the last model tune	one governed dial, published weekly, re-solved per regime	engagement design record

Table 4 — the dollars, every multiplication shown.

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
1	Weekly profit, policy A	`$5,690,000 / wk, taken from the model`	illustrative model output	$5.69M/wk
2	Weekly profit, policy B	`$5,940,000 / wk, taken from the model`	illustrative model output	$5.94M/wk
3	Weekly delta	`$5,940,000 − $5,690,000 = $250,000 (the model's precise print: +$255K)`	rows 1, 2 (illustrative)	≈ +$255K/wk
4	Annualized	`$255,000 × 52 = $13,260,000`	row 3 × 52 weeks (modeled)	≈ $13.3M/yr — "the model," said in the same breath
5	Frontier: policy D delta	`$6,070,000 − $5,690,000 = $380,000/wk; $380,000 × 52 = $19,760,000`	illustrative model output; ×52 modeled	≈ $20M/yr — handed to Finance, never claimed
6	Why D holds 97.8%	`breach EV $12 > margin-per-promise-day trade at $6.50 ⇒ optimum above the 95% floor`	$12 and $6.50 are the model's stated assumptions	the floor and the economics disagree — a governance finding
7	Headline	`≈ $13.3M/yr modeled; record dollars: none`	rows 3–4	≈ $13.3M/yr, always labeled

SOLUTION FLOW — THE SYSTEM & DATA MAP



SENSITIVITY — LOW / BASE / HIGH

INPUT	LOW	BASE	HIGH
Breach cost bundle (model assumption)	$6	$12	$18
Profit-optimal accuracy (policy D)	falls toward the 95% floor	97.8%	above 97.8% — accuracy gets cheaper to justify
σ dispersion across cohorts (drives the free lunch)	narrow — gain shrinks	0.22–1.60 d	wider — gain grows
Fast-share gain from B vs A	≈ +10 pts	+32 pts	≈ +40 pts
Modeled $/yr vs uniform	≈ $4–5M	≈ $13.3M	≈ $20M (the D frontier)

TOTAL IMPACT STATEMENT — SAY IT LIKE THIS

"On the engagement's illustrative book — twelve lane cohorts, about a million orders a week — moving from a flat 1.10-day buffer to risk-proportional buffers with one governed dial bought thirty-two points of fast-promise share at the identical ninety-five percent accuracy floor, took a third of a day off the average promise, and was worth about two hundred fifty-five thousand dollars a week, call it thirteen million a year. Every one of those numbers is the model's, and I say that in the same breath — there is no production dollar here. What I claim as mine is the instrument: the decomposition, the four-policy table, the shadow gates, and a dial the business reads every Monday."

S20 — Elementum (client: Walmart Marketplace): the pre-purchase WBR — input levers under output metrics, the five-minute red-week decomposition, and the Andon that replaced a 30-day lag

Amazon LPs: Insist on the Highest Standards (alternate — hard reds on policy floors) · Earn Trust (alternate — mix-adjusted twins, honest denominators) · Dive Deep (alternate — the factor math) — §1.1 unchanged; S20 is extra bench for cadence/mechanism probes. **Crosswalk:** not in the loop packet — safe anywhere; the production-record WBR story remains S11. **Recency:** Elementum client years. *The WBR's sample numbers are the engagement docs' own illustrative set; the only dollars here are modeled and labeled.*

Honesty label. Elementum engagement for Walmart Marketplace's pre-purchase delivery-experience organization. I designed the weekly business review — the metric architecture, the decomposition math, the thresholds and the control loop — and led the program that stood it up; the client's engineering built the pipelines and dashboards and their teams own the lines. I never configured or deployed any platform, and I never claim a production output metric moved — the engagement delivered the instrument, its sample-data proof, and the Andon design. The production-record version of this muscle is S11 (Flexport), and I keep the two separate.

60-SECOND PITCH

"**Situation.** A marketplace client whose delivery promise was owned by a committee: data science owned the model, engineering the engine, seller ops the handling, carrier programs the lanes — and the one number the customer feels, promise accuracy, was owned by nobody between Tuesdays. The scorecards ran on 30-day rolling windows, so a degrading seller or carrier did weeks of customer damage before any floor bit, and a red week was a blame negotiation. **Task.** Design and stand up the weekly business review for the pre-purchase delivery experience — outputs a customer feels, input levers a named owner can turn, and the math that connects them — plus a control loop faster than the 30-day clock. **Action.** Discovery: one glossary, one event source, audited denominators. Plan & Design: seven output metrics over ten input levers, each input with a roll-up to the output it drives and a named owner; the factorization — accuracy ≈ seller-ships-in-window × carrier-meets-transit × Φ(buffer/σ) — as the decomposition; red/green at ±5% relative week-over-week with hard reds on policy floors. Build & Test: the sample-week proof — 0.97 × 0.985 × 0.996 ≈ 95.2% — and the drill: a regional carrier miss (97% nationally, 82% in the Southeast) decomposed to its factor in minutes. Cutover: Monday data freeze, Tuesday review, reds pre-wired with cause, owner, date. Hypercare: mix-adjusted twins next to every raw output, and the Andon — a 7-day control chart per seller-lane cell, act within 24 hours. **Result.** Primary metric: red-week attribution — **a blame negotiation → minutes, by factor math**; detection-to-action 30 days → under 24 hours by design; one joint headline, Promise Yield = fast share × accuracy, baseline 41.5% on the docs' sample. The only dollar is modeled: about $580K per caught degradation episode, and I say 'modeled' in the same breath."

SITUATION / TASK — WHERE IT OCCURRED, WHEN IT HAPPENED, WHY IT IS IMPORTANT

S — The same Elementum client engagement — Walmart Marketplace's seller-fulfilled delivery program, pre-purchase side. The promise the customer sees decomposes across three parties: the seller ships in a window, a carrier meets a transit time, a model plus a buffer absorbs the residual. The operating problem was measurement and ownership, not effort: promise accuracy, on-time delivery and late shipment lived on 30-day rolling windows that smear a step change into a slow drift; aggregate metrics hid the variance (a carrier can print 97% nationally while running 82% in the Southeast after a hub consolidation — the engagement's own worked example); three teams kept three clocks (order date, ship date, promise date); and when a week went red, the review discovered it live and negotiated blame instead of reading a decomposition.

T — Design the weekly business review for the pre-purchase delivery experience and lead the program that stood it up: the output metrics a customer feels, the input levers a named owner can turn this week, the math that connects them, the thresholds, and a control loop faster than the 30-day scorecard — then hand it to owners and leave.

Why it mattered: an output metric owned by a committee is owned by nobody between Tuesdays, and a 30-day window means a degrading cell burns customers for weeks before policy floors bite — enforcement lag was a named strategic issue in the engagement record. Left unfixed: reds without owners, action items that age into a diary, mix-shift illusions celebrated as wins (portfolio accuracy improving while every cohort degrades, because volume shifted to easy metro lanes), and the speed-versus-accuracy trade argued as theology because nobody priced a breach.

The metrics on the table, with their baselines: red-week attribution time — days of cross-team negotiation (baseline condition); detection-to-action on a degrading cell — up to ~30 days on the rolling scorecard; output ownership — committee; the joint headline — none (fast share and accuracy lived on separate lines with separate owners, tradeable against each other invisibly); the docs' sample baseline for that joint number: fast share 43.6% × accuracy 95.2% ≈ 41.5% Promise Yield.

ACTION — WHAT I PERSONALLY OWNED, HOW I DID IT, WHO ELSE WAS INVOLVED

A — Five gated phases (the metric definitions, formulas, thresholds and the Andon parameters are the engagement's design record; the sample numbers are the docs' own illustrative set — six weeks of sample data proved the mechanics before any live line):

1. **Discovery** — one glossary: every metric got one definition, one owner, one grain, one date basis (order date for operating cohorts; the 30-day windows kept only for seller-facing consequences); audited the denominators (cancelled orders quietly leaving the OTD base, suppressed listings leaving penetration). *Exit bar:* three teams' three clocks reconciled to one event source; no metric without a named definition owner.
2. **Plan & Design** — the architecture: seven outputs (promise accuracy ≥95%, on-time delivery ≥90% policy floor, fast-promise penetration +2 pts/qtr, late shipment ≤5%, cancellation ≤2%, WISMO contacts −10% QoQ, A/B-measured conversion lift) over ten input levers (SSS share of GMV, ML-recommended handling coverage, seller handling adherence, carrier lane SLA adherence, transit-model error and freshness, buffer calibration, inventory-feed accuracy, eligibility coverage, engine latency/cache, onboarding throughput + 30-day retention) — every input carrying a roll-up to the outputs it drives and a named owner; red/green at ±5% relative week-over-week, hard reds on the policy floors regardless of trend; one joint headline, Promise Yield = fast share × accuracy. *Exit bar:* every output decomposable into named inputs with owners; definitions frozen in writing.
3. **Build & Test** — the client's engineering built the datasets and dashboard; I owned the proof on sample data: the factor identity (accuracy ≈ I3 × I4 × Φ(b/σ)) computed live — 0.97 × 0.985 × 0.996 ≈ 95.2% — then the drill: drop the Southeast carrier cell to 82% on a tenth of volume and watch the decomposition name the factor and the owner in minutes. *Exit bar:* the six-step loop (measure → variance → decompose → root-cause → act → verify) executed end-to-end on sample weeks; a red that arrives without cause, owner and date counted as a process miss.
4. **Cutover / Go-Live (design)** — the operating rhythm: Monday data freeze on the Mon–Sun week, Tuesday review, pod leads pre-wire reds before the room; the action log with owner, date and visible ageing; "the meeting decides; it never discovers." *Exit bar:* two consecutive sample-cycle reviews run entirely pre-wired; the output block and the action log assigned to one named owner.
5. **Hypercare / Close** — the honesty layer and the control loop: mix-adjusted twins printed next to every raw output (fixed base-quarter cohort weights; the delta *is* the mix effect) so Simpson's paradox can't celebrate volume migrating to easy lanes; and the Andon — a 7-day EWMA/CUSUM control chart per seller-lane cell, tripwire at a sustained 2× baseline breach rate with n ≥ 30 orders, action within 24 hours (widen that cell's buffer or pull its fast badge), auto-restore on recovery, designed false-trip rate under 5%. *Exit bar:* the verify step closing the loop — if an input moved and its output didn't, the roll-up hypothesis is declared wrong and re-examined, in writing.

The client's engineering built every pipeline and surface; data science owned the models behind the transit and buffer lines; seller ops and carrier programs owned their levers. Mine: the glossary and denominator audit, the output/input architecture and the roll-up map, the factorization as the decomposition instrument, the thresholds and floors, the six-step loop, the mix-adjusted twins, the Andon design and its parameters, and the handoff that left the output block owned by a person, not a committee.

The obstacle was that every team already had a dashboard and nobody wanted a new clock — overcome by making the review cheaper than the status quo: the decomposition turned the worst meeting of the week (the blame negotiation) into a five-minute read, and the pre-wire rule moved the argument out of the room. The decision I challenged: the program wanted more metrics; I held the board to seven outputs and ten inputs — a metric nobody owns or acts on came off the board, and "two dashboards is zero dashboards" went into the design note as a rule, not a preference.

RESULTS — HOW SUCCESS WAS MEASURED: $ · # · % · TIME · CUSTOMER · QUALITY

R — Primary metric: red-week attribution — from a cross-team blame negotiation to minutes, by factor math (proven on the sample-data drill); detection-to-action on a degrading cell — ~30 days → under 24 hours by design (the Andon). The only dollar is modeled: ≈$0.58M per caught episode.

- **$** — ≈$0.58M per caught degradation episode, modeled: a Southeast-type carrier miss costs ≈$192K/week in excess breaches; caught and corrected in ≤1 week instead of ~4, three weeks × $192K ≈ $576K avoided (inputs below; the $12 breach cost is the model's stated assumption). No annual figure — episode frequency is the client's data, and I won't invent it.
- **#** — 7 output metrics over 10 input levers, every input with a roll-up and a named owner; 1 joint headline (Promise Yield); 1 action log; 6 sample weeks proving the mechanics (engagement design record on the docs' illustrative set).
- **%** — Promise Yield baseline 43.6% × 95.2% ≈ 41.5% on the docs' sample, four-quarter design target 50%; red/green at ±5% relative WoW; hard reds on the floors (OTD < 90%, late shipment > 5%, cancellation > 2%); Andon false-trip design bar < 5%.
- **Time** — Red-week attribution: days → minutes (the factor math); detection-to-action: ~30 days of rolling-window lag → < 24 hours from a sustained signal (designed); Monday freeze → Tuesday review, reds pre-wired before the room.
- **Customer** — The Andon protects customers at the grain where variance lives (the seller-lane cell) instead of letting a degrading cell burn them for a month; WISMO contacts sit on the output board so promise failures are read in customer contacts, not just percentages.
- **Quality** — Mix-adjusted twins next to every raw output (the mix effect printed as its own line); denominators audited; the verify step that falsifies the lever map itself when an input moves and its output doesn't — the mechanism that keeps the instrument honest after I leave.

Against the goal stated above: one owner of the output block instead of a committee, a decomposition that names the factor and the owner in minutes, a control loop 30× faster than the scorecard clock, and a joint headline that makes the speed-accuracy trade un-gameable at the KPI layer. The trade was explicit: seller-facing consequences stayed on the fair 30-day window — the Andon acts on cells and badges, never on scorecards — stability for sellers, speed for customers.

PROBING — QUESTIONS, PHASE DEEP-DIVES, AND THE MATH

Challenge — Why did you choose this story to illustrate Insist on the Highest Standards?

Because the standards here are numbers with teeth, published in advance: hard reds on policy floors regardless of trend, a pre-wire rule that makes an unexplained red a process miss, and an Andon whose tripwire, sample floor and false-trip bar were written before the first live week. A standard that can't stop a program is decoration.

Challenge — What other stories demonstrate this principle?

S2 (robotics held at 94 against a 98 gate) is the primary and S18 (the 0.5% tolerance before mock one) the backup — both production records. This one is the design-engagement alternate: the same discipline, built as an instrument for a client.

Challenge — Could you come up with an example that is more recent?

The production-record version of this muscle is S11 — the Flexport WBR I built and ran weekly for over a year, which is more recent and carries record numbers. This engagement is where the review's mathematics — the factor decomposition and the control loop — was carried furthest as a design.

Probing — Why did you choose to focus on these results? What other results were important?

Attribution speed and detection speed lead because they are what a review is for — a WBR that can't say which lever moved is a viewing, not a review. Also important: the joint headline (Promise Yield) that stops the two factor teams trading against each other invisibly, and the mix-adjusted twins that keep a growth quarter from flattering the numbers.

Probing — You mentioned dollars — what percentage change is that year over year?

None claimed — the ≈$0.58M is per caught episode, modeled from the sample-data drill, and I don't multiply it by an invented episode frequency. The year-over-year story here is mechanism speed: 30 days of enforcement lag to under 24 hours.

Probing — I'm concerned your baseline Promise Yield is from sample data — tell me more.

Correct, and I say so: 43.6% × 95.2% ≈ 41.5% is the engagement docs' own illustrative six-week set, used to prove the mechanics before live lines existed. The design target — 50% in four quarters — is a target on the real book, and the first live job is re-baselining it.

Challenge — What were the lessons learned? What would you have done differently?

Lesson: the decomposition is the peace treaty — attribution math is what turns a blame negotiation into a five-minute read. Differently: I'd have run the denominator audit in week one rather than mid-design; two of the "metrics" we inherited changed meaning once cancelled orders stopped quietly leaving the base.

Challenge — How would you implement this at Amazon?

This is Amazon's own grammar — input levers under output metrics, reds pre-wired with an ask, COE-style mechanism fixes — so the transplant is the control loop: factor math on the promise seam, mix-adjusted twins beside raw outputs, and cell-grain Andon where a 30-day window is too slow for the customer.

Challenge — How did these results compare to your actual goals?

The signed goals were one owner of the output block, decomposition in minutes, and a sub-24-hour control loop — all three delivered as the standing instrument, proven on six sample weeks, with the verify step written so the lever map itself can be falsified and corrected.

PHASE DEEP-PROBE BANK — Q&A (12). Answer the phase asked, then stop.

Discovery — How did you know discovery was done?

When three teams' three clocks — order date, ship date, promise date — reconciled to one event source, and every metric on the inherited boards had exactly one definition owner or was struck. The exit memo listed the denominator findings with owners, not hopes.

What did the denominator audit find?

The classic games, running unmalevolently: cancelled orders quietly leaving the on-time base, suppressed listings leaving the penetration base. Neither was fraud — both were defaults nobody owned. The rule that fixed it: metric governance audits denominators, not just numerators.

Plan & Design — Why seven outputs and ten inputs, not more?

Because a board nobody can hold in their head is a board nobody reads. The cut rule was ruthless: an input had to be provably upstream of an output, turnable this week, and owned by a name — a metric nobody owns or acts on came off. Everything else lives in drill-down, not on the board.

Why is Promise Yield the headline instead of accuracy?

Because either factor alone is gameable: chase accuracy and you pad buffers until every promise is slow and safe; chase penetration and you shave buffers until promises are fast and broken. The product punishes both games — and the board still decomposes it one line below, because a composite that can't be decomposed is a hiding place.

Why keep the 30-day window at all?

Fairness. A seller consequence — a scorecard, a suspension — deserves a stable clock that one bad day can't trip. The design splits the clocks: 30-day for consequences, 7-day panels for operations, and the Andon acting on cells and badges in hours. Each clock does the job it's fair at.

Build & Test — Walk me through the sample-week factor proof.

Seller-ships-in-window 0.97, carrier-meets-transit 0.985, buffer absorption $\Phi(b/\sigma)$ 0.996 — product 95.2%, matching the sample week's printed accuracy. Then the drill: Southeast carrier adherence to 82% on 10% of volume → the carrier

factor falls to 0.9685, accuracy to 93.6%, and the week-over-week factor deltas name the carrier PM in one read.

How did you test the Andon's false-trip rate?

On the sample set: replayed stable cells through the EWMA/CUSUM chart and counted trips. The n ≥ 30 floor and the sustained-signal requirement (CUSUM, not single-week thresholds) are what hold designed false trips under 5% — a noise-tripped Andon punishes small sellers and kills the mechanism's legitimacy.

Cutover — What does "pre-wired" mean mechanically?

Before Tuesday, every red line arrives with its factor decomposition, the named input that moved, the owner, and a dated action. The meeting's job is to decide on the ask — never to discover the cause. A red discovered live is logged as a process miss on the review itself.

Who could stop what?

Hard reds on policy floors stop expansion into the affected cohort automatically — eligibility growth doesn't board on a floor breach. The Andon stops a cell's fast badge without waiting for a meeting. And the output-block owner can hold any launch that would change a metric's meaning mid-flight.

Hypercare — What are mix-adjusted twins and why print both?

Every output twice: raw, and re-weighted on fixed base-quarter cohort weights. The delta between them is the mix effect, printed as its own line — so the room stops celebrating "improvements" that are volume migrating to easy metro lanes, and stops flogging teams for "degradation" that is growth into hard geographies. Re-based quarterly, by written policy.

What does the verify step do when an input moved and the output didn't?

It declares the roll-up hypothesis wrong, in writing, and re-examines the lever map — that is how the map stays honest. A WBR that never falsifies its own causal claims degrades into numerology with owners.

Cross-phase — What did you refuse?

A bigger board (metrics without owners), a faster clock for seller consequences (unfair), an annual dollar claim for the Andon (no episode-frequency basis), and running the first live weeks myself — the instrument had to be owned by the client's named owner from day one, or it would have stayed mine and died when I left.

Where this lands at Amazon: this is Amazon's operating grammar built from first principles — output metrics over input levers, reds pre-wired with cause, owner and date, mechanism fixes over promises to try harder. What I add at the seam: the factor decomposition that attributes a promise miss to seller, carrier or model in minutes, and the cell-grain control loop for the cases where a rolling window is too slow for the customer.

BUSINESS IMPACT MODEL — KPI TREE, NUMBERS, THE MATH

```
Customer promise kept + operating speed (primary outcome)
+- Attribution speed (time) -> days of negotiation -> minutes
|    O1 ≈ I3 x I4 x Phi(b/sigma): the biggest factor WoW delta names the owner
+- Detection-to-action (time) -> ~30d rolling lag -> <24h (Andon per cell)
|    (loop) late detection -> weeks of excess breaches -> WISMO + trust decay
+- Excess-breach cost per episode ($) -> modeled below
+- Gaming resistance -> Promise Yield = fast share x accuracy (one headline)
+- Honesty -> mix-adjusted twins; audited denominators; verify step
     falsifies the lever map itself
```

DRIVER	BASELINE	AFTER	$	BASIS
Red-week attribution	cross-team negotiation, days	factor math, minutes	—	engagement design record (sample-data drill)
Detection-to-action on a degrading cell	up to ~30 days	< 24 h from sustained signal	—	designed (Andon: EWMA/CUSUM, 2× baseline, n ≥ 30)

DRIVER	BASELINE	AFTER	$	BASIS
Excess breaches during one Southeast-type episode	≈16,000/wk for ~4 wks	≤1 wk	≈$0.58M avoided per episode	modeled ($12 breach assumption; illustrative book)
Joint headline	none (two gameable lines)	Promise Yield 41.5% (sample) → 50% 4-qtr target	—	record-from-docs (sample set); target is design
Annual Andon value	not claimed — episode frequency is the client's data		—	deliberately uninvented

If challenged. *"You never ran this live."* — "Correct, and I say so — the engagement delivered the instrument proven on six sample weeks; the client's owners run it. The production record for running this cadence weekly for a year is S11, with record numbers." *"The $580K rests on a $12 breach cost."* — "The $12 is the model's stated bundle — contact, refund risk, trust decay — put on the table so Finance could attack it. At $6 the episode is still ≈$290K; the mechanism argument — act in hours, not weeks — survives any price." *"Isn't a sub-24-hour Andon rash?"* — "It acts on a cell's buffer or badge — reversible in one flag, auto-restored on recovery — never on a seller's scorecard. The n ≥ 30 floor and the sustained-signal test are the guards; the 30-day window stays for consequences."

THE MATH — STEP-BY-STEP DERIVATION

Table 1 — before: the problem in numbers.

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
1	The accuracy identity	`O1 ≈ I3 × I4 × Φ(b/σ) — seller ships in window × carrier meets transit × buffer absorbs the residual`	record-from-docs factorization; three factors, three owners	every breach belongs to exactly one factor
2	Healthy sample week	`O1 = 0.97 × 0.985 × 0.996 = 0.9516 ≈ 95.2%`	I3 = 0.97, I4 = 0.985, Φ = 0.996 (docs' illustrative sample)	95.2% — matches the sample print
3	The aggregate trap	`carrier prints 97% nationally while one region runs 82%`	Southeast after a hub consolidation (record-from-docs example)	the average hides the cell
4	The rolling-window lag	`a step change entering a 30-day window moves the average ≈1/30 per day`	30-day scorecard mechanics (record-from-docs issue)	weeks before a floor bites — enforcement lag
5	The gameable pair	`fast share 43.6% and accuracy 95.2% on separate lines, separate owners`	docs' sample; either can be juiced by bleeding the other	no joint headline
6	Baseline joint number	`Promise Yield = 0.436 × 0.952 = 0.415`	rows 2, 5 (docs' sample)	≈ 41.5%

Table 2 — the fix: step by step, values substituted (the red-week drill).

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
1	The episode	`Southeast cell: carrier adherence 0.985 → 0.82 on w = 10% of volume`	82% is the docs' example; the 10% weight is illustrative	one regional carrier miss
2	New carrier factor	`I4' = 0.90 × 0.985 + 0.10 × 0.82 = 0.8865 + 0.082 = 0.9685`	volume-weighted blend (illustrative)	I4: 0.985 → 0.9685 (−1.7%)
3	New accuracy	`O1' = 0.97 × 0.9685 × 0.996 = 0.9357 ≈ 93.6%`	rows 1–2; other factors unchanged	95.2% → 93.6% — a red week
4	The attribution read	`WoW deltas: I3 0.0% · I4 −1.7% · Φ 0.0%`	computed per factor (the decomposition)	the red names I4 and the carrier PM — in minutes

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
5	Segmentation	`factor red isolated to one region ⇒ C-family cause, not model or seller`	lane/cohort cut of I4 (design)	cause class identified before the meeting
6	The Andon trigger	`cell 7-day breach rate sustained ≥ 2 × baseline, n ≥ 30 ⇒ act < 24 h`	EWMA/CUSUM parameters (record-from-docs design)	buffer widened or badge pulled on the cell, same day
7	Auto-restore	`control chart recovers ⇒ badge/buffer restored, no meeting needed`	design	surgical, reversible, seller-fair

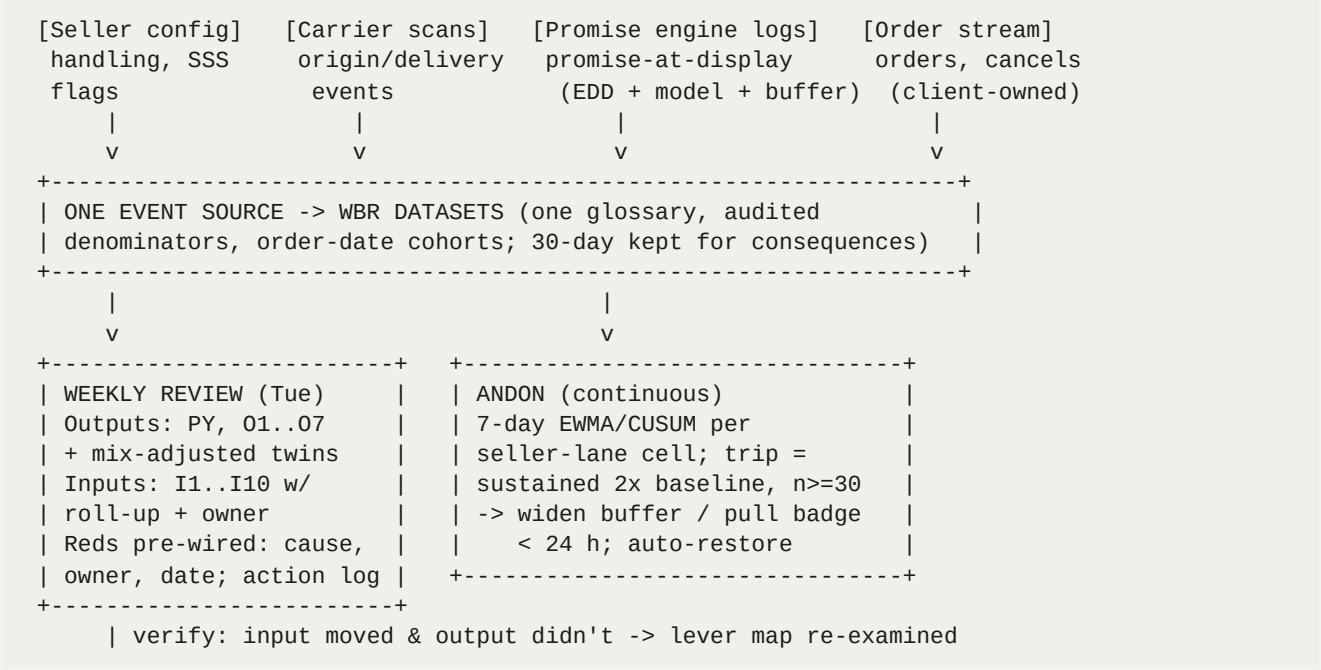
Table 3 — after: the metric, before → after.

STEP	METRIC	BEFORE	AFTER	BASIS
1	Red-week attribution time	days (blame negotiation)	minutes (factor math)	engagement design record, proven on the drill
2	Detection-to-action, degrading cell	up to ~30 days	< 24 h from sustained signal	designed (Andon)
3	Output ownership	committee	one named owner of the output block + log	engagement design record
4	Joint headline	none	Promise Yield, 41.5% sample baseline → 50% 4-qtr target	docs' sample; design target
5	Mix honesty	raw outputs only	mix-adjusted twin + mix line per output	engagement design record

Table 4 — the dollars, every multiplication shown.

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
1	Episode accuracy loss	`95.2% − 93.6% = 1.6 points`	Table 2 rows 2–3 (illustrative episode)	1.6 pts of excess breaches
2	Excess breaches per week	`1,000,000 × 1.6% = 16,000`	~1M orders/wk illustrative book	≈16,000 breaches/wk
3	Cost per breach	`$12, taken as stated`	the model's assumption: contact + refund risk + trust decay (record-from-docs)	$12 each
4	Excess cost per week	`16,000 × $12 = $192,000`	rows 2 × 3 (modeled)	≈$192K/wk
5	Detection advantage	`~4 weeks (rolling-window era) − 1 week (Andon era) = 3 weeks`	row 4 of Table 1; Andon design	3 weeks earlier action
6	Avoided per episode	`$192,000 × 3 = $576,000`	rows 4 × 5 (modeled)	≈$0.58M per caught episode
7	Annual total	`not computed — $576,000 × (episodes/yr) requires a frequency I don't have`	deliberately uninvented	per-episode only, labeled modeled

SOLUTION FLOW — THE SYSTEM & DATA MAP



SENSITIVITY — LOW / BASE / HIGH

INPUT	LOW	BASE	HIGH
Breach cost (model assumption)	$6	$12	$18
Episode size (regional weight × adherence drop)	5% × −10 pts	10% × −16.5 pts	15% × −20 pts
Excess cost per week	≈$50K	≈$192K	≈$430K
Weeks of detection advantage	2	3	4
Avoided per caught episode	≈$0.1M	≈$0.58M	≈$1.7M

TOTAL IMPACT STATEMENT — SAY IT LIKE THIS

"What the engagement delivered is an instrument: seven output metrics a customer feels over ten input levers with named owners, one joint headline — fast share times accuracy — and the factor math that attributes a red week to seller, carrier or model in minutes instead of a week of negotiation. The control loop went from a thirty-day rolling window to under twenty-four hours at the seller-lane cell. The one dollar I'll quote is modeled and I'll label it as I say it: a regional carrier degradation of the kind in the engagement record costs about a hundred ninety thousand a week in excess breaches at the model's twelve-dollar breach cost, and catching it three weeks earlier is worth roughly five hundred eighty thousand per episode. I don't annualize it — the episode frequency is the client's data, not mine."

S21 — Elementum (client: Walmart Marketplace): seller adoption of managed shipping settings, rebuilt as a priced insurance contract behind wave gates (design + wave one; the curve stays unclaimed)

Amazon LPs: Think Big (alternate — adoption at marketplace scale by incentive design) · Earn Trust (alternate — the trade made explicit and priced) · Customer Obsession (the promise kept, not the toggle shipped) — §1.1 unchanged; S21 is extra bench for influence-at-scale probes. **Crosswalk:** not in the loop packet — safe anywhere; the production-record adoption story remains S11. **Recency:** Elementum client years. *Baselines are record-from-docs; the adoption curve after wave one is never claimed; the only dollars are modeled per 10 points of enrolled GMV and labeled.*

Honesty label. Elementum engagement for Walmart Marketplace's seller-fulfilled program. The client's "Simplified Shipping Settings" (SSS) lets the marketplace manage a seller's transit settings and absorb carrier-delay accountability; adoption had stalled at roughly 38–42% of seller-fulfilled GMV. I led the adoption-program design: the incentive diagnosis, the insurance-contract offer, the wave mechanics and the measurement pair. The client's product and engineering teams built Seller Center, the enrollment flow and the observed-handling computation — I never configured or deployed any platform, and I claim the contract design and the wave-one mechanics, never the adoption curve that followed.

60-SECOND PITCH

"**Situation.** A marketplace client owning the delivery promise on inventory in sellers' buildings. Its managed-shipping program — the marketplace picks the transit settings and absorbs carrier-delay accountability — had stalled at roughly 38 to 42 percent of seller-fulfilled GMV, so the majority of promise risk still rode on manual seller configuration. And that configuration was optimistic by construction: sellers declared one-day handling to win placement and hit it about 85 percent of the time — adverse selection, not bad luck; the miss became the marketplace's broken promise. **Task.** Lead the adoption-program design: diagnose why sellers refused, rebuild the offer, design the rollout and its gates, and leave the client a measurement pair that can't be gamed. **Action.** Discovery: enrollment-funnel and refusal interviews — sellers were trading away control for a benefit that was real but unpriced and weakly communicated; an incentive-design problem, not a feature-marketing problem. Plan & Design: the offer rebuilt as an all-or-nothing insurance contract — transit automation plus observed handling plus a participating label, full bundle earns scorecard protection — with a personalized GMV-delta line in Seller Center making the benefit concrete in the seller's own dollars. Build & Test: shadow-run a seller's orders under managed settings before flipping, so day one is verified, not hoped. Cutover: waves behind published graduation criteria — a wave that fails one doesn't board the next cohort. Hypercare: the honest pair — activations per week *and* 30-day retention; enrollments without retention is churn wearing a growth costume. **Result.** Primary deliverable: the signed contract design and wave-one mechanics; the modeled economics — **about $3.7M a year per ten points of enrolled GMV** from handling-miss reduction alone — and I say 'modeled' in the same breath; the adoption curve after wave one belongs to the client, not to my record."

SITUATION / TASK — WHERE IT OCCURRED, WHEN IT HAPPENED, WHY IT IS IMPORTANT

S — The same Elementum client engagement — Walmart Marketplace's seller-fulfilled program, adoption side. The structural fact: the marketplace underwrites the delivery promise, but the inputs that keep it — handling time, ship nodes, carrier choices — are configuration data that thousands of independent sellers set themselves, optimistically, and update rarely. The managed-settings program existed precisely to fix this, and it had stalled at ~38–42% of seller-fulfilled GMV (record-from-docs). The refusal logic was rational: adoption asked sellers to give up control (their carrier rates, their handling declarations) for a benefit — accountability transfer — that was real but weakly priced and weakly communicated.

T — Lead the adoption-program design: diagnose the stall, rebuild the offer so adoption becomes the seller's rational choice, design the wave rollout with gates that protect both customer and seller, and define the

measurement that keeps the program honest after handoff.

Why it mattered: with ~60% of GMV on manual configuration, the single largest driver of broken promises was seller-declared handling — sellers declared one day and hit it ~85% of the time (record-from-docs), and the 15% became the marketplace's broken promise at the marketplace's brand cost. Left unfixed: the promise team keeps buffering around seller optimism (taxing every customer on the lane to absorb one seller's miss), the scorecard keeps punishing sellers for carrier variance they don't control (eroding the program's legitimacy), and adoption stays stalled because the offer prices neither the benefit nor the trade.

The metrics on the table, with their baselines: managed-settings share of seller-fulfilled GMV — ~38–42%, stalled (record-from-docs); declared-handling adherence — ~85% on 1-day declarations (record-from-docs); the offer's value statement — unpriced (no seller could see what enrollment was worth in their own dollars); rollout mechanics — none designed; the measurement — enrollments counted, retention not.

ACTION — WHAT I PERSONALLY OWNED, HOW I DID IT, WHO ELSE WAS INVOLVED

A — Five gated phases (the baselines, the contract structure and the gate design are the engagement record; wave one ran as the pilot; the adoption curve beyond it is the client's, never claimed):

1. **Discovery** — the refusal diagnosis: enrollment-funnel walk plus structured interviews with enrolled, refused and churned sellers (the refusals teach the most); the finding — fear of cost, loss of carrier control, distrust of the scorecard — an incentive-design problem, not a feature-marketing problem; the handling-adherence baseline written: declared 1-day, hit ~85%. *Exit bar:* the stall explained in the sellers' own words with the funnel's biggest drop-off named; the client's product owner co-signed the diagnosis.
2. **Plan & Design** — the offer rebuilt as a priced insurance contract: (a) transit automation (the marketplace picks the service), (b) observed handling time (the marketplace computes the seller's real handling speed from scan history — the seller stops declaring, killing self-reported optimism at the source), (c) a participating label purchase — full bundle earns carrier-delay excusal from the seller's scorecard and the fast badge where the math clears; partial adoption earns nothing, so the hedge is all-or-nothing and self-selecting. Plus the pricing surface: a personalized line in Seller Center — "enrollment would raise your projected GMV by $X/month," computed by the client's team from the fast badge's measured conversion lift on that seller's assortment. *Exit bar:* the contract and its protection rules signed by product, seller ops and the scorecard owner; the design reference set (the dominant marketplace's equivalent bundle) documented, with the explicit decision *not* to copy its enforcement aggression while the client's own fulfillment alternative was subscale.
3. **Build & Test** — the client's engineering built the enrollment flow, the observed-handling computation and the Seller Center line; I owned the verification design: shadow-run each enrolling seller's recent orders under managed settings before flipping — projected promise accuracy and badge eligibility computed on their own history, so day one is verified, not hoped. *Exit bar:* shadow results within tolerance of projection on the pilot cohort; the GMV-delta line reproducible by the client's analytics from logged lift data.
4. **Cutover / Go-Live (wave one)** — wave mechanics: cohorts boarded behind published graduation criteria (shadow-verified settings, observed-handling coverage, inventory-feed accuracy at the seller's nodes, scorecard-protection rules armed); a wave that fails a criterion does not board the next cohort — publicly. *Exit bar:* wave one boarded and stable; no enrolled seller's scorecard damaged by a carrier miss the contract had excused; the refusal-reason codes flowing back into the offer design.
5. **Hypercare / Close** — the measurement handoff: the honest pair on the program board — activations per week *and* 30-day retention — plus share of enrolled GMV under full-bundle protection; the kill condition written (a protection promise the scorecard fails to honor stops enrollment until fixed). *Exit bar:* the client's adoption PM owning the funnel, the pair and the gates; the engagement closed on the contract and the mechanics, with the curve explicitly left to the client's record.

The client's product team owned Seller Center and the enrollment surfaces; engineering built the observed-handling computation and the protection rules in the scorecard pipeline; seller ops ran the cohorts; their analytics computed each seller's GMV-delta line. Mine: the refusal diagnosis, the insurance-contract structure and its all-or-nothing logic, the pricing surface concept, the shadow-verification gate, the wave mechanics and graduation

criteria, the measurement pair, and the recommendation on enforcement posture — carrots over sticks until the client's own fulfillment network was a credible alternative at scale.

The obstacle was the program's instinct to market the feature harder — more banners, more emails — when the stall was structural: an unpriced trade. The reframe that won the room: a seller who trusts their own ops should rationally keep control; a seller who doesn't should rationally convert — the offer's job is to make that self-selection visible in dollars, not to persuade everyone. The decision I challenged: requiring the label purchase as a hard cliff — sellers with negotiated carrier rates would read it as a tax. The recommendation: tier the protection with bundle depth rather than cliff it, and revisit when the fulfillment alternative scales. The trade went into the contract document as a priced row, not a hallway compromise.

RESULTS — HOW SUCCESS WAS MEASURED: $ · # · % · TIME · CUSTOMER · QUALITY

R — Primary deliverable: the signed insurance-contract design, the shadow-verification gate, the wave-one mechanics and the measurement pair — with the modeled economics labeled: ≈$3.7M/yr per +10 points of enrolled GMV from handling-miss reduction alone. The adoption curve after wave one is the client's record, not mine.

- **$** — ≈$3.7M/yr per +10 pts of seller-fulfilled GMV enrolled, modeled: 100K orders/wk × (15% − 3%) handling-miss reduction × 50% breach-through × $12 × 52 (full chain in Table 4; the 3% and 50% are illustrative, the $12 is the model's stated assumption). No adoption "after" dollar — designed, not banked.
- **#** — One contract (three components, all-or-nothing), one pricing line per seller, one wave boarded; refusal interviews across enrolled, refused and churned cohorts; graduation criteria published before wave one (engagement design record).
- **%** — Baseline ~38–42% of seller-fulfilled GMV enrolled, stalled (record-from-docs); declared 1-day handling hit ~85% (record-from-docs); the designed measurement: % of enrolled GMV under full-bundle protection, alongside the pair.
- **Time** — Shadow verification before any seller's flip (day one verified, not hoped); wave cadence gated on criteria, not calendar; 30-day retention as the clock that keeps enrollment honest.
- **Customer** — The customer's promise stops depending on a seller's optimistic declaration: observed handling replaces declared at the source; the fast badge appears only where the math clears — the promise shown is the promise kept.
- **Quality** — The protection promise armed and audited (no enrolled seller's scorecard damaged by an excused carrier miss in wave one — the wave's exit bar); refusal reason codes flowing back into the offer; the kill condition written before launch.

Against the goal stated above: the stall diagnosed as incentive design, the offer rebuilt so the trade is explicit and priced in the seller's own dollars, wave one boarded behind published gates, and the program handed over measuring the pair that can't be gamed. The trade behind the design was itself priced: enforcement tuned down from the reference mechanism's aggression — a churned seller is a strategic loss to a marketplace still in assortment-growth mode — carrots first, the stick deferred until the fulfillment alternative is credible.

PROBING — QUESTIONS, PHASE DEEP-DIVES, AND THE MATH

Challenge — Why did you choose this story to illustrate Think Big?

Because the move is structural, not incremental: the program stopped marketing a settings toggle and started selling an insurance contract — accountability transfer, priced per seller, self-selecting at marketplace scale. Thinking big here meant redesigning the incentive so thousands of independent sellers each make the right call on their own.

Challenge — What other stories demonstrate this principle?

S12 (the roadmap flip, ~$5M avoided) is the primary and S9 (the 1,000-lane vision staged honestly) the backup — both production records. This is the design-engagement alternate for influence-at-scale questions.

Challenge — Could you come up with an example that is more recent?

The production-record adoption machine is S11 — Flexport, 7 → 68 behind published graduation gates — which is more recent and carries record numbers. This engagement is the same motion designed for sellers instead of enterprise customers, with the incentive layer carried further.

Probing — Why did you choose to focus on these results? What other results were important?

The contract design leads because it is the thing I can honestly claim — the diagnosis, the structure, the gates, the measurement. Also important: the enforcement-posture recommendation (carrots until the alternative scales) and the kill condition — a protection promise the scorecard fails to honor stops enrollment, because one broken protection kills the program's legitimacy faster than slow adoption does.

Probing — You mentioned dollars — what percentage change is that year over year?

None claimed — the ≈$3.7M/yr is per ten points of enrolled GMV, modeled from the handling baseline, and the curve itself is the client's record. I give the unit economics so the investment argument is quantitative; I don't book the outcome.

Probing — I'm concerned you're taking credit for a program you didn't run to completion — tell me more.

That's exactly why the claim is scoped: mine is the diagnosis, the contract, the gates, wave one and the measurement pair; the client's is the curve. I'd rather under-claim a design than over-claim an outcome — the same discipline as S17, where I refuse the award dollar on the carrier bid.

Challenge — What were the lessons learned? What would you have done differently?

Lesson: adoption stalls are usually incentive-design problems wearing feature-marketing costumes — price the trade and the funnel moves. Differently: start the refusal interviews before touching the funnel data; the sellers' own words located the drop-off faster than the analytics did, and I ran them second.

Challenge — How would you implement this at Amazon?

Amazon runs the reference version of this contract — observed handling, settings automation, protected labels, all-or-nothing accountability transfer. What I'd bring is the posture judgment: knowing when the contract's teeth fit the business (a mature fulfillment alternative, churn tolerance) and when they don't — and the measurement pair, because enrollments without retention is churn wearing a growth costume at any company.

Challenge — How did these results compare to your actual goals?

The engagement's signed goals were a diagnosed stall, a rebuilt offer, a gated wave one and an honest measurement handoff — all four delivered. The goal was never "move the share N points under my name"; that outcome belongs to the owners who run the waves.

PHASE DEEP-PROBE BANK — Q&A (12). Answer the phase asked, then stop.

Discovery — How did you know discovery was done?

When the stall had one explanation that survived the sellers' own words, the funnel data and the product team's review: a real benefit, unpriced and weakly communicated, traded against a visible loss of control. The exit memo named the funnel's biggest drop-off and the three refusal reasons in rank order.

Why interview churned sellers, not just refusers?

Because churn is the expensive refusal — someone who tried the program and left knows exactly which promise broke. Wave-one's protection-audit exit bar came directly from a churned seller's story about a scorecard ding the program should have excused.

Plan & Design — Why all-or-nothing instead of à la carte?

Because partial adoption keeps the adverse selection: a seller who keeps declared handling keeps the optimism that breaks promises. Cede all three controls and the marketplace can underwrite you; keep any one and you keep the risk. The bundle is what makes the contract insurable — and self-selecting.

Why does observed handling matter so much?

It kills self-reported optimism at the source. Declared handling is a bid for placement (declare one day, hit it 85%); observed handling is a measurement. Moving the promise onto what sellers actually do — rather than what they say — is the single largest accuracy lever in the seller-execution family.

What was the reference mechanism and what did you refuse to copy?

The dominant marketplace's equivalent: the same three-part bundle with full accountability transfer, enforced aggressively. We copied the contract structure and the grain. We did not copy the enforcement aggression or the churn tolerance — this client was still in assortment-growth mode, its fulfillment alternative subscale, so a churned seller was a strategic loss. Tiered protection instead of a cliff, revisited when the alternative scales.

Build & Test — What exactly did the shadow run verify?

Each enrolling seller's recent orders replayed under managed settings: projected promise accuracy, badge eligibility, and the scorecard-protection rules firing where they should. Day one verified on the seller's own history, not hoped — the same calibration-before-onboarding toll as my Flexport waves (S11).

How was the GMV-delta line computed and policed?

The client's analytics computed it per seller from the fast badge's measured conversion lift applied to that seller's assortment — their model, reproducible from logged lift data. My rule on it: the line is a projection and labeled as one in Seller Center; an unlabeled promise there would be the program lying to its own sellers.

Cutover — What were the wave-one graduation criteria?

Shadow results within tolerance of projection; observed-handling coverage on the cohort; inventory-feed accuracy at the seller's ship-from nodes (a stale feed causes cancellations — the worse defect); and the protection rules armed and audited. A wave that failed a criterion did not board the next cohort — publicly.

What was the kill condition?

A protection promise the scorecard fails to honor — an enrolled seller dinged for an excused carrier miss — stops enrollment until fixed. The contract's credibility is the product; one broken protection spends it faster than any growth banks it.

Hypercare — Why activations and 30-day retention?

Because enrollments without retention is churn wearing a growth costume. The pair is the honest read: throughput says the funnel works; retention says the contract kept its promise. A program graded on activations alone will board sellers it should have fixed first.

What flowed back into the offer after wave one?

Refusal and churn reason codes, as data: which component sellers feared (the label requirement led), which protection they doubted, where the GMV-delta line read as marketing instead of math. Overrides with reasons are data; silent refusals are a funnel you can't fix.

Cross-phase — What did you refuse?

An adoption target under my name (the curve is the client's), a hard label-purchase cliff (a tax on sellers with negotiated rates — tiered instead), enrollment pushes ahead of the shadow gate (day one must be verified), and counting activations without retention. Every refusal is in the contract document as a priced row.

Where this lands at Amazon: Amazon invented the reference version of this contract and runs adoption as incentive design at the largest scale there is. The transferable asset is the motion: diagnose a stall in the sellers' own words, price the trade in their dollars, gate the rollout on verified cohorts, and grade the program on throughput *and* retention — influence over thousands of independent actors through instruments, not authority, which is the seller-facing version of the same muscle as S11's 7 → 68.

BUSINESS IMPACT MODEL — KPI TREE, NUMBERS, THE MATH

```
Promise kept at the source + adoption that sticks (primary outcome)
+- Handling-miss rate on manual-config GMV (flow) -> declared ~85% kept
|    miss -> breach-through -> broken promise -> WISMO + trust decay
+- Enrolled GMV share (stock) -> ~38-42% stalled -> waves behind gates
|    (loop) priced offer -> rational enrollment -> observed handling
|           -> fewer misses -> badge earned -> conversion lift -> more GMV
+- Protection credibility (stock) -> kill condition; audit per wave
+- Measurement honesty -> activations/wk x 30-day retention (the pair)
+- Unit economics -> ~$3.7M/yr per +10 pts enrolled GMV (modeled)
```

DRIVER	BASELINE	AFTER (DESIGNED)	$/YR	BASIS
Enrolled share of seller-fulfilled GMV	~38–42%, stalled	waves behind gates; curve unclaimed	—	record-from-docs baseline

DRIVER	BASELINE	AFTER (DESIGNED)	$/YR	BASIS
Handling adherence on declared 1-day	~85%	observed handling replaces declared; ~97% modeled	inside the unit economics	85%: record-from-docs; 97%: illustrative
Unit economics per +10 pts enrolled GMV	—	≈6,000 breaches/wk avoided	≈$3.7M	modeled (Table 4; $12 assumption)
Offer pricing surface	benefit abstract	personalized GMV-delta line per seller	—	engagement design record (client-computed)
Program honesty	enrollments counted	activations/wk × 30-day retention + full-bundle share	—	engagement design record

If challenged. *"Your result is a design, not an outcome."* — "Yes — and I say exactly that. Mine is the diagnosis, the contract, the gates, wave one and the measurement pair; the adoption curve is the client's record. The production-record version of me running an adoption machine end-to-end is S11: 7 to 68 behind published gates." *"The $3.7M rests on two illustrative rates."* — "It does — the 3% post-fix miss rate and the 50% breach-through are labeled illustrative; the 85% baseline and the $12 are the engagement record's own figures. Halve both assumptions and the unit is still about a million a year per ten points — the argument survives." *"Why not just mandate enrollment?"* — "Because a mandate spends seller trust the marketplace hasn't earned yet — and a churned seller is a strategic loss in assortment-growth mode. The contract makes enrollment the rational hedge; mandates are for after the fulfillment alternative is credible."

THE MATH — STEP-BY-STEP DERIVATION

Table 1 — before: the problem in numbers.

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
1	GMV on manual configuration	`1 − enrolled share = 1 − 40% = 60%`	enrolled ~38–42% (record-from-docs); 40% midpoint used	≈60% of seller-fulfilled GMV on declared settings
2	Orders on manual config	`1,000,000 × 60% = 600,000 / wk`	~1M orders/wk illustrative book	≈600K orders/wk
3	Handling misses	`600,000 × (1 − 85%) = 600,000 × 15% = 90,000 / wk`	declared 1-day handling hit ~85% (record-from-docs)	≈90K seller-made misses/wk
4	Breach-through	`90,000 × 50% = 45,000 / wk`	a handling miss of a day typically exhausts the buffer; 50% breach-through (illustrative)	≈45K broken promises/wk from handling alone
5	Cost of the manual book	`45,000 × $12 = $540,000 / wk`	$12 breach bundle (the model's stated assumption)	≈$540K/wk of handling-driven breach cost
6	Why it persists	`declared handling = a bid for placement; adoption = control traded for an unpriced benefit`	adverse selection + unpriced trade (record-from-docs diagnosis)	a stall no banner fixes

Table 2 — the fix: step by step, values substituted.

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
1	The contract	`protection ⇔ (transit automation) AND (observed handling) AND (participating label)`	all-or-nothing bundle (engagement design)	full bundle → carrier-delay excusal + badge where the math clears

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
2	Observed handling effect	`miss rate 15% → ~3% on enrolled orders`	handling set from scan history, not declaration; 3% illustrative	−12 points of handling misses on enrolled GMV
3	Pricing the benefit	`seller line = conversion lift × seller's eligible assortment GMV = "$X/month"`	client-computed from measured badge lift (design)	the trade visible in the seller's own dollars
4	Self-selection logic	`trusts own ops ⇒ keeps control and risk; doesn't ⇒ converts`	incentive design, not persuasion	adoption as the rational hedge
5	Shadow gate	`replay seller's recent orders under managed settings; flip only within tolerance of projection`	verification design	day one verified, not hoped
6	Wave mechanics	`cohort boards ⇔ all published criteria pass; fail one ⇒ hold, publicly`	graduation gates (design; the S11 motion)	adoption earns its expansion
7	The honest pair	`program health = activations/wk AND 30-day retention AND full-bundle share`	measurement design	growth that can't be faked by churn

Table 3 — after: the metric, before → after.

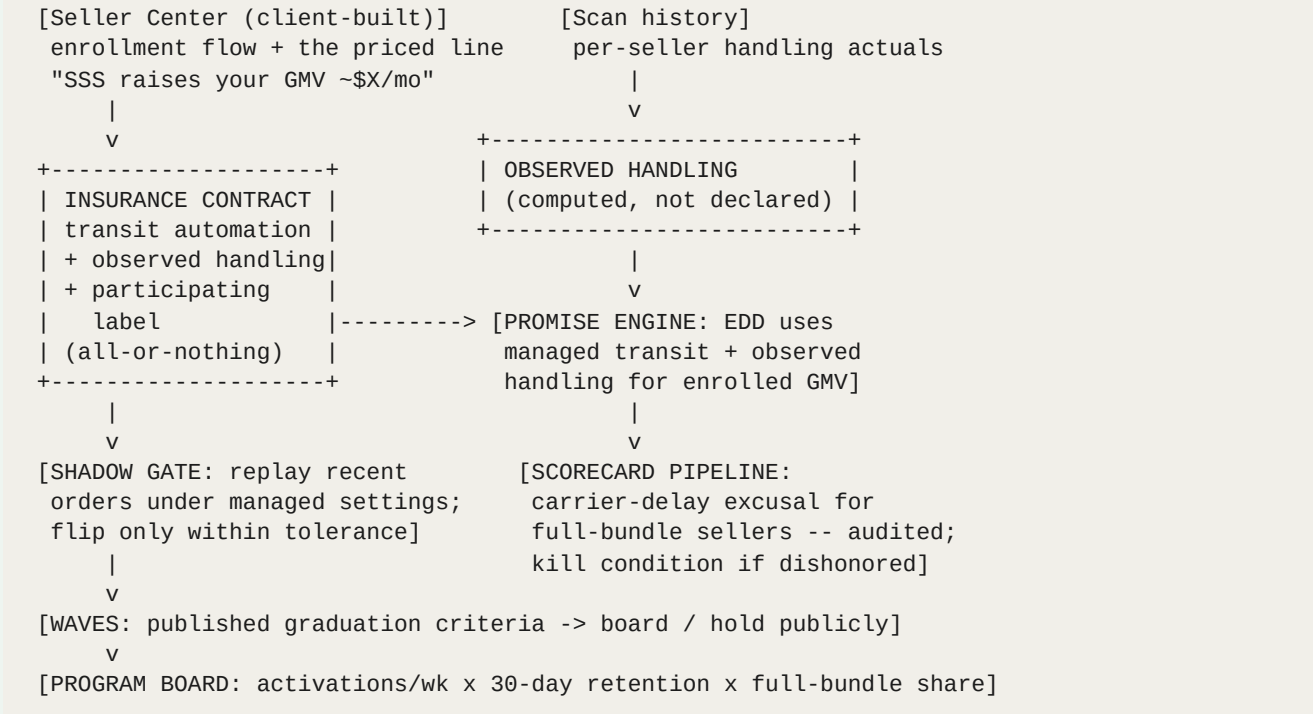
STEP	METRIC	BEFORE	AFTER	BASIS
1	The offer	settings toggle, benefit abstract	priced insurance contract, benefit in seller dollars	engagement design record
2	Handling input on enrolled GMV	declared (15% miss)	observed (~3% miss, modeled)	15%: record-from-docs; 3%: illustrative
3	Rollout mechanics	none (push marketing)	shadow gate + waves behind published criteria; wave one boarded	engagement design record
4	Program measurement	enrollments	activations/wk × 30-day retention × full-bundle share; kill condition armed	engagement design record
5	Enrolled GMV share	~38–42% stalled	curve unclaimed — the client's record	record-from-docs baseline; honesty rule

Table 4 — the dollars, every multiplication shown.

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
1	Orders in +10 pts of enrolled GMV	`1,000,000 × 10% = 100,000 / wk`	~1M orders/wk illustrative book	100K orders/wk move onto the contract
2	Handling-miss reduction	`15% − 3% = 12 points`	15% record-from-docs; 3% illustrative post-fix rate	12 pts fewer misses
3	Misses avoided	`100,000 × 12% = 12,000 / wk`	rows 1 × 2	≈12K misses/wk avoided
4	Breaches avoided	`12,000 × 50% = 6,000 / wk`	50% breach-through (illustrative)	≈6K broken promises/wk avoided
5	Weekly value	`6,000 × $12 = $72,000 / wk`	$12 breach bundle (the model's stated assumption)	≈$72K/wk
6	Annualized	`$72,000 × 52 = $3,744,000`	row 5 × 52 (modeled)	≈$3.7M/yr per +10 pts enrolled

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
7	Not claimed	`adoption curve × unit economics = the client's outcome`	honesty rule	unit economics only; conversion-lift upside also excluded

SOLUTION FLOW — THE SYSTEM & DATA MAP



SENSITIVITY — LOW / BASE / HIGH

INPUT	LOW	BASE	HIGH
Post-fix handling-miss rate (illustrative)	6%	3%	2%
Breach-through on a handling miss (illustrative)	30%	50%	70%
Breach cost (model assumption)	$6	$12	$18
Breaches avoided per +10 pts (/wk)	≈2,700	≈6,000	≈9,100
Modeled $/yr per +10 pts enrolled	≈$0.8M	≈$3.7M	≈$8.5M

TOTAL IMPACT STATEMENT — SAY IT LIKE THIS

"The honest version first: what I delivered is the contract, not the curve. The program was stalled around forty percent of seller-fulfilled GMV because it asked sellers to trade control for an unpriced benefit — so we rebuilt it as insurance: give up transit settings, declared handling and the label, and the marketplace absorbs carrier-delay accountability and shows you what the fast badge is worth on your own assortment, in dollars. The unit economics are modeled and I label them: on the engagement record's own baseline — sellers declaring one-day handling and hitting it eighty-five percent of the time — every ten points of GMV that enrolls is worth about three point seven million dollars a year in avoided broken promises at the model's twelve-dollar breach cost. Wave one boarded behind published gates; the adoption curve after that belongs to the client's owners, and I say so."

3. The Elementum design engagements — the twelve designs, each with the math and the probes

The GFiber packet's Part 3 carried twelve system-design prompts. Here they are told as what they were underneath: **design engagements from the Elementum client work** — supply-chain design for manufacturers and network operators running SAP, 3PLs, EDI and field fleets. Use these for Amazon's functional and "how would you design / approach…" questions; they are design stories, not behavioral stories. Every design is answered in the same six moves, said out loud so the interviewer can follow the map: **(a) stocks and flows** — every bucket a unit can sit in, every flow between them, with the driver and the document that records it; **(b) first principles** — the invariants (every unit is in exactly one stock; every flow has exactly one document; every number has exactly one owner); **(c) the design** — components, the data model at its true grain, integration events, controls and kill switches; **(d) sequencing** — phases ranked Reach × Impact × Confidence ÷ Effort, with the invariants deciding what may ship at all (shown in full on E1–E2; folded into the design elsewhere); **(e) the deep-dive probes**; **(f) one metric with a dollar** — illustrative, labeled. *Every dollar in this section is illustrative — a sizing method with stated assumptions. Where a pattern wasn't run end-to-end in production, the honest line says so.*

3.0 — How I'd run programs at Amazon (the one shared section — methodology lives here, never inside a story)

STEP	WHAT IT IS	THE GATE IT CREATES	FROM THE RECORD
1. Design doc	Context · goals · non-goals · the stocks-and-flows map · data model at grain · integration events · alternatives considered · cross-functional dependencies · risks	Architecture and the business owners sign the policy before any configuration starts	The FSD signed by three orgs before config (S1, S5); "a no that lives in a document survives"
2. Goals / metrics	One objective per program; 3 key results that are counts or rates, each with an owner and a denominator	A KR without a denominator is not a KR	Seven frozen metrics, honest denominators (S11); MAPE/bias published from month 2 (S6)
3. Milestones with exit criteria	Discovery → Plan & Design → Build & Test → Cutover → Hypercare, each with entry / graduating / exit stated in writing before the phase starts	A phase graduates on counted rows, never on confidence	Every story's Action in §2; the 0.5% tolerance published before mock 1 (S18)
4. Launch readiness review	Checklist: reconciliation clean N days running · monitors live · rollback rehearsed · control owners named · comms drafted	Go/no-go is a recommendation read from the checklist, taken by the forum as a decision	Go/no-go packs (S1, S2, S18); six graduation criteria published before wave 1 (S11)
5. Staged rollout + rollback	Canary (one site / one partner / one node) → wave → full; rollback trigger written as a number; kill switch per feed	A wave that misses a criterion does not board — publicly	Wave 3 held (S11); robotics throttle to ~30% (S2); RF paper-pick fallback (S18)
6. Blameless postmortem	Timeline · contributing causes · mechanisms that outlive the incident · owner per action	Every Sev produces at least one mechanism, not a reminder	Scheduler heartbeat + DST checklist + 3σ page rule (S13); the in-transit back-test rule (S4)
7. Release cadence	Fixed cadence; a regression matrix at the true grain is the gate; freeze around peak / close; emergency change with retrofit, never a manual table edit	A rule change rides the train or does not ship	4-week cadence with the QA lead owning the gate (S15); the policy fix released on cadence (S4)

Sequencing convention used throughout (illustrative): Reach = units, orders or people touched per month; Impact on the 0.25 / 0.5 / 1 / 2 / 3 scale; Confidence 50–100%; Effort in person-months. Score = Reach × Impact × Confidence ÷ Effort. **The score ranks; the invariants decide what is allowed to ship at all.** If interrupted at minute two: clarify the book of record and the primary metric in twenty seconds, then draw the stocks.

E1 — Elementum design engagement: contractor / truck-stock inventory for a field-install fleet

Use for: "design field/mobile inventory," "stock you own but don't hold," ambiguous-ownership questions. **Record behind it:** the owner-partitioned pool and audited acts (S1, S2), lower-of-two when two books disagree (S1, S18), reason-coded reconciliation (S14).

> **Honest line.** A design engagement for a network-operator client whose contractors' trucks carried client-owned install equipment; I owned the inventory model, the event contract and the controls — the client's SAP team configured. I have not run a full van program end-to-end; the volumes here are illustrative.

The problem as I restate it. Contractors' trucks carry client-owned equipment and consumables; issues from the 3PL are posted, installs post late or not at all, counts are sporadic, and Finance sees "unaccounted" growing. Design the inventory model, the events, and the controls. Book of record: the client's SAP inventory. Primary metric: unaccounted units per contractor per month, and their value.

(A) STOCKS AND FLOWS · (B) FIRST PRINCIPLES

```
[3PL SLoc] --issue (311, scanned)--> [Van SLoc] --install (601 by serial)--> [Installed]
                 |                          |-- swap-out (311) --> [Returns-untested]
                 |                          |-- van-to-van (311, both scan)
            weekly count --> [Adjust 701/702 + reason] --> [Chargeback only from a posted
count]
Unaccounted(van) = issued - installed - returned - counted_on_hand
Classify like out-of-balance: timing -> unposted install -> serial mismatch -> true loss
```

Invariants. Conservation: every owned unit is in exactly one stock at every instant — today the van is an *unmeasured* stock; "unaccounted" is not theft yet, it is a missing document. One document per flow: issue = scanned transfer; install = work-order close that posts the goods issue; count = a physical-inventory document — if a flow has no document, it does not exist and the design adds one before adding a report. One owner per number: Inventory Control owns the book; Field Ops owns counts; Finance owns the chargeback method; the contractor owns the physical; I own the definitions and gates.

(c) Design. A van is a storage location under the market plant (not a spreadsheet); serialized issues with forced scan on every flow; min/max replenishment per van by tech type, released only when last week's count is posted (**count-as-gate**); a reconciliation service computing issued − installed − returned − counted per van, reason-coded; governed screens for count, adjustment (role + reason) and chargeback review; the CRM reads only the assigned tech's van availability. Data grain: SKU × plant × SLoc(van) × serial × state, with an append-only serial-history ledger; every event idempotent on document + line, replays never double-post, reversals reference the original. Controls: no new issue to a van with an unposted weekly count (hard gate); days-of-cover cap with audited override; a serial issued to van A cannot be installed from van B without a posted transfer — the install is held, not lost; two weeks over tolerance ⇒ count-every-issue mode; kill switch = freeze issues to a contractor, 3PL fulfills directly.

(d) Sequencing (illustrative). P1 van-as-SLoc + serialized issue + count-as-gate (score 300) → P2 field-app install/swap events posting by serial (175) → P3 reason-coded reconciliation + Finance chargeback method (267 — ranks above P2 but needs its events; the invariant sequences it after) → P4 min/max + cap → P5 CRM same-day availability read. Rejected: a spreadsheet outside SAP — a second book.

(f) Metric. Unaccounted units per contractor per month → near zero, from posted documents only, published weekly per van with the reason mix. Illustrative dollar: 500 vans × 2 unaccounted/month × $150 average unit ≈ $150K/month ≈ $1.8M/yr of stock with no owner; the design targets the unexplained share — Finance sets the chargeback threshold, not me. Supporting: install events posted same-day ≥ 99%; vans with a posted weekly count = 100%; "van said yes, book said no" reschedules → 0.

Deep-dive probes.

Why a storage location and not consignment or subcontracting special stock?

The client owns the unit until install, so it is their stock at the contractor's location — a storage location under the market plant. Subcontracting stock is a manufacturing pattern (components consumed into a finished good); customer consignment implies the customer holds it. Neither describes a truck.

Movement types?

311 for 3PL → van (scanned both ends), 601 on the install (or 201 to a cost object if not order-based — decide once), 311 van → RMA for swap-outs, 701/702 for count differences with a reason code. 561 only for the initial load — never for "fixing" a count.

The field app is offline half the day — how does the book stay right?

Capture with a client-generated idempotency key and timestamp; replay in order per serial; reconciliation counts "captured, not posted" as its own reason code so timing is never read as loss.

Failure modes?

Install posted from the wrong van — hold, never auto-transfer. Duplicate install replay — idempotency key. Count with a phantom serial — quarantine the line. Chargeback from an unposted count — impossible by design; chargebacks read only physical-inventory documents.

Trade-offs?

Count-as-gate slows issues in week one and finds every uncounted van — accepted. Per-serial postings raise volume — sized. The days-of-cover cap frustrates long routes — override with a reason code, audited.

What if a contractor has no scanner?

Stage 1: SLoc + signed paper count uploaded; stage 2: scanning. Never skip the SLoc for the scanner.

SOLUTION FLOW — THE SYSTEM MAP (PROPOSED)

```
[Salesforce [BoR customer/order]]        [BI / BigQuery: van recon, aging]
  WorkOrder close (install/swap)                ^  MATDOC + install/count
  | event, real-time                           |  events, hourly
  v                                           |
[Field app: tech / van / install_event / return_event / count]
  offline capture, client idempotency key, replay in order per serial
  | install event -> 601 (or 201 cost ctr) by serial, real-time
  | swap event    -> 311 van -> RMA SLoc by serial, real-time
  | van count     -> PI document -> 701/702 + reason, weekly
  v
[SAP CPI / Integration Suite]  --retry / DLQ per hop, idempotent-->
  v
[SAP S/4HANA [BoR stock + finance]]
  plant = market · SLoc per contractor x truck · serials (MARC-SERNP)
  311 3PL SLoc -> van SLoc · 601/201 install · 651 customer return
  701/702 count adjust -> chargeback report (Finance)
  ^
  | 940 pick / 945 ship-confirm for the van issue, event
[3PL WMS [BoR bin]]  -- replenishment to van min/max, weekly wave
```

E2 — Elementum design engagement: serial-number tracking, vendor → warehouse → technician → customer → return

Use for: traceability, serialized inventory, state-machine design questions. **Record behind it:** state machines for inventory buckets with one owning system per transition (S1, S2); two books reconciled to a published tolerance (S18).

> **Honest line.** I designed and governed the state model and reconciliation; the client's SAP team configured the serial profile — I own the state list and which transitions are legal, not the IMG screens. Volumes illustrative.

The problem as I restate it. Every serialized unit the client ships — network hardware to customer premises — must be traceable by serial across its whole life, in SAP and in every system that touches it (3PL WMS, field app, CRM asset, RMA/refurb). Book of record for the serial state: SAP, with the 3PL and field app as event sources. Primary metric: serial-state accuracy — serials whose SAP state matches a physical or system observation.

(A) STOCKS AND FLOWS · (B) FIRST PRINCIPLES

```
[Vendor FG] --856 ASN w/ serial list--> [In-transit] --receipt by serial (101)-->
[3PL unrestricted] --issue (311)--> [Van] --install (601 + CRM asset)--> [Installed base]
[Installed] --return/disconnect--> [Returns-untested] --test--> refurb / RTV / scrap (E10)
```

Invariants. A serial never appears from nowhere — its first legal appearance is the ASN; a receipt for a serial the ASN did not carry is an exception, not a receipt; an install for a serial not in that van is a hold. Every flow has a reversal that references the original, so a replay or correction never manufactures a phantom serial. Minimal design forced: serial profile with stock check; the ASN carries serials; every downstream event carries the serial; one append-only serial-history ledger; a daily three-way reconciliation (SAP vs 3PL vs field app/CRM asset).

(c) Design. Serial profile with stock check on the serialized materials, so a goods issue for a serial not in that location fails loudly instead of posting silently; EDI/ASN ingest validating serial lists against the PO; receipts by serial; install/swap events by serial creating the CRM asset; an RMA module with test-result events and grade — refurb keeps the *same serial, same material*, with a grade attribute: never re-number, so the asset history stays continuous. Grain: serial × SKU × state × location × owning-system × as-of, plus the serial-history ledger (one row per transition: document, source, actor, timestamp, idempotency key). Controls: ASN serial count = PO quantity or the ASN quarantines; a receipt for an unknown serial goes to an exception queue with a 24-hour vendor SLA — never auto-create; installed serials with no CRM asset after 24 h alert; corrections only through a governed screen with role + reason — *never* a manual table edit, and never "fix SAP to match the 3PL" — lower-of-two for promising while the cause is found.

(d) Sequencing (illustrative). P1 serial profile + ASN serials + receipt by serial (~50K serials/mo, score 24,000) → P2 install/swap by serial + CRM asset (10,500) → P3 three-way daily reconciliation + governed correction screen (28,333 — after P1/P2; needs the events) → P4 RMA/refurb states → P5 scrap/RTV with certificates.

(f) Metric. Serial-state accuracy ≥ 99.5%, measured by the daily three-way reconciliation with a reason-coded exception list. Illustrative dollar: each mis-stated serial costs a truck roll or a lost unit; at 200K installed serials, 97% → 99.5% removes ~5,000 mis-states/yr × ~$120 blended ≈ $600K/yr — before warranty-recovery value (in-warranty failures returned to vendor by serial). Supporting: ASN-to-receipt serial match ≥ 99.5%; installs with an asset record within 24 h = 100%.

Deep-dive probes.

How do serials travel on EDI?

The 856 ASN carries serials in its hierarchical loops at item/pack level; the 945 ship advice carries serials for the van issue; adjustments must carry serials or they quarantine.

Idempotency?

Key = document + line + serial; the ledger rejects a duplicate transition and logs it; reversals reference the original document. A replayed install never creates a second asset.

The 3PL receives a serial the ASN didn't list?

Exception queue, 24-hour SLA to the vendor; the unit sits blocked and unpromisable until resolved. Never auto-create the serial.

Refurb identity — same serial, new material?

Same serial, same material, a grade attribute or refurb location — never re-number; the asset history stays continuous.

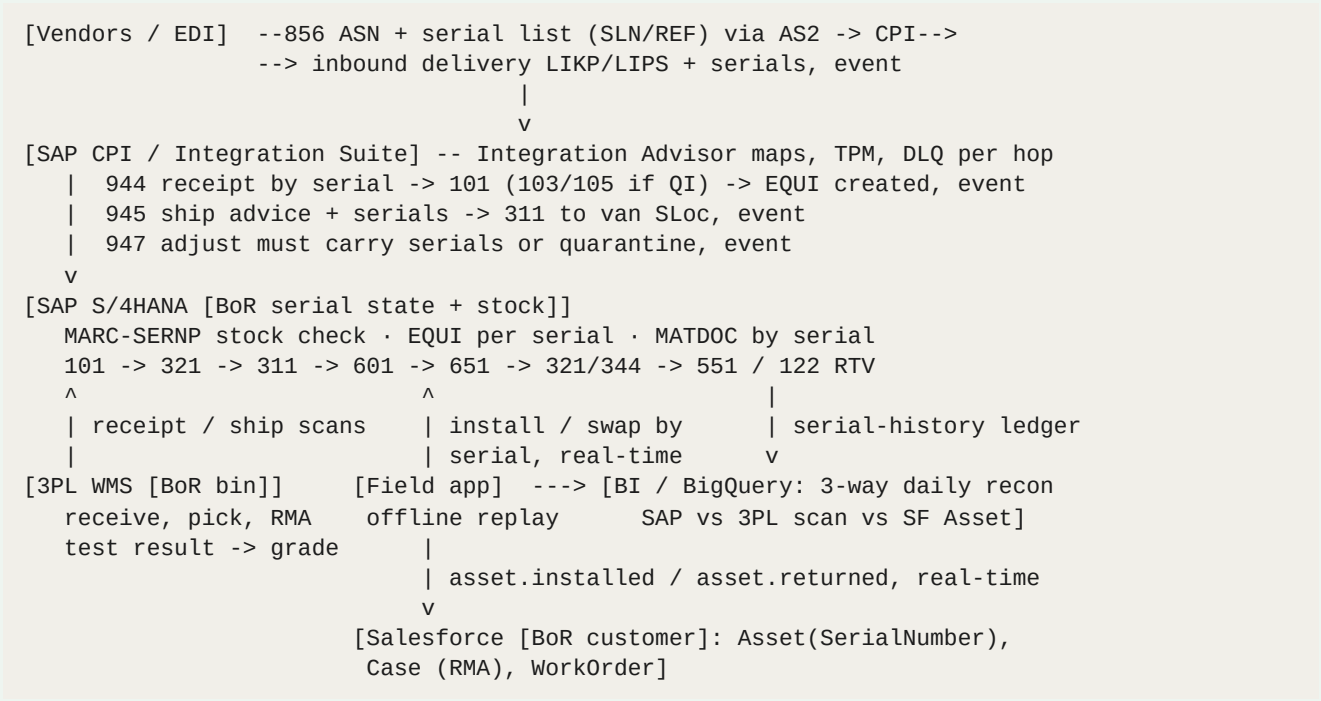
Failure modes?

Duplicate serials from a vendor — block at ASN validation. Typo serial from the field — scan-only with check digits. State fixed by a table edit — forbidden; correction screen with reason and role. Reconciliation that "fixes" SAP to match the 3PL — forbidden; lower-of-two while the cause is found.

Trade-offs?

Serialization slows receiving (a scan per unit) — offset by ASN pre-load. Three return states add screens — but an untested return that is promisable is a truck roll.

SOLUTION FLOW — THE SYSTEM MAP (PROPOSED)



E3 — Elementum design engagement: vendor EDI + VMI for long-lead network equipment

Use for: vendor integration, consignment/VMI, inbound-reliability design questions. **Record behind it:** 597 lanes of X12 EDI behind a ≥95% valid-ASN cohort gate (S9); lead times measured as distributions (S8).

> **Honest line.** The EDI onboarding is run experience (S9); the VMI/consignment mechanics are the SAP pattern, designed in the engagement but not run by me as a program — labeled.

The problem as I restate it. Long-lead equipment from a few strategic vendors; the client wants EDI for POs/ASNs/invoices and a VMI arrangement where the vendor replenishes to agreed min/max at the central 3PL. Design the transaction set, the stock-ownership model, and the controls. Book of record: SAP MM (PO, goods receipt, consignment stock). Primary metric: confirmed-date reliability — the vendor's receipts measured against its own confirmations — and valid-ASN rate.

(A) STOCKS AND FLOWS · (B) FIRST PRINCIPLES

```
[Vendor inventory] --850 PO out / 852 product activity (VMI)--> [Vendor committed (855 ack)]
  --ship (856 ASN; 214 status)--> [In-transit] --receive (101 K if consignment / 101 if
own)-->
[Consignment K at 3PL] --consume/transfer (411 K)--> [Own unrestricted] -> settlement (MRKO)
Reorder point ~ demand over p90(LT) + z x sigma_L  -- from measured distributions (S8),
  refreshed monthly with a drift flag; brochure lead times are inputs to nothing
```

Invariants. A confirmation without a date is not a confirmation. A receipt without an ASN is an exception, not a receipt. Ownership is explicit at every instant: vendor-owned consignment on our floor is special stock K, visible to planning, settled at consumption — or the vendor replenishes our own stock under blanket POs; who carries the working capital is a Finance decision, modeled both ways, not a default.

(c) Design. Transaction set: 850 PO out · 855 acknowledgment (confirmation, with date and quantity) · 856 ASN · 810 invoice · 997 functional acknowledgment on every interchange · 852 product activity for VMI · 861 receiving advice for discrepancies. Confirmation discipline: the 855's date becomes the planning date; "acknowledged" with no date = unconfirmed, a planner exception. ASN gate per vendor: ≥95% valid ASN before activation (the S9 gate, reused); vendors with weak data onboard on portal templates and graduate. Min/max set from measured lead-time distributions per vendor × lane, refreshed monthly, drift-flagged. Controls: gateway de-duplication on interchange control numbers; document idempotency on PO + line + confirmation sequence; out-of-order 855 after 856 — the ASN wins quantity, the ack updates only the date; a mis-parsing pack hierarchy quarantines, never partially posts.

(f) Metric. Confirmed-date reliability ≥ 95% — receipts on or before the vendor's own confirmed date, measured from month one (baseline unknown, so the metric starts honest). Illustrative dollar: if late equipment idles a build crew at ~$50K/week and confirmation discipline avoids ~20 such weeks/yr ≈ $1M/yr; plus working capital if consignment carries ~$2M at the vendor's expense (≈$160K/yr at 8%). Supporting: valid-ASN ≥95% per vendor before activation; receipts without ASN → 0.

Deep-dive probes.

Consignment vs own stock for VMI?

Special stock K: vendor-owned on our floor, visible to planning, settled at consumption. Alternative: vendor replenishes our own stock under blanket POs — simpler accounting, but we carry the inventory. Choose by who should carry the working capital; I model both and let Finance decide.

Idempotency and ordering?

Control numbers de-duplicate at the gateway; document-level idempotency on PO + line + sequence; an ASN received twice creates one inbound delivery.

Failure modes?

An 855 with "acknowledged" but no date — treat as unconfirmed. An ASN with an unparseable pack hierarchy — quarantine, not partial post. Unit-of-measure mismatch — a conversion table at the map (the classic mock-1 defect from S18). A looping 997 on a failed interchange — dead-letter after three.

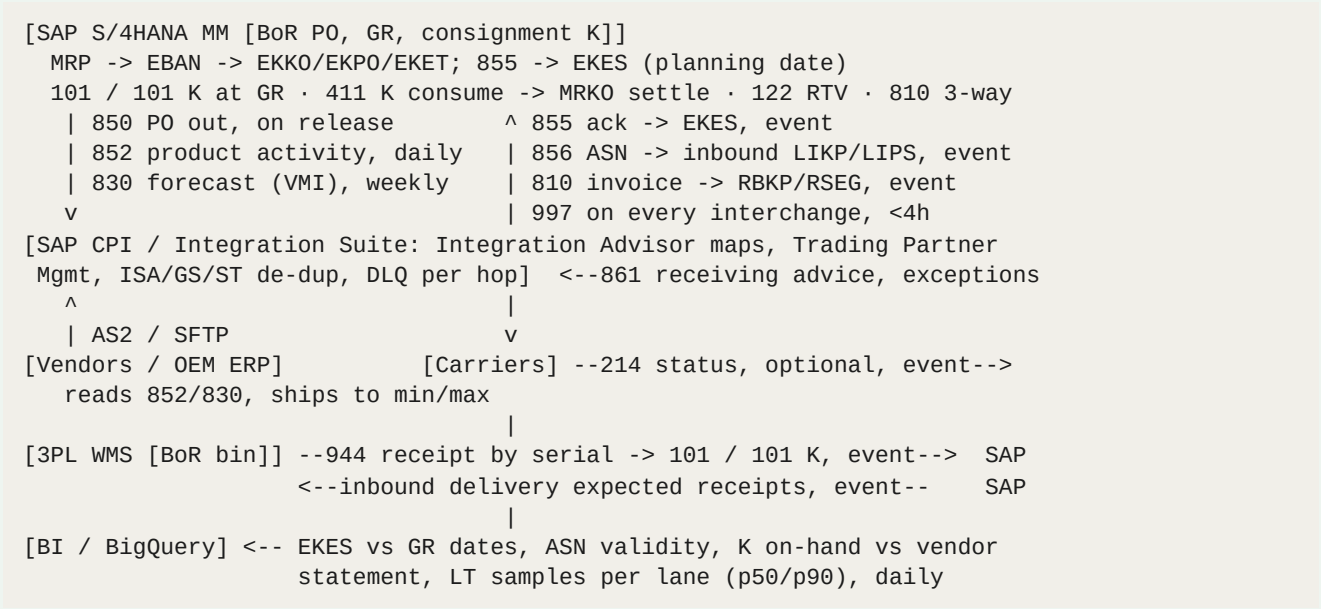
How do you set min/max?

From measured lead-time distributions per vendor × lane: reorder point ≈ demand over the p90 lead time plus safety stock from σ_L, refreshed monthly with a drift flag. Brochure lead times are inputs to nothing.

Trade-offs?

Confirmation control adds planner exceptions in month one — that is the point; it reveals the unreliable vendors. The ASN gate delays go-live for weak-data vendors — they onboard on the portal floor and graduate, which keeps the long tail inside the standard.

SOLUTION FLOW — THE SYSTEM MAP (PROPOSED)



E4 — Elementum design engagement: CRM order → SAP fulfillment integration with status sync

Use for: order-orchestration, system-integration, "orders are dropping" design questions. **Record behind it:** the OMS held to "read published availability, never compute" and the event chain across WMS/OMS/POS/S4 (S1, S3); scheduler and status-mapping failures lived through (S13, S14, and a ~$2.8M status-mapping root-cause at Flexport).

> **Honest line.** I have not implemented the client's CRM; I owned the integration contract — the event chain, idempotency, status truth and reconciliation — and the client's teams built both ends.

The problem as I restate it. A customer signs up in the client's CRM; the order must create the SAP fulfillment (a kit delivery from the 3PL or van), reserve stock, and keep status flowing back so the CRM, the technician and the customer see the same state — without orders dropping at month-end or double-shipping on retries. Book of record: CRM for the customer order; SAP for stock and fulfillment; **status truth per hop is the system that performed the hop**. Primary metric: orders reconciled across every hop — sent = created = delivered = status returned — 100% daily.

> **(A) STOCKS AND FLOWS · (B) FIRST PRINCIPLES**
>
> ```
> [CRM order created] --order.created (correlation id)--> [SAP order created]
>   --ATP against PUBLISHED availability (never computed in the CRM)--> [Reserved]
>   --install date - lead--> [Delivery created] --pick/ship/PGI--> [Shipped]
>   --install event--> [Complete] ... status flows BACK per hop, one writer per field
> ```
>
> **Invariants.** One creation path: deliveries are created only from the SAP order, never from the inbound event — a duplicate event cannot make a second delivery. One availability truth: the CRM reads published availability; it never computes its own. One writer per status field; a status the map does not know defaults to "unknown — investigate," never to "shipped." Every message carries a correlation id that lives on the SAP order, so a retry finds its own order.
>
> **(c) Design.** Durable messaging with retry and dead-letter queues per hop; an outbox table on the CRM side so order-create is at-least-once with no lost orders; idempotency at create (correlation id unique on the order — a second create returns the first order); version checks so an update arriving before its predecessor holds rather than overwrites; availability published to a cache with a staleness flag ("hold last-good, flag it") so CRM promising survives an SAP outage; daily hop-count reconciliation with an unreconciled queue cleared by 9 AM; a heartbeat on every scheduler in the chain (the S13 lesson — detect in one interval, not by a VP at 4 AM).
>
> **(f) Metric.** Orders reconciled across every hop daily → 100%, unreconciled count → 0 by 9 AM. Illustrative dollar: if 0.5% of 50K orders/month stall and each stalled install costs ~$75 in rescheduling plus a day of deferred revenue, ≈$19K/month ≈ $225K/yr direct — before churn on a bad first impression, which is the number that matters and which Finance, not I, should size. Supporting: dead-letter age p95 < 1 h; duplicate deliveries = 0; status latency p95 < 5 min.

> **Deep-dive probes.**
>
> **Durable documents or APIs?**
>
> Durable, monitorable documents (IDoc-class) for order create/change — retry with visible status; APIs for status back where latency matters. Either way the correlation id lives in a reference field on the order so a retry finds its own order.
>
> **How do you stop a retry from double-shipping?**
>
> Idempotency at create — the correlation id is unique on the order, so the second create returns the first order — and deliveries are created only from the SAP order, never from the inbound event.
>
> **Orders drop at month-end — where do you look?**
>
> Count reconciliation across hops for the last three month-ends; the message monitor and dead-letter queue by error class; job overlaps and lock waits at the drop times — close jobs locking customer or material masters, availability runs in the same window; the release calendar. Fix: retry with backoff plus idempotency immediately; re-sequence or throttle the contention as the cure; counters on a dashboard so it is visible before close.
>
> **Status mapping — what breaks?**

A new partner status not in the map (default to "unknown — investigate," never "shipped"); a mapping changed without regression (release train only); two hops writing the same CRM field (one writer per field). A ~$2.8M root-cause I ran at Flexport was exactly a status-mapping defect.

Failure modes?

Version 3 arrives before version 2 — version check holds the older. SAP created but the confirmation back was lost — reconciliation finds "created, unconfirmed" and re-emits. A ship-confirm for a cancelled delivery — exception; physical stock to be found. A scheduler dies silently — heartbeat page.

Trade-offs?

The CRM reading SAP availability adds an uptime dependency — mitigated by the published cache with a staleness flag. The outbox adds a table and a relay — the price of at-least-once with no lost orders.

SOLUTION FLOW — THE SYSTEM MAP (PROPOSED)

```
[Salesforce [BoR customer + order]: Account, Order/OrderItem, WorkOrder,
 Asset, status field (one writer per field)]
   | order.created/changed/cancelled via outbox, at-least-once, real-time
   | correlation_id = SF order + line + version
   v                                    ^ order.confirmed (SAP VBELN)
[SAP CPI / event bus]                   | status.changed(hop, ts, corr id)
   ORDERS/ORDCHG IDoc or API ·          | p95 < 5 min, per hop
   DLQ per hop · per-hop counters ·     | availability read (never
   status_map table (versioned) -------+ computed), cached + stale flag
   v
[SAP S/4HANA [BoR stock + fulfillment]]
   VBAK/VBAP (corr id in reference field) -> ATP -> VBBE requirement
   -> LIKP/LIPS delivery (install date - lead time) -> PGI 601 (MATDOC)
   | 940 pick, daily wave        ^ 945 ship confirm -> 601, event
   v                             |
[3PL WMS [BoR bin]] --or-- [Field app: van pick / job.completed]
                              | job.completed -> SAP install posting
                              | + SF Asset(SerialNumber), real-time
[BI / BigQuery] <-- hop_counter, orders-by-state aging, DLQ age, daily
   received = processed + failed + in_flight per hop, balanced by 9 AM
```

E5 — Elementum design engagement: 3PL inventory reconciliation — two books, one truth, Finance protected

Use for: reconciliation design, "two systems disagree," inventory-integrity questions. **Record behind it:** the lower-of-two basis and out-of-balance sign rule (S1), the 0.5% tolerance published with Finance and read by cause (S18), reason-coded reconciliation as a standing control (S14).

> **Honest line.** This is the seam I have owned repeatedly; the design is the generalized version of run experience. Volumes illustrative.

The problem as I restate it. The 3PL's WMS and SAP disagree on on-hand by SKU (and by serial); the delta drifts, promising reads SAP, Finance closes on SAP, and adjustments are posted ad hoc. Design the reconciliation so the two books converge, the customer is protected while they disagree, and Finance can trust the close. Books: SAP at storage-location grain; the 3PL WMS at bin/serial grain. Primary metric: reconciled delta — rows outside tolerance after reason-coding, trending to inside tolerance.

(A) STOCKS AND FLOWS · (B) FIRST PRINCIPLES

```
[In-transit (ASN)] --receive (944 -> 101)--> [3PL bin] = [SAP unrestricted]
[SAP unrestricted] --pick/ship (940 -> 945 -> PGI 601)--> [Shipped]
[WMS bin moves] -- internal, MUST NOT change SAP qty
[Damage/hold] --947 status adjust--> [SAP blocked] · [Count diff] --947 w/ id--> [701/702]
Daily: snapshot(WMS by bin, rolled to SLoc) vs MB52(SAP) per SKU x SLoc x state
Delta classified: timing (in-flight ledger) | mapping | status | physical -> only
  "physical" ever becomes an adjustment
```

Invariants. Neither side wins automatically: SAP is the book of record for Finance and promising; the 3PL is the physical truth; promising reads **the lower of the two** until the reason is known. Grain alignment is a contract: bins roll up to the storage location; WMS statuses map to unrestricted/QI/blocked through one owned table; units of measure and pack factors per SKU; serials compared as sets — any row that cannot be aligned is a mapping defect, not a variance. Adjustments only from documents: a 947 with an adjustment id, posting a referenced 701/702 — never a retroactive snapshot that rewrites history.

(c) Design. A daily snapshot comparison against the in-flight ledger (so timing never reads as shrink), reason-coded into timing / mapping / status / physical with an owner per class; a tolerance per value band published with Finance *before* the first comparison; serialized equipment at zero tolerance — every unexplained serial is a unit whose state is unknown, which is exactly the work; idempotent snapshots (date + warehouse) and adjustment ids so replays never post twice; the delta report read before every close, trending from "pipe" causes to "physical" causes as the pipes get fixed.

(f) Metric. Reconciled delta: SKU × location rows outside tolerance after reason-coding → 0 on consecutive days; serialized equipment: 0 unexplained serials. Illustrative dollar: with ~$20M of equipment at 3PLs and a 2% unexplained delta, ~$400K of inventory has no reliable owner; converging to 0.5% frees ~$300K of buffer Finance no longer reserves against — before the truck rolls avoided by not promising phantoms. Supporting: unposted ship-confirms >48 h → 0; adjustments with reason code = 100%; month-end close with zero silent true-ups.

Deep-dive probes.

Which side wins?

Neither, automatically. SAP is the book for Finance and promising; the 3PL is the physical truth. Promising reads the lower until the reason is known; Finance sees the reason-coded delta; only the "physical" class becomes an adjustment.

How do you align grains?

Bins roll to the storage location; statuses map through one owned table; unit-of-measure and pack factors per SKU — the classic mock-1 defect class; serials as sets. Unalignable rows are mapping defects, not variances.

Idempotency?

Snapshots idempotent on date + warehouse; adjustments carry the 3PL's id — a replay never posts twice; the book adjustment references the 3PL document.

Failure modes?

An inventory snapshot arrives before overnight ship-confirms post — timing; compare against the in-flight ledger, not zero. A status the map does not know — "unknown, hold," never unrestricted. The 3PL "fixes" a count by re-sending corrected history — reject retroactive snapshots; adjustments go through documents.

Trade-offs?

Lower-of-two under-promises while the delta stands — a planning variance accepted over a customer defect. Zero tolerance on serials creates more exceptions — each is a unit whose state is unknown, which is exactly the work.

SOLUTION FLOW — THE SYSTEM MAP (PROPOSED)

```
[Vendors / EDI] --856 ASN -> CPI -> inbound delivery LIKP/LIPS, event-->
                                                                  |
[3PL WMS [BoR bin / serial scan]]                                 v
   bins, WMS status, cycle counts, holds      [SAP S/4HANA [BoR SLoc stock
   | 944 receipt -> 101, event                  + finance]]  (#2: IM grain)
   | 945 ship confirm -> PGI 601 + serials       MARD / MCHB / MSKA by
   | 947 adjust (qty or status) -> quarantined    MATNR x WERKS x LGORT
   |     until reason-coded -> 701/702 or 344     x stock type; EQUI serials
   | 846 stock snapshot at SKU x whse x status    MATDOC movement trail
   |     x serial, daily by 06:00 local           ^ 940 pick, daily wave
   v                                              |
[SAP CPI / Integration Suite]  ---- IDoc status WE02 (51 / 64) --------+
   idempotent on date + warehouse (846), 947 adjustment id, 945 id
   |                                              |
   v                                              v
[BI / BigQuery: recon_fact] <-- MARD/MCHB + MATDOC + EQUI, daily 06:30 --
   align bin->SLoc, status->stock type, UoM/pack; Δ by reason;
   in-flight ledger (940 w/o 945, ASN w/o 944, 945 not posted)
   | min(SAP, WMS) per SKU x node while Δ != 0, hourly
   v
[Salesforce / promising] reads the lower basis  ·  [Finance] reads Δ list
```

E6 — Elementum design engagement: PLM ECN → SAP material/BOM handoff, with obsolete-part disposition

Use for: engineering-change, BOM-integrity, obsolescence questions. **Record behind it:** the contract manufacturer's BOM made safe to plan on — phantoms, revision overlaps, one supply equation, per-site reconcile gate (S7) — the same problem from the SAP side.

> **Honest line.** I have not implemented a PLM system; I have run the part of the PLM→ERP handoff that hurts — effectivity, phantoms and where-used are what break planning. Volumes illustrative.

The problem as I restate it. Engineering releases change notices in the client's PLM (new revision, substitute component, end-of-life). SAP must get the material-master and BOM changes with correct effectivity; planning must not buy the obsolete part; stock of the obsolete part — say 5,000 units — needs a disposition; and the field must know which revision to install. Books: PLM for engineering design; SAP for the planning/costing BOM and inventory. Primary metric: ECN-driven exceptions — POs placed for obsolete parts, and revision overlaps in the planning BOM → 0.

(A) STOCKS AND FLOWS · (B) FIRST PRINCIPLES

```
[PLM in-work] --ECN released--> [Released revision] --transfer (change master,
  valid-from)--> [SAP material + BOM, new rev] --effectivity switch--> [MRP re-plans]
[Old-part stock, 5,000 units] --disposition--> run-out | rework | return-to-vendor | scrap
Purchasing BLOCKED at ECN approval, not at effectivity -- the gap between the two is
  where obsolete buys happen
```

Invariants. Two revisions are never valid on the same date — reject at transfer. Effectivity is never set in the past — planning would re-plan history. The BOM never arrives before its material — order the transfer or stage the BOM. A phantom assembly explodes through to components, so the reconcile gate must explode through phantoms or it will pass a wrong structure (the S7 lesson). Disposition is a signed sequence — Engineering → Planning → Finance — and purchasing is blocked at approval while issuing continues for run-out.

(c) Design. The ECN transfer carries the change master with valid-from; date effectivity for planning (MRP needs one switch), serial or lot effectivity for the field only where revisions are not interchangeable; a where-used check drives the disposition decision tree: still consumable in a valid assembly → run-out (push valid-from to the run-out date, if not safety-related); small modification → costed rework order; vendor accepts → return against the PO; otherwise scrap with Finance's write-down — approved *before* effectivity, with the workflow posting the status change itself so "approved but not blocked" cannot exist. Field protection: asset-record effectivity plus a bulletin so an old revision cannot be installed after the switch.

(f) Metric. ECN exceptions — POs placed for obsolete parts plus revision overlaps in the planning BOM → 0 per month, measured from open-PO data against material status and from the reconcile-gate log. Illustrative dollar: one avoided obsolete buy of 5,000 units × $40 = $200K; at ~10 obsoletes/yr with even 20% previously mis-bought ≈ $400K/yr, plus the write-down Finance no longer takes by surprise. Supporting: ECN → SAP lead time ≤ 1 business day; dispositions approved before effectivity = 100%; field installs of a superseded revision → 0.

Deep-dive probes.

Date vs serial effectivity?

Date for planning — MRP needs one switch; serial or lot for the field when old and new revisions must be distinguished at install. Both recorded on the ECN; the planning BOM uses the date.

5,000 units on hand — how do you decide?

Where-used first: is the part still consumable in any valid assembly? Run-out if yes and the change is not safety-related; rework if it's a small costed modification; return if the vendor accepts; scrap with the write-down otherwise. The signature order is Engineering → Planning → Finance, and purchasing is blocked at approval, not at effectivity.

How do phantoms behave?

A phantom explodes through to its components in planning; an ECN on a phantom changes the parent's effective components — the reconcile gate must explode through phantoms or it passes a wrong structure.

Failure modes?

Two revisions valid the same date — reject at transfer. Effectivity in the past — reject; planning would re-plan history. BOM before material — stage it. Disposition approved but status unchanged — the workflow posts the status itself. Field installs the old revision after effectivity — asset effectivity plus the bulletin.

Trade-offs?

Blocking purchasing at approval can starve a run-out if Planning misjudged — mitigate by allowing issue while blocking purchase, with a dated review. Serial effectivity costs field data discipline — use it only where revisions are not interchangeable.

SOLUTION FLOW — THE SYSTEM MAP (PROPOSED)

```
[Siemens Teamcenter [BoR engineering design]]
   Item / ItemRevision / BOMView / ChangeNotice (ECN), release status
   | ECN released -> T4S (or CPI/API, #4): MATMAS then BOMMAT, ordered,
   | with change master AENR + valid-from; per ECN, <= 1 business day
   v
[T4S / SAP CPI]  -- staging queue: hold BOM until material exists;
                    reject two revisions valid on one date; idem = ECN id
   v
[SAP S/4HANA [BoR planning BOM, stock, cost]]
   MARA/MARC (MMSTA/MSTAE status) · MAST/STKO/STPO under AENR
   CS15 where-used · MD04/MD06 MRP · EKPO open POs · MARD on-hand
   | where_used.snapshot per ECN (BOMs, POs, on-hand, installed), event
   | disposition approved (Eng -> Planning -> Finance) -> status change
   | PO cancel / convert -> 850 change -> 855 from vendor, per ECN
   | 122 RTV · 551 scrap · rework order · run-out (valid-from pushed)
   v                                    v
[Vendors / EDI] <- 850/855 ->     [Field app + Salesforce Asset]
                                    serial effectivity -> which rev to
                                    install; service bulletin, per ECN
[BI / BigQuery] <-- ECN log, reconcile-gate results, EKPO vs status,
                    revision overlaps, daily
```

E7 — Elementum design engagement: equipment supply & demand forecasting for a new-market build

Use for: forecast-to-buy, cold-start demand, launch-planning design questions. **Record behind it:** forecast-accuracy measurement and input integrity (MAPE/bias, 70→85%, the both-errors release rule, S6); lead-time distributions (S8); need floors so one channel cannot starve another (S1).

> **Honest line.** I have not owned a market-launch forecast end to end; the forecast number belongs to Planning — I own the equation, the measurement and the triggers. Newsvendor and safety-stock formulas are the MIT MicroMasters toolkit, labeled. Volumes illustrative.

The problem as I restate it. The client launches market N: build sites ramp by month from the build plan; demand follows adoption curves with no local history; kits and long-lead equipment have vendor lead times of weeks to months. Design the forecast-to-buy chain so installs are never delayed by stockouts and stranded stock stays inside a band. Books: the project system for the build plan; Planning for the adoption curve; SAP MRP for supply. Primary metric: install delays caused by stockouts → 0, with days-of-cover inside band.

(A) STOCKS AND FLOWS · (B) THE MATH

```
[Build plan] --milestones--> [Sites passed] --adoption curve x marketing-->
[Install demand] --kit BOM explosion--> [Component need by SKU x week, as PIRs]
--MRP (measured lead times, confirmed dates)--> [Planned orders -> PRs -> POs]
--856/101--> [Central 3PL pool] --STOs as sell-through proves the curve--> [Markets]
Newsvendor first buy: CR = Cu / (Cu + Co)
  missed install ~ $300 (re-roll + deferred revenue); holding ~ $3/mo
  CR ~ 0.99 -> order the 99th percentile of month 1-3 demand for the core unit;
  lower-Cu extras sit at the 70th (illustrative inputs; the method is the toolkit)
Re-forecast trigger: actual take-rate vs curve +/-20% for two consecutive weeks ->
  new version, re-explode kits, MRP on confirmed dates; bias sign published weekly
```

Invariants. Demand enters planning exactly once, through the engine — manually loaded requirements beside it are blocked. Lead times are measured distributions, never brochures (the S8 rule): the planned delivery time is the measured p50 with a drift flag, and the *confirmed* date drives the receipt, not the requested date. The first buy is held central: under maximum uncertainty one pool covers several markets' variance (risk pooling); stock forward-deploys only as sell-through proves the curve. Launch reserves carry need floors with hold-until dates — one market cannot steal another's launch; a floor breach is a governed screen with a reason code, not an email.

(f) Metric. Install delays caused by stockouts → 0 in the first 90 days of a launch, measured from reschedule reasons joined to the supply view; days-of-cover inside band per SKU × market; MAPE and bias published from month 2. Illustrative dollar: 2,000 installs/month × 3% stockout-caused reschedules × ~$300 ≈ $18K/month ≈ $216K/yr per market, plus stranded-stock avoidance — holding 20% of a $500K first buy central rather than in-market keeps ≈$100K from stranding. The client's take-rate and unit costs belong to its Planning and Finance — never invented.

Deep-dive probes.

Why central-first?

The first buy happens under maximum uncertainty; holding it central collapses the distribution — one pool covers several markets' variance — and stock transfers forward-deploy as the curve proves out.

Show the newsvendor logic.

Critical ratio = underage cost over underage plus overage. A missed install at ~$300 against ~$3/month of holding gives ~0.99 — order the 99th percentile of early-month demand for the core unit; accessories with a lower miss cost sit at the 70th. Illustrative inputs; the method is the toolkit, labeled.

Re-forecast trigger?

Actual vs curve ±20% for two consecutive weeks → new forecast version, kits re-exploded, planning on confirmed dates. Bias sign publishes weekly so Planning sees direction before magnitude.

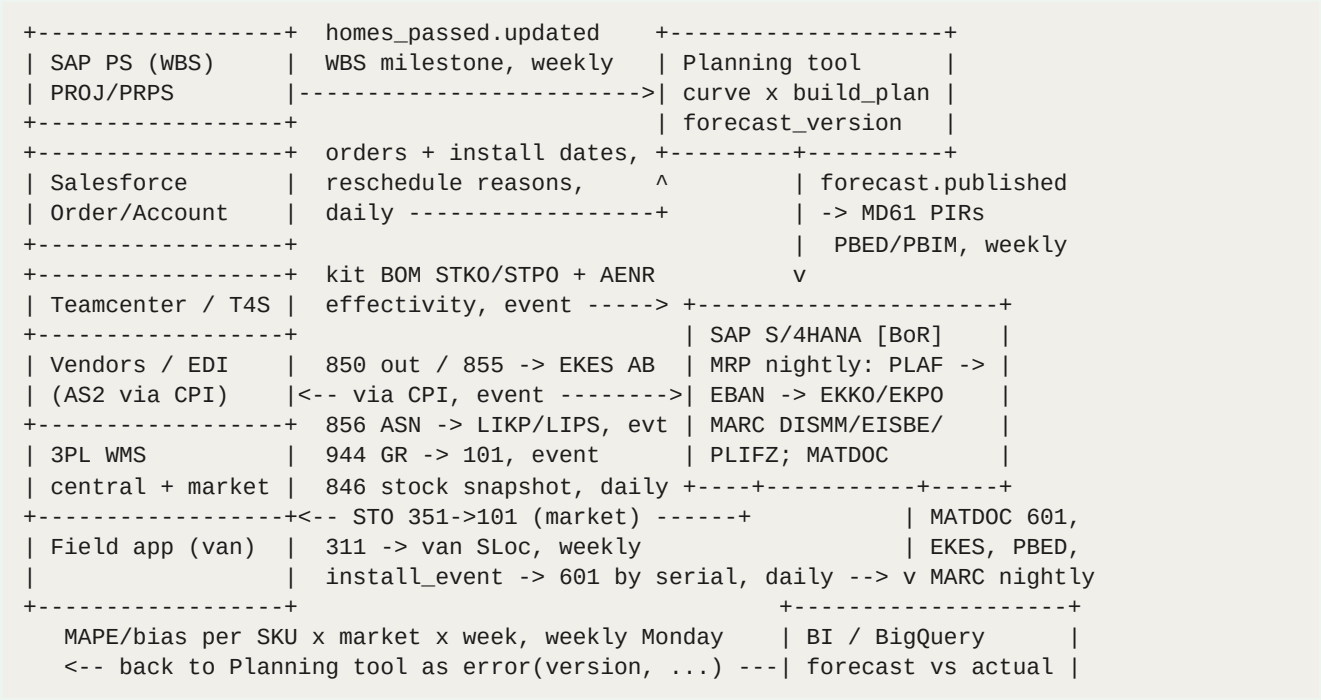
How do you stop one market stealing another's units?

Need floors and hold-until dates on launch reserves; only the uncommitted pool is reallocatable; a floor breach is a screen with a reason code.

Failure modes?

Requirements loaded manually beside the engine — block. A brochure lead time in the master — replace with measured p50, flag drift. Kit BOM effectivity undated — the ECN process (E6). An adoption curve borrowed from a non-analog market — archetype review with Marketing.

SOLUTION FLOW — THE SYSTEM MAP (PROPOSED)



E8 — Elementum design engagement: the monthly release / regression strategy for SAP + integrations

Use for: release management, regression strategy, "a release broke production" questions. **Record behind it:** run experience — the 4-week cadence with the QA lead owning the gate (S15); node × channel × transition regression matrices as the gate (S1); rule changes riding the cadence with a kill switch (S4).

> **Honest line.** This one is experience generalized, not adjacency. Volumes illustrative.

The problem as I restate it. SAP configuration, custom objects, integration maps, the CRM, the 3PL interface and the field app all change; a release broke MRP last month. Design the release train and the regression strategy so changes ship monthly with no surprise on the floor. Book of record: the transport/change log and the release manifest. Primary metric: release-caused Sev 1–2 incidents per release → 0, with the cadence held.

(A) THE TRAIN · (B) FIRST PRINCIPLES

```
[Backlog] --commit (freeze T-15d; definition of ready cites a regression cell)-->
[Release scope] --build--> [DEV transports/packages] --promote T-10d--> [QA integrated]
--regression matrix + E2E chain, T-5d--> [QA verified] --go/no-go T-2d (QA lead owns
the gate; Blocker+Critical = 0; rollback plan; freeze check)--> [PROD] --hypercare,
numeric exit--> [closed]
Rules: scope swapped, never absorbed · config and code in separate transports ·
emergency changes counted and 100% retrofitted · freeze around peak and close
```

Invariants. A change rides the train or does not ship. The regression matrix at the true grain is the gate, and every cell has an owner. The manifest is frozen at T−15; new work is estimated and swapped, never absorbed. Transports import to QA in the same sequence as production; nothing goes "just this once" outside the manifest. Hypercare exits on a number, never a calendar.

(c) Design. The MRP-protection cell: run MRP in QA on a production-like snapshot before and after the transport, diff the exception counts and planned-order totals by plant — unexplained deltas fail the build. Integration maps regression-tested with golden files per transaction set replayed against the new map version, payloads asserted field by field, contract tests against the partner's staging endpoint, dead-letter queue empty after replay. The CRM and field app ride the same manifest and the same go/no-go; if an app-store lag decouples them, SAP-side changes deploy dark and enable when the app reaches the field.

(f) Metric. Release-caused Sev 1–2 per release → 0, cadence held (12 trains/yr, no skipped month), measured from the incident ledger joined to the deploy log by object. Illustrative dollar: one MRP-breaking release that mis-plans a week of equipment can idle a build crew (~$50K/week) and strand or delay ~$200K of POs; avoiding two such incidents a year ≈ $300–500K/yr — before the trust cost of a floor that stops believing the calendar. Supporting: emergency changes ≤1 per release, 100% retrofitted; hypercare exit ≤ T+5; regression cells owned = 100%.

Deep-dive probes.

Why did the release break MRP, and what cell catches it?

Typical causes: a changed MRP type, lot size or planning strategy; a BOM effectivity landing early; a planning-file change; a job variant. The cell: run MRP in QA on a production-like snapshot before and after, diff exception counts and planned-order totals by plant; unexplained deltas fail the build.

How do you regression-test integration maps?

Golden files per transaction set replayed against the new map version; the produced payload asserted field by field; contract tests against the partner's staging endpoint; the dead-letter queue empty after replay.

Transport discipline?

A managed change list; QA imports in production sequence; configuration and code in separate transports so a config rollback doesn't drag code.

Failure modes?

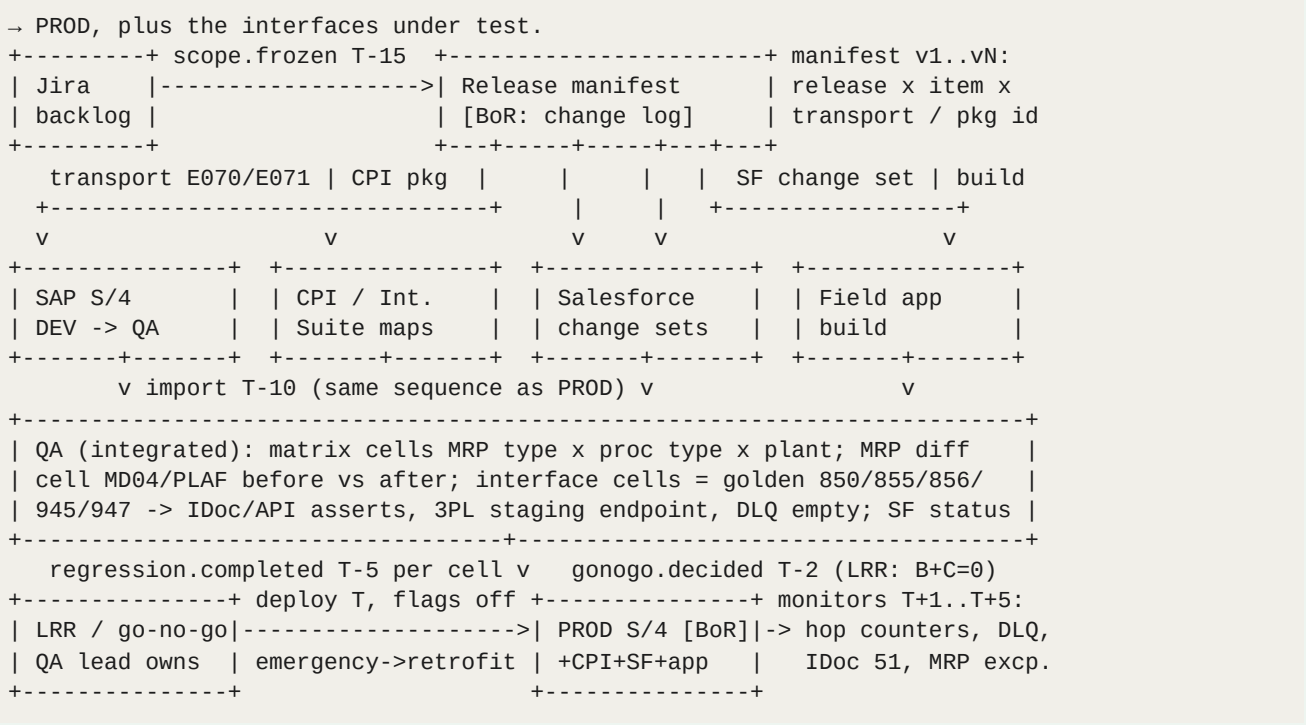
Scope creep after freeze — swap, never absorb. Regression passed on synthetic data missing the failing case — production-like snapshot, anonymized. A transport imported out of order — sequence check. Hypercare exited on the calendar — numeric exit only. Emergency changes not retrofitted — counted in the postmortem, retrofit ticket mandatory.

Trade-offs?

A 4-week train delays small fixes — the emergency path exists and is counted. Automation is expensive up front — build the manual matrix first, automate the cells that fail most.

SOLUTION FLOW — THE SYSTEM MAP (PROPOSED)

The "data" here is the release artefact: transport → regression cell → interface test cell → go/no-go (LRR)



E9 — Elementum design engagement: the ERP cutover from a legacy WMS — tolerance, mocks, freeze, one promise publisher

Use for: cutover/migration design questions — the designed generalization of S18's run experience. **Record behind it:** the 0.5% tolerance published before mock 1, mocks 7 → 1.8 → 0.3 read by cause, the 72-hour freeze, the RF fallback (S18); zero-then-publish (S2); book-of-record discipline (S1, S3).

> **Honest line.** The design is the generalized playbook of a migration I ran; mechanics from the record, volumes illustrative.

The problem as I restate it. A DC (or a 3PL site) runs a legacy WMS with its own inventory book; the client wants the ERP as book of record, with no double-promising during the switch and a Finance-grade reconciliation at go-live. Book of record: the legacy WMS until the switch instant; the ERP after. Primary metric: cutover reconciliation variance vs a published tolerance, and zero double-promises.

(A) STOCKS AND FLOWS · (B) FIRST PRINCIPLES

```
[Legacy on-hand, bin x status] --extract + map (UoM, location, status vocab)-->
[Staging, SKU x SLoc x stock type x serial] --load (561 / migration cockpit,
control totals)--> [ERP initial stock]
[In-flight receipts/shipments] --72-h freeze + their own batch--> [ERP in-transit]
[Open orders] --convert / re-create at cut-off--> [ERP deliveries]
[Both books] --reconcile by cause vs published tolerance--> go/no-go
Promise switch: zero the legacy availability FIRST, then publish the ERP's --
  never the reverse; consumers read one publisher at a time
```

Invariants. The tolerance is published with Finance before the first mock, so "good enough" is arithmetic, not negotiation. A mock is read by cause class, never by headline — a total that shrinks without a named cause is not evidence. The initial load has one door (migration postings with control totals); nothing is "fixed" afterwards outside the governed path. In-transit gets its own batch behind the freeze — the mock-2 lesson. Serials load only after their masters exist, at zero tolerance.

(c) Design. Three mocks on full production data, each a rehearsal of the real cutover including the failback; mapping sets as data (units of measure, location names, status vocabularies — the mock-1 defect classes); the cut-off list that catches late receipts posted against legacy orders; device readiness as a program row with a rehearsed manual fallback (the S18 RF lesson); hypercare with a numeric exit (<2 blockers/day sustained). If the 3PL keeps its WMS, the ERP integrates at storage-location grain over the standard document set (E5); if the client runs the floor, an embedded warehouse module with bin mapping — decided by who runs the floor, not by preference.

(f) Metric. Cutover reconciliation variance vs the published tolerance — e.g., 0.3% against 0.5% at go-live, 0 unexplained serials — with zero double-promises during the switch, measured by the reconciliation-by-cause report at mock 3 and at T+1. Illustrative dollar: the cost of missing the date is the program's own number — in my record it was a $20M pipeline at ~$2.5M/month of slip; for a new client I ask Finance for the equivalent rather than invent it. Supporting: hypercare exit ≤ T+3 weeks; manual stock corrections post go-live outside the governed path = 0.

Deep-dive probes.

Load mechanics?

Migration cockpit or initial-entry postings with control totals; batch and serial masters pre-loaded before balances; value flows to the migration clearing account and Finance reconciles it. The migration posting is the only door — never "fix" with count adjustments afterwards without a cause.

What produced the 7% in mock 1 and how did it get to 0.3%?

Units of measure, location names that mapped to nothing, and order-status vocabulary — fixed in the mapping set (1.8%); then in-transit timing — fixed with the 72-hour freeze and a separate batch (0.3%). Each cause had an owner; the number was read by cause every time.

How do you avoid double-promising during the switch?

Zero the legacy promise source first — legacy availability reads zero to the order systems — then publish the ERP's availability. Never the reverse; consumers read one publisher at a time.

Failure modes?

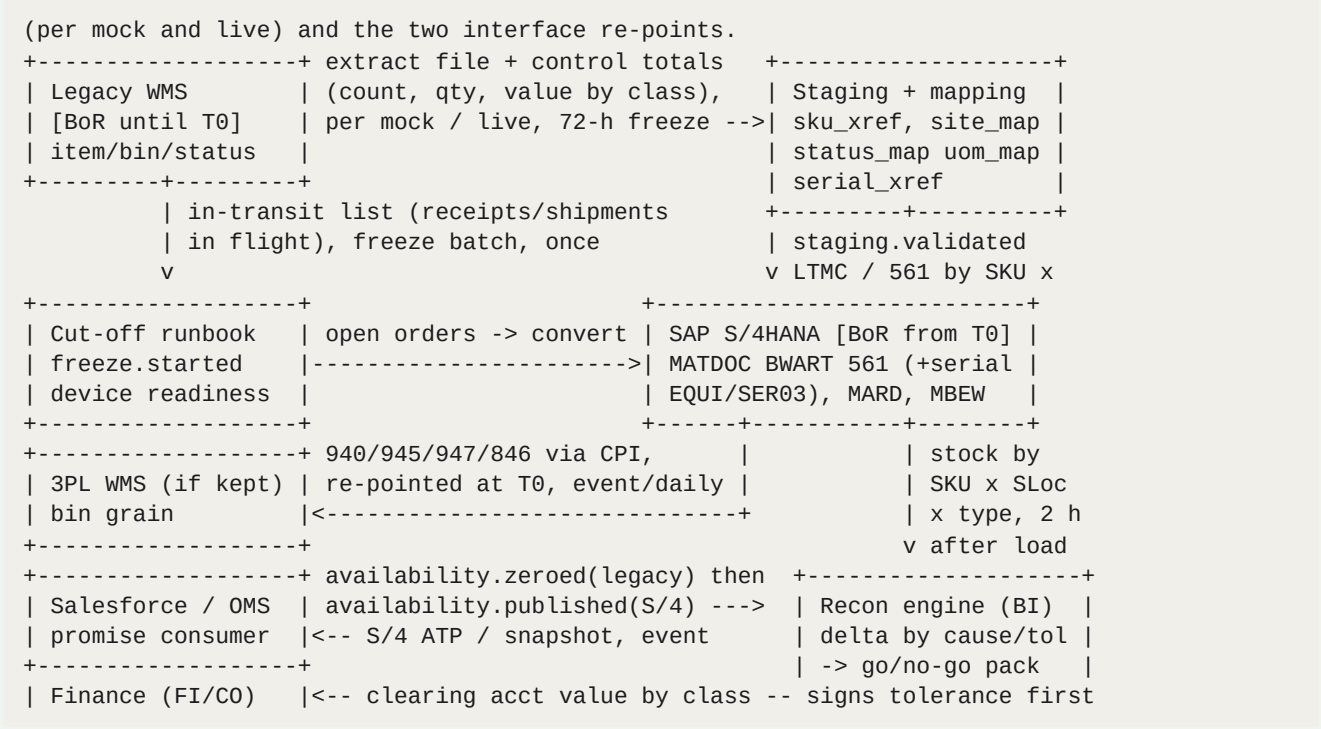
Control totals match but value doesn't — valuation/price mapping, the Finance row. Late receipts posted against legacy orders — the cut-off list. Serials without masters — pre-load. Devices on old screens — readiness row, rehearsed fallback. The one extract expert on vacation — documented, rehearsed by a second.

Trade-offs?

Three mocks cost weeks — they are the weeks that would otherwise be spent live. A 72-hour freeze delays receipts — priced with operations, scheduled off-peak. Zero tolerance on serials produces exceptions — each is a real unit whose state is unknown.

SOLUTION FLOW — THE SYSTEM MAP (PROPOSED)

Before the switch instant the legacy WMS is [BoR] ; after it, S/4 is. The diagram shows the load path once



E10 — Elementum design engagement: the returns / RMA flow — premise → test → refurb / return-to-vendor / scrap

Use for: reverse-logistics, warranty-recovery, disposition design questions. **Record behind it:** bucket state machines with one owning system per transition (S1, S2), reason-coded reconciliation (S14), serial-grain reconciliation at cutover (S18).

> **Honest line.** No RMA program run end to end; the design generalizes the serialized state-machine discipline. Volumes illustrative.

The problem as I restate it. Customers disconnect or swap; techs bring units back; some go straight to a returns bin and are never tested; refurb yield is unknown; warranty recovery from vendors is leaking. Design the reverse flow so every returned serial is tested, graded and dispositioned, and in-warranty failures go back to the vendor. Books: SAP for serial state and stock; the RMA system for reason and test result. Primary metric: returned units dispositioned within SLA, and the warranty recovery rate.

(A) STOCKS AND FLOWS · (B) FIRST PRINCIPLES

```
[Installed base] --disconnect/swap (RMA w/ reason code)--> [Return-in-progress]
--pickup / tech return (651 by serial)--> [Returns-untested, market RMA SLoc]
--weekly consolidation (STO)--> [Central test center] --test-->
  [Refurb-good A/B (321)] | [Repairable] | [Failed (344 blocked)]
Failed + in-warranty --> RTV w/ vendor RMA number (122 / return PO) -> credit
Failed + out-of-warranty --> scrap (551, with certificate)
No test, no grade -- an untested return that is promisable is a truck roll
```

Invariants. Every returned serial passes through test — no grade, no availability. Refurb identity: same serial, same material, a grade attribute — never re-numbered, so the asset history stays continuous. Warranty eligibility computes from the serial's own receipt or install event per the vendor's contract term — never from a person's memory of a PO date. A return-to-vendor without the vendor's RMA number is blocked — the credit would be denied. Scrap of an in-warranty serial is prevented by rule; the exception needs Finance approval.

(c) Design. Return order with reason code at initiation; receipt by serial into returns-untested; weekly consolidation to one central test bench — one grading standard beats faster-but-unreliable grading on the truck; test-result events drive the state machine; graded availability published with the grade so the order system can decide which customers receive grade B (policy, not technology); the RTV credit reconciliation as a standing report. Idempotency: serial + event type + document; a repeated test result replaces only if newer and flags the earlier; a repeated return receipt for the same RMA is rejected.

(f) Metric. Dispositioned within SLA ≥ 95% within 14 days; warranty recovery on in-warranty failures ≥ 80%. Illustrative dollar: 48K returns/yr × 75% refurb yield × $120 replacement cost avoided ≈ $4.3M/yr of new-unit purchases displaced (net of ~$25 test cost on all returns ≈ $3.1M net); warranty recovery lift 40% → 80% on 8K failures × 60% in-warranty × $150 ≈ $290K/yr. Supporting: returns-untested aging p95 < 14 days; refurb field-failure rate ≤ new-unit rate + margin; scrap with certificate = 100%.

Deep-dive probes.

Movement types and documents?

Return delivery to returns stock (651; to QI if inspecting at receipt — never straight to unrestricted); QI → refurb location on pass; unrestricted → blocked for fails; stock transfers for consolidation; return against the original PO for RTV; scrap with reason; a write-off for a non-return only with a case reference.

Warranty start — ship or install?

Contract-dependent; store the term and start event per vendor and compute eligibility from the serial's receipt or install event. Never a person's memory.

Failure modes?

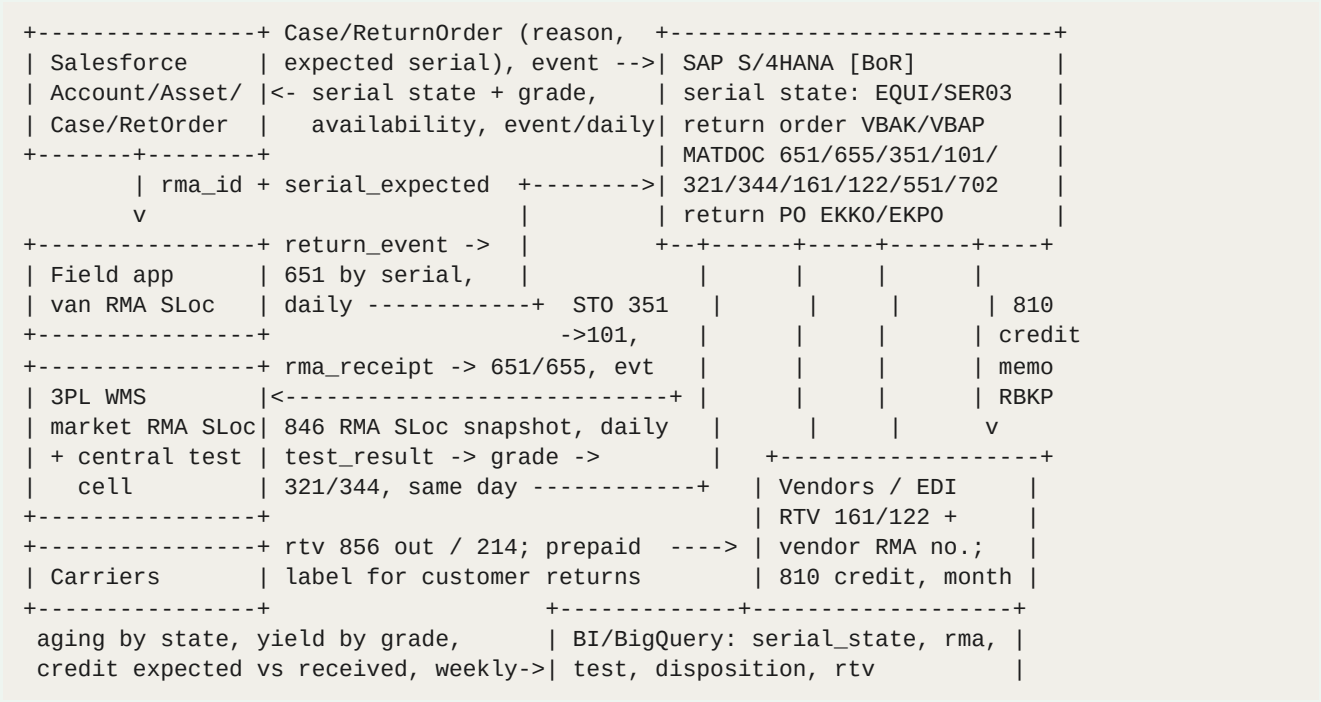
A tech "returns" a unit that never posts — that's E1's unaccounted; the return is a swap event on the van. A test result for a serial not in returns-untested — hold; the state machine refuses. RTV shipped without a vendor RMA number —

blocked. Scrap posted for an in-warranty serial — rule prevents it.

Trade-offs?

Centralized testing adds a hop and days — but one grading standard. "No test, no grade" means refurb stock lags during a surge — better than a truck roll on a bad unit.

SOLUTION FLOW — THE SYSTEM MAP (PROPOSED)



E11 — Elementum design engagement: direct procurement of long-lead equipment — PO lifecycle, confirmations, ranked expediting

Use for: procurement design, vendor-reliability, long-lead-time questions. **Record behind it:** SAP purchasing data domains and EDI confirmations (S6, S9); lead-time distributions (S8); the carrier bid's constraint rules (S17).

Honest line. The indirect-procurement platform boundary is a knowledge pattern — I have not configured one; the boundary design is how I would frame the decision, not a claim. Volumes illustrative.

The problem as I restate it. Core equipment carries 12–40-week lead times, allocation risk, and multi-site delivery; today POs are placed off spreadsheets, confirmations live in email, and expediting is heroic. Design the procure-to-receive lifecycle in SAP for direct materials, define the boundary with the indirect-spend platform, and make confirmed-date reliability a managed metric. Primary metric: on-time-to-confirmed-date receipts, and PO lines without a vendor confirmation → 0.

(A) STOCKS AND FLOWS · (B) FIRST PRINCIPLES

```
[Demand: build plan / MRP need by week] --requisition (release strategy)--> [PR approved]
--source (source list / info record / contract)--> [PO sent, 850 out]
--confirm <= 48 h (855 -> confirmation record, MRP-relevant)--> [PO confirmed]
--at risk? expedite ranked by milestone-at-risk, not by who shouts--> [re-confirmed]
--ship (856) / receive (101) / invoice (810, three-way match)--> [closed]
OTIF measured BOTH ways: to-request (our planning discipline) and to-confirmation
  (vendor reliability) -- managed to confirmation; the gap published separately
```

Invariants. A confirmation without a date is not a confirmation — after 48 hours the line is a planner exception. Planning runs on the *confirmed* date, not the requested one, so the plan reflects reality rather than hope. Every urgent buy goes through a requisition — spreadsheets creeping back is a failure mode, blocked by the release strategy. A PO changed after confirmation without re-confirmation is flagged.

(c) Design. Confirmation control made MRP-relevant so the vendor's date replaces the schedule line in planning; expedite list ranked by the milestone at risk × value, refreshed weekly; for allocated long-lead items: outline agreements with scheduled releases, dual sourcing where qualified, and a strategic buffer sized by the p99 lead-time tail against the build plan's variance (toolkit, labeled). The direct/indirect boundary as a category table owned with Procurement and Finance: items that become stock or project stock → SAP MM; services, consumables expensed at receipt, tools, software → the indirect platform, which posts to the GL via the standard integration and never creates stock.

(f) Metric. Receipts on or before the vendor's confirmed date ≥ 95%; PO lines lacking a confirmation after 48 h → 0 — measured per line, monthly per vendor. Illustrative dollar: a late shipment that idles a build week costs ~$50K in crew and deferred activations; confirmation discipline plus ranked expediting avoiding ~15 such weeks/yr ≈ $750K/yr; the measured-lead-time buffer that prevents one allocation stockout protects a quarter's plan — the client Finance's number, not mine. Supporting: requested-vs-confirmed gap published; receipts without ASN → 0 for EDI vendors; spreadsheet POs → 0.

Deep-dive probes.

Confirmation control — how does it change planning?

With an MRP-relevant confirmation category, the confirmed date and quantity replace the schedule line for planning; the supply view shows the confirmed receipt. A PO with no confirmation shows the requested date — which is why the 48-hour exception exists.

OTIF to confirmation or to request?

Measure both, manage to confirmation: it isolates vendor reliability from our own over-asking. The requested-vs-confirmed gap is its own planning-discipline metric.

Allocated items with 40-week lead times — what do you do?

Outline agreements with scheduled releases; dual source where qualified; a strategic buffer sized from the p99 lead-time tail and the build plan's variance — toolkit, labeled; expediting ranked by the milestone at risk, not by who shouts.

Where exactly is the direct/indirect boundary?

Items that become stock or project stock live in SAP MM; services, expensed consumables, tools and software live on the indirect platform, which posts accruals to the GL and never creates stock. The boundary is a category table owned with Procurement and Finance — decided once.

Failure modes?

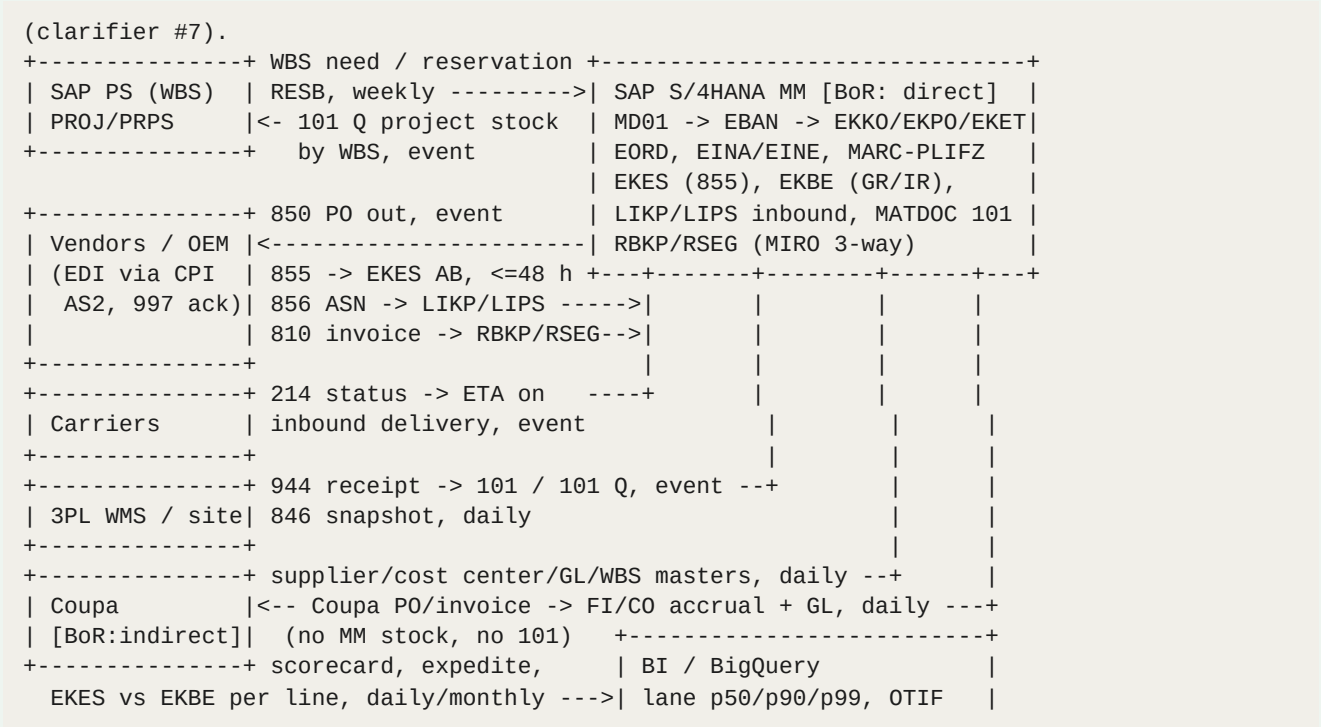
"Acknowledged" with no date — unconfirmed. A PO changed after confirmation without re-confirmation — block or flag. A receipt posted to the wrong plant or project — account-assignment check. A confirmation for a closed PO — exception. Spreadsheets creeping back for urgent buys — the release strategy requires a requisition.

Trade-offs?

Confirmation discipline creates exceptions in month one — that is the diagnostic; it reveals the unreliable vendors. Project stock improves site accountability but fragments the pool — use it only when the site is fixed.

SOLUTION FLOW — THE SYSTEM MAP (PROPOSED)

Direct (becomes MM or project stock) lives entirely in S/4; Coupa handles indirect and never creates stock



E12 — Elementum design engagement: smart vending machines — the consignment ledger, and why a tech van is a vending machine

Use for: open-ended system design, event-ledger design, consignment/working-capital questions — the showpiece design.
Record behind it: the real-time decrement ledger with idempotent events and daily reconciliation to a settlement book (S1); one unit in one bucket, counts change the book only through posted documents (S1, S14); lower-of-two when ledger and physical disagree (S1, S18).

> **Honest line.** I have not built a vending network or run a consignment program; the par-level and safety-stock formulas are the MicroMasters toolkit, labeled; every number is illustrative.

The problem as I restate it. A network of connected vending machines sells product with no human at the counter: slots with a planogram, dispenses on payment, trucks on refill routes, machines sometimes offline. Design the inventory ledger, the payment flow, the replenishment and the reconciliation so the operator, the supplier and Finance agree on what was sold, what is on the shelf, and who owns it. Books: the dispense-event ledger for units, the ERP as the settlement book; the payment processor for money. Primary metric: fill rate (dispenses served / attempted) with shrink % inside tolerance.

Why this design matters beyond vending. It is the consignment question in its purest form: if the supplier owns the product on our shelf until it is dispensed, that is special stock K — receipt creates custody, not liability; the dispense is the consumption that creates the liability; month-end is a settlement, not an invoice. **And a field tech's van is a vending machine**: the van is the machine, the install is the dispense, the weekly restock is the refill, the van kit is the planogram. E1 is the own-stock version of this design; E12 is the version where the vendor owns the unit until the dispense — for a capital-hungry operator, a working-capital lever: hundreds of vans of equipment stop sitting on the balance sheet until the day of install.

(A) CLARIFIERS · STOCKS AND FLOWS · (B) THE MATH

```
Clarifiers first (five): scale (default 500 machines x 40 slots x ~12 dispenses/day
  ~ 6,000 events/day) · ownership (own stock vs consignment K -- K is the interesting
  case) · connectivity (assume a machine can be dark for hours) · payment (offline auth
  allowed to a small cap) · replenishment (our drivers, fixed weekly routes, handheld)
[Depot (own or K)] --load route truck--> [In-transit per route x stop] --refill event
(counted_before + qty_added + planogram version)--> [Slot on-hand] --dispense (payment
auth -> motor -> capture; event_id = machine x slot x seq)--> [Sold / consumed (411 K)]
Monthly: settlement = Sigma consumption x agreed price (MRKO -- a settlement, not an
  invoice), three-way matched: consumption vs delivery notes vs refill counts
Par per slot = mu_d*(R + L) + z*sigma_d*sqrt(R + L), capped at slot capacity C
  e.g. mu=1.2/day, sigma=0.8, R=7, L=1, z=1.65 -> 9.6 + 3.7 ~ 13; capacity 15 holds;
  if capacity were 10, the slot is capacity-bound -> the ROUTE INTERVAL moves, not the par
Refill trigger: on_hand < Par - mu_d x (days to next scheduled visit)
Route ranking: Sigma_slots P(stockout before next visit) x margin per dispense
Shrink_slot = ledger_before_refill - counted_before; shrink % reason-coded per route
Worked network: 20,000 slots · ~6,000 events/day (~250/hr, 10x lunch peak ~2,500/hr --
  correctness is the hard part, not throughput) · weekly routes -> ~71 visits/day ->
  3 trucks x ~25 stops · shrink 1.5% of ~42,000 refilled units/wk x $1.50 ~ $50K/yr
```

Invariants. Every unit is in exactly one stock; the dispense event is minted at the edge (journaled before the motor turns) so a crash between dispense and send still produces exactly one event. Money is stricter than units: auth is synchronous or offline-capped; capture is at-least-once with idempotent ids; a refund is a compensating event referencing the original — never an edit. The count changes the book only through a posted refill count; the settlement is reproducible from the event store or it is not trusted.

(f) Metric. Fill rate ≥ 98% with stockout slot-hours published per route; shrink ≤ 1.5% (illustrative tolerance). Illustrative dollars: 500 machines × 12 dispenses × $2.00 ≈ $12K/day of revenue; a 2-point fill-rate lift on a network stocking out 5% of the time recovers ≈ $85K/yr; the van-consignment analogue defers cash on ~500 vans × 30 units × $150 ≈

$2.25M of stock off the balance sheet until install — Finance's number to confirm, not mine. Supporting: dispense → ledger latency p95 < 60 s online, backlog drained < 10 min after reconnect; payment success ≥ 99%; offline exposure loss < the margin offline enables.

Deep-dive probes.

Event ordering with clock skew?

Order by the machine's monotonic sequence number per machine × slot, never by device wall-clock; carry both device and received timestamps. Cross-machine order doesn't matter — each machine is its own stream; device time is corrected from the heartbeat offset for reporting only.

Idempotency key design?

machine × slot × sequence, minted at the edge and persisted in the journal before the motor turns — a crash after the dispense and before the send still produces one id. The server keeps a per-machine high-water mark plus a short dedupe window.

Offline authorization limit?

Per machine: a cap on count and amount per offline hour, decaying as the outage lengthens; per card: one offline auth per machine per day; a floor limit on price. Set the cap where expected decline loss < the margin the offline window enables.

How do you detect a jammed slot from data?

Auth without a capture on the same slot; dispense events with a jammed outcome; a slot whose ledger never moves while neighbours do; a refill count equal to the ledger plus a run of voided auths. Two together raise the alert before the customer calls.

Reconciliation cadence?

Per dispense: auth ↔ capture ↔ event, continuous. Per refill: count vs ledger per slot — the physical inventory. Daily: captures vs the processor settlement file. Monthly: consumption vs supplier delivery notes vs refill counts — the three-way match behind the settlement. Unit-ledger tolerance zero; shrink tolerance negotiated in the consignment agreement.

Consistency model and scale?

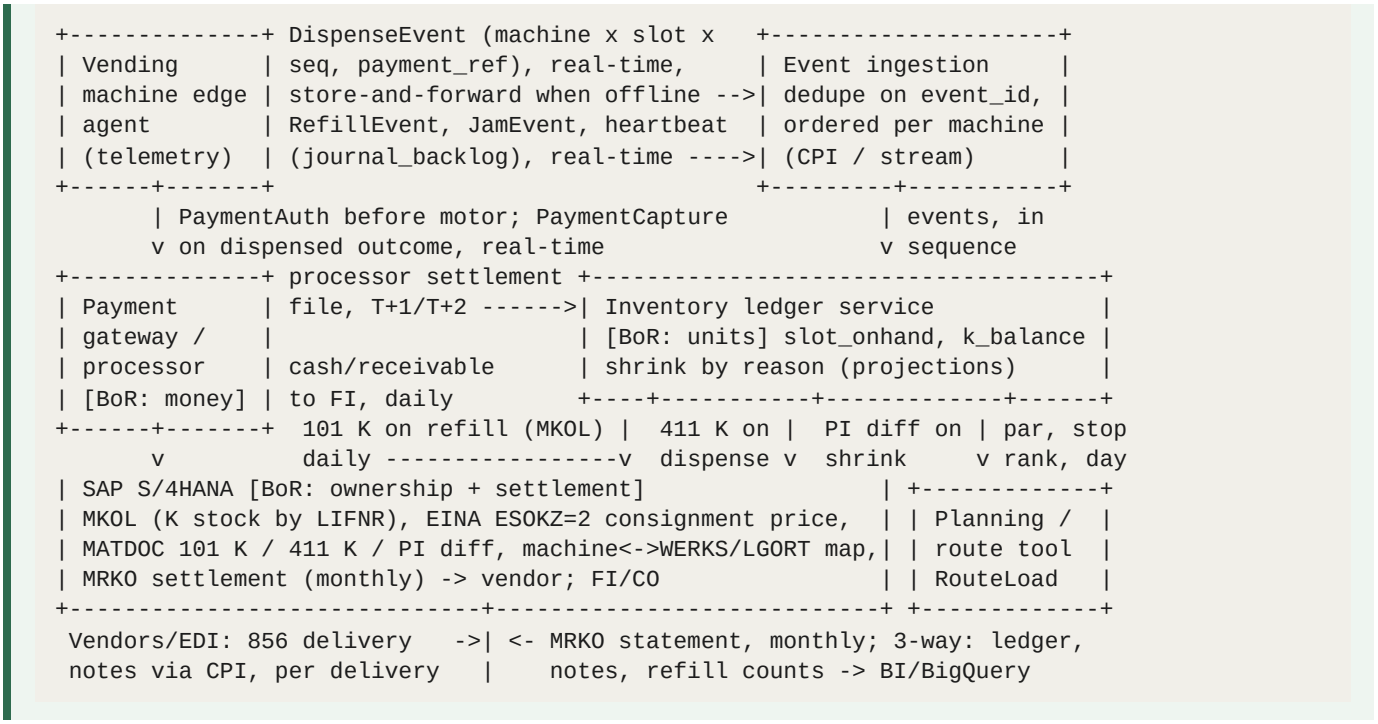
Eventually consistent projections over a strongly ordered per-machine event log; partition by machine id; consumers idempotent on event id; projections rebuilt from the log on demand. Six thousand events a day is small — the design holds the same shape at 50,000 machines; only the partition key count changes.

Fraud?

Repeated offline auths on one card across machines; refund-outcome clusters on one machine; shrink concentrated on one route or driver; door-open events without a refill event. Controls: offline caps, remote lock, driver-level shrink reporting, refund rate per machine on the daily dashboard.

SOLUTION FLOW — THE SYSTEM MAP (PROPOSED)

and settlement (special stock K, MRKO), the payment processor is [BoR] for money. In the GFiber analogue the machine is the tech van and the dispense is the install.



3.99 — Cross-cutting: how I open any "design X" prompt (say this in the first minute)

1. **Book of record and metric first** — "SAP is the book; the metric is unaccounted units per contractor per month — agree?"
2. **Stocks on the whiteboard** — every bucket a unit can sit in; then the flows, each with the document that records it.
3. **The invariant that is broken today** — a unit in two stocks, a flow with no document, a number with two owners.
4. **Minimal design the invariants force** — then components, grain, events, controls, NFRs.
5. **Sequence the phases** (Reach × Impact × Confidence ÷ Effort) — and say which invariant overrides the score.
6. **One metric, one dollar** — illustrative, labeled, measured from posted documents.
7. **How I'd run it** — point to §3.0 once; never re-explain the method per prompt.

Employer block 4 — Manhattan Associates (back-running)

S13 Cyber Monday 4 AM allocation · S14 the $2.5M OMS↔WMS discrepancy

The oldest records — prep them last; they back-run the loop as the peak-incident and Earn Trust anchors (S13 is still the Customer Obsession / Bias for Action primary in §1.1).

S13 — Manhattan: Cyber Monday, 4 AM — allocation across 3 DCs and 100 stores when the scheduler died

Amazon LPs: Bias for Action (primary) · Customer Obsession (primary backup — the promise ranked first) · Deliver Results. **Crosswalk:** loop packet S14 — consistent. *Older story — if asked for recent, pair with the UDC robotics throttle (S2): same muscle, 2024–25. Never "$2MM baseline."*

Honesty label. OMS consultant on allocation, promising and the OMS–WMS handshake; engineering fixed the scheduler once I isolated it; the post-mortem mechanisms were mine to drive. $8M/$6M/99% are record; the margin and cost figures are my model at benchmark rates.

60-SECOND PITCH

"**Situation.** Manhattan OMS consultant on allocation and promising across 3 DCs and 100 stores; 4 AM Cyber Monday the VP called — backorders piling with inventory present, ~$6M of sales at risk against a 95% fill-rate goal. **Task.** Own the response before the root cause was known. **Action.** Discovery: the contradiction (ATP > 0, backorders rising) was the entry bar to declare an incident. Plan & Design: I ranked by promise-at-risk, not FIFO; node choice = the promise — near 0.85 days vs far 4.60. Build & Test: this was an incident — one bridge, every rebalance timestamped; reason-coded alerts I had already built. Cutover / clear: rebalanced so no DC drowned; status every two hours until 9 PM. Hypercare / close: root cause — the allocation scheduler died silently at the daylight-saving change; post-mortem shipped a heartbeat, a DST checklist, a 3σ page rule, and a rehearsed rebalance playbook. Lesson: decide reversibly, log it, find the cause later. **Result.** Primary metric: peak fill rate — 95% goal → **99% delivered**. $ impact: ~$8M of peak sales protected against ~$6M at risk (record); my model puts that at ≈$3.2M of gross margin at a 40% benchmark margin; structural stockouts −20% over the engagement."

SITUATION / TASK — WHERE IT OCCURRED, WHEN IT HAPPENED, WHY IT IS IMPORTANT

S — OMS consultant on order allocation, promising and the OMS–WMS handshake for a strategic retailer — 3 DCs, 100 stores — on call 6 AM–9 PM for the holiday promotion. 4 AM Cyber Monday, the VP of Supply Chain on the phone: backorder lines piling up with plenty of inventory; promises aging; ~$6M of sales at risk against a 95% fill-rate goal.

T — Nobody had written a clearing rule for failure. I owned the response.

The goal: clear the right orders first and hold the promise through the peak. The challenge: act before the root cause was known, without making it worse. Every hour of waiting cost promises — the queue was aging while inventory sat idle; a wrong intervention could corrupt allocation; and peak was the one window where the retailer could not absorb a miss.

The metrics on the table, with their baselines: fill rate — the 95% goal, with ~$6M of sales at risk against it at 4 AM; time-to-detection — a VP phone call, the baseline the post-mortem's heartbeat existed to retire; structural stockouts — the standing run-rate the engagement later cut.

ACTION — WHAT I PERSONALLY OWNED, HOW I DID IT, WHO ELSE WAS INVOLVED

A — Five gated phases — the incident's phases (hours), plus the post-mortem program (weeks; durations illustrative):

1. **Discovery (04:00–05:00)** — triage read: symptom (allocation stalled), exposure (~$6M vs the 95% goal), what I did not know (root cause); decided to act before the cause. *Exit bar:* bridge open; exposure sized; the clearing question framed as "which orders first," not "what broke."
2. **Plan & Design (first hour)** — the clearing rule (rank by promise-at-risk, not FIFO), the rebalance plan across three DCs (near node ~0.85 d vs far ~4.60 d), the two-hourly comms cadence onto the bridge log. *Exit bar:* rule on the log; VP agrees to the ranking.

3. **Build & Test (morning)** — manual clears in promise-at-risk order; rebalanced allocation so no node drowned while others idled; read my reason-coded alerts for the cause. *Exit bar:* backorder queue draining; alerts point to the scheduler.
4. **Cutover (root-cause fix)** — handed the isolated DST scheduler death to engineering with the evidence; verified automated allocation resumed; timestamped the log. *Exit bar:* scheduler restored; manual clears stopped; fill trending to 99%.
5. **Hypercare / Close (post-mortem)** — blameless post-mortem off the log timeline; mechanisms driven: scheduler heartbeat, DST checklist, 3σ page rule, rehearsed rebalance playbook, surge-scaling rebuilt. *Exit bar:* mechanisms live before the next peak; "the next peak ran boring."

The response ran mine end to end — the triage read, the exposure sizing, the ranking rule, the rebalance plan across three DCs, the two-hourly cadence and the post-mortem mechanisms; engineering fixed the scheduler once I isolated it, and the client's teams executed the clears. The contribution: the clearing rule itself — promise-at-risk, not FIFO — written on the bridge log in the first hour so nobody re-argued it at noon, and the isolation of the cause through my own reason-coded alerts: the allocation scheduler had died silently at the daylight-saving change.

The obstacles: the VP wanting the fastest clear — one sentence on the difference between fastest and best, then the rule written in the log; acting without corrupting allocation — checked with engineering first, every rebalance timestamped; and carrying the first stretch solo, which is one of my do-differentlys. Promise-at-risk order set the priorities — the orders whose miss costs most, first; buy-in was the VP agreeing the ranking in the first hour, it going on the log, and two-hourly status with the same numbers keeping leadership steady until 9 PM. The decisions challenged: FIFO — the default everyone reaches for — because it processes cheap orders while at-risk promises age; and “wait for the root cause,” because waiting had an hourly price and a reversible, logged intervention did not.

RESULTS — HOW SUCCESS WAS MEASURED: $ · # · % · TIME · CUSTOMER · QUALITY

R — Primary metric: peak fill **99% vs the 95% goal**; ~$8M protected vs ~$6M at risk (record); structural stockouts −20% over the engagement.

- **$** — ~$8M of peak sales protected vs ~$6M at risk (record pair); ≈$3.2M gross margin at a 40% benchmark margin (modeled); structural-stockout margin run-rate ≈$1.3M/yr modeled on an illustrative $400M retailer.
- **#** — 3 DCs and 100 stores; the backorder queue drained in promise-at-risk order, every rebalance timestamped on one log (record).
- **%** — Peak fill 99% vs the 95% goal (record); structural stockouts −20% over the engagement (record).
- **Time** — Node choice = the promise: near node ~0.85 days vs far ~4.60 (record); status every two hours until 9 PM; detection moved to one interval via the heartbeat, not a VP at 4 AM.
- **Customer** — Orders cleared by promise-at-risk, not FIFO — the promises whose miss costs customers most, first (record policy).
- **Quality** — Blameless post-mortem off the log timeline; mechanisms shipped — scheduler heartbeat, DST checklist, 3σ page rule, rehearsed rebalance playbook — and the next peak ran boring (record).

Against the goal stated above: peak fill 99% against the 95% goal; ~$8M of peak sales protected against the ~$6M sized at risk at 4 AM; detection moved from a VP phone call to one interval via the heartbeat; structural stockouts −20% over the engagement — and the next peak ran boring. The trade: clearing by promise-at-risk meant some early, cheap orders waited — the explicit price of protecting the promises whose miss cost most.

PROBING — QUESTIONS, PHASE DEEP-DIVES, AND THE MATH

Challenge — Why did you choose this story to illustrate Bias for Action?

Because I decided before the cause was known — reversibly, logged, with the cause hunted in parallel. The clearing rule went on the bridge log in the first hour; manual clears ran in promise-at-risk order while my reason-coded alerts isolated the scheduler.

Challenge — What other stories demonstrate Bias for Action?

S18 — the rehearsed fallback that fired while the date held — is the backup; S14's two days to the executive review is the third.

Challenge — Could you come up with an example that is more recent?

Manhattan years — the oldest era in the set. The same muscle in 2024–25 is S2's UDC robotics throttle: a pre-built dial and a published gate — decide on a rule written before the crisis.

Probing — Why did you choose to focus on these results? What other results were important?

99% fill against a 95% goal and ~$8M protected against ~$6M at risk are the record pair — the event's result. The post-mortem's result matters as much: structural stockouts −20% over the engagement, and the next peak running boring.

Probing — You mentioned dollars — what percentage change is that year over year?

Not YoY — the ~$8M is one-time peak sales protected (sales that didn't die), never a saving; the margin translation (≈$3.2M at a benchmark 40%) and the stockout run-rate are my models, labeled.

Probing — I'm concerned about the stockout run-rate — tell me more.

Agreed — it is the biggest number and the softest: a 4% lost-sales-to-stockout benchmark on an illustrative $400M retailer; at 2–6% it runs $0.4M to $2.7M. I would measure it from the retailer's own out-of-stock lines times unit margin, before and after the playbook went in.

Challenge — What were the lessons learned? What would you have done differently?

Decide reversibly, log it, find the cause later. Differently: the DST checklist should have existed before that Monday, and I'd page on-call engineering sooner instead of carrying the first stretch solo.

Challenge — How would you implement this at Amazon?

Peak is sacred and the scheduler always dies at the worst hour — rank the clear by promise-at-risk, keep every intervention reversible and timestamped on one log, communicate on a fixed cadence with the same numbers, and leave a heartbeat behind so the next failure is detected in one interval, not by a VP at 4 AM.

Challenge — How did these results compare to your actual goals?

The goal was 95% fill; the event delivered 99%, with ~$8M protected against the ~$6M at risk sized at 4 AM — and the mechanisms (heartbeat, DST checklist, 3σ page rule, rehearsed playbook) made the next peak run boring.

PHASE DEEP-PROBE BANK — Q&A (12). Answer the phase asked, then stop.

Discovery — How did you decide without knowing the cause?

The queue was aging while inventory sat idle; waiting for a root cause would have cost promises by the hour. I ranked by promise-at-risk first and hunted the cause in parallel.

How did you size the exposure?

Backorder lines against the 95% fill-rate goal — about $6M at risk. That number went into the first status and every one after.

Plan & Design — Why promise-at-risk and not FIFO?

FIFO processes cheap orders while at-risk promises age. Promise-at-risk clears the orders whose miss costs most first — the ranking is the policy.

How did you handle the VP wanting the fastest clear?

One sentence on the difference between fastest and best, then the rule written in the log so nobody re-argued it at noon.

Build & Test — How did you rebalance without making it worse?

Checked with engineering that manual clears would not corrupt allocation, then moved load by node distance — near ~0.85 days versus far ~4.60 — logging each rebalance with a timestamp.

How did you find the root cause?

Through my own reason-coded alerts: the pattern pointed at the allocation scheduler, which had died silently at the daylight-saving change. Engineering fixed it once isolated.

Cutover — How did you know automated allocation was safely back?

Manual clears stopped, the scheduler processed a batch on its own, and fill trended to 99%. The log shows the handoff timestamp.

What was your "rollback"?

None available — the equivalent was the rehearsed rebalance playbook the post-mortem produced, so the next failure had a tested manual mode.

Hypercare — How did you run the post-mortem?

Blameless and from the log: timeline, contributing causes, mechanisms with owners — heartbeat, DST checklist, three-sigma page rule, rehearsed playbook, surge-scaling rebuilt.

How did you know the mechanisms worked?

The next peak ran boring: the heartbeat paged nothing because nothing died, the DST checklist ran before the change, and structural stockouts fell 20% over the engagement. The VP's appreciation email was nice; the boring peak was the evidence.

Cross-phase — What did you escalate and to whom?

Two-hourly status to internal and client leadership until 9 PM — same numbers each time.

This was years ago — anything more recent? · Do differently?

The UDC robotics throttle (S2) is the same muscle in 2024–25: a pre-built dial and a published gate — decide on a rule written before the crisis. Differently: the DST checklist should have existed before that Monday, and I'd page on-call engineering sooner instead of carrying the first stretch solo.

Where this lands at Amazon: peak is sacred and the scheduler always dies at the worst hour. The transferable mechanics: rank the clear by promise-at-risk, keep every intervention reversible and timestamped on one log, communicate on a fixed cadence with the same numbers — and leave behind a heartbeat so the next failure is detected in one interval, not by a VP at 4 AM.

BUSINESS IMPACT MODEL — KPI TREE, NUMBERS, THE MATH

```
Peak sales protected ~$8M vs ~$6M at risk (record pair)
+- Peak fill rate 95% goal -> 99% delivered (record)
|    +- backorder queue (stock) drained by promise-at-risk, not FIFO
|    +- node choice = the promise: near 0.85 d vs far 4.60 d (record)
|         load_n = queue_n / capacity_n -> re-point -> loads level (balancing loop)
|    +- scheduler liveness: heartbeat + DST checklist + 3-sigma page ->
|         detect in 1 interval, not by a VP at 4 AM
+- Gross margin on protected sales (= $8M x margin, modeled)
+- Structural stockouts -20% over the engagement (record) -> margin run-rate (modeled)
```

DRIVER	BASELINE	AFTER	$	BASIS
Peak fill rate	95% goal	99%	—	record
Peak sales protected	~$6M at risk	~$8M protected	one-time event	record
Gross margin protected	—	≈$3.2M	one-time	modeled ($8M × 40%; benchmark 35–45%)
Cancel / CS contacts avoided	≈40K × $6.5	0	≈$0.26M	modeled ($6M ÷ $150 AOV; CS $5–8)
Expedite premium avoided	—	near-node sourcing	≈$0.13M	modeled (benchmark $5–12/order)
Structural stockouts	—	−20%	≈$1.3M/yr	modeled margin run-rate, illustrative $400M retailer

If challenged. *"The $8M — protected how?"* — "It's the record pair: ~$6M at risk at 4 AM, ~$8M of peak sales through the event at 99% fill. The margin translation is my model at a benchmark 40%. I never count the $8M as savings — it's sales that didn't die."

THE MATH — STEP-BY-STEP DERIVATION

Table 1 — before: the problem in numbers.

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
1	4 AM state	`scheduler dead; backlog = 30,000 orders; ~$6M of sales at risk vs the 95% fill goal`	record risk pair; backlog illustrative	the bridge call
2	FIFO failure	`FIFO clears by arrival — cheap, distant-promise orders first while at-risk promises age`	record rule problem	value-blind clearing
3	Naive node loads	`DC1: 22,000 / 10,000 = 2.2 days; DC3: 2,000 / 6,000 = 0.33 days`	capacities 10K / 8K / 6K orders/day (illustrative)	worst node 2.2 days behind while another idles
4	No liveness detection	`a dead scheduler was found by a VP at 4 AM, not by a page`	record problem	detection by seniority

Table 2 — the fix: step by step, values substituted.

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
1	Promise-at-risk ranking	`risk = value × urgency(promise − now); clear descending`	on the bridge log in hour 1 (record rule)	at-risk promises first, not first-in
2	Node choice = the promise	`T = handling + transit: near 0.85 d vs far 4.60 d; 4.60 − 0.85 = 3.75 d`	record pair	−3.75 days per re-pointed order
3	Load levelling	`rebalance 30,000 → 12,500 / 10,000 / 7,500: 12,500/10,000 = 1.25; 10,000/8,000 = 1.25; 7,500/6,000 = 1.25 days`	load = queue / capacity (illustrative)	worst slip 2.2 → 1.25 days
4	Liveness page (post-mortem)	`page if x < μ − 3σ = 1,200 − 3 × 150 = 1,200 − 450 = 750/hr`	μ = 1,200 allocations/hr, σ = 150 (illustrative); a dead scheduler reads 0/hr	caught in one interval, not by a VP

Table 3 — after: the metric, before → after.

STEP	METRIC	BEFORE	AFTER	BASIS
1	Peak fill rate	95% goal	99% delivered	record
2	Peak sales	~$6M at risk	~$8M protected	record pair — one-time event
3	Worst-case promise slip	2.2 days	1.25 days	illustrative walk
4	Detection	a VP at 4 AM	one interval via the 3σ page	post-mortem design
5	Structural stockouts	—	−20% over the engagement	record

Table 4 — the dollars, every multiplication shown.

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
1	Margin protected	`$8,000,000 × 40% = $3,200,000`	benchmark margin 35–45% (modeled)	≈$3.2M one-time
2	Orders at risk	`$6,000,000 / $150 = 40,000 orders`	AOV benchmark	40,000 orders
3	Cancels / CS avoided	`40,000 × $6.50 = $260,000`	CS contact $5–8 (benchmark; modeled)	≈$0.26M

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
4	Expedite premium avoided	`40,000 × 40% = 16,000 re-pointed; 16,000 × $8 = $128,000`	re-pointed share and premium midpoint of the $5–12 benchmark (both illustrative)	≈$0.13M
5	Structural-stockout run-rate	`$400M × 4% = $16M lost sales; × 20% = $3.2M recovered; × 40% = $1,280,000`	illustrative retailer × lost-sales benchmark (2–6%) × −20% record × margin	≈$1.3M/yr modeled

SENSITIVITY — LOW / BASE / HIGH

INPUT	LOW	BASE	HIGH
Retail gross margin (benchmark)	35%	40%	45%
Margin protected on $8M (one-time)	$2.8M	$3.2M	$3.6M
CS contact $ (benchmark)	$5	$6.5	$8
CS / cancel avoided (one-time)	$0.2M	$0.26M	$0.32M
Lost-sales-to-stockout share (benchmark)	2%	4%	6%
Illustrative annual revenue	$300M	$400M	$500M
Stockout −20% margin run-rate	$0.4M	$1.3M	$2.7M
Total modeled $/yr	$0.4M	$1.3M	$2.7M

TOTAL IMPACT STATEMENT — SAY IT LIKE THIS

"About eight million of peak sales protected against six million at risk, and ninety-nine percent fill against a ninety-five goal; those three numbers are the record. Underneath them my model says roughly three point two million of gross margin at a forty percent retail margin, about a quarter million of cancellations and service contacts we never handled, and a bit over a hundred thousand of expedite we never paid because the near node was the promise. Then the twenty percent cut in structural stockouts, which is record, models to about a million three a year of margin on an illustrative four-hundred-million retailer. So: the eight, the six, the ninety-nine and the twenty percent are measured; the margin, the contact cost and the run-rate are mine."

If challenged — the full answers.

How can you protect eight million when only six was at risk?

Because the six was the backlog on the phone at four in the morning and the eight is what shipped on promise through the whole event, including the four points above goal. Reconciling them implies about fifty million of event order value, which is my derivation, not a record number; I would confirm it against the retailer's Cyber Monday order total.

The stockout run-rate is the biggest number and the softest.

Agreed, it is the hardest one. It rests on a four percent lost-sales-to-stockout benchmark and an illustrative four-hundred-million revenue base; at two to six percent it runs from four hundred thousand to two point seven million. I would measure it from the retailer's own out-of-stock lines times unit margin before and after the playbook went in.

Forty percent margin and a hundred-fifty AOV are assumptions.

They are, and I labeled them benchmarks. Thirty-five to forty-five percent moves the protected margin from two point eight to three point six million; the AOV only moves the contact- cost line. I would swap both for the retailer's finance numbers, which is exactly what I would ask for on day one of the post- mortem.

S14 — Manhattan: the $2.5M OMS↔WMS discrepancy my manager wanted to fix quietly — transparency as the reconciliation control

Amazon LPs: Earn Trust (primary — the flagship integrity story) · Have Backbone (backup — refusing the quiet fix) · Ownership · Bias for Action (third — two days to the review). **Crosswalk:** not in the loop packet — safe anywhere. *Pair with S12 for a recent upward disagreement; S4 for the one where I was wrong. The $10M initiative is never counted as savings.*

Honesty label. The census, the risk argument, the customer pack and the war room were mine; the consultants traced the handshake, wrote and verified the per-line patch, and built the permanent fix.

60-SECOND PITCH

"**Situation.** Two days before an executive review at a Manhattan customer, OMS and WMS did not reconcile — about two hundred orders, $2.5 million — and a $10 million initiative rode on that review; my manager and the room favored a quiet back-end correction. **Task.** I was the consultant who would run whichever path was chosen. **Action.** Discovery: I built the discrepancy census by cause on a sample of the two hundred before proposing any path, and named the fear — losing the initiative — as legitimate. Plan & Design: I disagreed on risk, not principle — a cross-table script written in two days could not be verified, and an IT audit would turn a data issue into a reputation issue — and offered to run the war room personally so the cost of my position landed on me. Build & Test: customer pack with root cause, plan and timeline; a per-line, reversible patch method. Cutover: the customer prioritized their own orders; we patched that set with before/after evidence per order and briefed them before the review. Hypercare: a clean root-cause fix to the handshake and a standing reason-coded reconciliation report. Lesson: silent corrections are loans against trust at a terrible interest rate. **Result.** Primary metric: OMS↔WMS discrepancy — **~200 orders / $2.5M → zero** on the customer's prioritized set before the review, then zero through the clean fix. $ impact: the $10M initiative won (record, not counted as savings)."

SITUATION / TASK — WHERE IT OCCURRED, WHEN IT HAPPENED, WHY IT IS IMPORTANT

S — Two days before an executive review, OMS and WMS didn't reconcile — ~200 orders / $2.5M — and a $10M initiative rode on the review. My manager and the room favored a quiet back-end correction.

T — I was the consultant who would run whichever path was chosen; I owned the recommendation and then the war room.

The goal: the discrepancy to zero and a review the relationship could survive, without a silent correction. The challenges: a two-day clock, and a legitimate fear — losing the initiative — that had to be named, not dismissed. The risks of the quiet path: a rushed cross-table script corrupting data (the record's own illustration: expected ≈$1.5M of corruption), a near-certain audit finding turning a data issue into a reputation issue on a $10M relationship, and the customer surprised at the review.

The metrics on the table, with their baselines: the OMS↔WMS discrepancy — ~200 orders / $2.5M; target zero with before/after evidence per order. New discrepancies — unmonitored; target zero on a standing daily report.

ACTION — WHAT I PERSONALLY OWNED, HOW I DID IT, WHO ELSE WAS INVOLVED

A — Five gated phases — two days plus the war room (durations illustrative):

1. **Discovery (day 1, morning)** — the discrepancy census by cause (which handshake step, which order states); the exposure statement (review in two days; $10M initiative); both options named honestly — quiet script vs transparency. *Exit bar:* the room agrees the facts before the path; the fear of losing the initiative named as legitimate.
2. **Plan & Design (day 1, afternoon)** — the options memo on risk, not principle: a two-day cross-table script is error-prone; an IT audit turns a data issue into a reputation issue; transparency = root cause + action plan + I run the war room. *Exit bar:* manager accepts, hesitantly; my labor attached to my position.

3. **Build & Test (day 2)** — the customer communication pack (root cause, plan, timeline — no apology-first); the war-room plan (bridge, log, cadence); the verification method for any patch (per line, reversible, before/after recorded). *Exit bar:* customer briefed before the review; war room staffed.
4. **Cutover (review + patch)** — took the customer's prioritized order list; ran the bulk patch per line with verification; scheduled the clean root-cause fix. *Exit bar:* patch verified; discrepancy to zero on prioritized orders; review held with the facts on the table.
5. **Hypercare / Close** — the permanent OMS–WMS handshake fix; the reconciliation report as a standing control read before every executive review. *Exit bar:* $10M initiative won; handshake reconciles; new discrepancies = 0 on the daily report.

The census, the risk argument, the customer pack and the war room were mine; the consultants traced the handshake, wrote and verified the per-line patch, and built the permanent fix. The contribution was making the argument about risk, not virtue — a cross-table script written in two days cannot be verified, and an IT audit turns a data issue into a reputation issue — and attaching my labor to my position by offering to run the war room personally, so the cost landed on me.

The obstacle was my manager favoring the quiet fix — told privately first, with the census; he accepted, hesitantly, once the cost of the honest path landed on me; the two-day clock was handled by building the census by cause on a sample of the ~200 orders before proposing any path. The customer set the priority — their own prioritized order list got patched first, with before/after evidence per order — and buy-in ran facts before path: the room agreed the census before the options, and the customer was briefed before the review, not in it. The decision challenged was the quiet back-end correction itself — on risk, with the expected-loss comparison; had he said no, I would have documented the risk assessment in writing and asked for the decision recorded with its owner — I would not have run the silent script without that record.

RESULTS — HOW SUCCESS WAS MEASURED: $ · # · % · TIME · CUSTOMER · QUALITY

R — Primary metric: discrepancy **~200 orders / $2.5M → 0**; the $10M initiative won (record, not counted as savings); the standing reconciliation control outlived the war room.

- **$** — ~200 orders / $2.5M discrepancy → zero (record); the $10M initiative won — a record fact about the relationship, never counted as savings; ≈$0.1M/yr modeled from the standing reconciliation (range $55–160K).
- **#** — ~200 orders patched per line, reversible, with before/after evidence per order, in the customer's priority order (record).
- **Time** — Two days to the executive review; the customer briefed before the review, not in it; the clean root-cause fix scheduled after (record).
- **Customer** — The customer prioritized their own orders, asked for the bulk patch on that list, and gave time for a clean fix — the review held with the facts already on the table (record).
- **Quality** — Per-line reversibility — no bulk script across related tables; the standing reason-coded OMS↔WMS reconciliation read before every executive review; new discrepancies = 0 on the daily report (record).

Against the goal stated above: the ~200-order / $2.5M discrepancy to zero on the customer's prioritized set before the review; the review held with the facts already on the table; new discrepancies at zero on the daily report — and the $10M initiative was won, a record fact about the relationship, never counted as savings. The honest path's costs were priced onto me deliberately: my time in a war room, the risk of being wrong in front of a customer, and a clean root-cause fix scheduled after the review rather than a rushed everything-now script.

PROBING — QUESTIONS, PHASE DEEP-DIVES, AND THE MATH

Challenge — Why did you choose this story to illustrate Earn Trust?

Because trust was built the expensive way: the customer told first, with root cause and a plan; the cost of my position priced onto me — I ran the war room; and a standing control left behind so the courage is rarely needed again. It is my flagship integrity story.

Challenge — What other stories demonstrate Earn Trust?

S1 — one signed “available” — is the mapped backup, with S3 (Finance and Ops on one number) and S5 (the council) carrying the same LP in my current role.

Challenge — Could you come up with an example that is more recent?

Manhattan years — older. For recent upward disagreement, S12 (flipping my manager's roadmap) and S4 (the one where I was wrong) are the records; for current-role Earn Trust, S1 and S5.

Probing — Why did you choose to focus on these results? What other results were important?

The discrepancy to zero with before/after evidence per order is the operational record; the $10M initiative won is the relationship record — and I never count it as savings. The control that outlived the war room is the result that keeps paying.

Probing — You mentioned dollars — what percentage change is that year over year?

Not YoY — the $2.5M is order value made consistent in the open, not money earned; the $10M is a won initiative, never in a savings column; the only recurring dollar is ≈$0.1M/yr modeled from the standing reconciliation, labeled.

Probing — I'm concerned you're counting the $10M initiative as impact — tell me more.

I am not counting it — the initiative was won in the review the facts survived; I claim the control and the posture, not the contract value. The ≈$1.5M expected corruption from the silent script is avoided risk, not a dollar earned, and I label both.

Challenge — What were the lessons learned? What would you have done differently?

Silent corrections are loans against trust at a terrible interest rate. The transferable control is not the courage — it is the reason-coded reconciliation read before every review, so the next discrepancy surfaces early.

Challenge — How would you implement this at Amazon?

The purest Earn Trust pattern — tell the customer first, price the honest path's cost onto yourself, and leave a standing control so the courage is rarely needed again.

Challenge — How did these results compare to your actual goals?

The need was a review the relationship could survive — delivered: discrepancy to zero on the customer's prioritized set before the review, the review held with facts on the table, the $10M initiative won, and new discrepancies at zero on the daily report.

PHASE DEEP-PROBE BANK — Q&A (12). Answer the phase asked, then stop.

Discovery — How did you establish the facts under a two-day clock?

Reconciliation report by cause on a sample of the ~200 orders — which handshake step, which order states — before proposing any path. Facts first made the options argument about risk, not virtue.

Who did you tell first?

My manager, privately, with the census. The room heard the options after he had.

Plan & Design — What was in your risk register?

A silent fix found by IT audit, a rushed cross-table script corrupting data, and the customer being surprised at the review. Mitigations: transparency, per-line reversible patch, brief them first.

How did you disagree with your manager?

On risk: an unreliable script plus an audit exposure versus a customer told the truth with a plan. Then I offered to run the war room so the cost of my position landed on me. He accepted, hesitantly.

What if he had said no?

I would have documented the risk assessment in writing and asked for the decision to be recorded with its owner — disagree and commit, with the tripwire (any audit query on those orders) named. I would not have run the silent script without that record.

Build & Test — What was your definition of done for the patch?

Every line verified against both OMS and WMS, reversible, executed in the customer's priority order. No bulk script across related tables without per-line verification.

How did you prepare the customer conversation?

Root cause, action plan, timeline — no apology-first. The pack went to their IT and finance before the executive review, not in it.

Cutover — What did the customer do?

Prioritized their own orders, asked for a bulk patch on that list, and gave us time for a clean root-cause fix. The review happened with the facts already on the table.

What was your rollback?

Per-line reversibility on the patch. A script across every related table would have had none — that was the core of my risk argument.

Hypercare — What control outlived the war room?

The OMS↔WMS reconciliation report as a standing check, read before every executive review rather than two days before one. The handshake fix removed the cause; the report made sure the next one surfaced early.

Cross-phase — What did you refuse and what did it cost?

A silent correction. Cost: my time in a war room and the risk of being wrong in front of a customer. Benefit: a relationship that survived an audit and won the initiative.

Anything more recent where you disagreed upward?

S12 — flipping my manager's roadmap — and S4, where I was the one who was wrong. Same posture: disagree on evidence, carry the consequence.

Where this lands at Amazon: the purest Earn Trust answer in the set — trust built by telling the customer first, pricing the honest path's cost onto myself, and leaving a standing control so the courage is rarely needed again. The transferable control is not the courage; it is the reason-coded reconciliation read before every review.

BUSINESS IMPACT MODEL — KPI TREE, NUMBERS, THE MATH

```
Customer trust that survives an audit -> $10M initiative won (record)
+- OMS<->WMS discrepancy ~200 orders / $2.5M -> 0 (record)
|    +- recovery: $2.5M made consistent in the open; per-line reversible patch,
|         before/after evidence per order; customer-prioritized set first
|    +- prevention: standing reason-coded reconciliation (daily)
|         exposure at detection: $2.5M/60 d -> ~1 day (loop keeps the stock near 0)
+- Risk not taken: silent multi-table script
|    +- P(script error) x blast radius + P(audit) x reputation = avoided expected loss
+- Trust loop (+): tell first -> customer sets priority -> gives time ->
     clean fix -> initiative won -> more latitude next time
```

DRIVER	BASELINE	AFTER	$	BASIS
OMS↔WMS discrepancy	~200 orders / $2.5M	0	—	record
$10M initiative	at risk	won	—	record — **never counted as savings**
Standing reconciliation (prevention)	war rooms	daily report	≈$0.1M/yr	modeled (manual fixes + stuck cash + war rooms avoided; range $55–160K)
Silent script not run	—	—	≈$1.5M expected corruption + audit finding	modeled — avoided risk, not a saving

If challenged. *"You're claiming the $10M?"* — "No — the initiative was won in the review the facts survived; I claim the control and the posture, not the contract value. The only recurring dollar I model is the ≈$0.1M/yr the standing reconciliation saves, and it's labeled."

THE MATH — STEP-BY-STEP DERIVATION

Table 1 — before: the problem in numbers.

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
1	The discrepancy	`~200 orders / $2.5M; $2,500,000 / 200 = $12,500 average per order`	record pair; average derived	~200 orders, ≈$12.5K each
2	The stakes	`$10M initiative riding on the executive review two days out`	record	two days
3	The tempting path	`a quiet multi-table script written in under two days`	P(script error) high; P(audit discovery) ≈ 1 over an IT-audit horizon (record argument)	fast, silent, and wrong

Table 2 — the fix: step by step, values substituted.

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
1	Census by cause	`Δ grouped by handshake step × order state, on a sample of the ~200`	built before proposing any path (record)	facts before path
2	Quiet path priced	`blast = 2 × $2.5M = $5.0M; E[loss] = 0.3 × $5.0M = $1.5M + a near-certain audit finding`	P(script error) = 0.3; blast radius 2× the affected orders (illustrative)	≈$1.5M expected corruption
3	Honest path priced	`a few days' delay on a subset of $2.5M, customer-controlled`	record argument	bounded, one-time
4	Patch walk	`customer-prioritized set first; per-line, reversible; evidence rows = set size`	before/after evidence per order (record)	0 on the set before the review
5	Standing control	`match rate = orders with OMS status == map(WMS status) / orders in flight; Σ 4 reason classes = all of Δ`	reason codes: status mapping · timing · failed message · manual edit (record design)	a row with no class is an investigation, not a bucket

Table 3 — after: the metric, before → after.

STEP	METRIC	BEFORE	AFTER	BASIS
1	OMS↔WMS discrepancy	~200 orders / $2.5M	0, with evidence per order	record
2	New discrepancies	war rooms	0 daily on the standing report	record design
3	Exposure at detection	~$2.5M built over ~60 days	~1 day's worth	record design
4	$10M initiative	at risk	won in the review the facts survived	record — never counted as savings

Table 4 — the dollars, every multiplication shown.

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
1	Expected corruption avoided	`2 × $2,500,000 = $5,000,000 blast; × 0.3 = $1,500,000`	illustrative probabilities	≈$1.5M avoided risk — not a saving
2	Standing reconciliation value	`manual fixes + stuck cash + war rooms ≈ $0.1M/yr`	modeled; range $55–160K	≈$0.1M/yr
3	$10M initiative	`won — and never counted as savings`	record fact about the relationship	excluded by rule

SENSITIVITY — LOW / BASE / HIGH

INPUT	LOW	BASE	HIGH
Discrepancies arising per month (illustrative)	50	100	150
Cost per manual fix (benchmark)	$20	$30	$40
Manual-fix labor avoided	$10K	$29K	$58K
Cost of capital (benchmark)	8%	10%	12%
Order-to-cash carry avoided	$24K	$30K	$36K
Repeat war rooms avoided per year (illustrative)	1	2	3
War-room cost avoided	$22K	$43K	$65K
Total modeled $/yr	≈$55K	≈$0.10M	≈$0.16M

TOTAL IMPACT STATEMENT — SAY IT LIKE THIS

"Two and a half million of order value made consistent in the open, with before-and-after evidence on every order, and the ten-million initiative won; both of those are the record, and neither is a saving I claim. What the standing reconciliation is worth going forward is small and honest: about a hundred thousand a year in my model, roughly thirty in manual fixes we no longer do, thirty in cash that is not stuck behind mismatched orders, and forty in war rooms we do not run. The bigger number is the one we did not lose: the silent script had an expected one and a half million of corruption plus an audit finding on a ten-million relationship, and that is risk math, not money in the bank. So the two point five and the ten are measured; the hundred thousand and the one point five are modeled."

If challenged — the full answers.

A hundred thousand a year is small for a story you lead with.

Yes, and I would rather say small and true. The story is about the control and the choice, not the run-rate. The hardest input is how many discrepancies arise per month; I assumed the two hundred built up over about sixty days. I would firm it up by reading the daily match report's reason-code counts for a quarter and pricing exactly what surfaced.

You are counting the ten million as impact.

I am not. The ten million was won at the review; it is a record fact about the relationship, and I never put it in a savings column. The one point five million is the record's own illustration of the expected loss from the silent script, so it is avoided risk, not a dollar earned. To firm it up I would want the actual error rate of comparable rushed scripts from that practice.

Thirty dollars a fix and ten percent cost of capital are assumptions.

Both are benchmarks and I ranged them: twenty to forty a fix and eight to twelve percent move the total from roughly fifty-five to a hundred sixty thousand. The exception-handling minutes I would measure from the war-room log itself, which timestamped every per-line patch; the cost of capital I would take from the customer's finance team.

4. Numbers and never-say — one canonical value per metric

Carried from the GFiber v8 ledger, kept as-is per the transform instruction. A record dollar is said flat; a modeled dollar carries the word "modeled" in the same breath and its basis lives in the story's Business impact model. Where §1.3 marks a conflict with the loop packet, this table is this document's canon.

4.1 One canonical value per metric

METRIC	VALUE TO SAY	DO NOT SAY	MODELED IMPACT (BASIS IN THE STORY)
Sephora ISM spec (S1)	FSD v1.1 (3/5/25) → v1.2 (5/16/25) → signed 6/5/25 by Supply Chain Strategy, Inventory Management, FSC Inventory Accounting	—	S1: ≈$2.5M/yr = $1M MCS support retired (user anchor) + ≈$1.5M modeled (oversell ≈$0.9M, recon labor ≈$0.33M, change premium ≈$0.3M); range $1.7–5.2M; upside not in headline: shrink ≈$0.3M, time-to-market ≈$0.2M
UDC (S2, S5)	Zero customer-facing incidents; +15% processing efficiency; 836-row issues log 85.4% closed; robotics gate 98%, week 2 at 94%, ~30% throttle one week; a dozen core systems, 20+ touched; 60+ stakeholders (~80 names across ~15 functions); council 5/15 → sign-off 7/31 → FSD 8/14; rollback triggers sync <99% for 4 h / p95 > 2× for 1 h; cutover 343 tasks / ~815 hrs	"12 systems" alone or "20 systems" alone; a single stakeholder count	S2: ≈$1.6M/yr modeled + ≈$2.2M one-time √n release · S5: ≈$0.5M one-time modeled · S3: ≈$0.6M/yr modeled · S4: ≈$0.5M/yr modeled, oversell +$0
RP → JDA (S18)	8 months; ~$1.8M, ~3% under (~$54K); mocks 7% → 1.8% → 0.3% vs 0.5% tolerance; $20M pipeline (~$2.5M/month); 72-hour freeze; legacy read-only 30 days; hypercare exit week 3 (<2 blockers/day); RF ~100 of ~400 devices; 1.5-week training slip	"400 devices failed"	S18: record stays headline; modeled ≈$0.1M less inventory in dispute (0.3% vs 0.5% on illustrative $50M), ≈$45K fallback cost; ~$875K critical-path slip avoided is record math, not banked
Supply/demand view (S6)	Forecast accuracy 70% → 85% on scoped class-A SKUs; $2M safety stock released; shortage alert live in 3 months; planner effort −50%; eight domains	Pasting "3 months" onto the six-site ingest	$2M is record (one-time); modeled recurring ≈$0.8M/yr (carry ≈$0.2M, expedites ≈$0.3M, planner capacity ≈$0.3M; range $0.5–1.2M)
Six-site CM ingest (S7)	Six sites, three waves, 5 months; hardest site 6 weeks vs 3-week average; zero auto-loaded revision overlaps	"6 NA sites"; "500+ product families"; 60→95% inside this story	≈$0.3M/yr modeled (recon labor, phantom expedites, overlap rework; range $0.2–0.7M) — $1.5M/580K stays with S8
Lead-time program (S8)	200 lanes, 99% lead-time accuracy, 580K units / ~$1.5M released	Adding it to the $2M into one number	$1.5M is record (one-time); modeled recurring ≈$0.6M/yr (carry ≈$0.3M, expedites ≈$0.3M; range $0.3–1.4M)

METRIC	VALUE TO SAY	DO NOT SAY	MODELED IMPACT (BASIS IN THE STORY)
EDI (S9)	1,000 routes asked → 597 first = 80% of spend; completeness 60% → 95%; adoption 8×; 100% visibility on high-value lanes; freight −10% ≈ $2M in 6 months; ≥95% valid-ASN cohort gate; ~200 next wave	Cohort sizes or the number of cohorts	$2M is record; modeled decomposition compliance ≈$0.8M / mode ≈$0.6M / expedite-detention ≈$0.6M (they SUM to the record); modeled add-on ≈$0.4M/yr document labor (range $0.2–0.9M)
Austria routing guide (S10)	True DQ ~55% (reported 40%) → ~99% in 3 months; manual 8–10 → 1–2 hrs/wk; $500K/yr (10% of ~$5M); 25 EU sites / 400+ lanes ≈ $8M; ~8% of lanes need Incoterm in the key	"$1M"; "99% in two months"; 1–2 hrs as the before	$500K → ~$8M is record; roll-up shown (~$320K/site); modeled add-on ≈$0.5M/yr analyst capacity (range $0.2–0.7M)
Carrier bid (S17)	6 weeks → ~3 (~50%); pilot 3 weeks early; <5% mismatch gate; 10-day stale banner; no $ figure	"$500K/yr"; "−25% planning effort"; any award dollar	≈$0.19M/yr modeled analyst hours returned (range $35–420K) — never award savings
Manhattan (S13)	3 DCs + 100 stores; ~$6M at risk → ~$8M protected; 99% fill vs 95% goal; stockouts −20%	"$2MM baseline"	$8M / $6M / 99% are record; modeled ≈$3.2M margin (40% GM), ≈$0.26M CS/cancel, ≈$0.13M expedite (one-time); stockouts −20% ≈ $1.3M/yr margin (range $0.4–2.7M)
Manhattan discrepancy (S14)	~200 orders / $2.5M; $10M initiative won	The $10M counted as savings	Modeled prevention run-rate ≈$0.1M/yr (range $55–160K); the silent script's expected ≈$1.5M corruption is avoided risk, not a saving
Flexport WBR (S11)	Automation 40% → 90%; CTS $0.66 → $0.03 (−95.5%); accuracy 60% → 90%; rollout unlocked 2 months early; 1.5–2 → ~4 hrs/wk by hand; 99.4% week discarded; customers 7 → 68 (9.7×); 170 MAU target	"$1.00 → $0.03"; "70% baseline"; "10×"	CTS record; modeled direct ≈$50K/yr at 32K shipments (range $30–80K) + ≈$0.25M one-time pull-forward — the value is scale without headcount; ~$1.5M pre-prod is a separate canonical
Pre-prod measurement (S11)	300 containers, ±1.6% at 95%; accuracy 85% → ~98%; rollout 1.5 months vs 6; ~$1.5M saved; 500 containers / 20 min / 99.6%	Merging the $1.5M with CTS	—
Roadmap flip (S12)	18 months → value in 6; 9 vs 6 months re-estimates; −25% vs −50% CTS; 66% less effort; ~$5M cost avoidance (say once); 339.5 min/shipment; 2,200 interactions; 48 of 53 migrated	"$5M annual cost reduction" in the same breath as the avoidance	~$5M avoidance is record (said once); modeled ≈$2.9M eng avoided + ≈$1.5M pulled forward (≈$4.4M, within ~12%); run-rate ≈$1.5M/yr (range $1.1–2.3M)
Snowflake (S16)	4 systems → 1; >$100K/yr priced; parity within 1%; 2 weeks lost to the security review; accuracy 60% → 90%; 4% ticket-sync drop caught	The $100K as claimed savings	>$100K/yr is a priced baseline (record of the pricing); modeled ≈$90K/yr inside it (hours ≈$30K, decision risk ≈$70K → ≈$50K retired, bus factor ≈$15K)
Agile conversion (S15)	20-person team; 6 weeks; critical defects −50%	A direct-report headcount	≈$0.13M/yr modeled (escaped criticals ≈$90K, late-land rework ≈$40K; range $80–270K); no headcount

METRIC	VALUE TO SAY	DO NOT SAY	MODELED IMPACT (BASIS IN THE STORY)
SAP tenure	S/4HANA integration program lead 3+ years (Apr 2023 – present); client-SAP data integration at Elementum 2017–19; 15 years full-lifecycle SCM systems	"10 years of SAP configuration"; "SAP functional owner"	—
Quant toolkit	"MIT Supply Chain MicroMasters + Supply Chain Network Design certificate — acquired so I could design what I'd been implementing"	"I build supply chain models for a living"; any story where modeling is the day job	The modeling formulas across §2–3 are this toolkit, labeled in the cover voice block and the §3 preface, and per story where used

4.2 Never-say list (one glance before every conversation)

- Never "I built the model" as a day job — "I owned the requirement/definition/rule; the data team or engineers built; the toolkit is from MIT."
- Never "I configured S/4," "I deploy SAP," or "I wrote ABAP" — "I led the integration program and owned the data contract and definitions; the SAP team / SI configured and coded to the spec I signed." The only platform I configured myself is Manhattan DOM.
- Never claim Salesforce, Coupa, Teamcenter, VMI or serial-profile implementation experience — "adjacent, bridge honestly," and name the pattern. The §3 designs carry their honest lines for exactly this reason.
- Never: "6 NA sites" · "500+ product families" · 60→95% inside the CM-ingest story · a $ figure for the carrier bid · "$1M routing guide" · "$1.00 → $0.03" · "10×" · "400 devices failed" · cohort sizes · $2M + $1.5M added together · $5M "avoidance" and "annual reduction" as two results.
- Never present a modeled number as record: the SNB drift dollars (S3), the bid cycle-time value (S17), the S1 oversell dollars, the S2 labor dollars are modeled — say "modeled from these inputs" and give the basis. Numbers with no basis stay uninvented: direct-report headcounts, carrier-bid award savings, realized dollars on the $10M initiative. Say "I don't have a canonical figure and won't invent one; what I can defend is…"
- Never "we" for an action that was mine; name the team's part once, explicitly.
- Never volunteer only successes — S4 (the rule I got wrong), S18 (the RF miss), S16 (the security-review miss) and the UAT-crash bench story are the failure stories; Amazon explicitly probes for lessons from mistakes.
- **Amazon-specific:** never tell the same story to two interviewers — the §1.1 map exists so each LP has a primary, backup and third. Never carry a §1.3-conflicted number into the room before resolving which document is right. Every modeled dollar gets its basis label out loud; the loop packet's "my model, Finance's rates" discipline applies to every figure here.

4.3 Bench stories (not core headings — pull only if asked for "another example")

BENCH STORY	WHERE IT LIVES
Status-mapping root cause, ~$2.8M (Flexport)	Referenced in E4's probes — the status-map defect class
UAT crashed under load; numeric entrance criteria ever since	The second failure story if S4/S16/S18 are spent (loop packet S16)
110%-capacity disagree-and-commit with a tripwire	Backbone backup — disagree, commit, name the tripwire
Two programs, one team, war room	Prioritization under conflict

BENCH STORY	WHERE IT LIVES
"Too tactical" → fever chart, 40 integration points (NA scope, 1 month)	Never "25 integration points"
LLM triage agent — eval set first	Recent AI judgment; unit economics before scale (loop packet S13)
Controlled release bench: 30% of lanes ≈ 90% volume; lead times 7–30 days; 98% ASN first 3 days; APAC ≈ $2M	Never "LATAM"; never "20–30 days"

4.4 Addendum — the Walmart-client Elementum engagements (S19–S21): canonical numbers

Added with stories S19–S21; **no row above this heading changed.** Source: the Walmart Marketplace pre-purchase delivery-experience engagement record (the R-2596147 war book and the Tech Ops WBR deep-dive). Basis vocabulary for this addendum: *record-from-docs* — stated in the engagement documents, say it with its own caveat · *illustrative* — from the engagement's illustrative model (twelve lane cohorts, ~1M orders/week, σ 0.22–1.60 days), always said with "the model" in the same breath · *modeled* — computed here from stated inputs · *benchmark* — a published external rate, named. **No number in S19–S21 is a production record; never promote one.**

METRIC	VALUE TO SAY	DO NOT SAY	BASIS
Promise buffer policy (S19)	Uniform 1.10-day buffer: accuracy 95.8%, average promised day 3.09, fast share (≤2-day) 7.6%, weekly profit $5.69M → σ-proportional b = 1.645σ: 95.0% floor held exactly, 2.75 days, 39.6% fast share (+32 pts), $5.94M (+$255K/wk ≈ $13.3M/yr); profit optimum (policy D) 97.8% accuracy / $6.07M (≈ +$20M/yr vs uniform) — handed to Finance, never claimed; breach cost $12 vs $6.50 contribution margin; Q4 regime σ ×~1.45, frozen summer dial prints 91.9% — below the floor	Any production dollar; "the buffer model saved $13M" without "the model" in the same breath	All from the engagement's illustrative model (the four-policy table is record-from-docs *of the analysis*); $12 / $6.50 are the model's stated assumptions; annualizations modeled (×52)
Pre-purchase WBR (S20)	Decomposition O1 ≈ I3 × I4 × Φ(b/σ); sample week 0.97 × 0.985 × 0.996 ≈ 95.2% (illustrative); Promise Yield = fast share × accuracy = 43.6% × 95.2% ≈ 41.5% baseline, 4-quarter target 50%; the aggregate trap: carrier 97% nationally, 82% in the Southeast after a hub consolidation; red/green at ±5% relative WoW, hard reds on policy floors (OTD < 90%, late shipment > 5%, cancellation > 2%); Monday data freeze, Tuesday review; Andon design: 7-day EWMA/CUSUM per seller-lane cell, tripwire sustained 2× baseline with n ≥ 30, act < 24 h, false-trip < 5%; ≈$0.58M per caught episode (modeled)	Claiming a production output metric moved; an annual Andon dollar (episode frequency is the client's data)	Formulas, floors, cadence, Andon parameters, Southeast example, Promise Yield sample: record-from-docs (the WBR sample data is the docs' own illustrative set); episode dollars modeled from stated inputs

METRIC	VALUE TO SAY	DO NOT SAY	BASIS
SSS adoption (S21)	SSS at ~38–42% of seller-fulfilled GMV (stalled); sellers declare 1-day handling and hit it ~85% of the time — adverse selection, not bad luck; the fix: an all-or-nothing insurance bundle (transit automation + observed handling + participating label → scorecard protection) plus a personalized GMV-delta line in Seller Center, rolled out in waves behind published graduation criteria, measured on activations/week *and* 30-day retention; ≈$3.7M/yr per +10 pts of enrolled GMV (modeled: 100K orders/wk × (15% − 3%) handling-miss reduction × 50% breach-through × $12 × 52)	An adoption "after" number (the engagement delivered the contract design and wave one, not the curve); any record dollar	Baselines and the contract/wave mechanism: record-from-docs; the 3% post-fix miss rate and 50% breach-through: illustrative; the $12: the model's stated assumption; the $3.7M/yr: modeled

Voice rule for S19–S21: Elementum engagement lead for a marketplace client — I owned the decomposition, the policy instruments, the gates and the governance design; the client's engineering and data-science teams built the promise engine, the models and the seller tools. Never claim configuring or deploying any platform — the only personally configured platform in this packet remains Manhattan DOM.

> **Final rule of the packet.** This document and the loop packet are one system: this one carries the depth (gated STARs, phase probes, impact models, the design round); the loop packet carries the loop mechanics (interviewer tracker, the 128-question AIQB bank, writing defense, video execution). Rehearse from both; resolve the §1.3 conflicts before the loop; and in the room, say less than you know — the pitch ends on the primary metric, and the probes are how the interviewer fills their worksheet in your favor.